

Monocular Surface Reconstruction

Single-image geometry, metric height and semantic analysis

Technical Documentation — Status as of 28 September 2026

Version 0.1.1 • Active research prototype



TEAM SEIFUKU

Independent research project

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How to use this handbook

This is a dated technical snapshot of an active research prototype. Current chapters explain the implemented system, scientific boundaries and available evidence. The historical section preserves protocols, outcomes and decisions that led to that state. A failed experiment is retained as evidence; a planned feature is not presented as complete.

Read Overview and User guide for the system; Scientific contracts and Validation for interpretation; Results, Experiments and Decisions for what was learned; Setup and Release for reproduction. The source repository is the editable companion to this PDF.

Historical records use normalised public names and paths. Exact old source hashes describe their original archive, not renamed files. Full private data/run archives are not bundled. See the source inventory and evidence directory for the included records and verification scope.

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01 Project overview and current scope

SOURCE docs/OVERVIEW.md

Technical documentation — status as of 28 September 2026. Version 0.1.0 is the initial public research package; the development history predates this release.

The problem

Overhead imagery contains visible clues about roofs, vegetation, shadows and terrain, but a single view does not uniquely determine physical height. Reliable surface measurements normally require additional observations or reference data. This project explores a lower-input alternative: estimate an above-ground surface from one optical image, preserve its scientific limitations, and give the user a connected environment for inspection, calibration, comparison and export.

The goal is an inspectable research pipeline. A visually convincing mesh does not establish metric accuracy, and a georeferenced image alone does not supply the ground elevation needed for an absolute surface model. The system exposes those distinctions in its outputs and documentation.

Intended users and workflow

The supported audience is researchers, students and technical reviewers working with overhead imagery and geospatial surface estimates. An analyst can load a saved demonstration, reconstruct a supported RGB image, add a terrain datum or ground-control points where available, inspect local values and compare against a supplied reference. Planning, forestry and infrastructure analysis are potential downstream uses; no operational benefit in those applications has been established.

The complete workflow is image acquisition, input validation, relative-depth extraction, learned height estimation, optional calibration, interactive exploration, and export with provenance. Semantic identification is a separate RGB-only branch. It does not silently rewrite the height field to fit class labels.

What this project contributes

The project integrates an overhead-image height decoder and protected surface refinement with a pretrained relative-depth prior. It adds independent validity masks, geospatial alignment and calibration, scoped evaluation, strict candidate comparison, and a browser workspace that distinguishes source measurements from display geometry. The dated experiments document unsuccessful designs as well as working components, including comparisons against matched controls.

Depth Anything V2, ConvNeXt V2, SegFormer, PyTorch, rasterio and Three.js are upstream work and are credited as such. This project does not claim to have invented monocular depth estimation, semantic segmentation, or 3D rendering. Its research contribution is the implemented integration, training/evaluation work, failure analysis and traceable handling of scientific outputs.

Current delivery

Area	Status in this snapshot
Height reconstruction	Implemented; protected model retained because later candidates did not satisfy all acceptance gates.
Six-class identification	Automatically available for supported inputs when the classifier is installed; regional weaknesses remain disclosed.
Geospatial products	Grid preservation, DEM composition, control-point calibration and scoped reference comparison implemented.
3D exploration	Orbit, fly, walk, map, class selection, source-image reference, metric grid and surface-profile tools implemented.
Roof geometry research	Experimental modules preserved; fitted/relief roof changes are inactive in the current viewer.
Illustrative tree trunks	Reverted from the current interface; not a delivered accuracy improvement.
External metric-height validation	A preregistered Bay of Plenty package is prepared but unconsumed.
Scientific acceptance	Internal numerical and engineering evidence exists; independent expert acceptance and broad geographic reliability remain open.

Reading the record

Current chapters take precedence when explaining what this release does. Dated historical records preserve the state at the time of each experiment. The September 13 head-only classifier work is different from the later independent RGB V1/V3 classification pipeline. Likewise, later user-interface integration is not a retroactive pass of an earlier model-safety gate.

Status labels mean implemented, measured in a stated scope, exploratory, rejected, or planned. They are not interchangeable. The release report records the actual package verification; the results chapter reports experiments without treating old test counts as a current acceptance certificate.

02 Team Seifuku

SOURCE docs/TEAM.md



Team Seifuku logo

Monocular Surface Reconstruction is an independent research project by **Team Seifuku**, based in Pune, Maharashtra, India. The team details and logo in this edition were confirmed on 28 September 2026.

Members and confirmed responsibilities

Member	Confirmed responsibility
Adwita Kurle	Team leader
Gaurav Ranade	Website creator and operator
Anushree Dixit	Team member
Spruha Kurle	Team member
Hrishikesh Tarade	Team member
Tanisha Natrajan	Team member

Contact and affiliation

Public contact: ranadegaurav30@gmail.com.

Location: **Pune, Maharashtra, India.**

The project has no university affiliation or university involvement in its development. No additional member-specific engineering roles, mentors or institutional endorsements are claimed. The logo identifies the team; it does not imply endorsement by a dataset provider, model author or software dependency.

Technical Documentation — Status as of **28 September 2026**.

03 Using the research workspace

SOURCE docs/USER_GUIDE.md

Saved demonstrations are frozen historical products with their own provenance; they are not recomputed when opened. For example, Copenhagen's saved metadata identifies an earlier landscape checkpoint. New uploads use the released protected height model. Compare model identities and protocols before comparing their values.

First investigation

1. Start the viewer and select a bundled reference scene. The neutral start screen does not itself initiate image inference.
2. Rotate the surface and compare its texture with the original image. Use the source-image reference when an oblique perspective makes a feature ambiguous.
3. Inspect a point. Read its units and product type before interpreting the number. Above-ground height and absolute elevation answer different questions.
4. Enable six-class identification when the local API and classifier are available. Select Ground, Buildings, Water, Roads, Low vegetation or Trees to inspect their spatial coverage. A class highlight is not a corrected height measurement.
5. Explore with fly or walk navigation. The map shows camera position and direction; it is a navigation aid, not a separately surveyed map.
6. For metric terrain, draw a profile and inspect elevation, distance and slope. Download the profile CSV or the source raster product as appropriate.
7. If an eligible reference is present, inspect the signed error and scoped metrics. Preserve the scene identifier and reference description with any exported figure.

Choosing inputs

Ordinary JPG/PNG imagery supports a relative relief interpretation. A TIFF file can contain optical imagery, an elevation raster, or another numeric product; its file extension alone is insufficient to establish meaning. The import path uses raster metadata, dimensions and band properties. An image-only TIFF still needs inferred heights or an external terrain source.

The API accepts JPG, JPEG, PNG, TIFF, JP2, WebP and BMP imagery subject to decoding support. Automatic classification currently requires 8-bit RGB. Higher-bit-depth height processing and six-class support are separate capabilities. Unsupported classification must produce an explicit limitation rather than a fabricated label.

Large image inputs are read onto a bounded processing grid. A smaller processing grid preserves geographic bounds but loses spatial detail. The metadata records both original and processing sizes; inspect it before making a fine-resolution claim. The API's 512 MiB file limit does not mean every image of that byte size is fast or accurate.

Calibration and reference comparison

Choose one calibration route: a supplied terrain DEM, supplied GCP CSV, or optional automatic public terrain. A DEM must describe the intended ground datum and align through a supported CRS transformation. A GCP table needs `elevation_m` and either map coordinates `x, y` or image coordinates `row, col`.

Automatic public terrain is convenient for a demonstration, but its coarse grid does not become high-resolution ground truth when resampled. The provider product can represent a surface rather than bare earth. The user is responsible for the datum and acquisition compatibility of a manually supplied reference.

A validation reference is distinct from a calibration input. Select nDSM when the reference measures object height above ground, or DSM when it measures absolute surface elevation. The application aligns it to the prediction grid and masks invalid support. Never evaluate against the same values used to fit a calibration and describe the result as independent validation.

Display controls and measurements

Terrain Source/Smooth modes and vertical exaggeration change the displayed mesh. Smooth/Soft is the current default for compatible terrain, with a gentle 1.5x display multiplier for non-flat metric terrain. The source elevations remain the basis of numerical inspection, profiles and exports. The source-mode view makes that difference directly inspectable.

Building roofs remain flat optical-textured representations with model-derived building-height estimates. Detailed roof-height estimation is explicitly unconfigured. Facade textures and reconstructed solids should not be interpreted as recovered architectural detail. Current scenes do not include the reverted illustrative tree-trunk experiment.

Class labels may disagree with the geometry. The RGB classifier does not control the protected height-routing masks or collision geometry. Use the model-details panel to read accuracy boundaries; pending, failed and unsupported identification are meaningful states, not alternate class predictions.

Export checklist

Question	What to retain
What quantity is this?	Product label, units, relative/above-ground/absolute interpretation
Where is it located?	CRS, transform, source bounds and processing-grid information
Which model produced it?	Model identity and artifact hash from metadata
Was it calibrated?	Terrain or GCP source, method and fit information
Was it evaluated?	Reference identity, mask, alignment and metric scope
Was the view exaggerated?	Keep display settings with screenshots; use raw values for numerical claims

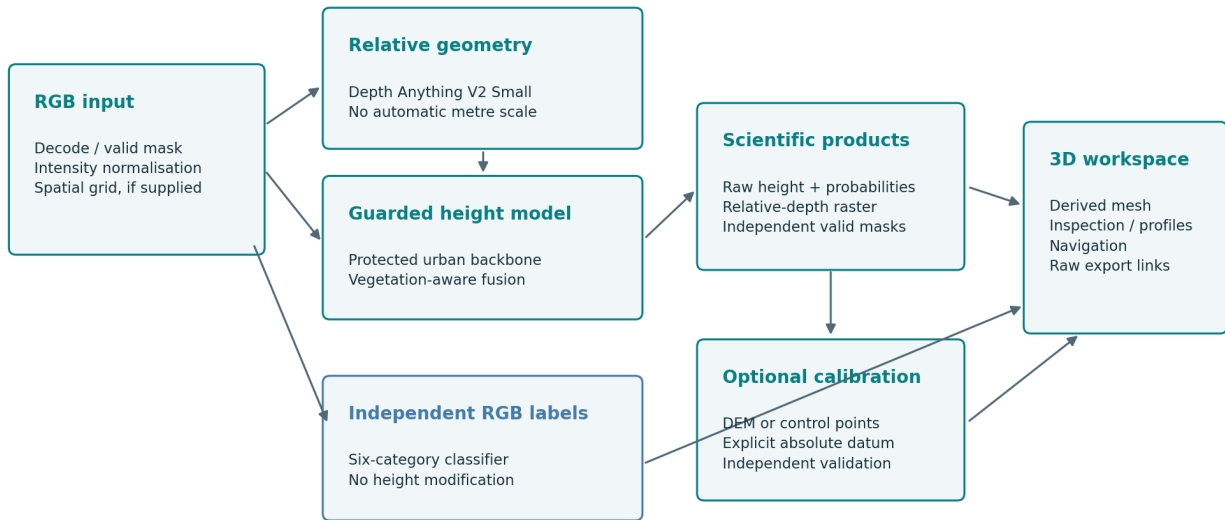
Saved API jobs are stored in the local `outputs/web_jobs/` directory. There is no cloud account or promised durable backup. Copy the artifacts you need before cleaning local outputs. See the security chapter for retention and access limits.

04 Architecture and data lifecycle

SOURCE docs/ARCHITECTURE.md

From one image to inspectable surface products

Model outputs, calibration, classification and display remain distinct.



Reference data evaluates a frozen output. Display styling never rewrites the raw height product.

System architecture

Component boundaries

Component	Implementation	Responsibility
Raster input	<code>src/msr/io/raster.py</code> , <code>data/radiometry.py</code>	Decode imagery, track validity, bound interactive resolution and preserve the spatial grid.
Relative prior	<code>inference/relative_depth.py</code>	Run Depth Anything V2 Small and produce scale-ambiguous geometry.
Height model	<code>models/height_net.py</code> , <code>models/domain_surface_net.py</code>	Estimate above-ground height through a multiscale decoder and guarded refinement.
Image inference	<code>inference/predict.py</code> , <code>inference/tiling.py</code>	Load models, normalize image tiles, blend overlapping predictions and preserve masks.
Calibration	<code>geospatial/calibration.py</code> , <code>geospatial/gcps.py</code>	Align terrain, fit control points and compose absolute products.
Independent identification	<code>models/rgb_segmenter.py</code> , <code>inference/rgb_preview.py</code>	Predict six RGB semantic categories and class coverage.
API	<code>api/app.py</code> , <code>api/classification.py</code>	Validate requests, serialize model access and publish job artifacts locally.
Presentation geometry	<code>inference/viewer_mesh.py</code> , viewer geometry helpers	Prepare a bounded display mesh and building solids separate from source measurements.
Browser	<code>viewer/app/terrain-workspace.tsx</code> and focused helper modules	Coordinate uploads, scenes, rendering, interaction, analysis and downloads.
Evaluation	<code>evaluation/</code> and evaluation scripts	Define fixed supports, compare models, report metrics and enforce promotion gates.

Paths in the implementation column are relative to `src/msr/` unless a different root is shown. The viewer follows the Next.js App Router component structure and runs through `vinext/Vite`. `Three.js` owns the interactive scene; `React` coordinates application state and controls. The public package uses a Node build without the original host-specific preview bindings.

Prediction lifecycle

An uploaded image receives a random job identifier. The API streams the upload to its job directory, validates requested calibration/reference combinations and invokes the runtime. The runtime lazily loads the height and foundation models and uses a lock around GPU work. This limits simultaneous model execution within that process; it is not a distributed queue or admission-control system.

The RGB reader selects supported bands, derives image validity and creates a bounded processing grid when needed. Relative geometry is computed from the image. The protected height model runs over overlapping normalized tiles. Its numerical outputs are saved separately from the mesh representation. Calibration, if requested, derives the appropriate additional product. Reference comparison uses the declared prediction type and an independently supplied raster.

The API returns URLs for source-height, relative-depth, texture, probability, presentation and validation products, plus structured metadata. The browser loads these artifacts and constructs the scene. Once reconstruction completes, eligible scenes can request the separate RGB classifier. The UI guards against stale class maps during scene switches and caches a bounded number of successful maps in the browser session.

Independent semantic branch

The six-class model receives image data and image-validity support. It does not receive reference heights, reference classes, calibration targets or the height prediction as an input. It returns class IDs and coverage statistics. The final class map may change an overlay or inspection label but does not mutate the height raster, collision field or building-solid geometry.

The older three-group ground/building/vegetation mechanism remains part of height fusion. It is not the same classifier as the visible six-category identification. This separation is deliberate: the experiments did not establish that replacing protected height behavior with semantic-guided corrections was consistently safe.

Artifact lifecycle

Training runs retain configuration, source bindings, epochs, metrics and checkpoint identities. Development comparison is distinct from final testing. Release packaging extracts inference-only tensors and the configuration needed to restore the model; optimizer/RNG payloads and machine-specific paths are omitted. Tensor equality and output equivalence are checked separately from file-hash equality.

The public source package has its own identity and history. Renaming modules and normalizing locators changes source bytes, so it cannot truthfully inherit every historical cryptographic source seal. Historical reports remain evidence of those earlier runs; new experiments need their own source/configuration bindings.

Operational boundaries

The packaged service is intended for local use. The frontend defaults to an API on the loopback interface. There is no user authentication, multi-tenant job isolation, durable cloud storage or production GPU scheduler. A hosted version would require a separate deployment design. Source publication is not a hosted inference service.

05 Scientific contracts

SOURCE docs/SCIENTIFIC_CONTRACTS.md

These contracts define what the application's quantities and evidence mean. They take precedence over the visual appearance of a scene.

Quantities and units

Quantity	Meaning	What it does not establish
Relative depth / rDSM	A dimensionless ordering or relief score inferred from one image.	Absolute metres, a surveyed height or an elevation datum.
nDSM / object height	Estimated surface height above local ground.	Absolute elevation or hidden ground under an object.
DEM / DTM input	A supplied terrain-elevation reference with its own grid and datum.	Fine object height or guaranteed bare-earth quality for every provider product.
Absolute DSM	Aligned terrain elevation plus nonnegative inferred object height.	Accuracy beyond the terrain and inference sources used to construct it.
GCP-calibrated surface	A scene-level affine mapping fitted from supplied control elevations.	Independent validation on those same fitting points.
Semantic class	A predicted RGB category from the six-class classifier.	Water depth, vegetation species, a verified outline or correctness probability.
Presentation height	A mesh-oriented height field adjusted for readable geometry.	An additional measurement or a better numerical prediction.
Signed error	Prediction minus eligible reference, in matched units.	Error outside the evaluated support or against an incompatible datum.

Validity is task-specific

Image validity, semantic-label availability and height-label availability are separate masks. A usable optical pixel can lack a class label or a height label. That missing label must not be replaced with zero. Semantic road or water pixels can teach identification while invalid height artifacts at the same locations are excluded from height regression.

HighBuild primary height evaluation uses measured-height support intersected with eligible annotated buildings. Estimated annotations are separated. Unknown background is not evidence for zero-height ground, and a positive-only building inventory cannot support exhaustive counting precision.

OpenCanopy supervision respects source validity and LiDAR class limitations. Source acquisition dates and mosaic overlap matter even where tile IDs are different. Derived strata such as low/medium/high vegetation are height groups, not botanical labels.

Spatial alignment

Paired RGB, target and masks must match shape and their declared spatial grid, or an explicit alignment operation must produce a new derived raster. Reprojection must retain a record of interpolation and support. An array with no CRS cannot be silently assigned an authoritative map scale from its filename.

The interactive reader can reduce resolution while preserving full geographic bounds through an adjusted affine transform. Pixel-coordinate control points must be transformed to that processing grid. Geographic-coordinate control points remain interpreted in the declared spatial reference. A resampled raster does not gain information from its finer output pixel spacing.

Evaluation definitions

For N eligible pairs, let $e_i = \text{prediction}_i - \text{reference}_i$.

- $\text{MAE} = \text{sum}(\text{abs}(e_i)) / N$.
- $\text{RMSE} = \sqrt{\text{sum}(e_i * e_i) / N}$.
- $\text{Bias} = \text{sum}(e_i) / N$; positive bias means overestimation.
- $\text{R-squared} = 1 - \text{sum}(e_i * e_i) / \text{sum}((\text{reference}_i - \text{mean_reference})^2)$.
- Correlation describes association, not absence of scale or offset error.
- Precision, recall, F1 and IoU come from class-specific confusion counts.
- Six-class macro scores average the six final categories equally. A merged five-group vegetation statistic is a different result.

Degenerate cases need explicit treatment rather than an invented finite score. Pooled pixel metrics weight pixels; per-scene means weight scenes; equal-region scores weight regions. Those aggregation rules can rank the same candidates differently. A published selector must not be changed after seeing results.

Data separation and claim scope

Training diagnostics, reused development validation and a previously untouched external test are distinct evidence. A city excluded from a newly trained head can still have been seen by the inherited system. Unknown upstream pretraining coverage prevents claims of absolute global novelty.

Consumed holdouts remain consumed even after a poor result or a process crash. New model selection in response to an external test needs another reserved test for a fresh final claim. A prepared/downloaded holdout is not a measured result.

Visualization and provenance

Source arrays support measurements, slope/profile calculations and exports. Smoothing, facades, vertical exaggeration and illustrative solids support display. Source/presentation separation is tested because a plausible 3D view can otherwise make an inaccurate field look authoritative.

Each reported metric should identify the model/checkpoint, input contract, geography, split, reference type, validity support, grid, aggregation rule and evaluation date. Hashes identify bytes; they do not themselves establish label quality, licensing, independence or scientific validity.

06 Model and geospatial methods

SOURCE docs/METHODS.md

Relative geometry

Depth Anything V2 Small provides a pretrained relative-depth prior. The wrapper handles image processing, inference and interpolation to the processing grid. The robust normalization uses eligible finite pixels and the 2nd/98th percentiles to form a clipped $[0,1]$ score. A near/far convention is explicit, degenerate spans produce a zero relative score, and invalid pixels remain missing.

This prior is target-independent. Height-label availability must not influence normalization or the cached image geometry. Earlier cache audits locked current RGB/prior/model identities and reproduced five deterministic examples, while acknowledging that creation-time hashes were absent from the original cache.

Base height network

HeightNet uses timm's ConvNeXt V2 Tiny multiscale encoder and a U-Net-like decoder. At each level, the decoded field is bilinearly upsampled and concatenated with the corresponding encoder feature map. Convolution blocks use GroupNorm and GELU. A final softplus head produces nonnegative height; an auxiliary building logit head provides a separate spatial signal.

Training uses geographic manifests, synchronized image/label transformations, masked losses, mixed precision, gradient accumulation and resumable artifacts. The initial full urban run was followed by error analysis rather than interpreting one all-pixel score as sufficient evidence of useful building reconstruction. Tall structures and background/support definitions were important failure sources.

Protected surface refinement

DomainGatedSurfaceNet retains the base network and learns a compact adapter from image, relative prior, base height and building evidence. Its three domain probabilities sum to one: ground, building and vegetation. It also predicts a bounded building residual, canopy height, refinement strength and a variance-like quantity. The three groups are internal height-routing domains.

The active protected-vegetation mode preserves base building height within its mixture while allowing a vegetation correction. In simplified notation, the intermediate height is $\max(0, h_{\text{base}} + \alpha_{\text{effective}} * (h_{\text{gate}} - h_{\text{base}}))$. The effective strength is reduced by the building-protection term. A separately configured vegetation-expert contribution is applied only at its declared gate. The exact computation is centralized in `fuse_surface_endpoint`.

This distinction explains an important diagnostic: improving a separate building residual head need not improve the final height if that head is not used in the active fusion formula. Likewise, a weak effective gate can transmit only a small part of a supervision gradient. Head-level metrics are not substitutes for final predicted-height metrics.

Later height experiments

A residual decoder initialized at zero tested a richer correction pathway while preserving the original prediction at initialization. Class-assisted experiments tested whether frozen semantic probabilities helped an equally trained height model beyond a neutral control. Low-surface losses and height-band sampling tested specific tradeoffs. A semantic-expert design mixed six learned corrections using frozen class probabilities.

These are preserved research alternatives. None established a candidate satisfying all the required benefit and safety gates. The release keeps the protected height model; the results chapter does not combine the best strata from different runs.

Independent RGB classification

The visible classifier is a SegFormer B0-based six-class model initialized from the ImageNet-pretrained MiT-B0 encoder. Raw RGB is scaled by 255 and normalized inside the model. Class outputs are ground, buildings, water, roads, low vegetation and trees. The active inference path uses native tiles up to 1024 pixels with 128-pixel overlap, averages softmax probabilities and then takes argmax. Invalid image pixels use class ID 255. No reference target is used at inference.

The V3 training package adds regional data/augmentation to improve transfer. It is not an isolated causal test of any one augmentation. Its selected development checkpoint improves OEM transfer but regresses on some GAMUS classes and road boundaries. Automatic app integration does not change those results.

Tiled height inference

Overlapping tiles are blended with spatial weights to reduce seams. Input normalization, tile size, overlap, precision and padding policy are part of the evaluation contract. They can change GroupNorm context and therefore output. The full-native research benchmark and the app's 512/128 path must be labelled separately. The optional minimal-padding audit improved canopy error but worsened forest-ground error, so it was not silently enabled globally.

Calibration and validation

DEM composition reprojects the supplied ground field to the prediction grid and adds nonnegative object height where both supports are valid. Control-point calibration estimates a scene-level affine scale/offset with robust trimming; its fit error describes the supplied controls. Calibration does not solve hidden ground, vertical-datum mismatch or acquisition-time change.

The reference evaluator aligns supported raster types, reports valid support and computes signed errors and summary statistics. Instance evaluation matches available building annotations and distinguishes measured from estimated heights. Boundary and counting interpretations depend on the completeness of reference annotations. The benchmark and scorecard histories give the exact acceptance rules.

Presentation reconstruction

The viewer constructs terrain and building geometry for interaction. Terrain smoothing uses positive normalized Gaussian weights, supports rectangular pixel spacing, preserves missing data, and keeps source values separate. Fine-resolution standalone DSMs without a ground reference have additional protection to avoid flattening building structure. Map, collision and highlight behavior follow the displayed geometry; exported source products retain their scientific meaning.

07 Implementation map and maintenance guide

SOURCE docs/CODE_MAP.md

The Python package is `msr`; its source is under `src/msr`. The public package uses a fresh repository history and scientific naming. This chapter describes the implementation families so that a reviewer can trace a claim to the responsible code and its tests without needing the private development archive.

Data and learning

`data/` implements raster loading, normalisation and validity handling. RGB arrays are converted to channel-first float arrays, with the selected intensity scale and ImageNet channel normalisation. Reference validity is an independent array; unknown height is never made valid merely because its numeric value is zero.

`models/height_net.py` defines the urban encoder/decoder; `domain_surface_net.py` wraps the protected urban model and guarded vegetation fusion. The public loader reconstructs the base architecture from the packaged configuration and loads all of its tensors. Training does not occur during loading. `residual_height.py` and the semantic/class-assisted experiment modules are research alternatives, not the released default predictor.

The scripts directory contains preparation, training, fixed-protocol evaluation, error analysis, checkpoint sealing and demonstration tooling. Configurations retain experiment names and protocol choices. A config is not proof that a run finished; consult the matching outcome record and checkpoint identity.

Inference and calibration

`inference/predict.py` owns checkpoint construction, tiled metric prediction, valid masks and optional semantic products. `inference/tiling.py` determines tile starts and positive blending weights. A complete raster can involve many model calls; average overlapping predictions only on supported pixels.

Foundation relative depth and learned height are distinct products. Calibration code operates on explicitly supplied DEM or control-point evidence. Presentation meshes and building solids are derived for viewing; they do not become the raw prediction simply because their appearance is smoother.

API boundary

`api/app.py` owns request parsing, runtime loading, serial model access, job directories and response manifests. The endpoint writes actual output rasters and returns URLs under `/results/`. `scripts/serve_api.py` configures Uvicorn, device, checkpoint and output directory. The documented default binds locally.

`scripts/download_models.py` validates release sizes and SHA-256 before replacing files. The foundation repository revision and required files are pinned in the same manifest. Acquire artifacts before starting a production workflow; a health response alone does not prove a model has successfully loaded.

Viewer boundary

`viewer/app/terrain-workspace.tsx` coordinates scene state, loading, Three.js, navigation, controls and output links. Smaller modules isolate scientific and interaction contracts: `surface-query.ts`, `terrain-analysis.ts`, `terrain-presentation.ts`, `six-class-layer.ts`, `six-class-request.ts`, `image-import.ts`, `flight-navigation.ts`, `walk-navigation.ts`, `fullscreen-navigation.ts`, `metric-grid.ts`, `selection-region.ts` and `explore-map-math.ts`. Their Node tests execute deterministic geometry and state logic without requiring a GPU or browser.

`landscape-evaluation.tsx` reports selected demonstration measurements. `validation-comparison.tsx` visualises aligned prediction/reference/error rasters. The restricted US3D imagery is excluded, and its historical row cannot request an unavailable image comparison. The other public demonstrations remain usable.

Safe extension points

For a new predictor, declare its input range, channel order, output units, normalisation, valid-mask behavior and checkpoint identity before integration. Evaluate it under the existing matched protocol; do not change the baseline while measuring a candidate. For a new terrain workflow, establish horizontal CRS and vertical datum compatibility and retain the original surface as a separate export.

For a new viewer effect, operate on a presentation copy. Add numerical tests only where the change can affect units, coordinates, classification, geometry or state transitions. Use browser checks for visible interaction and an API check for actual generated products. Preserve a reproducible failing example before adjusting a scientific rule.

08 Data, provenance and geographic coverage

SOURCE docs/DATA.md

Dataset roles

Source	Role in this project	Principal boundary
HighBuild-1M	Urban height training, building annotations and measured-building evaluation.	Mixed source licences; sparse/estimated annotations and missing per-tile geospatial metadata require explicit filtering.
Open-Canopy	Forest RGB and LiDAR-derived canopy supervision; retained official split information.	Mosaic/acquisition overlap, LiDAR-class masks and image/reference date differences matter.
GAMUS	Height/semantic experiments and six-category classification training/development.	Local HDF5 lacks spatial metadata; height units remain assumed in the older height experiments.
OpenEarthMap	External V1 classification testing and later multi-region V3 development.	Regional licence and ontology mapping; development regions are not final tests.
LINZ/BOPLASS Bay of Plenty	Reserved external metric-height package from orthophoto, DSM and DEM.	Prepared and sealed, but not yet evaluated.
Sentinel-2 / Copernicus / public terrain	Georeferenced hilly demonstration and terrain datum.	Coarse terrain, datum and surface-versus-bare-earth differences limit interpretation.
US3D / DFC2019 demonstration	One selected urban RGB/LiDAR comparison case.	A disclosed selected demonstration, not a population-level validation set.
Roof-reconstruction sources	Prepared research inputs for roof geometry experiments.	Experimental modules are retained; detailed roof reconstruction is not enabled.

Source links, attribution and model terms are collected in [References](#) and [Third-party notices](#). The source package does not redistribute the full training archives.

HighBuild preparation

The recorded initial full-data preparation used a city-disjoint, conservative source-licence filter: 25,446 training tiles from five cities, 1,921 validation tiles from four held-out cities and 2,310 test tiles from five held-out cities. Those are counts for the pinned local preparation, not a claim about the latest upstream dataset size. The acquisition script pins revision 2f5d76f8c5b5b4b7e925871d46db7aae64fd3daa.

The later corrected height benchmark is a different support: 160 complete urban scenes with strict measured-height building support, plus 120 forest scenes. Annotations marked estimated are excluded from the primary measured-building score. COCO footprints and height validity are independently represented. Training crops are selected to contain useful support, while evaluation uses the complete fixed native grid.

The first urban error analysis showed a background-dominated aggregate and strong tall-building underestimation. That motivated support correction and stratified reporting. Earlier zero-filled/all-pixel scores remain historical results and cannot substitute for the corrected measured-building baseline.

Open-Canopy preparation

The project range-streamed selected image/LiDAR chips rather than downloading the entire release. Stored decimetre heights are converted to metres. Chips retain region, source image, acquisition dates, GSD and source URLs. The accepted expanded subset contains 600 training, 120 validation and 120 test chips.

The provenance audit reports accepted samples rather than every rejected cached candidate. It preserves official spatial-split information and distinguishes coarse-region overlap from shared source mosaics. No single overlap statistic is treated as proof of complete geographic independence. LiDAR unclassified/building support must not be mistaken for valid forest height supervision.

GAMUS preparation and exposure

The locked quality index contains 5,001 approved training tiles, 859 validation tiles and 2,860 official-test tiles. All 8,720 RGB/prior pairs were inventoried and hash-locked. Five sampled cache recomputations reproduced 5,242,880 float16 values. That verifies the sampled computation and current identities, not missing creation-time provenance for the entire cache.

For a later paired head experiment, DC/Philadelphia contributed 3,837 training tiles and 859 development-validation tiles; 1,164 NYC tiles were excluded from that head's training. New York was already known to the inherited project, so the experiment does not establish system-unseen geography. The DC grid audit found all 359 validation tiles adjacent to training under an eight-neighbour rule. This split is useful for development but is not a geographically isolated final test.

The published nominal resolution is 0.33 m, but local HDF5 files contain no CRS, transform or per-file pixel-size proof. The raw AGL unit was not independently confirmed in the reviewed project record. Older GAMUS height scores must therefore retain their metre-assumed label. Extreme values above 200 were inventoried before changing any cutoff; they include implausible class/height combinations.

Holdout ledger

Dataset / region	Recorded state	Allowed interpretation
GAMUS official test	Consumed by earlier Stage-3 work.	Historical audit only; not an untouched test for later tuning.
NYC head-exclusion partition	Excluded from specified fresh heads, but prior system exposure exists.	Head-transfer scope, not complete-system novelty.
Christchurch OEM V1	All 49 publicly labelled pairs evaluated once by the frozen V1 classifier.	A completed V1 external classification transfer test; now consumed.
OEM Duesseldorf, Dhaka, Chiang Mai challenge	Twelve label-enriched scenes inspected and evaluated.	Diagnostic mixed-class challenge; not random global sampling.
OEM Melbourne/Vienna V3	Used for development selection.	Development transfer scores.
OEM Rosario/Monrovia	Recorded as reserved in the V3 outcome.	Not a published final result. Recheck consumption before any future use.
Bay of Plenty metric-height package	Sealed manifest exists; no evaluation plan or consumption marker at release preparation.	Prepared external evidence path, no measured model result.

Bay of Plenty protocol

The fixed orthophoto scene is BC36_1000_2314 from 2018–2019 urban imagery. Paired 1 m LiDAR DSM/DEM products use item BC36_10000_0302. Their difference defines the reference above-ground surface. The source grid is EPSG:2193. Capture ranges overlap, but pixel-level simultaneous acquisition is not established.

Three supports were fixed before inference: all-valid, object reference above 2 m, and tall-object reference above 5 m. Negative reference artifacts are excluded; there is no upper-height cutoff. The sealed preparation records 326,925 all-valid pixels, 133,567 object pixels and 124,217 tall-object pixels.

This package is not used for routine development, calibration or scene selection. A future evaluation must freeze candidate identities first and publish all predeclared supports regardless of whether the result is favourable.

Reproduction and redistribution

Public scripts describe acquisition and preparation, but downloading a dataset does not grant uniform rights to all its imagery. Keep provider and regional attribution with derivative figures. The two older Chicago smoke assets are omitted from the public package because the source inventory excludes their imagery from redistribution. Demonstration assets and restricted research model weights retain separate notices. No universal commercial-data licence is claimed.

US3D/DFC2019 raw imagery and reference rasters are excluded from this public package because the source terms prohibit redistribution. The historical selected-scene numerical result remains documented; its visual comparison is unavailable in the public viewer. The Copenhagen urban reconstruction demo is separate and remains available.

09 Validation protocol and evidence interpretation

SOURCE docs/VALIDATION.md

Separate four kinds of evidence

Research evaluation measures a named model against independent reference data. Application verification checks whether a implemented workflow behaves correctly. Publication verification checks that a portable release preserves identities, notices, links and model values. Presentation inspection checks that the visible experience communicates those products clearly. Passing one does not imply the others have passed.

Metric definitions

For valid paired prediction p and reference y , error $e = p - y$. MAE is the mean of $|e|$, RMSE is $\sqrt{\text{mean}(e^2)}$, and signed bias is $\text{mean}(e)$. R-squared is $1 - \frac{\text{sum}(e^2)}{\text{sum}((y - \text{mean}(y))^2)}$, undefined when the reference variance is zero. Pearson correlation measures centered linear association, not metre accuracy. Report support counts, masks, aggregation and units alongside every score.

Pooled RMSE is the square root of pooled squared error per valid pixel; it is not the mean of per-scene RMSE values.

Pixel-weighted and equal-scene summaries answer different questions. Object metrics require an explicit matching rule and aggregation within each building. An object-only score omits unmatched objects and therefore cannot replace detection or all-eligible pixel evaluation.

Semantic scores begin from a confusion matrix with a documented ontology. Class precision is $TP/(TP+FP)$, recall $TP/(TP+FN)$, F1 $2TP/(2TP+FP+FN)$, and IoU $TP/(TP+FP+FN)$. Macro averages weight classes equally; region-balanced averages also prevent a large or easy city from dominating the selector. Missing classes and nodata need declared treatment. V1 and V3 results cannot be transferred across checkpoint identities.

Reference isolation

Reference nDSM/DSM is accepted for evaluation only after the inference input is fixed. It must not alter inference, select a visually favourable crop, fit a post-hoc scale/offset, or repair the prediction. A terrain DEM used as an absolute datum cannot simultaneously be presented as an independent validation source for that same calibration. The hilly demonstration therefore uses separate SRTM for its disclosed coarse comparison.

Matched height protocol

Freeze model hashes, full scene lists, preprocessing, tiling, precision, units, masks and aggregation before comparing candidates. HighBuild measured-building support excludes estimated annotations and unknown background. OpenCanopy splits vegetation, ground and low vegetation using its fixed LiDAR-derived supports. Record tallest-height bands, bias, centered spread, agreement and safety guards.

Development subsets are allowed for fast iteration, but must be labelled as such. The paired control receives the same budget and non-semantic inputs. A candidate is eligible only when the registered selection and retention requirements hold; an improved average cannot erase a failed safety support. Rejected experiments remain valuable records of what was learned.

Geographic and temporal dependence

City exclusion for a new head is not necessarily complete-system novelty. Adjacent tiles, overlapping mosaics and prior tests can create dependence even when file names differ. Record image date, LiDAR date, provider and geographic overlap. Tile bootstrap intervals describe within-support uncertainty; they do not prove global geographic generalisation. Label-enriched challenges are diagnostic rather than random population samples.

Reproducibility ladder

The public package supports source inspection, deterministic logic tests, saved scene inspection and inference with identified release weights. It also includes historical protocols/configurations needed to reconstruct research workflows after lawful data acquisition. Full training replay additionally requires original data revisions, caches, sampled indices, hardware/software settings and run archives. The release does not claim bitwise training replay from a small inference bundle.

Release tensor equality and demo-raster equality are recorded under docs/evidence. The Python suite reports historical archive dependencies explicitly. No sealed holdout is consumed merely to verify packaging.

10 Results and interpretation

SOURCE docs/RESULTS.md

All scores below are preserved research measurements with their stated protocols. They are not newly rerun full-dataset evaluations for the source publication. Release packaging checks are reported separately in [Release](#).

Canonical corrected height baseline

The protected model was evaluated on 160 complete HighBuild scenes and 120 OpenCanopy scenes using matched inputs and valid reference support. These are development results. HighBuild uses strict measured-height masks, not unknown background filled with zero.

Population	Support	RMSE (m)	MAE (m)	Correlation	R-squared
Pooled strict urban + forest	31,011,260 pixels	10.16	4.91	0.482	0.203
Measured building pixels	13,855,690 pixels	13.24	5.36	0.186	-0.066
All valid forest-scene pixels	17,155,570 pixels	6.71	4.55	0.659	0.422
Vegetation-only pixels	7,852,823 pixels	8.36	6.51	0.411	-0.174
Forest-ground pixels	9,302,747 pixels	4.89	2.90	0.502	-0.088
Matched measured buildings	328 objects	11.15	4.42	0.244	0.037

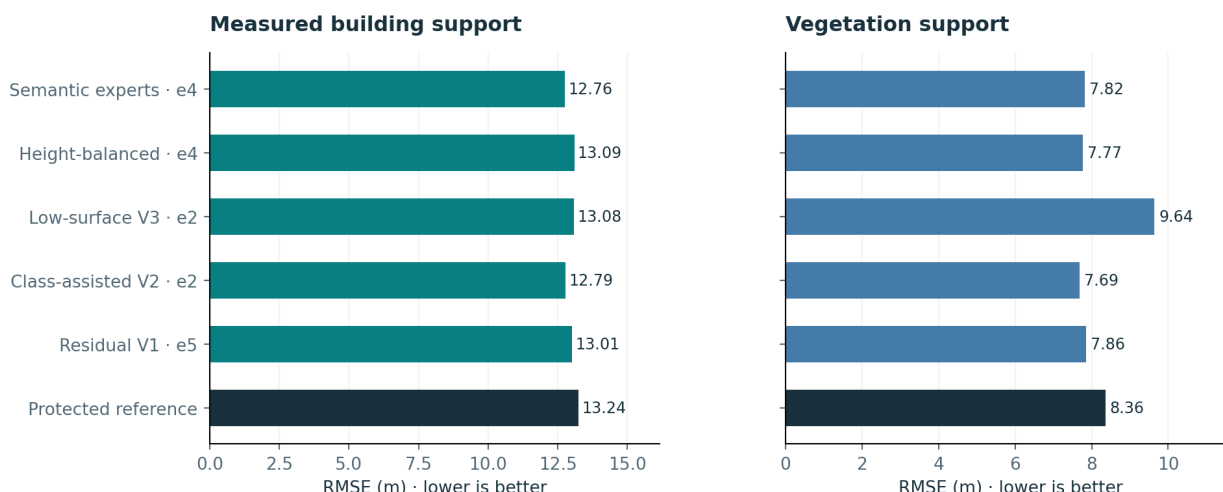
Pixel and object rows answer different questions. The object row uses one median prediction per matched building and excludes unmatched references; it is not a replacement for the all-eligible measured-building pixel score. A weak or negative R-squared remains visible even when the mean error looks modest.

Why later height candidates were not promoted

Candidate / selected comparison	Building RMSE (m)	Canopy RMSE (m)	Decision
Protected reference	13.2426	8.3610	Retained
Residual V1, epoch 5	13.0121	7.8565	Rejected: not all registered guards pass
Class-assisted fast V2, epoch 2	12.7883	7.6885	Rejected: ground and short vegetation worsen
Low-surface V3, epoch 2	13.076	9.637	Rejected: canopy loss offsets low-surface gains
Height-band-balanced, epoch 4	13.094	7.770	Rejected: safety and comparison tradeoffs
Six-class semantic experts, epoch 4	12.7558	7.8238	Rejected: no eligible paired-benefit/safety outcome

These rows share the corrected development context but are different experiments and checkpoints. They are not a ranking that replaces each run's predeclared selector. The matched neutral control is essential: in the semantic-expert run, control epoch 4 scores 12.7649 m buildings and 7.7913 m canopy. The class-guided candidate is only 0.0091 m better for buildings and worse for canopy. One paired run does not establish a reliable benefit from class guidance.

Lower averages did not establish an eligible replacement



Development comparisons. Different selected experiments; their paired controls and safety guards still govern promotion. The protected model remains the released height predictor. Source protocols are indexed in Results and Experiments.

Height experiment comparison

Tall structures dominate building error. In the residual-height analysis, reference heights of at least 20 m account for 17.64% of measured building pixels but 89.31% of baseline squared error. Reduced bias can improve RMSE without improving centered error spread or shape agreement. Those diagnostics motivate better supervision and context checks rather than unsupported accuracy claims.

Classification results belong to named models

Model and evaluation	Six-class macro F1	Scope
RGB V1, DC/Philadelphia development	83.83%	Reused model-selection validation
RGB V1, Christchurch	61.63%	One completed external test on 49 labelled pairs
RGB V1, 12-scene all-class OEM challenge	54.02%	Label-enriched diagnostic selection
RGB V3 epoch 1, GAMUS	83.20%	Development comparison; some retention regressions
RGB V3 epoch 1, OEM pooled	73.29%	Pooled development confusion counts
RGB V3 epoch 1, OEM equal-region	69.42%	Predeclared region-balanced selector

The V3 package is the active app classifier because its integration was authorized as product behavior. It is not certified by V1's external result and did not pass all V3 model-safety gates. Epoch 4 had higher pooled OEM F1 (74.47%) but lower equal-region F1 (68.56%); switching the selector afterward would misrepresent the experiment.

Christchurch V1 per-class F1 is 24.78% ground, 72.57% buildings, 87.38% water, 62.27% roads, 53.77% low vegetation and 69.03% trees. Its macro IoU is 47.23%. The tile-bootstrap macro-F1 interval of 58.43–63.83% is within-city descriptive uncertainty; it does not account for all neighbouring-tile dependence or global geographic variation. Road-boundary F1 is only 23.07% at the stated tolerance.

Demonstration cases

Case	Recorded RMSE / MAE	Interpretation
Sparse Amsterdam	1.48 / 0.71 m	One selected held-out scene under its reference contract
Urban US3D	2.54 / 1.30 m	One selected 262,144-pixel RGB/LiDAR comparison
OpenCanopy forest	4.22 / 2.86 m	One 147,456-pixel official-test chip
Manali terrain comparison	15.03 / 10.24 m	Absolute product versus separately aligned coarse SRTM; datum/resolution effects remain

These cases demonstrate supported workflows. They must not replace the broader corrected benchmark, be averaged into a new overall score, or be presented as randomly sampled independent final evidence. High terrain correlation can be dominated by broad relief while meaningful local elevation errors remain.

Inference-protocol sensitivity

On matched development support, app tiling changed building RMSE from 13.243 to 13.442 m and vegetation RMSE from 8.361 to 11.882 m relative to full-native evaluation. A separate fixed-precision padding comparison reduced vegetation RMSE to 8.366 m but worsened forest-ground RMSE from 4.297 to 4.885 m. It is therefore incorrect to report the favourable padding number as a universal model upgrade.

What is still unmeasured

The reserved Bay of Plenty metric-height scene has no final model score. Broad cross-sensor transfer, calibrated correctness probabilities, precise individual tree height, comprehensive building counting and independent expert validation remain unestablished. Relative error reductions are not percentages of correct predictions. The project is useful as a transparent research prototype while those questions remain open.

US3D/DFC2019 raw imagery and reference rasters are excluded from this public package because the source terms prohibit redistribution. The historical selected-scene numerical result remains documented; its visual comparison is unavailable in the public viewer. The Copenhagen urban reconstruction demo is separate and remains available.

11 Experiment register

SOURCE docs/EXPERIMENTS.md

This register links the research question to the intervention, result and decision. Dates identify development records. A source release does not repeat training or retroactively change the original acceptance rules.

ID / period	Question and intervention	Observed outcome	Decision / source
E01 · Aug 26	Can a geographically separated urban pipeline train and resume reliably? Small reviewer pilot, then a full ConvNeXt V2 height run.	Training/resume worked; early all-pixel scores were strongly affected by reference support.	Retain plumbing evidence; use corrected masks for current claims. <code>history/PROGRESS.md</code>
E02 · Aug 28	Can one external pretrained height model transfer without adaptation? FusedSeg-HE evaluation adapter.	FP16 non-finite predictions invalidated the first run; FP32 was finite but poor on the selected tall-building sample.	Domain/scale compatibility is essential; not a universal model ranking. <code>history/PROGRESS.md</code>
E03 · Aug 28–29	Add canopy estimation without erasing useful urban behavior. Protected urban expert plus guarded surface refinement.	A protected vegetation checkpoint was retained; later quality checks exposed remaining tall-height weaknesses.	Current protected height family. <code>history/PROGRESS.md</code>
E04 · Sep 12–13	Expand with GAMUS specialists, replay and routing.	Router confidence and endpoint tradeoffs did not establish a safe universal replacement.	Retain bounded experiments and inspect final outputs, not only expert heads. <code>history/EXPANDED_GAMUS_PLAN.md</code>
E05 · Sep 13	Joint height and six-class extension.	Height pilot and joint candidate underperformed protected forest results; water recall collapsed in the first joint classifier.	Reject promotion. <code>history/GAMUS_SIX_CLASS_EXPERIMENT_STATUS.md</code>
E06 · Sep 13	Preserve all height tensors while training a linear semantic head; address class imbalance.	Water improved, but ground/low-vegetation/tree retention gates failed.	Reject; inherited tensor identity verified. Same status record.
E07 · Sep 13	Adjust decisions through additive class bias on fixed outputs.	Development tradeoff improved; original class floors still missed.	Not probability calibration and not enabled as a certified improvement. <code>history/GAMUS_SIX_CLASS_DECISION_BIAS_CALIBRATION.md</code>
E08 · Sep 13–15	Compare spatial head and hierarchical vegetation head on a matched city split.	Corrected replay macro F1: spatial V3 56.48%, hierarchical V4 55.11%.	Hierarchy did not improve the combined six-class result. <code>history/ACCURACY_RECOVERY_2026_09_15.md</code>
E09 · Sep 15	Diagnose stalled height learning: tiny trainable control and gated correction pathway.	The 98-parameter control showed no eligible gain; train diagnostics found weak effective correction and gradient flow.	Test a richer residual decoder; diagnostics are not validation scores. Same recovery record.
E10 · Sep 15	Audit full-native/app inference and small-image padding.	Strong forest padding sensitivity with opposing canopy and ground effects.	Keep protocols separate; no global padding switch. Same recovery record.
E11 · Sep 15	Train a zero-initialized spatial residual-height decoder.	Five completed epochs; modest aggregate improvement, persistent tall-object errors and failed guards.	Reject every candidate. <code>history/RESIDUAL_HEIGHT_V1_OUTCOME.md</code>
E12 · Sep 15	Train independent RGB-only six-class segmentation.	V1 improved development identification; reproducibility replay verified epoch 8.	Preserve model-specific limits and test external transfer. <code>history/RGB_SEGMENTER_V1_OUTCOME.md</code> , <code>history/RGB_SEGMENTER_V1_TEST_REPORT.md</code>
E13 · Sep 15	Freeze V1 and test Christchurch once.	61.63% macro F1, with strong water and weak ground/road boundaries.	Publish complete result; mark city consumed. <code>history/RGB_SEGMENTER_V1_CHRISTCHURCH_RESULTS.md</code>

ID / period	Question and intervention	Observed outcome	Decision / source
E14 · Sep 15	Does targeted ground/low-vegetation crop sampling beat an equally trained control?	Final targeted advantage was small; key benefit and tree-retention rules failed. Earlier epoch remained exploratory.	No proven sampling improvement. history/RGB_SAMPLING_V2_OUTCOME.md
E15 · Sep 19	Test six categories together in diverse mixed scenes.	54.02% macro F1 across 12 label-enriched OEM scenes; ground/low vegetation/roads weak.	Diagnostic challenge, not general accuracy. history/OEM_ALL_SIX_CLASS_TEST_2026_09_19.md
E16 · Sep 19	Add regional RGB training/augmentation.	V3 improved OEM transfer but regressed on some GAMUS classes and Philadelphia road boundaries.	Best development epoch 1; no fully safety-passing checkpoint. history/RGB_MULTIREGION_V3_OUTCOME.md
E17 · Sep 19	Does class-assisted height correction beat neutral inputs at matched budget?	Fast V2 reduced building/canopy errors but damaged short surfaces; neutral control explained most gain.	Reject release. history/CLASS_ASSISTED_HEIGHT_FAST_V2_OUTCOME.md
E18 · Sep 19	Protect ground and short vegetation more strongly.	Short-surface errors improved, canopy suppressed too much.	Reject; investigate selective geometry instead of only loss reweighting. history/CLASS_ASSISTED_HEIGHT_LOW_SURFACE_V3_OUTCOME.md
E19 · Sep 19	Balance height-band crop exposure.	Some tall-height gains; weak paired canopy advantage and worse other groups.	No accepted replacement. history/HEIGHT_BAND_SAMPLING_V1_OUTCOME.md
E20 · Sep 19	Mix six learned correction experts with frozen class probabilities.	Class routing affects outputs, but no clear overall advantage over matched neutral routing.	Reject; do not claim class-guided height success. history/SEMANTIC_EXPERT_HEIGHT_V1_OUTCOME.md
E21 · Sep 19–20	Expose independent six-class identification in the application.	Preview and then automatic class identification delivered; raw height artifacts preserved.	Authorized product integration with model limitations intact. history/SIX_CLASSES_DEFAULT_INTEGRATION_2026_09_20.md
E22 · Sep 19	Fit more detailed roof geometry.	The cited Strasbourg example yielded zero accepted fits out of 12 footprints under the experimental gates.	Flat roofs restored; modules retained inactive. history/ROOF_RECONSTRUCTION_STAGE1_2026_09_19.md
E23 · Sep 26–27	Improve terrain imports, smoothing, profile analysis and exploration.	Source-preservation and interaction checks recorded across specified scenes.	Delivered presentation/tooling work; no new height-model accuracy claim. Dated terrain/viewer records.

How to add the next experiment

Record the hypothesis before running it. Specify the changed variable, unchanged control, initialization, data exposure, training budget, evaluation grid and masks, selector, safety thresholds and stop condition. Bind source/configuration/data identities, save every completed epoch and disclose failures or restarts.

Compare final deployed outputs rather than an internal head that the deployed fusion does not use. Include height strata and per-region/per-class results. Describe a one-seed effect as a one-seed observation. If several recipe elements change together, call the result a package comparison rather than an isolated causal ablation. Preserve negative results because they explain future design.

No new training or reserved-geography evaluation was launched for this publication.

12 Technical decisions and boundaries

SOURCE docs/DECISIONS.md

The decisions below consolidate evidence already recorded in source, protocols and outcome reports. They explain the current package rather than inventing a design history after the fact.

ID	Decision	Reason and consequence	Evidence
D01	Keep relative, above-ground and absolute products separate.	A monocular image does not supply an absolute datum. Product tags and units must express the actual computation.	Raster/calibration code; scientific contracts.
D02	Preserve a protected height checkpoint.	Later experiments can improve one aggregate while degrading another population. The newest checkpoint is not automatically the best release.	Height scorecard; residual and class-assisted outcomes.
D03	Evaluate final fused height.	Improvements to an internal expert do not matter if the active fusion does not use its output or heavily attenuates its gradient.	Accuracy recovery; fusion implementation.
D04	Use independent image, class and height masks.	Missing or invalid supervision is not zero. Classification and regression can have different valid support.	GAMUS preparation audit; surface datasets.
D05	Make corrected measured support the canonical urban result.	Sparse/estimated building references and unknown background distort all-pixel claims. Preserve older scores under their original protocol labels.	Benchmark protocol; HighBuild instance evaluation.
D06	Keep geography and exposure explicit.	Different IDs or excluded-head cities can still share neighbourhoods or inherited exposure.	Geographic holdout audit and consumed-test history.
D07	Do not tune on a final reserved geography.	A one-shot evaluation loses its status once inspected or used for model selection.	Christchurch protocol; external-height holdout.
D08	Use matched neutral controls for class-assisted height.	Better error than the old model does not prove semantic information caused the gain.	Fast V2, low-surface, sampling and expert-height outcomes.
D09	Keep RGB identification separate from protected height geometry.	The recorded controlled experiments did not justify changing height from six-class guidance. Labels can disagree with geometry and must say so.	Six-class default integration.
D10	Distinguish user-authorized integration from metric acceptance.	V3 identification can be available while its safety failures remain disclosed. Interface availability is not certification.	V3 outcome and September 20 integration.
D11	Preserve the original selector.	Epoch 4 wins pooled OEM F1 but epoch 1 wins the declared equal-region selector. Changing criteria afterward biases the reported result.	RGB multi-region V3 outcome.
D12	Audit inference context separately from learning.	Tile size, precision and padding can change predictions, especially with GroupNorm. A padding gain is not retraining evidence.	Matched app/native and padding audits.
D13	Bound interactive image resolution.	Compressed file size is a poor proxy for decoded memory. Overview reads preserve map bounds while recording lost resolution.	Raster reader; large-raster recovery record.
D14	Separate source measurement from presentation geometry.	Smoothing, exaggeration and solids improve readability but can move the displayed surface substantially. Inspection/export must retain the source contract.	Terrain presentation and source-preservation checks.

ID	Decision	Reason and consequence	Evidence
D15	Restore flat roofs and remove illustrative tree changes.	Experimental geometry did not meet the desired visual/coverage outcome. Inactive code remains available as research history.	Roof reconstruction status and subsequent product rollback record.
D16	Use a new public package and scientific identity.	A reproducible source release needs portable paths, a coherent entry point and an explicit artifact inventory. Original experiments retain their own immutable identities.	Release manifest and model-packaging checks.
D17	Preserve third-party restrictions and attribution.	Model code availability does not grant unrestricted commercial model rights; imagery also has provider-specific terms.	Upstream model notices and data inventory.
D18	Date the documentation and identify open work.	A research snapshot should distinguish delivered behavior, measured scope and planned investigation.	Status date: 28 September 2026.

Alternatives that remain open

A stronger encoder, more epochs, new data, different loss weights, per-instance supervision and explicit class-guided experts are possible interventions, not guaranteed improvements. Future work should isolate which problem each change addresses and preserve the comparator. Broad retraining before resolving label support, context and error concentration risks repeating the same failure mode.

More detailed roof or tree geometry is also separate from learning reliable metric heights. Reintroducing a visual layer would require its own labels, interaction checks and interpretation boundary. It must not silently change the meaning of exported scientific products.

13 Current feature register

SOURCE docs/FEATURES.md

Status refers to the source package as of 28 September 2026. Automated checks support specific contracts; they do not establish every possible input or device.

Feature	Implementation and evidence	Current boundary
RGB reconstruction	<code>api/app.py</code> , <code>inference/predict.py</code> ; prediction and API tests	Model artifacts and supported decoding are required; domain shift remains.
Bounded GeoTIFF reading	<code>io/raster.py</code> ; raster/import tests	Interactive processing can downsample; original detail is not recovered.
Relative geometry	<code>inference/relative_depth.py</code> ; normalization tests	Dimensionless until a justified metric mapping exists.
Terrain alignment and absolute DSM	<code>geospatial/calibration.py</code> ; calibration tests	Datum/source quality determines what the composition means.
GCP calibration	<code>geospatial/gcps.py</code> ; GCP and affine-fit tests	Fit residual is not independent validation.
Optional public terrain	<code>geospatial/public_terrain.py</code> ; terrain tests	Network-dependent, coarse and not guaranteed bare-earth truth.
Reference comparison	<code>evaluation/raster_validation.py</code> , viewer comparison	Requires a compatible independent reference and explicit support.
Six-class identification	<code>api/classification.py</code> , <code>inference/rgb_preview.py</code> , viewer class helpers	8-bit RGB; V3 regression failures disclosed.
Class coverage and export	<code>coverage_report</code> , classification raster outputs	Areas require supported projected scale; pixel share otherwise.
Orbit, fly and walk	Viewer navigation helpers and tests	Mouse/keyboard-oriented research workspace; broad device coverage is unverified.
Point inspection and scanning	<code>inspection-readout.ts</code> , <code>surface-query.ts</code> ; navigation records	Source/class/geometry disagreement is possible and displayed.
Class focus and selection glow	<code>class-focus.ts</code> , <code>six-class-layer.ts</code> , <code>selection-glow.ts</code>	Highlighting does not alter physical measurements.
Navigation map	<code>explore-map.tsx</code> , <code>explore-map-math.ts</code>	Scene-relative navigation aid; bounded update rate.
Original image reference	<code>image-reference-view.ts</code> ; image-reference tests	Side/top context is the input image, not an independent measurement.
Metric grid and scene scale	<code>metric-grid.ts</code> , <code>scene-scale.ts</code>	Requires justified metric scale; display exaggeration remains visible.
Surface profile and CSV	<code>terrain-analysis.ts</code> , <code>terrain-profile.tsx</code>	Missing samples stay missing; unknown scale cannot produce metre distances.
Source/smoothed terrain	<code>terrain-presentation.ts</code> ; source-preservation tests	Smoothness is a presentation change, not an accuracy score.
Building solids and facades	<code>inference/viewer_mesh.py</code> , <code>building-facades.ts</code>	Illustrative geometry; fine roof configuration remains unestablished.

Feature	Implementation and evidence	Current boundary
Roof fitting	<code>roof-reconstruction.ts</code> , <code>roof</code> dataset/evaluation modules	Experimental and inactive in the current viewer.
Height/semantic candidate gates	<code>evaluation/height_scorecard.py</code> and protocol scripts	Historical sealed identities require their original artifacts.
Local evidence artifacts	GeoTIFF/JSON/CSV outputs and metadata	Local storage has no automatic durable backup or expiry policy.

Implementation paths without a viewer prefix are under `src/msr/`. The current release report identifies which checks ran on the public package. The historical appendix preserves additional dated UI checks and known harness limitations.

14 Model cards and artifact identity

SOURCE docs/MODEL_CARDS.md

Protected above-ground height model

Purpose: estimate surface height above local ground from supported overhead RGB imagery and a target-independent relative-depth prior. Intended for research, demonstration and evaluation, not certified survey or safety-critical decisions.

Architecture: a ConvNeXt V2 Tiny feature encoder with a multiscale height decoder and auxiliary building head, retained inside a guarded ground/building/vegetation surface model. The active fusion is the protected-vegetation variant.

Development lineage: urban training on a geographically separated, source-filtered HighBuild preparation, followed by multidomain work using OpenCanopy. The protected August 29 checkpoint remains the application model. Later residual and class-assisted experiments did not earn a replacement.

Canonical evidence: corrected complete-scene development evaluation gives 13.2426 m RMSE on measured building pixels and 8.3610 m on valid vegetation pixels. These results do not establish a universal error bound or unseen-geography acceptance. Refer to the results chapter for support counts and other metrics.

Known limitations: tall-height compression, domain shift, sparse/estimated urban reference labels, source-date mismatch, pixel-scale uncertainty and inference-context sensitivity. Shape agreement can remain weak after a reduction in aggregate RMSE. Roof detail, individual-tree measurement and hidden-ground recovery are not established capabilities.

Restrictions: the underlying ConvNeXt V2 pretrained weights are distributed under CC BY-NC 4.0. The packaged height derivative remains research/non-commercial; it is not described as an unrestricted commercial model. Training-source terms and attribution remain separate. See the upstream model card and retained notice.

RGB six-class classifier

Purpose: identify ground, buildings, water, roads, low vegetation and trees from 8-bit RGB. It supports overlays and inspection independently of height.

Architecture: SegFormer B0 segmentation model initialized from the ImageNet-pretrained `nvidia/mit-b0` encoder. The active artifact is V3 epoch 1, selected by equal-region OEM development macro F1. Labels use IDs 0–5 and nodata 255.

Input contract: raw 8-bit RGB, scaled to [0,1], with ImageNet normalization inside the model; image validity is independent of class/height labels. Native tiles, up to 1024 pixels with 128-pixel overlap, are combined in probability space.

Evidence: 69.42% equal-region OEM macro F1, 73.29% pooled OEM macro F1 and 83.20% GAMUS macro F1 for the selected V3 development checkpoint. Some class/city and road-boundary retention gates failed. Product integration did not certify it. V1's Christchurch 61.63% macro F1 belongs to that older frozen artifact only.

Limitations: ground/low-vegetation confusion, region-dependent road quality, merged boundaries, sensor/radiometry shift and unknown upstream pretraining coverage. Class probability is not a calibrated probability of correctness. Water labels do not establish water depth; tree labels do not identify species.

Restrictions: the NVIDIA SegFormer licence limits use to non-commercial research/evaluation and requires its notices with redistribution. The complete upstream licence is retained in `licenses/SEGFORMER.txt`.

Relative-depth foundation

Depth Anything V2 Small supplies relative geometry. The upstream Small model is Apache-2.0; this does not make the separate height/classification artifacts or all imagery unrestricted. The downloader pins the upstream revision and checks local file hashes. The prior's output is scale-ambiguous and is not independently a metric elevation estimate.

Public packaging

`model-artifacts.json` is the authoritative inventory of public inference files, their expected sizes, download locations and SHA-256 values. Optimizer, RNG and machine-specific training payloads are removed. The height artifact carries its base architecture so it does not depend on a nested training-checkpoint path.

The packaging audit verifies exact equality of all 240 height tensors (31,428,489 values) and all 210 classifier tensors (3,716,205 values). A separate numerical check compares model outputs. File hashes change when packaging changes; the new hash is not presented as the original training-checkpoint hash. No retraining, quantization or claimed accuracy improvement occurs in this step.

Historical source artifact identifiers remain in the dated evidence. Their existence does not make private archive paths downloadable from this repository. Only artifacts explicitly included in the publication manifest are part of the public release.

15 API and output contracts

SOURCE docs/API.md

The supported default is a local FastAPI service on 127.0.0.1:8000. The viewer uses this endpoint unless configured otherwise. The service exposes OpenAPI at /openapi.json and interactive API documentation at /docs when running.

Endpoints

Method / path	Input	Result
GET /api/health	None	Status and configured device. A ready response does not prove model files have been loaded.
POST /api/predict	Multipart image and optional calibration/reference fields	Job ID, product URLs and metadata after processing.
POST /api/classify	Exactly one completed job_id or allowed demo_id	Six-class map, coverage, raster/metadata URLs and model information.
GET /results/{job}/{file}	Generated job path	Saved image, GeoTIFF, JSON or other output file.

Prediction request

Field	Type	Contract
image	Required file	JPG/JPEG/PNG/TIF/TIFF/JP2/WebP/BMP; supported decoding and nonempty content required.
dem	Optional file	Terrain GeoTIFF; mutually exclusive with GCP and automatic terrain.
gps	Optional file	CSV with elevation_m and map X, y or pixel row, col coordinates.
reference	Optional file	Independent reference GeoTIFF for supported comparison.
reference_kind	ndsm or dsm	Must match the reference quantity.
auto_dem	Boolean	Request automatic terrain; incompatible with manual DEM/GCP.
auto_dem_if_georeferenced	Boolean	Attempt automatic terrain for a georeferenced input under the declared route.

Each file is limited to 512 MiB in the application streaming loop. The image processing grid is limited to 3072 by 3072 dimensions/pixel budget. These are application limits, not a front-proxy upload quota or a promise about every decoder.

Response and files

The response includes job_id, uploaded_bytes, texture_url, height_url, raw_height_url, rdsm_url, relative_depth_url, structures_url, optional probability/semantic/calibration/reference URLs, and both embedded metadata and metadata_url. Optional products are null when unavailable; their absence must not be silently presented as zero.

Product	Interpretation
height_above_ground_m.tif	Learned above-ground estimate, with its model and grid contract.
rdsm_relative.tif	Dimensionless relative surface score.
relative_depth product	Target-independent foundation prior.

Product	Interpretation
<code>presentation_mesh_height_m.tif</code>	Display-oriented surface; not used as a new accuracy claim.
<code>dsm_absolute_m.tif</code>	Aligned terrain plus nonnegative object-height estimate.
<code>gcp_calibrated_surface_m.tif</code>	Surface mapped with supplied control elevations.
<code>validation_signed_error_m.tif</code>	Prediction minus eligible aligned reference.
Probability/class products	Explicitly named internal or six-class outputs; these are different branches.
<code>metadata.json</code>	Input and processing grids, product type, units, calibration, validation and warnings.

Use response URLs rather than assuming every optional filename exists. Relative URLs are resolved against the API origin, not the frontend origin.

Classification request

Uploaded-scene identifiers must match 32 lowercase hexadecimal characters and refer to a completed job with an available original image. Demo identifiers are allowlisted as `urban`, `sparse`, `hilly` and `forest`. Clients do not supply an arbitrary filesystem path or source URL.

The classifier returns `labels_url` for row-major bytes, `raster_url` for class GeoTIFF and `metadata_url`, plus dimensions, coverage and scope. Class IDs are 0 ground, 1 buildings, 2 water, 3 roads, 4 low vegetation, 5 trees; 255 is invalid. Area is reported only when the grid has a supported projected CRS and unit scale.

Errors and operational behavior

Unsupported image/reference suffixes return 415; invalid option combinations, empty inputs or reference kinds return 400; oversized files return 413; invalid processing/classification inputs use 422; unexpected runtime failures use 500. The client distinguishes unavailable API, failed prediction, unsupported identification and retryable classification failure.

The model runtime is lazy and serializes GPU execution in one process. There is no external work queue, cancellation guarantee or multi-user authentication. Successful jobs persist locally; failed prediction jobs are removed by the error path. Classification failure cleanup and parser-level resource control have separate limitations described in the security review.

Changing the public package's metadata prefix to `MSR_` is a publication naming change. Source numerical arrays and their scientific interpretations must be preserved and checked independently from metadata byte identity.

16 Installation, operation and reproduction

SOURCE docs/SETUP.md

Supported setup

The checked development host uses Windows x64, Python 3.10.11, Node.js 22.x and an NVIDIA RTX 4070 Laptop GPU with about 8 GiB VRAM. Python declares support for 3.10–3.12. CPU inference is available but has not been given the same performance claim as GPU execution. Linux/macOS commands are provided as portable equivalents, not as claims of a completed unrelated-host installation.

Clone the public repository and work from its root. Do not copy a `.venv` from another machine. Create the environment and install a compatible Torch pair:

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install torch==2.7.0 torchvision==0.22.0 --index-url
https://download.pytorch.org/whl/cu128
.venv/Scripts/python.exe -m pip install -r requirements.txt
.venv/Scripts/python.exe -m pip install -e . --no-deps
```

For a CPU-only installation, use the same Torch versions with the PyTorch CPU wheel index instead of `cu128`, then run with `--device cpu`. On Linux/macOS, replace `.venv/Scripts/python.exe` with `.venv/bin/python`. The direct application dependency versions are recorded in `requirements.txt`; `pyproject.toml` describes package extras. The frontend has an npm lockfile.

Acquire model artifacts

Read the third-party notices before acquiring the restricted research weights. Then run:

```
.venv/Scripts/python.exe scripts/download_models.py --include-foundation
.venv/Scripts/python.exe scripts/download_models.py --include-foundation --verify-only
```

The manifest pins the packaged height/classifier SHA-256 values and a foundation revision. Downloads use temporary files, verify size/hash and replace the target only after verification. The downloader does not deserialize model payloads. Training archives are not required for normal inference. They are required for reproducing full training or historical dataset-dependent analyses.

Start the application

In the first terminal:

```
.venv/Scripts/python.exe scripts/serve_api.py --device cuda
```

In a second terminal:

```
npm --prefix viewer ci
npm --prefix viewer run build
npm --prefix viewer run start -- --hostname 127.0.0.1 --port 3000
```

Open `http://127.0.0.1:3000`. Saved sample scenes work without downloading model weights, but automatic classification and new-image reconstruction require the API. `/api/health` reports the configured device; the first prediction still has to load and verify the actual model path.

The default API and viewer use ports 8000 and 3000. `--port` selects the API port; `NEXT_PUBLIC_MSR_API_URL` selects the frontend's API base. `MSR_VIEWER_ORIGINS` configures the API's comma-separated allowed viewer origins for a deliberately changed local port. Public hosting is outside this release's verified scope.

Verify and inspect evidence

```
.venv/Scripts/python.exe -m pip install -r requirements-dev.txt
.venv/Scripts/python.exe -m pytest tests -q
npm --prefix viewer run test
npm --prefix viewer run typecheck
npm --prefix viewer run build
.venv/Scripts/python.exe scripts/verify_publication.py
```

The public test suite explicitly skips 26 historical checks bound to original source hashes or unbundled experiment archives. They passed in the original development workspace, but the renamed source package is not that exact sealed artifact.

--run-historical-seals attempts them only when their original prerequisites are intentionally available; it is not required for the public inference workflow. Numerical and application assertions are not loosened.

Reproduce research responsibly

Historical configs and scripts preserve experiment structure, but some locators point to historical data/experiment arrangements. Start from the corresponding protocol, acquire the exact allowed source revision, prepare the manifests and validate shapes, units, masks and geographic overlap. Create new source/config/data bindings for a new run rather than editing an old seal to imply identity.

Use the declared comparator, budget, masks, selector and safety gates. Keep full scene and app-tiling evaluation separate. Do not repurpose a consumed external test as new validation. The reserved Bay of Plenty package is not needed for setup, smoke tests or this release verification and remains untouched.

Rebuild the handbook

Install the documentation dependencies and run the included builder:

```
.venv/Scripts/python.exe -m pip install -e ".[docs]"
.venv/Scripts/python.exe scripts/build_documentation.py
```

The chapter order is explicit in the builder. Current chapters precede dated history; the source inventory and evidence paths remain available. The builder creates a contents page and PDF bookmarks from those sources. Figures use actual recorded values or clearly labelled architecture diagrams.

Recovery

If the viewer loads but classification fails, check the local API and classifier artifact hash. If inference fails, check the configured device, height and foundation files, supported input type and server log. A relative image cannot be made geographically metric by changing a display setting. Do not select a new training checkpoint merely to clear an error.

Local outputs/web_jobs/ stores uploads and derived products. Back up selected results explicitly. Stop only processes you started; do not terminate unrelated Python/Node processes on a shared development machine.

17 Performance and application verification

SOURCE docs/PERFORMANCE.md

Checked environment

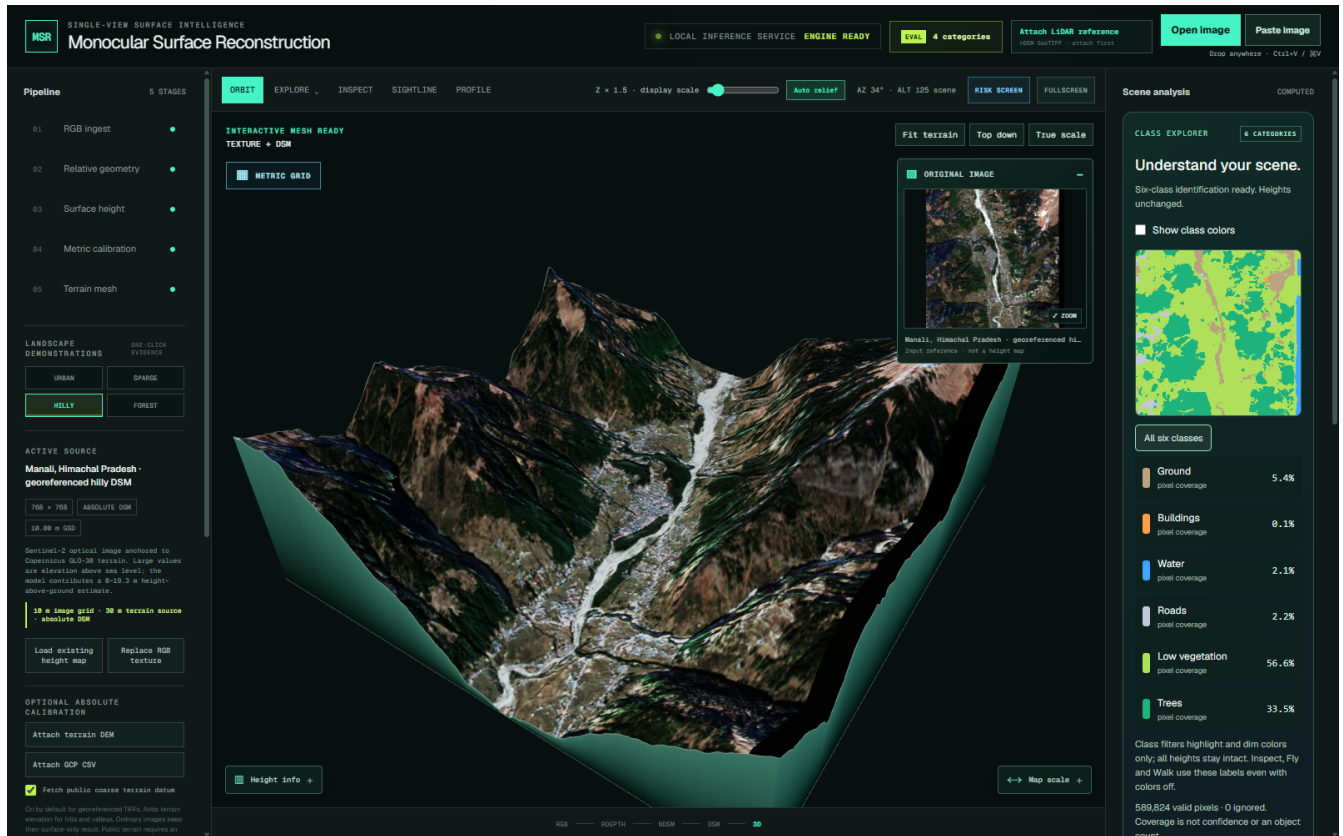
Release checks ran on Windows with Python 3.10.11, an NVIDIA RTX 4070 Laptop GPU (about 8 GiB VRAM), PyTorch 2.7.0 with CUDA 12.8, and Node.js 22. The frontend dependency versions are pinned by viewer/package-lock.json. They were refreshed for publication; model weights and scientific selectors were not changed by that dependency update.

Actual workflow check

The browser loaded the production SSR handler and hydrated the real React/Three.js application. API calls were routed through the real FastAPI TestClient and local release weights. This opened no application listening socket. Urban, hilly and forest saved scenes rendered; a Copenhagen RGB JPEG was then uploaded and produced new raw height, relative depth, probability, presentation and metadata products. Automatic classification ran on the actual image. Result URLs returned data. A missing-upload request returned 422. The 390-pixel viewport had no horizontal overflow. Browser console/page errors: zero.



Packaged urban workspace



Georeferenced hilly workspace

The bridge measured the following sequential request durations during that check:

Request	In-process elapsed	HTTP result
/api/classify	3.076 s	200
/api/classify	0.201 s	200
/api/classify	0.336 s	200
/api/predict	3.896 s	200
/api/classify	0.401 s	200

These are individual local observations, not percentiles, throughput, a production latency promise or an internet-network benchmark. The first classifier request includes lazy loading; the later requests use different scene sizes/content and must not be treated as a controlled speed comparison. The prediction includes the first height/foundation load of this process. Browser render time, driver startup, transport and other hardware can change the user-visible duration.

Numerical integrity and exports

The release bundles preserve 240 height-model tensors and 210 classifier tensors exactly. A seeded CPU float32 comparison on a synthetic 128 x 160 input produced 20 exactly equal model outputs between original and packaged height loaders. Eight newly generated GeoTIFF exports were reopened and had finite values on their valid support. These checks support packaging integrity, not geographic accuracy.

Test and build outcome

- Python public package: 826 passed, 26 explicit historical-archive skips, zero failures.
- Original workspace: 852 passed, zero failures.
- Viewer deterministic tests: 155 passed, zero failures.

- TypeScript and production build: passed.
- Browser: 10 completed checks, zero console/page errors.

The production build emits a large-bundle notice and vinext cannot statically classify every route. Mesh downsampling bounds interactive geometry; full raster resolution still consumes browser memory and upload processing time. Performance work should measure decode, inference, transfer, mesh construction and frame time separately rather than attributing every delay to the model.

Advisory review and scope

The reviewed npm dependency tree reports 0 critical, 0 high, 4 moderate and 1 low advisories after updates. The remaining moderate chain is vinext's OG-rendering dependency path through Satori/fflate; the low advisory concerns esbuild's Windows development server. This viewer does not implement OG image generation and is scoped to local research use. The counts are the scanner's dated observations, not a general security certification. See the machine-readable summary for package names and advisory links.

Reproduce the browser scope

Build the viewer, install its Playwright browser, and point MSR_PYTHON to the environment containing this package and models. From viewer/, run `node scripts/verify-browser.mjs`. Set `MSR_TEST_DEVICE=cpu` for CPU or supply `MSR_BROWSER_CDP` to attach to an existing controlled Chromium instance. The script records evidence and screenshots under docs/; review them before publishing an updated report. Model weights and the pinned foundation are required. No sealed research holdout is used. This does not replace a future network deployment test.

18 Security review and data handling

SOURCE docs/SECURITY_REVIEW.md

This is a scoped source and test review of a local research application. It is not an independent penetration test or a certification for internet-facing deployment. The public repository does not itself operate an inference service.

Reviewed boundaries

Boundary	Implemented safeguard	Remaining limit
File submission	Allowed filename suffixes; nonempty checks; streaming byte counters.	Multipart parsing precedes some application checks; filenames are not proof of safe file content.
Decoded image size	Bounded raster processing dimensions and pixel budget.	Native image/GDAL parsers are not sandboxed; byte limits alone do not bound every decoding resource.
Calibration options	Mutually exclusive DEM/GCP/automatic terrain, explicit reference kind.	Compatibility depends on real input provenance, datum and content.
Classification source	Allowlisted demo IDs, fixed job-ID format, resolved directory checks.	This is path validation, not authenticated ownership.
Model execution	Runtime lock serializes GPU work within one process.	Requests can queue; there is no distributed quota, rate limiter or bounded public admission queue.
Model acquisition	Release downloads are size/hash checked before replacement.	Only documented trusted model files should be loaded; legacy training loaders can deserialize Python checkpoint payloads.
Browser origins	Explicit local CORS origin list, configurable by the operator.	CORS is not authentication and does not prevent all non-browser requests.
Outputs	Random job identifiers; static file service rooted at results directory.	Anyone with access to the service and a result URL can read it; there is no per-user authorization.

Data lifecycle and privacy

Prediction uploads are saved under a random local job directory together with derived products and metadata. Successful jobs are retained until the operator removes or archives them. Failed prediction jobs are removed in the documented error path. No automatic retention deadline is implemented. Classification failures can have their own output/cleanup behavior and should be inspected during operational maintenance.

The API serves generated files from the result root. That directory can include the uploaded image and supplied reference/control files, not only the final mesh. Treat the entire directory as potentially sensitive. There is no cloud account, consent database, multi-tenant access model or encrypted object-storage service.

Automatic terrain uses a configured provider URL template rather than arbitrary client-provided URLs. It still requires network access and discloses the requested geographic area to the provider through tile requests. Raw model input is not sent to an external inference provider by the documented local prediction path.

Publication checks

The source package excludes environment files, credentials, hosting-account state, runtime logs, full training datasets and private development history. Its publication scan checks text, file paths, PDF text/metadata and bundled data metadata against the release's allowed public identity. Required third-party attribution remains.

The release also records numerical equality of bundled raster values after metadata normalization, and exact equality of model tensors after inference-only packaging. Metadata cleanup is not described as byte-identical preservation of those files.

Before a hosted service

A separate hosted design needs authentication or deliberate public access policy, request admission/quotas, upload deadlines at the ingress layer, parser isolation, error-detail handling, storage retention, per-user result authorization and a GPU work queue. HTTPS and deployment/platform limits would need their own verification. These are identified missing controls, not features claimed by this local release.

Dependency manifests record the checked versions. They do not establish absence of vulnerabilities. Do not infer internet-facing security from a passing build or from source publication alone.

19 Release engineering and verification

SOURCE docs/RELEASE.md

Snapshot identity

Version 0.1.1 is a research prototype snapshot with technical documentation as of 28 September 2026. Source, documentation and inference artifacts are versioned together. The Git tag identifies the published source; the artifact manifest independently identifies weight files and the pinned foundation revision.

This is a fresh publication repository under the scientific name Monocular Surface Reconstruction. Original development history is not imported. Public historical records carry a banner explaining name/path normalisation; they are not represented as untouched original source seals.

Publication transformations

The package namespace is `msr`. Frontend metadata, UI branding, environment names, Python imports, commands, notebook-independent scripts and current documentation use the scientific identity. The development application remains separately available in its original workspace. Generated outputs, private data, model caches, local environments and credentials are excluded from Git.

The height bundle embeds the base architecture and all protected tensors so its loader does not require an old workstation checkpoint path. The classifier bundle retains its architecture, inference contract and weights. Optimizer/RNG and unused training state are not required by normal inference. State tensor equality is checked for 240 height tensors (31,428,489 values) and 210 classifier tensors (3,716,205 values). This is a packaging check, not new accuracy evidence.

Public GeoTIFF metadata is normalised while raster arrays, validity masks, CRS and affine transforms are compared exactly. Restricted source imagery is omitted rather than relicensed. Upstream model licences and data attribution remain.

Test scope

The original workspace completed 852 Python tests. The portable package completed 826 tests and explicitly skipped 26 cases that require exact historical source seals or experiment archives not included here. The skip manifest names each affected test; numerical assertions are not weakened. Skips are visible in test output and can be attempted with `--run-historical-seals` when their prerequisites exist. The two counts describe different verification contexts.

The viewer has 155 deterministic tests covering navigation, terrain analysis, classification state, image import and geometry behavior. TypeScript and a production build are checked. Dependency versions and the final advisory count are in the machine-readable verification summary.

The browser harness exercises the actual production request handler and actual FastAPI TestClient with local model files. It does not open an application listener. This validates real rendering, request handling and generated products under an in-process transport; it does not establish network deployment, TLS, reverse-proxy, authentication or multi-user service behavior.

Remaining operational boundaries

The Python checks use the established environment with the public package source on its import path. This release does not claim a fresh GPU-driver installation on every supported OS. The application is a local research tool with a serialized inference runtime. Public hosting, concurrency hardening and storage lifecycle controls remain roadmap work. Large browser bundle notices are retained as a performance limitation, not suppressed to imply a clean optimisation result.

Run `scripts/verify_publication.py` for source/PDF identity, link, size and manifest checks. Rebuild documentation from its included source before changing a dated release. Never edit a historical result merely to make a check pass.

The first clean hosted CPU check exposed an omitted direct Requests dependency: 824 tests passed and two backbone-acquisition import checks failed. Version 0.1.1 declares Requests explicitly. The per-commit GitHub workflow checks the corrected installation independently of the original workstation environment.

20 Ongoing work and research roadmap

SOURCE docs/ROADMAP.md

Status as of 28 September 2026. The items below are research directions and acceptance requirements, not promises that more training will automatically solve the measured weaknesses.

Priority 1: reliable object height

Audit tall-building reference support, per-instance consistency, acquisition alignment and physical pixel scale. Test context-matched training and per-instance supervision as separate interventions. Retain fixed native-resolution development support and report short/medium/tall groups so one aggregate cannot hide a regression. The large concentration of squared error in tall structures justifies this priority.

An eligible development candidate must improve the declared object-height metric while preserving bias, MAE, correlation, shape/instance behavior and other groups under the applicable scorecard. The canonical gate uses bounded RMSE/MAE/bias regressions, no decrease in correlation/R-squared and at least one material gain. Passing development is still not final external acceptance.

Priority 2: geographic classification transfer

Improve ground/low-vegetation separation and road boundaries while retaining behavior in the original cities. Use matched replay/teacher-regularization or another explicitly defined intervention with the same budget and selection rule. Multiple independent seeds would help distinguish small effects from run variation. Keep class ontology mapping and image radiometry fixed or report their change as part of a package experiment.

Freeze the candidate before evaluating reserved geography. Do not reinterpret Christchurch or the inspected all-class scenes as untouched. V3's development improvement and authorized app integration do not close the external evidence gap.

Priority 3: final metric-height transfer

After an eligible candidate and inference protocol are frozen, seal a one-shot evaluation plan for the prepared Bay of Plenty package. Report all-valid, object-only and tall-object results regardless of outcome, with dates, source identities, mask counts and datum/acquisition limitations. A failed or interrupted attempt still consumes the test and must be disclosed.

Priority 4: usability and operational quality

Measure unfamiliar-user task completion on the actual released workflow: choose a scene, identify units, distinguish class from geometry, inspect a profile, compare a reference and export evidence. Obtain domain review of the interpretation, not only visual approval. Add unrelated physical-device checks before making broad browser/GPU claims.

If public hosting is desired, design it as a separate service project with authentication/admission, workload limits, result isolation and retention. The present local package is not a validated multi-user deployment.

Deferred geometry extensions

Detailed roof shapes and tree-object geometry remain research modules rather than certified features. Reintroduce them only with useful coverage, reference evidence, clear source/display separation and interaction verification. Flood simulation, snow depth, species recognition and before/after change detection would require their own labels, methods and validation; a current class map does not provide them.

Documentation maintenance

Update the status date when a new release is actually checked. Keep the current overview short enough to assess, preserve dated experimental records, and add new evidence instead of overwriting inconvenient outcomes. Every future release should state its exact model/source identity and which conclusions changed.

21 Glossary and product dictionary

SOURCE docs/GLOSSARY.md

Term	Meaning in this project
DSM	Surface elevation in an absolute datum, including objects above terrain.
DEM / DTM	Terrain elevation used as a datum; provider details determine what it represents.
nDSM / AGL	Height above local ground; zero is local ground, not sea level.
Relative depth	A scene-relative depth ordering/shape product without automatic metric scale.
GSD	Ground sampling distance per pixel; distinguish horizontal axes and projected/geographic units.
CRS	Coordinate reference system for horizontal coordinates; does not by itself establish a vertical datum.
Valid mask	Pixels supported by input/reference observations and the declared evaluation rules.
Nodata	Explicit absence of a valid value; not equivalent to zero metres or semantic ground.
Semantic class	A categorical predicted label; not a measurement of height.
Confidence	Model-derived score/probability; not a calibrated probability that height is correct.
Promotion	Acceptance as the new scientific/default model under a predefined protocol.
Integration	Activation of a model in the application; separate from benchmark promotion.
Protected model	Retained urban/guarded-vegetation baseline whose measured behavior is not silently replaced.
Presentation surface	Derived mesh suitable for display; retain the raw scientific export separately.
Development	Data used to choose models, thresholds, sampling or implementation behavior.
External test	Predeclared data with appropriate novelty, untouched until a frozen evaluation; then consumed.
Sealed artifact	Files bound to recorded hashes/configuration; renaming source means it is no longer the same source seal.
Macro F1	Equal-class mean F1 under the specified absent-class policy.
Equal-region F1	Average across named regions, preventing sample-rich regions from dominating selection.

Output interpretation

Raw height TIFF contains the learned height-above-ground product on the working grid. Relative-depth TIFF preserves a separate foundation prediction. A DEM-aligned absolute DSM combines compatible terrain and above-ground estimates. A GCP surface depends on supplied point coverage and residual fit. Probability TIFFs and the six-category preview have their own class/validity conventions. A displayed solid building or smoothed hill is an interpretation of these products, not a new ground-truth observation.

See [API](#) for the response fields and [Scientific contracts](#) for requirements governing units, alignment and export behavior.

22 References and upstream resources

SOURCE docs/REFERENCES.md

References identify methods, upstream software, datasets and applicable source terms. A citation does not imply endorsement or waive the linked licence. Access review: 28 September 2026. Original project measurements are traceable through the dated records in [Source inventory](#).

Methods and models

1. Woo et al., *ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders*, CVPR 2023. [Paper](#), [upstream implementation](#), [specific timm weight card](#).
2. Xie et al., *SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers*, NeurIPS 2021. [Paper](#), [NVIDIA upstream](#), [MIT-B0](#).
3. Yang et al., *Depth Anything V2*, 2024. [Paper](#), [small Hugging Face model](#). This release pins the small model, whose licence differs from some larger family members.

Data

1. [HighBuild-1M dataset and source inventory](#). Mixed imagery sources; use the supplied source inventory rather than a blanket imagery licence.
2. Fogel et al., *Open-Canopy: Towards Very High Resolution Forest Monitoring*, 2024. [Paper](#), [dataset](#).
3. [GAMUS paper](#), [EarthNets multimodal segmentation repository](#). Retain local-unit and spatial-provenance limitations described in this handbook.
4. Xia et al., *OpenEarthMap: A Benchmark Dataset for Global High-Resolution Land Cover Mapping*. [Paper](#), [dataset release](#), [regional attribution](#).
5. [LINZ aerial imagery](#) and [LINZ elevation](#). Bay of Plenty products retain BOPLASS and LINZ attribution and acquisition metadata.
6. [Copernicus Sentinel data access](#) and [Copernicus DEM](#). Optical and terrain resolution are not interchangeable.
7. [USGS SRTM](#). Used as the separate coarse hilly comparison reference.
8. [US3D / 2019 IEEE GRSS Data Fusion Contest](#). Raw source imagery and reference rasters are omitted from this public package because redistribution is prohibited by the supplied terms.
9. [GeoDanmark spring orthophoto service](#) and [Danish data-use terms](#).
10. [PDOK aerial imagery service](#). See provider metadata and the retained HighBuild inventory for the Amsterdam source.

Software and publication tooling

The implementation uses [PyTorch](#), [timm](#), [Transformers](#), [Rasterio](#), [FastAPI](#), [NumPy](#), [SciPy](#), [Three.js](#), [GeoTIFF.js](#), [React](#), [Next.js](#), [vinext](#), [Vite](#), [Playwright](#) and [ReportLab](#). Package lockfiles and dependency metadata identify the checked versions. Each dependency retains its own licence.

23 Research source inventory

SOURCE docs/SOURCE_INVENTORY.md

The documentation is grounded in implementation, tests, run protocols, outcome records and explicit decision notes. The following dated records preserve what was attempted, what was measured, what was rejected and what became app behavior. They are grouped in the handbook after the current technical chapters.

Historical records have public names and paths normalised. A local artifact path mentioned inside a record is an archive locator, not a promise that a large training output is committed. Exact public bytes are indexed in docs/evidence/source-inventory.json. New current chapters resolve conflicts in favor of the state documented on 28 September 2026 while retaining older outcomes.

Record	Subject	Words
ACCURACY_RECOVERY_2026_09_15	Accuracy recovery: 15 September 2026	2085
BENCHMARK_PROTOCOL	Monocular Surface Reconstruction locked benchmark protocol	995
CLASS_ASSISTED_HEIGHT_FAST_V2_20260919	Faster class-assisted height comparison	667
CLASS_ASSISTED_HEIGHT_FAST_V2_OUTCOME	Class-assisted height comparison: completed, not promoted	709
CLASS_ASSISTED_HEIGHT_LAUNCH_20260919	Class-assisted height comparison: launch handoff	365
CLASS_ASSISTED_HEIGHT_LOW_SURFACE_V3_OUTCOME	Low-surface protection outcome — not promoted	758
CLASS_ASSISTED_HEIGHT_LOW_SURFACE_V3_PROTOCOL	Low-surface protection: bounded matched experiment	627
CLASS_ASSISTED_HEIGHT_PILOT_PROTOCOL	Class-assisted height pilot — controlled experiment	1239
DEMO_RUNBOOK	Monocular Surface Reconstruction research Prototype Runbook	408
EXPANDED_GAMUS_PLAN	Monocular Surface Reconstruction expanded GAMUS plan	2441
EXPLORE_MAPPING_2026_09_27	Explore mapping — 27 September 2026	551
EXPLORE_WALKTHROUGH_2026_09_19	Explore: Fly and Walk	725
EXTERNAL_METRIC_HEIGHT_HOLDOUT	External metric-height holdout: Bay of Plenty v1	824
FULLSCREEN_ESCAPE_2026_09_19	Two-step fullscreen Escape	281
GAMUS_DIRECT_HEIGHT_PROTOCOL	GAMUS direct-height experiment protocol	731
GAMUS_GEOGRAPHIC_HOLDOUT	GAMUS classifier-head geographic exclusion protocol	1175
GAMUS_HEIGHT_SAFE_V2_PROTOCOL	GAMUS height-safe v2 pilot	441
GAMUS_PREPARATION_AUDIT	GAMUS preparation audit v1	624
GAMUS_SIX_CLASS_DECISION_BIAS_CALIBRATION	GAMUS six-class decision-bias calibration	853
GAMUS_SIX_CLASS_EXPANSION_PROTOCOL	GAMUS six-class expansion protocol	1069
GAMUS_SIX_CLASS_EXPERIMENT_STATUS	GAMUS six-class experiment status	1771
GAMUS_SIX_CLASS_V4_HIERARCHICAL_HEAD_DESIGN	GAMUS six-class v4: isolated hierarchical vegetation head	1583
HEIGHT_BAND_SAMPLING_V1_HANDOFF	Active height-band sampling experiment	457

Record	Subject	Words
HEIGHT_BAND_SAMPLING_V1_OUTCOME	Sampling comparison: no accepted replacement	395
HEIGHT_BAND_SAMPLING_V1_PROTOCOL	Height-band sampling: fixed four-epoch matched experiment	701
HEIGHT_PROMOTION_SCORECARD	Monocular Surface Reconstruction metric-height scorecard v1	1017
HIGHBUILD_INSTANCE_EVALUATION	HighBuild per-building evaluation	667
HIGHBUILD_RGB_CLASSIFICATION_AUDIT_2026_09_19	HighBuild classification: what works and what needs improvement	1083
INSPECT_CLASSES_AND_WALK_POLISH_2026_09_19	Inspect class isolation, faster Walk and stable import grid	592
METRIC_GRID_VIEW_2026_09_19	Metric grid view — 19 September 2026	718
OEM_ALL_SIX_CLASS_TEST_2026_09_19	All-six-class mixed-scene test — completed	946
OPEN_CANOPY_RICH_LABELS_AND_HOLDOUTS	OpenCanopy richer-label and holdout contract	981
OPENEARTHMAP_CHRISTCHURCH_EXTERNAL_HOLDOUT	External six-class holdout: Christchurch OpenEarthMap v1	642
PROGRESS	Monocular Surface Reconstruction progress and recovery record	3676
RESIDUAL_HEIGHT_POST_RUN	Residual-height evaluation handoff	535
RESIDUAL_HEIGHT_V1_OUTCOME	Residual-height V1: completed outcome	1238
RGB_MULTIREGION_V3_LAUNCH_2026_09_19	Multi-region V3 launched — 19 September 2026	332
RGB_MULTIREGION_V3_OUTCOME	RGB multi-region V3: completion and error analysis	471
RGB_MULTIREGION_V3_PROTOCOL	Multi-region RGB classification V3	938
RGB_SAMPLING_V2_OUTCOME	RGB sampling V2: completed experiment and decision	797
RGB_SAMPLING_V2_PROTOCOL	RGB V2: controlled crop-sampling continuation	894
RGB_SEGMENTER_V1_CHRISTCHURCH_PROTOCOL	RGB segmenter V1: Christchurch external-transfer protocol	771
RGB_SEGMENTER_V1_CHRISTCHURCH_RESULTS	RGB segmenter V1: completed Christchurch transfer test	764
RGB_SEGMENTER_V1_DC_VEGETATION_AUDIT	RGB segmenter V1: DC low-vegetation audit	369
RGB_SEGMENTER_V1_OUTCOME	RGB-only six-class pilot: completed outcome	755
RGB_SEGMENTER_V1_PROTOCOL	Independent RGB six-class identification experiment	1181
RGB_SEGMENTER_V1_TEST_REPORT	Independent saved-checkpoint test	572
ROOF_RECONSTRUCTION_STAGE1_2026_09_19	Roof reconstruction: stage 1, 19 September 2026	1228
ROOFTOP_RELIEF_2026_09_19	Predicted rooftop relief — 19 September 2026	647
SCENE_IMAGE_REFERENCE_2026_09_27	Original-image reference in the 3D viewport	280
SELECTION_GLOW_2026_09_27	Active inspection boundary	380
SEMANTIC_EXPERT_HEIGHT_V1_HANDOFF	Active six-class guided height experiment	442
SEMANTIC_EXPERT_HEIGHT_V1_OUTCOME	Six-class-guided height experiment: completion and error analysis	711
SEMANTIC_EXPERT_HEIGHT_V1_PROTOCOL	Six-class soft-routed height corrections: prospective experiment	532
SIX_CLASS_APP_PREVIEW_2026_09_19	Six-class application preview — 19 September 2026	704

Record	Subject	Words
SIX_CLASS_DEFAULT_INTEGRATION_2026_09_20	Six-class default integration — 20 September 2026	868
SIX_CLASS_OFFLINE_EXPORT	Offline six-class export	581
SIX_CLASS_SCANNER_AND_HEIGHT_AUDIT_20260919	Six-class scanning and height-training audit — 19 September 2026	733
SMOOTHNESS_AND_ACCURACY_REVIEW_2026_09_26	Smoothness and accuracy review — 26 September 2026	858
TERRAIN_IMPORT_AND_PROFILE_2026_09_26	Terrain imports and measured profiles — 26 September 2026	1409
TERRAIN_VIDEO_STYLE_2026_09_26	Terrain presentation matched toward the supplied video	831
VIEWER_CONTROLS_2026_09_19	Viewer controls update — 19 September 2026	568
VIEWPORT_DETAILS_ZOOM_2026_09_27	Compact scene details and original-image zoom	290

24 Third-party notices and redistribution boundaries

SOURCE THIRD_PARTY_NOTICES.md

Status as of 28 September 2026. This repository is a research source release. No blanket open-source or commercial-use licence is granted for the original project code, model derivatives or aggregated data. Public visibility is not a universal licence. Third-party components retain their own terms.

Model weights

Artifact	Upstream basis	Terms
Height model	ConvNeXt V2 tiny FCMAE pretrained weights distributed through timm	CC BY-NC 4.0; research/non-commercial use. The original model licence is retained in <code>licenses/CONVNEXT_MODEL.txt</code> .
Six-category classifier	NVIDIA MIT-B0 / SegFormer	NVIDIA non-commercial research/evaluation licence; see <code>licenses/SEGFORMER.txt</code> . Redistributed derivative weights retain these restrictions.
Relative-depth foundation	Depth Anything V2 Small, Hugging Face conversion	Apache License 2.0 for this specific small model; acquired separately from the pinned upstream revision.

The height and classifier release assets are inference-only derivatives, with original learned tensors preserved and source-machine/training metadata removed. Their hashes are in `model-artifacts.json`. This work modified task heads, training, fusion and packaging; it does not claim authorship of upstream architectures or pretrained weights. Keep this notice and the original model licences with copies of release weights. The unrestricted licence of an implementation library does not replace the more restrictive licence of its weights.

Included demonstration assets

Public paths	Attribution and source
<code>viewer/public/demo/copenhagen_*</code> , urban original image	Contains GeoDanmark orthophotography, Danish public data providers, distributed through the HighBuild source inventory; CC BY 4.0. Cropped/reprocessed and accompanied by model-derived products.
<code>viewer/public/demo/landscapes/sparse/</code> , sparse originals	Contains PDOK / Dutch aerial imagery, CC BY 4.0 under the retained source inventory. HighBuild annotations and project model-derived products are identified separately.
<code>viewer/public/demo/forest_validation/</code>	Open-Canopy / AI4Forest; SPOT optical imagery and IGN LiDAR-derived reference under the dataset's Etalab Open Licence 2.0. Fogel et al., 2024. Modified for the selected demonstration.
Hilly landscape, hilly validation and hilly originals	Contains modified Copernicus Sentinel data 2025; Copernicus DEM GLO-30 terrain. Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. Separate SRTM reference: USGS/NASA. Reprojected/resampled; modified model products are not official terrain products.

The public package excludes Chicago/Esri smoke imagery and US3D/DFC2019 raw imagery/reference rasters. Historical numerical measurements remain labelled as historical selected-scene results. Their presence does not grant access to the restricted source data. Full HighBuild, GAMUS, OpenEarthMap, Open-Canopy, LINZ and roof-research archives are not redistributed.

The upstream HighBuild source inventory is preserved unchanged in `licenses/HIGHBUILD_SOURCE_INVENTORY.md`. Its project/release statements refer to the upstream dataset. It is source evidence, not a new instruction or claim that every source listed there is included here. See `docs/DATA.md` for actual roles.

Data acknowledgements

We acknowledge the creators and providers of HighBuild-1M, Open-Canopy, GAMUS, OpenEarthMap, GeoDanmark, PDOK, IGN, Copernicus, USGS, NASA, LINZ and BOPLASS. We acknowledge Johns Hopkins University Applied Physics Laboratory and IARPA for the historical US3D data, and IEEE GRSS IADF for the associated dataset initiative. Provider links and scientific citations appear in `docs/REFERENCES.md`.

Software

Python and npm dependencies are installed from their upstream distributions and retain embedded licence notices. No `node_modules`, Python environment, proprietary system font or unrelated source tree is bundled. Fonts embedded in the generated handbook are used for document rendering; the font software is not separately distributed. Contact the repository owner for original-project reuse permissions.

25 Accuracy recovery: 15 September 2026

SOURCE docs/history/ACCURACY_RECOVERY_2026_09_15.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

What the audit actually found

The application still uses the protected 29 August model. Later experiments have not earned a release. Changes to the app interface and evaluation tooling are useful work, but are not evidence that height accuracy improved.

On the corrected full native scene development benchmark, the protected model has 13.243 m measured-building RMSE and 8.361 m vegetation RMSE. The older crop scores used different pixels and label rules. They cannot be substituted for these scores.

The fair DC + Philadelphia classification replay records six-class macro F1 of 56.48% for spatial V3 and 55.11% for hierarchical V4. The hierarchy did not improve the combined six-class result. V4 remains rejected.

Why the small height runs stalled

The four-epoch height-safe control changed only 98 of 31,428,489 parameters. Its pooled GAMUS validation RMSE moved from 7.1211 m to 7.1276 m. No epoch passed the improvement criteria.

A diagnostic on four fixed GAMUS **training** crops found:

- The canopy/semantic adapter sees RGB, scalar relative depth, base height, and building probability. It has a nominal 5 by 5 pixel receptive field, with no direct access to the backbone's richer encoder/decoder features.
- Mean effective correction strength was 0.0266.
- The optional hard canopy gate fired on about 0.085% of the true vegetation pixels in these training crops.
- The final vegetation loss passed about 1.45% of its direct height gradient through to the canopy output, approximately 69 times weaker.
- In the protected-vegetation fusion formula, the separate building residual expert does not enter the final building height. Improving this expert can therefore leave final building predictions unchanged.

These are diagnostic observations, not validation accuracy claims. Evidence:
outputs/diagnostics/height_learning_bottlenecks_v1/report.json.

Data and processing corrections

1. **HighBuild unknown pixels stay unknown.** Urban regression uses measured height support intersected with building footprints. Estimated heights are kept out of the primary metric. No height label means no height loss.
2. **Supervised urban crops.** Some buildings cover under 1% of a tile. The new training recipe samples a crop containing measured support. Validation still covers the complete fixed scene.
3. **Independent prior maps.** Cached Depth Anything V2 maps use their stored 0-1 values, independent of where height labels happen to exist.
4. **Independent masks.** Image validity, class availability, and height availability remain distinct. Road and water height artifacts are excluded while their classification annotations are retained.
5. **Equal contributions to the loss.** Sources and object types with sparse labels must not be outweighed just because forest/ground have more pixels.

6. Inference protocol audit. Compare the full-native benchmark with the actual app's 512/128 tiling on the identical reference pixels. GroupNorm makes crop size/context matter. This audit changes inference protocol and precision; it is not a claim of learned model improvement.

New experiment: residual_height_v1

App padding audit: a measured processing issue

The paired full-native/app audit evaluated all 160 urban and 120 forest scenes on the same 31,011,260 reference pixels. App tiling raised measured-building RMSE from 13.243 to 13.442 m and vegetation RMSE from 8.361 to 11.882 m.

A separate controlled forest comparison then held fp16 precision fixed and changed only small-image padding. All 120 scenes and 17,155,570 valid pixels were retained. Minimal padding reduced vegetation RMSE from 11.882 to 8.366 m and forest RMSE from 8.639 to 6.707 m. However, forest-ground RMSE worsened from 4.297 to 4.885 m; 89 scenes improved and 31 worsened. Therefore the optional `padding_policy="minimal"` is tested but **not enabled globally**. The API and backward-compatible `fixed_tile` default remain unchanged. This is an inference correction, not a learned-model improvement. Evidence: `outputs/evaluation/open_canopy_padding_v1/report.json`.

Training correction

The new model retains the entire protected model as a frozen feature source. A separate copy of its multi-scale decoder learns a correction to the final height. Its final correction layer starts at zero, so the initial prediction is exactly the protected prediction. The correction receives the encoder's image features, RGB, cached relative depth, and the original predicted height. Reference masks are used only by the loss and evaluator.

This increases learnable capacity from 98 numbers to a full spatial decoder while keeping the classification computation frozen. A height error directly teaches this decoder without passing through the earlier expert gates.

Training uses QC-approved GAMUS training data plus corrected measured HighBuild and OpenCanopy replay. First perform a 48-step feasibility check on four fixed training samples. Require at least 15% loss reduction, nonzero decoder gradients, and unchanged protected tensors/classification. A successful feasibility check only shows that optimization works; it is not generalisation. Restart from the zero correction before the real run.

The initial run has up to six epochs and saves atomic resumable checkpoints. All validation is paired with the same frozen baseline, image inputs, and masks. Track RMSE, MAE, bias, correlation and R2 by source and object type. Keep the full-native and app-tiling reports separately named.

What counts as a useful result

First milestone: at least a 0.50 m or 5% reduction in building or vegetation RMSE, while the other measured domains regress by no more than 0.15 m and MAE/bias/correlation checks remain acceptable. The classifier must produce the same results from the same image. Longer-term building RMSE below 5 m and vegetation RMSE below 7 m remain targets, not promised outcomes.

The frozen full-scene scorecard and additional per-building checks still govern release review. Candidate success on development validation alone cannot establish new-location accuracy or justify a production switch.

The development selection score gives equal weight to HighBuild building RMSE, OpenCanopy vegetation RMSE, and GAMUS's three-domain macro RMSE. Ground-only improvement cannot satisfy the required material improvement. Undefined correlation on constant targets/predictions is labelled non-comparable; it is not rewritten as zero or as perfect agreement.

Proposed competitive product targets (not research cutoffs)

The SAC reference README was rechecked on 15 September 2026: `project-brief-reference`. It specifies 50% DSM accuracy and 50% rendering/UX, but no numerical pass marks or winning scores. The targets below are engineering ambitions, not promised results or published industry certification thresholds.

Measure	Ambitious product target
Measured building heights	RMSE 3-5 m; MAE at most 2-3 m
Canopy heights	RMSE 4-6 m; MAE at most 2-3 m
Short vegetation (0-5 m reference band)	MAE at most 1 m; report separately
Correlation / R2	Aim for at least 0.8 / 0.7 per non-flat landscape
Systematic height bias	Absolute bias at most 1 m per evaluated category
Terrain / absolute DSM	Beat a registered DEM-only baseline on independent reference pixels; report all four landscapes and vertical datum
Additional six-class identification	Macro F1 at least 0.80 as a later independent target, never a replacement for height validation
Interactive usability	At least 30 FPS on a declared scene/mesh/device; 20 consecutive completed upload-to-export regression tests

Use independent geography and the actual deployed inference pipeline. Publish sample counts, resolution, height bands, per-scene spread and uncertainty of the measured scores. Pixel counts are not independent sample counts. Report above-ground heights separately from terrain elevation and absolute DSM. Calibration references must be separate from evaluation references. A good pooled R2 over a mountain scene can hide poor local building/tree reconstruction.

Reaching building RMSE below 5 m requires about 62% improvement from the current 13.243 m corrected development score. One passing optimization check or a small pilot cannot establish that such a reduction will occur.

Geographic evidence

GAMUS DC validation tiles are often adjacent to training tiles; prior work also exposed NYC. OpenCanopy development regions differ, but source mosaics overlap. These are development checks, not proof of geographic independence.

The Bay of Plenty RGB + metric LiDAR DSM/DEM package is sealed separately. It remains unconsumed until a candidate and a comparison protocol are fixed. Its one-shot result must not be used to select thresholds or train the model.

Credit and compute policy

Run one GPU experiment at a time. Use persistent local logs/checkpoints and an hourly scheduled check during training. Notify on completed epochs, a meaningful failure, or completion. Resume analysis after training; inspect failed outcomes before authorizing another experimental recipe. Keep source code, data contracts, rejected experiments and production artifacts saved.

Local run and recovery

- Launch: `powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_residual_height_v1.ps1`. A matching passing training-only feasibility report is mandatory.
- Resume: the same launcher with `-ResumeCheckpoint <absolute checkpoint_latest.pt>`. Resume uses the experiment's saved config and verifies protected artifacts.
- View: `powershell -NoProfile -File scripts/watch_residual_height_v1.ps1`. Closing the viewer does not stop training.
- Persistent log: `outputs/orchestration/residual_height_v1.log`.
- Results: `experiments/*_residual_height_v1/metrics.jsonl`, `status.json`, and atomic epoch/latest checkpoints. Only an eligible epoch may create `checkpoint_best_guarded.pt`; it does not change the app's active model.

Resume restores model, optimizer, scheduler and main-process RNG states at an epoch boundary. Multiworker random-crop streams are statistically resumable, not guaranteed bitwise identical to an uninterrupted run.

Execution handoff: 15 September, 17:30 IST

- Full Python regression suite: **576 passed**, 12 dependency/georeferencing warnings; no test failures.
- Final sealed training-only feasibility proof: `outputs/diagnostics/residual_height_v1_overfit/report.json` (proof run ID 20260915T115808Z; source snapshot is versioned by this ID). Loss fell 7.4554 to 5.3215 (28.62%) in 48 steps on four training examples. Every source contributed, copied-decoder gradients were nonzero, protected semantic outputs were exactly equal before/after, and inherited weights did not change. These are optimization checks, not validation accuracy.
- Fresh full run started: `experiments/20260915T115844Z_residual_height_v1`. Its first stage is paired full-scene baseline validation. No feasibility weights are reused. Up to six epochs follow, with the declared guards.
- Background launcher PID at startup: 10496. The virtual-environment Python launcher forwards to the actual interpreter; two related Python process entries do not mean two competing trainers.
- Hourly heartbeat: `msr-residual-accuracy-check`. It checks meaningful progress/failures and resumes the evidence-led evaluation work after training.
- RTX confirmed active during baseline validation; no app model change.
- Do not modify the source files listed in the experiment's source manifest while this run is active. The trainer authenticates its frozen source/config.

Recovery: 15 September, 17:55 IST

The first run failed at 17:34 IST during baseline validation, before training an epoch or writing any learned checkpoint. The common evaluator unconditionally computed both expert metrics even on HighBuild-only data, which has no valid vegetation reference pixels. This raised `No valid pixels were accumulated`. The original experiment, source snapshot, log, and a `failure_recovery.json` record are preserved; this failure is not a failed model-accuracy result.

The evaluator now reports unsupported experts as unavailable (not zero), skips zero-support regional summaries, and still rejects a wholly unlabeled suite. Regression tests cover urban-only, vegetation-only, ground-only, missing-region support and entirely missing truth. Per-suite batch progress and terminal failure status are now recorded by the residual trainer.

The recovery regression suite passed **584 tests**. A real CUDA validation smoke test processed two complete native samples from each source and passed:

`outputs/diagnostics/residual_height_v1_validation_smoke/20260915T122952Z/report.json`. It verified HighBuild's building-only support and OpenCanopy's ground/vegetation support without requiring nonexistent class labels. Architecture, training recipe and acceptance criteria are unchanged. A new source-bound training-only feasibility proof is required before the fresh restart.

Windows status-lock recovery

The second run `20260915T123138Z_residual_height_v1` finished GAMUS and HighBuild baseline passes but failed during OpenCanopy baseline when Windows denied the atomic replacement of a status file held open by a reader. It also trained no epoch. Its failure summary is preserved.

Atomic JSON/text/checkpoint replacement now retries only transient Windows sharing/access errors 5/32/33, with a bounded delay; permanent errors still raise and preserve the prior file. A real Windows `CreateFileW` sharing-lock integration test passed. The compact viewer opens status and metrics with `FileShare.ReadWrite` | `FileShare.Delete` so it cannot block these renames.

Smoke test `20260915T123837Z` and resealed 48-step proof `20260915T123910Z` passed. The new active run is `experiments/20260915T124056Z_residual_height_v1`, launched at 18:10 IST. The hourly monitor follows this run; both failed attempts remain available as evidence. Model architecture, recipe, data and guards did not change during these runtime recoveries.

Confirmed training handoff: 18:17 IST

All three baseline suites completed in the active 20260915T124056Z run. Epoch 1 was directly observed at batch 375/1500 with finite loss 5.8906; RTX utilization was 86%, temperature 63 C. No epoch-validation improvement is yet claimed. The final full regression suite passed **593 tests**, including the real Windows sharing-lock test. Hourly monitoring is active and follows this run. See docs/RESIDUAL_HEIGHT_POST_RUN.md for the remaining evaluator compatibility work required before a formal candidate release decision.

26 Monocular Surface Reconstruction locked benchmark protocol

SOURCE docs/history/BENCHMARK_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Why this evaluator exists

scripts/analyze_multidomain_checkpoint.py is useful during training, but it loads a centred 384 x 384 crop. The urban controls are 1024 x 1024, so that path scores only 14.06% of each urban tile. It is not equivalent to an upload through the application.

scripts/evaluate_locked_benchmark.py instead uses the application read limits, recomputes the Depth Anything V2 prior, and runs the production height predictor over the complete raster with 512 px tiles and 128 px overlap. References are loaded only after the input grid is established and are never passed to either model.

The report includes:

- RMSE, MAE, bias, correlation, and R2 for the raw upload;
- landscape and ground/building/vegetation breakdowns where labels permit them;
- the same metrics for deterministic stretch, JPEG, gamma, exposure, and channel-gain variants;
- prediction-versus-raw deltas, which expose sensor/radiometry shortcuts;
- manifest, dataset, checkpoint, and foundation-model hashes;
- train/validation overlap checks and a before/after checkpoint hash;
- source-licensing caveats.

Two-pass locked workflow

First produce a read-only integrity report. This obtains the exact hashes without running either neural network:

```
.\venv\Scripts\python.exe scripts\evaluate_locked_benchmark.py `
--checkpoint experiments\20260829T203146Z_multidomain_surface_v2_guarded_vegetation\check
point_best_guarded_vegetation.pt `
--manifest data\multidomain_v2\manifests_with_priors\test.csv `
--deny-manifest data\multidomain_v2\manifests_with_priors\train.csv `
--deny-manifest data\multidomain_v2\manifests_with_priors\validation.csv `
--benchmark-id msr-legacy-regression-v1 `
--benchmark-status development `
--integrity-only `
--output outputs\benchmarks\msr-legacy-regression-v1\integrity.json
```

Copy the four hashes from that report into the release invocation:

```
.\venv\Scripts\python.exe scripts\evaluate_locked_benchmark.py `
--checkpoint <CHECKPOINT> `
--manifest <FROZEN_MANIFEST> `
--deny-manifest <TRAIN_MANIFEST> `
--deny-manifest <VALIDATION_MANIFEST> `
--benchmark-id <RELEASE_ID> `
--benchmark-status locked_release `
--expected-manifest-sha256 <HASH> `
--expected-dataset-sha256 <HASH> `
--expected-checkpoint-sha256 <HASH> `
--expected-relative-model-sha256 <HASH> `
```

```
--device cuda `
--output outputs\benchmarks\<RELEASE_ID>\report.json
```

Never tune on a locked_release report. Candidate selection and threshold changes belong on validation/development data; run a new named locked release only after those choices are frozen.

Current evidence status

The existing combined test manifest has 200 HighBuild-derived urban controls and 120 Open-Canopy forest controls. It is geographically disjoint from the current train and validation manifests and all 3,300 RGB/reference files have no exact cross-split byte duplicates. It has nevertheless been evaluated repeatedly, so it is a **legacy regression suite**, not a fresh blind result. It also contains no true plains, hilly, or mixed-labelled scene.

A future manifest should use project-relative paths and contain `sample_id`, `region`, `landscape`, `rgb_path`, `surface_path`, `target_kind`, `dtm_path`, `building_mask_path`, `vegetation_mask_path`, `valid_mask_path`, `gsd_m`. A mixed scene requires both building and vegetation masks. Hilly absolute-DSM results need a calibration datum and a genuinely independent evaluation reference; coarse terrain R2 must not be presented as building/tree accuracy.

Corrected HighBuild protocols

HighBuild's `building_height_m.tif` is a sparse building-label raster, not a complete surface nDSM. A valid benchmark row must therefore declare `target_kind=building_height` and provide two distinct aligned masks:

- `building_mask_path` contains every official COCO building footprint and is semantic supervision only;
- `valid_mask_path` states which footprint pixels have height supervision under the named evaluation protocol.

The supported protocols are `highbuild_all_annotated_positive_v1` (measured and estimated positive annotated heights) and `highbuild_measured_coco_intersection_v1` (positive, non-estimated annotations only). Outside the footprint is unknown, never zero-height ground. Reports from these protocols must remain separate and cannot be compared directly with the historical zero-filled HighBuild metrics. Manifest loading and the locked benchmark now reject the ambiguous historical target contract.

Licence and provenance guardrails

- **Open-Canopy:** Etalab Open Licence 2.0. Keep attribution, retrieval date, upstream revision, and per-file hashes. Existing `resolve/main` URLs are not a revision lock. Its 120-chip test set is already inspected. The 5 km regions are disjoint, but many chips across splits originate from the same larger SPOT acquisition, so it is not an acquisition-scene-disjoint sensor test.
- **HighBuild-derived controls:** the local city split is conservative and the imagery inventory is source-specific, but the separate provenance/licence chain for height geometry and labels is not frozen locally. Use internally; do not redistribute chips or call every annotation measured ground truth yet. The upstream metadata marks some annotations as estimated, so final reports should stratify measured/non-estimated and estimated subsets.
- **DFC19 / US3D:** keep the current local copy internal. The secondary CC-attribution note conflicts with restrictive official DFC19 redistribution terms. Do not publish or package these assets until written permission and chain-of-title are confirmed.
- **Sentinel/Copernicus/SRTM hilly evidence:** usable with the required source notices. Copernicus is the calibration datum and SRTM is a coarse cross-check, not LiDAR or object-height truth.

This document records an engineering audit, not legal advice. Benchmark code does not grant redistribution rights to its inputs.

Pre-registered unused HighBuild candidates

The following tiles are absent from the current 320-row combined test manifest. They were selected without model predictions by SHA256 ordering of `msr-lock-v1|<sample_id>`. Source rows and members are in `data/highbuild_full/msr_splits/test.csv` and archives under `data/highbuild_full/msr_splits/shards/test/` (physically on D: through the junction). Extract and hash the frozen list before inference.

Dense urban controls (`num_annotations >= 10`, five per held-out city):

- Amsterdam: `grid_11377_z19`, `grid_18227_z19`, `grid_01605_z19`, `grid_01225_z19`, `grid_07725_z19`
- Frankfurt: `grid_00007_z19`, `grid_00040_z19`, `grid_00051_z19`, `grid_00047_z19`, `grid_00030_z19`
- Marseille: `grid_00091_z19`, `grid_00169_z19`, `grid_00066_z19`, `grid_00032_z19`, `grid_00150_z19`
- Odense: `grid_00026_z18`, `grid_00086_z18`, `grid_00023_z18`, `grid_00028_z18`, `grid_00048_z18`
- Vancouver: `grid_00092_z18`, `grid_00131_z18`, `grid_00140_z18`, `grid_00132_z18`, `grid_00081_z18`

Sparse-building controls (`num_annotations <= 3`; this is sparse urban, not a true plains benchmark):

- Amsterdam: `grid_00326_z19`, `grid_00695_z19`, `grid_00305_z19`, `grid_01318_z19`, `grid_15358_z19`, `grid_01156_z19`, `grid_08700_z19`, `grid_14699_z19`
- Marseille: `grid_00021_z19`, `grid_00023_z19`, `grid_00155_z19`, `grid_00035_z19`
- Vancouver: `grid_00146_z18`, `grid_00048_z18`, `grid_00101_z18`, `grid_00152_z18`, `grid_00137_z18`, `grid_00126_z18`, `grid_00149_z18`, `grid_00150_z18`

Use the full `webdataset_key` from the source CSV when materializing the final manifest; the short grid names above are grouped by city for readability.

No equivalent unused local set currently exists for forest, hilly object height, plains, or mixed building-plus-vegetation scenes. The next honest lock therefore requires new Open-Canopy test regions, independent references for the four unvalidated hilly candidates, a true plains set, and an openly redistributable mixed RGB+nDSM+semantic source. The existing 15 DFC19 windows cannot fill the mixed slot: all were ranked after evaluation, they overlap along one Omaha strip, and their redistribution rights are unresolved.

27 Faster class-assisted height comparison

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_FAST_V2_20260919.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Why this version exists

The user explicitly requested stopping and rerunning the slow pipeline using the laptop more effectively. Stopped only the registered trainer, after control A finished epoch 1 and saved epoch_01_batch_0600.pt; validation was interrupted. The original experiment remains intact and must not be restarted by its monitor.

Parent: D:/MSRData/experiments/class_assisted_height_development_v1/20260919T162009394247Z. Active: D:/MSRData/experiments/class_assisted_height_fast_v2/20260919T164305030475Z.

Measured bottleneck and change

The old loop reloaded full native rasters, decompressed six probability grids, constructed supervision tensors, and cropped them synchronously for each batch. The GPU waited while the CPU did this work.

A training-only 12-batch check compared the same tensor values and order:

Loader	Measured batches/s	Input tensor identity
Original, 16 CPU tensor threads, serial loading	4.71	Reference
One CPU tensor thread, serial loading	4.76	Exact
One CPU tensor thread, four loader workers	12.28	Exact

This is a ~2.6x loader benchmark gain, **not** a promised end-to-end speedup or accuracy improvement. Measurement includes a small, cache-warmed sample; real epoch timings remain the relevant whole-run evidence.

Initial live throughput check: successive 100-batch checkpoint intervals took about 34 seconds in the old A epoch 1, versus 11.6 seconds in fast A epoch 2 (first five checkpoints). This is roughly 3x faster observed training throughput, not a same-input whole-run benchmark; validation/checkpoint phases still vary.

Fast execution uses four ordered training loader workers with four prefetched batches, two validation workers/images, and pinned nonblocking GPU transfers. Memory is bounded; this 16 GB laptop had only about 4 GB RAM free when inspected. No overclock, temperature-limit change or unrelated-process shutdown was done. Original numerical model/loss/gradient loop, batch 2 + accumulation 2, optimizer, crop coordinates, sample order, precision and evaluation rules are unchanged.

Preserved training and integrity

The new run explicitly imports the authenticated A epoch-1 model, optimizer and RNG state. It finishes that validation, trains A epoch 2, then B epochs 1 and 2. The original checkpoint is never rewritten. continuation_origin.json records the old bindings and checkpoint hash; the imported checkpoint is a new artifact bound to the faster execution source/config. This is a declared continuation, not a silent override of an old checkpoint's resume checks.

All 1,330 original cached inputs are reused read-only and reauthenticated against original files and receipt hashes. DAV2 predictions and classification maps are not regenerated. Original baseline metrics/support are reused with pinned hashes. Both arms share the same initial tensors and matched sampling plans, checked at completion. Protected production model and pointer are unchanged.

Six dashboard/prefetch tests passed. The executable preflight authenticated 18 source files and the parent checkpoint. Loader equality was checked on 12 real training batches. These are implementation checks, not new accuracy evidence.

Watching and recovery

- Dashboard: `.venv/Scripts/pythonw.exe scripts/class_assisted_dashboard.py`.
- Logs: `outputs/orchestration/class_assisted_height_fast_v2.log` and `.err.log`.
- Registration: `outputs/orchestration/class_assisted_height_active.json`.
- Detailed reports: `outputs/reports/class_assisted_height/<run ID>/.`
- Reports include text and full-group CSV; no scores are fabricated while waiting.
- Hourly heartbeat `msr-residual-accuracy-check now monitors fast v2 only`.

After inspecting an actual failure and confirming no trainer remains:

```
.\scripts\run_class_assisted_height_fast.ps1 -Resume
D:\MSRData\experiments\class_assisted_height_fast_v2\20260919T164305030475Z
```

Never alter the active bound source/config or bypass a hash mismatch. The old and fast trainers share one lock. Checkpoints remain versioned and save every 50 optimizer updates. Do not replay the old run, proof or completed updates.

Honest interpretation

This pilot covers corrected urban building labels and forest/canopy/ground labels on the fixed development set. It does not certify broad hills/plains, water/road heights, all sensors/resolutions, or genuinely new geography. GAMUS height units remain unresolved. Require both legacy safety and matched benefit; neither is automatic app-release permission. Same-epoch improvement on building OR canopy must exceed both 0.5 m and 5%, beat the control, and pass all other safety checks. No final reserved geography is consumed by this run.

28 Class-assisted height comparison: completed, not promoted

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_FAST_V2_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Run: D:/MSRData/experiments/class_assisted_height_fast_v2/20260919T164305030475Z. Completed 2026-09-19 16:50:51 UTC. Two epochs per arm, including the authenticated continuation of control A epoch 1. No candidate passes both benefit and safety.

Fixed development results

RMSE in metres, lower is better. All rows use the same native-resolution 160 HighBuild / 120 OpenCanopy development images and reference-valid pixels.

Model	Buildings	Canopy	Ground	Short vegetation	Legacy safety
Protected app model	13.2426	8.3610	4.8944	5.2049	Baseline
A control, epoch 1	12.9630	8.0529	5.1278	5.5979	Fail
A control, epoch 2	12.7829	7.7268	5.3143	5.9793	Fail
B class-assisted, epoch 1	12.9553	8.0071	5.1252	5.6380	Fail
B class-assisted, epoch 2	12.7883	7.6885	5.2600	5.9624	Fail

Best unrestricted mean building/canopy RMSE is B epoch 2. Best individual building RMSE is A epoch 2. Best eligible candidate: **none**. Both B epochs fail safety; B epoch 2 passes the separate canopy-benefit rule, not release.

B epoch 2 versus protected: building RMSE falls 3.43%, canopy 8.04%; ground RMSE rises 7.47% and short-vegetation RMSE rises 14.55%. These are relative changes in error, not percentages of correct predictions.

What the matched comparison actually establishes

B epoch 2 beats A epoch 2 on canopy by only 0.0383 m (about 0.50%), while its building RMSE is 0.0054 m worse. Most improvement versus the protected baseline also occurs without informative six-class probabilities. One seed and two epochs do not establish a robust, material benefit from class conditioning. Nor do they prove that classification can never help height in a different controlled design.

Error pattern and limitations

- Underestimated tall-object heights improve somewhat: B epoch-2 tall-building RMSE 29.80 -> 28.60 m; tall-canopy RMSE 10.49 -> 9.01 m. Errors remain large.
- The gain comes with excessive positive height on shorter surfaces. Ground bias rises from +1.56 to +2.06 m; short-vegetation bias +2.56 to +3.08 m. This is consistent with an insufficiently selective upward correction, not proof of a single architectural cause. Both arms show the same broad tradeoff.
- B epoch-2 building correlation/R2 are 0.220 / 0.006; canopy 0.450 / 0.008. Lower RMSE does not make this a high-accuracy reconstruction system.
- Per-region safety checks also fail. A failed-check count is not an image count or a statistical significance estimate; multiple metrics/groups overlap.

- No fresh final geography, GAMUS height safety, road/water height validation, broad hill/plain stability or cross-sensor universal accuracy was established.

Justified next experiment, not launched

1. Inspect fixed development examples where ground/low vegetation rises alongside improved canopy; stratify by reference height, boundaries and class/geometry disagreement. Keep dataset split and input-validity contracts unchanged.
2. Define a bounded test of stronger ground/short-vegetation protection in the residual loss, paired against the same control. Do not hard-force roads or water to zero, use reference classes at inference, or hide failed categories.
3. Independently address tall-building training support and missing physical pixel scale before expecting a major building gain. More epochs alone are not proof these problems will disappear.
4. If a new development candidate passes predeclared guards, then run per-building, actual app-path and appropriately reserved geography checks before promotion. Do not loosen this completed run's rules or consume final tests while tuning.

Saved and protected

Both final committed checkpoints and evaluation hashes were reverified after completion. All 18 bound source files match; fixed support and protected semantic output identity checks pass. Production height checkpoint, live pointer, V1 classifier and V3 conditioning-classifier hashes match their protected values. The original slow run and its checkpoint/cache remain preserved. No trainer is active. The faster run took roughly 7m46s including cache authentication and four validations; this excludes the already-completed parent training epoch.

Readable report, full-group CSV and snapshot JSON are under `outputs/reports/class_assisted_height/20260919T164305030475Z/`. The run monitor is paused after completion reporting. No further training or app promotion was started as part of this status check.

29 Class-assisted height comparison: launch handoff

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_LAUNCH_20260919.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Launched 2026-09-19 16:20 UTC. This is an experiment, not an app-model release.

Saved runs

- Passed training-only proof: D:/MSRData/experiments/class_assisted_height_proof_v1/20260919T161049828458Z.
- Active development comparison:
D:/MSRData/experiments/class_assisted_height_development_v1/20260919T162009394247Z.
- Protocol: docs/CLASS_ASSISTED_HEIGHT_PILOT_PROTOCOL.md.
- Registration: outputs/orchestration/class_assisted_height_active.json.
- Logs: outputs/orchestration/class_assisted_height_v1.log and .err.log.

What is running

First authenticate and numerically replay DAV2 priors and cache frozen RGB-only classification predictions for 1,050 training and 280 development images. Then run two fresh, matched residual-height learners for two epochs each: uniform class inputs versus predicted six-class inputs. All other settings and sample orders match. Development evaluation stays at native resolution on fixed pixels.

The 64-update proof passed for both arms, with approximately 60% lower balanced training loss. This is **not a 60% accuracy improvement**. Its near-identical arm results do not establish a benefit from classification. Ground training loss increased, so ground regression checks are especially important.

Fifteen targeted tests passed. The live production height checkpoint and pointer were rehashed after launch and still match their protected values. Neither the app model nor existing experiments were replaced.

Monitoring and recovery

Hourly heartbeat msr-residual-accuracy-check is active for this exact run; it reports meaningful changes and pauses after completion. No automatic promotion or additional expensive training is authorized by the monitor.

For a compact local progress display, run:

```
.\scripts\watch_class_assisted_height.ps1
```

Checkpoints save every 50 optimizer updates and at epoch boundaries. Inspect failure evidence before attempting a compatible resume; never edit bound source or configuration, bypass hashes, or overwrite orphan artifacts.

```
.\scripts\run_class_assisted_height.ps1 -Resume
D:\MSRData\experiments\class_assisted_height_development_v1\20260919T162009394247Z
```

Only resume after confirming the original trainer is no longer running. A venv wrapper plus its Python child is one job, not duplicate training.

Interpretation limits

Report protected-baseline, uniform-control and predicted-class results separately. Safety and benefit are separate requirements. The 280 images are development validation, not a fresh final test. GAMUS height safety remains untested because units are unresolved. Per-building, app-path and final geography checks remain prerequisites for any release, even if this bounded comparison succeeds.

30 Low-surface protection outcome — not promoted

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_LOW_SURFACE_V3_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Completed 19 September 2026, 17:14:17 UTC. Run:

D:/MSRData/experiments/class_assisted_height_low_surface_v3/20260919T170601414242Z. Two fresh epochs per arm; approximately 8 minutes 16 seconds total including input authentication. Production and both frozen classifiers remain unchanged.

Result

All rows below use identical native-resolution development support, not final unseen tests. RMSE in metres; lower is better.

Category	Protected	Previous class-assisted epoch 2	New class-assisted epoch 2
Buildings	13.243	12.788	13.076
Canopy	8.361	7.689	9.637
Ground	4.894	5.260	4.144
Short vegetation, reference 0–2 m	5.205	5.962	3.233
Tall buildings, reference ≥ 20 m	29.797	28.600	29.041
Tall canopy, reference ≥ 15 m	10.486	9.012	12.577

The new class-assisted model reduces ground RMSE by 15.32% and short-vegetation RMSE by 37.89% from protected, but worsens canopy RMSE by 15.26%. These are error changes, not accuracy percentages. Errors of 3.23 m on vegetation known to be 0–2 m high remain too large for precise individual-object measurement.

Ground bias improves +1.555 $\rightarrow$ -0.031 m and ground MAE 2.896 $\rightarrow$ 1.931 m. Canopy bias worsens -3.517 $\rightarrow$ -6.161 m. The new objective has exchanged the previous upward spillover for excessive suppression of taller vegetation. It does not solve selective geometry estimation.

The matched uniform-input epoch-2 control has building/canopy/ground/short vegetation RMSE 13.085 / 9.619 / 4.155 / 3.316 m. Class information therefore adds only small mixed differences: approximately -0.009 m buildings, +0.018 m canopy, -0.011 m ground and -0.083 m short vegetation. This single paired run does not establish robust height improvement from six-class conditioning.

All four epochs fail original safety gates. Neither candidate epoch meets the priority benefit gate. Best unrestricted building/canopy average within the new run is **control A epoch 2**. Best eligible checkpoint: **none**. Earlier class-assisted epoch 2 still has better building/canopy numbers but remains rejected for ground/short-vegetation damage. Never combine category-wise minima from different checkpoints into a claimed single-model result.

Additional train-only context diagnostic

On the original fixed 20 feasibility examples (10 urban + 10 forest), changing urban input context from full 1024 px to its same-pixel 384 px training crop changes protected building heights by 0.774 m mean absolute difference; after excluding a 32 px crop border the difference remains 0.753 m. Forest examples already have native 384 px size and give identical predictions. This is a limited training diagnostic, not a population accuracy score or proof that context is the dominant failure. It cannot explain the forest tradeoff.

Decision and next justified work

Do not promote, prolong this recipe automatically, relax its gates, or consume reserved geography. Keep six-class identification independent of height.

The next bounded experiment should address object-height selectivity rather than merely raising/lowering the loss weight again:

1. Derive a training-only height-band exposure plan, retaining true labels and original masks, to separately cover shorter/taller measured buildings and shorter/taller vegetation. The saved HighBuild audit has 9,841 measured/not- marked-estimated outlines, overlap caveats, and missing GSD on all 450 tiles. Do not silently treat every outline or inferred pixel scale as authoritative.
2. Test height-band-balanced crop exposure as one intervention with fixed optimizer budget and matched control; retain full native validation and explicitly report short, medium and tall strata so aggregate gains cannot conceal the same failure again. Predeclare new experimental settings first.
3. Treat per-instance supervision, context-matched training and recovered physical pixel scale as separate tests, not simultaneous unexplained changes.
4. Only an eligible development candidate advances to per-building, app-path, and genuinely reserved-geography evaluation. No all-image accuracy promise.

Saved evidence and checks

- `outputs/reports/class_assisted_height_low_surface_v3/REPORT.md`: all epochs, historical controls, RMSE/MAE/bias/correlation/R2 and failed-check summary.
- `outputs/reports/class_assisted_height_low_surface_v3/verification.json`: protected hashes, source snapshots, committed artifact checks and paired initialization/sample order verified. A reporting-only serialization check was corrected to authenticate original config bytes plus saved JSON semantics; no training binding, checkpoint, evaluation or gate was changed.
- `outputs/diagnostics/class_assisted_low_surfaces_v1.json`: all 280 development images from the preceding candidate, stratified errors and correction direction.
- `outputs/diagnostics/height_crop_context_train_v1.json`: train-only context check.
- 49 targeted tests passed across loss, masking, protocol, resume, frozen model, gates, data provenance, dashboard, reporting and Windows atomic replacement.

All code and checkpoints are saved. No trainer remains active. The existing monitor remains paused; no extra recurring checks or new training were started.

31 Low-surface protection: bounded matched experiment

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_LOW_SURFACE_V3_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Authorized by the user's request to continue on 19 September 2026. This protocol is written before training. It does not change any previous experiment's verdict.

Evidence and hypothesis

The complete 280-image DEVELOPMENT audit of the previous class-assisted epoch 2 found that 99.13% of supported short-building pixels and 99.67% of tall-building pixels were raised. Short buildings (<20 m) RMSE worsened 4.77 -> 4.83 m, while tall buildings improved 29.80 -> 28.60 m. Forest ground received +0.50 m mean correction and short vegetation +0.52 m. Of labelled short-vegetation pixels, 75.45% had larger absolute errors. Medium vegetation (2-15 m) also worsened (5.90 -> 6.30 m), while tall canopy improved (10.49 -> 9.01 m).

Thus, the aggregate benefit concealed errors in shorter groups. These are pixel statistics on fixed development data, not percentages of wrong images, an independent final test, or proof of a single architectural cause.

Hypothesis: the existing vegetation loss averages short and tall vegetation; explicit short-vegetation supervision and a per-pixel low-surface non-regression penalty may reduce that tradeoff. This may also reduce the earlier canopy gain. No improvement or acceptance is guaranteed.

The one declared training change

Keep the original source/domain-balanced Huber + 0.02 MSE loss. Add:

- 0.5 times separate mean short-vegetation supervision (reference $0 < h \leq 2$ m).
- 1.0 times a low-surface regret penalty, averaged over present sample/group means for reference-labelled ground and short vegetation. For each pixel, $\text{regret} = \max(0, |\text{prediction-reference}| - |\text{protected-reference}|)$; penalty is $\text{regret} + 0.02 \text{ regret}^2$.

The frozen prediction is a comparator, not the truth. Improving it is allowed; making its absolute error larger costs extra loss. Actual labels are retained: short vegetation is NOT assigned zero. Missing labels are excluded before arithmetic. Classification, image-validity and height-validity remain distinct. No reference labels/masks are passed to inference. No water/road zero rule.

Controlled comparison and resources

Both arms restart from the same protected initialization and seed, not failed candidate weights. A receives uniform probabilities; B frozen RGB-predicted six-class probabilities. Both receive the SAME new loss. Compare each with its historical same-epoch old-loss result, and B with its new same-epoch A.

Exactly two epochs/arm, 600 source-paired batches per epoch: all 450 corrected HighBuild training scenes plus 150 urban repeats, and all 600 OpenCanopy scenes. Original batch 2, accumulation 2, 384 px crops, bf16, optimizer, learning rates, sample order and frozen model/semantics remain unchanged. Four-worker ordered CPU prefetch and pinned transfers preserve the measured faster input path. One GPU experiment only; no overclocking, cooling overrides or unrelated kills.

All 160 HighBuild and 120 OpenCanopy development images are evaluated at native resolution after every epoch on the same supported pixels. Reuse only fully authenticated probability/prior receipts. New immutable experiment on D; source/config/data/software binding, optimizer/RNG checkpoints every 50 updates, and atomic commits. Resume refuses changed bindings. The existing progress window follows the new active registration automatically.

Acceptance and stopping

Original safety thresholds are UNCHANGED, including regional groups and correlation/R2 checks. Candidate benefit still requires ≥ 0.5 m AND $\geq 5\%$ building or canopy RMSE reduction from protected, beating same-epoch matched control. Report safety and benefit separately, all four epochs, and best unrestricted versus eligible checkpoint. No silent adjustment to gates after seeing results. No automatic additional epochs, repeated hyperparameter search, or promotion.

GAMUS height units remain unresolved. No reserved geography is consumed, no universal accuracy claim, and no height validation claimed for roads/water. App-path, per-building and final-geography checks remain separate prerequisites if a candidate eventually passes. The protected production model, live pointer, V1/V3 classifiers and all earlier runs are preserved and hashed.

32 Class-assisted height pilot — controlled experiment

SOURCE docs/history/CLASS_ASSISTED_HEIGHT_PILOT_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **paired training-only proof passed; bounded development run launched**. Executable bindings are in `configs/class_assisted_height_proof_v1.json` and `configs/class_assisted_height_development_v1.json`. No candidate is promoted. Do not apply this recipe to an older experiment.

Question and scope

Does independently predicted six-class information help a height learner beyond the same RGB/DAV2/protected-height inputs? Better class F1 alone is not evidence.

Keep the protected production model, app pointer, V1 classifier, V3 classifier and previous residual-height experiment immutable. The app continues using the protected height model. Use a new experiment directory on D for large caches. Do not train six disconnected height models or force water/roads to zero.

Preparation gates

1. Authenticate exact training/development manifests, corrected COCO masks, protected model, frozen V3 epoch-1 classifier, DAV2 weights/preprocessor, code snapshot and software versions. Preserve the older failed residual run as historical comparison, **not** the matched ablation baseline.
2. Record height and semantic validity separately from image validity. Missing labels remain ignored. HighBuild supervises measured building support, not unlabeled background. OpenCanopy supervises available above-ground labels.
3. Use HighBuild and OpenCanopy for the first matched pair. GAMUS height units remain unverified, so do not silently treat its height values as trusted new supervision. Omitting GAMUS is common to both arms: results cannot isolate the effect of dropping GAMUS relative to the previous pilot.
4. Verify every cached prior used by the small proof against its image and frozen DAV2 recipe; record new evidence bindings without rewriting old caches. The completed 10-example replay is a useful check, not full certification.
5. Six-class inputs must be **model predictions from RGB**, never reference masks or true heights. Cache probabilities, invalidity and source/weights hashes. Keep per-pixel alignment under crop/rotation; do not color-augment imagery while treating unaltered cached class/depth predictions as freshly inferred.
6. Inspect representative training tall/short buildings and forest/ground transitions. The current 450-tile subset has missing explicit GSD: preserve native pixel handling, disclose the scale gap, and do not invent metre scale from filenames. Scale-aware augmentation requires recovered source metadata.
7. Resolve overlap policy for instance supervision: overlapping annotations and tiny supports cannot be silently treated as independent clean full buildings. Keep pixel support unchanged in this first experiment; add instance loss as a separate later experiment after versioning its support contract.

Matched arms

- Both start from the same protected height model and copied residual decoder. The protected backbone, original heads and six-class classifier remain frozen.

- Both use the same extra six-channel input capacity. **Control A** receives uniform class probabilities on valid RGB; **candidate B** receives the frozen classifier's soft probabilities. Both receive the identical image-validity channel. Unknown pixels are not assigned a semantic class or height.
- Concatenate conditioning to the learned residual context; do not hard-gate the protected height output by class. Initialize the final correction to zero.
- Same seed, sample sequence, batches, augmentations, optimizer steps, loss, decoder capacity and schedule in both arms. Same native development pixels. Log actual sampled IDs/transforms so the comparison is auditable.
- Preserve independent semantic outputs byte-for-byte. A later intentional feature/backbone change would require rechecking classification separately.

Bounded stages

1. Train-only feasibility: fixed examples spanning five HighBuild training cities and OpenCanopy training sources; maximum 64 optimizer updates per arm. Check finite gradients, zero-initialization identity, frozen tensors, valid masks, memory and at least 10% loss reduction. This is an optimization check only. Discard proof weights and never describe this training loss as accuracy.
2. Only after gates pass, freeze executable config and a maximum two-epoch paired development run. Same protected start in both arms; no final reserved geography. Save checkpoints, optimizer/RNG state, inputs/hashes and resumable atomic commits.
3. Report per-source/per-city RMSE, MAE, bias, correlation and R^2 ; separate buildings, canopy and ground. Include ≥ 20 m buildings, short vegetation, boundaries and six-class/legacy-geometry disagreements. Missing supported categories are N/A.
4. A useful candidate must reduce a priority-object RMSE by both ≥ 0.5 m and $\geq 5\%$ versus protected production **and** beat the matched control on that same metric. Report all other changes, including if the control improves more elsewhere.
5. Preserve the previous pilot's safety checks; no retrospective loosening. At minimum no protected group RMSE regression > 0.15 m, MAE > 0.10 m or absolute bias > 0.10 m; retain its correlation/ R^2 and tall-object checks. Define exact machine-readable rules before training, not after seeing results.
6. Passing this pilot does not release a model. Next: residual-compatible native per-building scorecard, app-tiled paired inference and a frozen one-shot genuinely new-geography test. No default-model change without explicit review.

Stop conditions

Stop on changed bindings, label/prior misalignment, unknown unit conversion, non-finite loss, unsupported frozen-tensor changes or failed optimization proof. If class conditioning offers no protected benefit, keep the scanner feature and prioritize height-data/scale/tall-object coverage instead of repeating the run. No confidence probability or height improvement is promised in advance.

Executed feasibility and locked development details

Proof: D:/MSRData/experiments/class_assisted_height_proof_v1/20260919T161049828458Z. Twenty training examples: one short and one tall-support tile from each of five HighBuild cities, plus ten OpenCanopy training region groups with both ground and vegetation supervision. Every proof prior was replayed against its source; cached soft classifications were inferred from RGB only. Group names do not imply ten independent forest geographies.

Both arms used batch 2, two-batch accumulation, 64 optimizer updates and identical initial tensors, crop coordinates and sample order. Balanced training-only loss fell from 17.1686 to 6.8253 (uniform) / 6.8178 (predicted), about 60.25% / 60.29%. The two results are nearly identical: **this is feasibility, not evidence that classification improves height accuracy**. Ground training loss increased in both arms. That is a reason to examine ground protection carefully in development, not to release either proof checkpoint. Frozen weights and semantic outputs remained identical. Proof checkpoints are preserved but never reused for training.

Development uses all 450 urban and 600 forest training records. Each of two epochs contains 600 source-paired batches: every forest tile once, every urban tile once plus 150 urban repeats. Each arm receives identical crop seeds and initialization. No

color/spatial augmentation; full-image class predictions and DAV2 priors are cropped with all other inputs. Probabilities are stored in float16 then renormalized at loading; original RGB, priors and targets remain unchanged.

Validation uses all 160 native HighBuild + 120 native OpenCanopy development tiles, with the same support/datum as the corrected benchmark. Each used prior must pass a numerical replay check before training. Evaluation includes source, domain, region/domain, tall objects (20 m building / 15 m canopy), short vegetation (0–2 m), boundaries and classifier/legacy-geometry disagreements. Guards apply to every supported reported group; correlation/ R^2 tolerate only 1e-6 numerical decrease. Material benefit requires BOTH 0.5 m and 5%, plus beating the matched same-epoch control. Report both epochs rather than switching selectors afterward.

Scope clarification: GAMUS heights are neither supervised nor evaluated in this two-source pilot because their units remain unresolved. Report **legacy safety**, not universal/GAMUS safety. GAMUS safety and the existing per-building/app-path/ external checks remain release prerequisites. Every artifact says release ineligible.

Recovery: `scripts/run_class_assisted_height.ps1 -Resume <exact run directory>`. An OS file lock prevents duplicate paired trainers. Source/config/software, proof evidence, original data, cached probabilities, baseline, and committed checkpoints are authenticated. Save every 50 optimizer updates and every epoch. Orphaned or changed artifacts fail closed for inspection, never bypass integrity checks or silently change the batch. New snapshots are on D; active app unchanged.

33 Monocular Surface Reconstruction research

Prototype Runbook

SOURCE docs/history/DEMO_RUNBOOK.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Pre-demo preparation

1. Keep the laptop connected to power and select its performance GPU mode.
2. Run `scripts/preflight_demo.ps1` from the repository root.
3. Select only a reviewed held-out checkpoint with `scripts/select_showcase_checkpoint.ps1`.
4. Build the viewer with `npm --prefix viewer run build`.
5. Start the stable local build with `scripts/start_prototype.ps1 -Production -OpenBrowser`.

The launcher records its API and viewer PIDs under `outputs/runtime`. Use `scripts/stop_prototype.ps1` to stop only those recorded process trees.

Recommended reviewer flow

1. Open the included sample to demonstrate orbit, fly, optical texture, height colours, wireframe, point height, and local slope without relying on network access.
2. Upload a normal JPG/PNG and explain that its output is a dimensionless rDSM; no absolute elevation is claimed because the image has no spatial datum.
3. Upload a GeoTIFF and attach a known terrain DEM, GCP CSV, or enable the public coarse-terrain fallback. Download the resulting absolute DSM and metadata.
4. If reference LiDAR/nDSM is available, attach it before inference and show the aligned RMSE, MAE, bias, correlation, R2, and signed-error raster.
5. Use raw exported height for numerical claims. Vertical exaggeration and the experimental tree-object layer are visualization controls, not extra model accuracy.

Product interpretation

- `height_above_ground_m.tif`: predicted object/surface height above local ground (nDSM).
- `rdsd_relative.tif`: relative surface relief for non-georeferenced imagery.
- `dsm_absolute_m.tif`: terrain elevation plus non-negative predicted surface height, produced only when an absolute terrain datum exists.
- `gcp_calibrated_surface_m.tif`: relative geometry affinely calibrated from supplied elevation control points.
- `validation_signed_error_m.tif`: prediction minus reference in metres.
- `metadata.json`: processing grid, checkpoint, calibration, public-terrain attribution, reference validation, and warnings.

Honest limitations to state

Single-view RGB cannot guarantee survey-grade height everywhere. Fine roofs, small bushes, shadowed canopy, unseen ground under dense forest, and domains far from training data remain the hardest cases. Public coarse terrain supplies the absolute ground datum but does not replace LiDAR object truth. The prototype is an integrated, measurable estimation system—not a substitute for certified survey products.

Recovery

- API error: inspect `outputs/runtime/api_stderr.log`.
- Viewer error: inspect `outputs/runtime/viewer_stderr.log`.
- Clean restart: run `scripts/stop_prototype.ps1`, then the production launcher.
- Failed new model: leave the protected pilot selected; never point the demo at `checkpoint_latest.pt` merely because it is newest.

34 Monocular Surface Reconstruction expanded GAMUS plan

SOURCE docs/history/EXPANDED_GAMUS_PLAN.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: protocol predeclared before training; implementation and run status are updated below without changing the frozen comparison rules.

Non-negotiable baselines

- The production pointer remains unchanged. Its protected checkpoint SHA-256 is e2d507fbe180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144.
- The existing three height-routing-group (ground/building/vegetation) direct-height pilot remains the height-focused comparison. It is not redefined after results are seen.
- The expanded model is a separate candidate. It cannot auto-promote.
- GAMUS train uses the full QC-approved split: 5,001 of 5,004 official tiles. All 859 approved validation tiles stay separate. The official GAMUS test was consumed once by the older Stage-3 final audit and is permanently excluded from all new training, tuning, threshold selection, model selection, and repeat evaluation.
- NYC is excluded from the new DC+Philadelphia classifier heads, but it is not system-unseen: earlier GAMUS experiments and the inherited HighBuild-trained checkpoint exposed New York data. A separate external geography remains the final generalisation holdout.

Work and status

Requirement	Current state	Next proof required
Full usable GAMUS data	Implemented: immutable QC index, 5,001 train and 859 validation	Treat the already-consumed official test as immutable historical evidence; never reuse it
Three-group metric-height machinery	Implemented and checkpoint-compatible; frozen comparison early-stopped after epoch 4 and is hash-sealed. “Three-group” means height-routing domains (ground/building/vegetation), not a three-class identification score.	Keep its raw result as the control; it is not promotion-eligible
Six GAMUS categories	Implemented and evaluated as a separate head: ground, building, water, road, low vegetation, tree; the isolated candidate early-stopped after epoch 4	Improve water recall and ground/building/tall-canopy guards before any app exposure
Ignore background/unknown	Implemented in grouped and six-class losses/metrics with ignore index 255	Verify full-validation support counts
Independent validity masks	Implemented and audited: explicit image-valid, class-valid, and height-valid tensors; regression intersects all required evidence	Keep colour transforms tied only to image validity
Water/road identification without unreliable height	Implemented and tested: both are classification-only and excluded from regression; v1 development metrics exposed a near-total water collapse	Compare V4 with paired V3 on the same DC+Philadelphia split, then report water/road metrics on a newly reserved external geography
Height protection	Production checkpoint and pointer protection implemented; the corrected legacy triplet completed and rejected both experimental height systems versus the protected model	Finish the independent tensor audit for the active head-only correction; do not promote automatically
Corrected HighBuild supervision	Implemented and run: two immutable pixel protocols plus per-building height, outline/boundary, support, and measured/estimated slices across 160 validation scenes	Improve annotated-building recall and obtain exhaustive negative/instance labels before claiming precision or count accuracy
OpenCanopy provenance and richer vegetation labels	Provenance implemented: accepted 600/120/120 manifests have authenticated dates, regions, source scenes and URLs; code and a smoke sample preserve upstream low/medium/high vegetation IDs; a 477/123 region+source-scene holdout plan is sealed	Run the full sidecar backfill and use the stronger holdout before richer-class training
GAMUS unit/pixel-scale audit	Implemented in <code>gamus_preparation_v1</code> : nominal 0.33 m pixels documented; local georeferencing absent; AGL metres remain an explicit unverified assumption	Obtain publisher/source-product confirmation before claiming absolute unit proof
DAV2 prior authentication	Implemented: all 8,720 approved pairs passed; current bytes are hash-locked; five train/validation samples reproduced exactly	Future caches must embed source/model/config hashes at creation time
Six-class evaluation	Implemented and reported on development validation: per-class precision/recall/IoU/F1, exact 6 x 6 confusion matrix, road boundaries, and water-vs-dark proxy	Recompute macro scores from all six final classes; do not substitute the easier five-group score formed by merging low vegetation and trees
Reviewer/app outputs	Six-map tiled inference support exists, but no GAMUS candidate is enabled in the active app	After validation: wire a six-class overlay, height summaries, coverage, and scale-aware area
Confidence	Not calibrated	Do not show probability-of-correctness until calibrated on held-out geography
Flood and snow modules	Future, explicitly separate	Flood needs terrain/hydraulics; snow cover needs labels; snow depth needs metric supervision

Common preparation contract

Both experiments must use the same approved IDs, radiometry, crop policy, cached relative priors, masks, height units, and validation code. Each sample has three independent masks:

1. `image_valid_mask`: pixels with usable optical input;
2. `classification_valid_mask`: pixels with a supported class in 1..6;
3. `height_valid_mask`: finite, physically accepted metric AGL labels.

The regression mask is `height_valid_mask` intersected only with ground, building, low-vegetation, and tree classes. Water and road remain valid for classification and invalid for height regression. Missing/ignored labels never become zero-height evidence. Colour augmentation stays disabled until this contract is verified.

The current six-epoch development protocol visits every one of the 5,001 approved training tiles once per epoch, using one deterministic-random 384 px crop from each tile. It does not claim that every source pixel is seen in six epochs. Validation processes all 859 approved 1024 px tiles.

Experiment order

1. Complete unit, pixel-scale, >200 m, prior-integrity, and validity audits.
2. Run the unchanged three-height-routing-group pilot and freeze its artifacts.
3. Initialize a separate six-class candidate from the same protected checkpoint.
4. Train it with the same height objective plus a six-class auxiliary loss.
5. Evaluate the protected model, height pilot, and expanded candidate with identical masks and inputs.
6. Run corrected HighBuild per-instance/boundary checks and OpenCanopy/forest regression checks. **Complete: the candidate failed against the protected model, so promotion remains blocked.**
7. Run the isolated six-class-head-only correction and verify bit-for-bit that every inherited height/shared tensor remains equal to the protected model.
8. Train paired V3 and V4 classifier heads on the same sealed DC+Philadelphia learning partition. A one-time NYC evaluation may measure transfer by those new heads, but must be labelled classifier-head-excluded rather than system-unseen.
9. Evaluate a separately reserved external geography that has not influenced the shared model, height pipeline, classifier, thresholds, or case selection.
10. Only after all classification, height, external-domain and genuinely new geography gates pass should app exposure be considered.

Required reports

Height is reported in metres as RMSE, MAE, bias, correlation and R-squared for overall, city, ground, building, vegetation, tall building and tall vegetation. The expanded candidate must not exceed the predeclared regression limits used by the height pilot, and it must be compared directly with the height pilot, not merely with historical scores.

Identification is reported independently for every one of the six categories: precision, recall, IoU, F1, support and the full confusion matrix. Water-shadow confusion and road boundary quality receive named slices. Empty or unsupported classes are reported as unavailable, never silently treated as perfect. The six-class macro values are recomputed from the exact final 6 x 6 matrix. Combining low vegetation and trees creates a five-group analysis and must be labelled separately; it is never a replacement for six-class macro F1.

HighBuild is evaluated per building as well as per pixel: matched-instance height MAE/RMSE/bias, outline/boundary score, detection/count precision and recall, and separate measured versus estimated-height annotations. Square-metre coverage is emitted only when CRS/pixel scale is authenticated.

Promotion rule

The expanded candidate result is complete but **not promotion-eligible**. The live model changes only after corrected HighBuild/OpenCanopy and genuinely new external-geography evidence is supplied, every required guard passes, and a human explicitly approves the change. NYC alone cannot satisfy that requirement.

Frozen height-focused result

The control early-stopped after epoch 4. Its best raw validation result was RMSE 4.471 m, MAE 2.646 m, bias +0.320 m, correlation 0.781 and R-squared 0.608. Domain RMSE was 2.497 m ground, 4.542 m building and 5.037 m vegetation; tall-building and tall-vegetation RMSE were 12.393 m and 10.367 m. It is not promotion-eligible: ground RMSE regressed beyond the frozen limit and building identification F1 was 0.631 versus the 0.815 protected baseline. The exact epoch-4 result remains reproducible through `checkpoint_latest.pt`; the production pointer was verified byte-identical after completion.

Expanded six-class result

The isolated candidate early-stopped after epoch 4. Its GAMUS development result was RMSE 4.442 m, MAE 2.604 m, bias +0.166 m, correlation 0.783 and R-squared 0.613. Against the height-focused control it improved overall RMSE by 0.029 m, ground RMSE by 0.113 m and vegetation RMSE by 0.030 m; building RMSE regressed by 0.020 m and tall-vegetation RMSE regressed by 0.286 m. The latter exceeds the frozen 0.20 m cap.

Six-class macro F1/IoU were 0.467/0.330. Per-class F1 was 0.402 ground, 0.690 buildings, effectively 0.000 water, 0.517 roads, 0.573 low vegetation and 0.621 trees. Road-boundary F1 was 0.257. The near-zero water recall is a clear failed category, and the low water-vs-dark false-positive proxy is not evidence of good water detection because the model almost never predicts water.

The comparison report is saved at `outputs/evaluation/gamus_six_class_comparison_v1/report.json`. A strict reporting check initially rejected harmless independent-CUDA accumulation drift (maximum 9.9e-6); it was corrected to a recorded 1e-4 fail-closed numerical tolerance, covered by 375 passing tests, without resuming training.

Current comparison limitations

The official GAMUS train and validation IDs do not overlap, but both cover DC and Philadelphia. In DC, 350 of 359 validation tiles are directly adjacent to a training tile in the four-neighbour grid, and all 359 are adjacent under an eight-neighbour test. Current validation is therefore a useful development comparison, not the final genuinely new-geography claim. NYC exclusion is also only a classifier-head check because other inherited system components have New York exposure.

The production baseline is evaluated with its deployed `protected_vegetation` fusion while both experimental candidates use the same `legacy` fusion. This is correct for deciding whether an entire candidate system is safer than the live system, and the height-pilot versus six-class ablation remains controlled. It does mean that protected-versus-height-pilot gains cannot be attributed to fine-tuning alone without an additional untrained legacy-fusion baseline.

Corrected HighBuild/OpenCanopy triplet result

The full corrected comparison is complete at `outputs/evaluation/corrected_legacy_triplet_v1/report.json`. Its status is failed because the acceptance gates failed, not because execution crashed. All three models used identical full-scene inputs and supervision. This report did not use official GAMUS test data; the older Stage-3 final audit had already consumed that test, so it is forbidden for reuse. The report is complete and the live pointer/checkpoint stayed unchanged.

Under the primary strict-measured protocol, pooled `protected/height-pilot/joint-` candidate RMSE was 10.161/15.460/15.463 m. OpenCanopy all-valid forest RMSE was 6.707/16.919/16.910 m, OpenCanopy vegetation-only RMSE was 8.361/10.893/10.802 m, and corrected HighBuild strict-measured building-only RMSE was 13.243/13.435/13.457 m. The joint candidate was approximately neutral versus the height pilot but failed every protected-system aggregate/forest guard and also exceeded the protected HighBuild building-RMSE allowance by 0.064 m beyond the permitted +0.15 m regression.

The protected model's canonical corrected full-scene baselines are therefore 13.243 m for HighBuild building-only, 8.361 m for OpenCanopy vegetation-only, and 6.707 m for OpenCanopy all-valid forest. These scopes have different masks and supports and

are not interchangeable. Historical crop-based, zero-filled, inclusive-annotation, or pooled metrics remain useful only under their original labels.

The evaluator also produced per-building and boundary evidence. The protected model matched 400 of 2,581 rasterizable annotations versus 282 for the joint candidate; annotated recall was 0.155 versus 0.109, boundary F1 0.488 versus 0.456, and matched-outline IoU 0.738 versus 0.709. Precision and count error remain deliberately unavailable because HighBuild supplies positive outlines, not exhaustive negative/instance truth.

Isolated classification correction result

`configs/multidomain_surface_gamus_six_class_head_only_v2.yaml` completed after epoch 6 with epoch 3 as the best development checkpoint. It was initialized from the protected checkpoint, permitted only `fine_semantic_head`. to train, set every height and three-height-routing-group loss to zero, and used a hash-sealed water-aware sampler plus class-weighted focal loss.

The independent audit at `outputs/evaluation/gamus_six_class_head_only_v2/checkpoint_best_audit.json` proved exact equality for all 240 inherited tensors (31,428,489 values). Best macro F1 was 0.506; class F1 was 0.379 ground, 0.833 buildings, 0.220 water, 0.617 roads, 0.518 low vegetation and 0.467 trees. Ground, low-vegetation and tree retention failed, so it was not promoted.

An external additive-bias development calibration then improved full DC+Philadelphia macro F1 to 0.547 and class F1 to 0.413/0.842/0.324/0.622/ 0.522/0.557 in the same class order. This changes argmax decisions only, not height or probability calibration. It still misses the original low-vegetation/tree retention targets and is not unseen-geography evidence.

The sealed spatial-refined head comparison completed after epoch 6. Epoch 2 was best at macro F1/IoU 0.554/0.400, with class F1 0.409/0.840/0.386/0.630/ 0.483/0.575 for ground/buildings/water/roads/low vegetation/trees. Its exact audit proves all 240 inherited tensors and 31,428,489 inherited values stayed bitwise equal to the protected model. It improves the isolated classifier but still fails the original low-vegetation and tree retention gates, so it remains inactive.

A decision-only calibration protocol is sealed for that best checkpoint. It can adjust only six external argmax offsets, not weights, heights or confidence, and requires a no-regression sampled gate before one authoritative full DC+Philadelphia pass. A spatial DC+Philadelphia-only learning recipe has also passed preflight for 3,837 learning tiles, 859 epoch-selection tiles and one locked 1,164-tile NYC evaluation; no NYC imagery was opened by preflight. That run provides a fair classifier-head comparison and, at most, a one-shot classifier-excluded NYC result. It cannot support a system-unseen claim or replace the separate external-geography evidence required for app promotion.

OpenCanopy provenance result

`outputs/data_audits/open_canopy_v2_provenance_v1.json` passes with no accepted cross-split region overlap. It indexes 600 training, 120 validation, and 120 locked-test samples, including official source identity, EPSG:2154 location, SPOT imagery date, LiDAR acquisition date, source URLs, GSD and metadata-file hash. Extra materialized candidates left behind during streaming selection are reported separately and are not treated as accepted training/evaluation rows. The accepted splits do share many large SPOT source mosaics: 57 train/validation, 49 train/test and 44 validation/test source identities overlap. Therefore this is geographic-region isolation, not source-acquisition isolation. A deterministic region-plus-source-scene plan now produces 477 learning and 123 held-out samples and is the recommended stronger comparison. Exact upstream classification IDs can also be written to separate, versioned sidecars without changing the current height masks; low, medium and high vegetation are LiDAR height strata, not tree species. A real 384-pixel smoke sample passed. Full backfill and training remain pending so the current OpenCanopy artifacts stay untouched.

HighBuild instance-contract result

The per-building evaluator verified all 160 corrected validation tiles and 2,588 COCO annotations. Seven valid outlines smaller than the raster pixel grid are disclosed and excluded rather than fabricated; 1,133 annotations without the optional estimated flag are also disclosed under the frozen contract. Matched-building height and outline metrics are now supported. Because the current HighBuild background is unknown rather than exhaustively labelled non-building, detection precision and count error deliberately remain unavailable until an independent exhaustive mask or instance reference exists.

35 Explore mapping — 27 September 2026

SOURCE docs/history/EXPLORE_MAPPING_2026_09_27.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Delivered

- Fly and Walk replace the original-image reference card with a live, heading-up minimap of the same source image.
- A high-contrast glowing player arrow, direction cone and subtle pulse show horizontal position and facing direction. Reduced-motion preferences disable the pulse.
- The image scrolls and rotates with the camera; Walk uses closer context than Fly. Coverage expands with travel speed and remains bounded by the scene footprint. Changes in coverage ease smoothly.
- M toggles a native modal covering the complete browser viewport, including inside the app's fullscreen mode. The whole image is visible with the player arrow and a dashed outline of the minimap coverage.
- Movement, jumping, look and scanner updates pause while the full map is open. M or the close button resumes Explore at the same location. Esc closes the map without also exiting Explore; click the scene to recapture the mouse if needed.
- Outside-image positions are explicitly marked in amber, with the arrow pinned at the map edge.
- Leaving Explore restores the original-image reference and existing enlargement controls.

Position and scale

The arrow uses camera X/Z in Fly and foot X/Z in Walk, not the crosshair's intersection with a roof or hill. Walk head bob, altitude and display exaggeration cannot move the horizontal marker.

Both positive horizontal pixel scales are required for metre labels. Otherwise the map uses pixel distances and explicitly says the geographic scale is unknown. Full-map orientation is image-up, not assumed north. No additional height inference or geolocation is fabricated.

Typical known-scale local coverage starts at 180 m for Walk and 420 m for Fly, with a lower limit of 24 source cells on coarse imagery and up to four seconds of horizontal travel visible. Unknown-scale coverage uses a bounded fraction of the image footprint. Small scenes cap coverage near their complete footprint.

Performance and implementation

The React performance review kept transient camera poses in refs. Small canvas views render at a bounded rate (approximately 30 Hz), cap pixel density at 2, skip drawing in hidden tabs, and clean up animation frames and ResizeObservers. No new dependencies, backend changes, classifier changes or height-model changes were needed.

Implementation: `viewer/app/explore-map.tsx`, `viewer/app/explore-map-math.ts`, and the existing `terrain/workspace/styles`.

Verification

- All 155 viewer unit tests pass, including 9 new map geometry/scale tests.
- TypeScript no-emit check passes.
- Targeted ESLint check covers the changed TypeScript/React implementation.
- Dedicated browser verification passes: Orbit shortcut isolation; image-to-camera registration; look rotation and movement tracking; complete M-screen coverage; paused navigation; M-repeat protection; close-button behavior; restored original

reference; Walk context; paused jumping; Esc priority inside fullscreen; Urban, Sparse, Hilly and Forest scene mapping; and full-map layout at 390 px width.

- No page exceptions in the successful integration run.
- Evidence: `outputs/runtime/explore_map/verification.json` and adjacent screenshots. Repeatable harness: `scripts/verify_explore_map.mjs` using the dedicated agent-browser session's CDP endpoint.

The mapping feature reuses each scene's RGB preview, including the backend-produced preview for TIFF uploads. This verification exercised the four existing landscape demonstrations; it did not start a new upload/inference job or alter saved predictions.

36 Explore: Fly and Walk

SOURCE docs/history/EXPLORE_WALKTHROUGH_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Follow-up update: The movement settings below describe the initial implementation. Normal pace is now 4.3 m/s with a game-style step bob (up to 4.5 cm downward dip), quicker acceleration/deceleration and the same G boost. See INSPECT_CLASSES_AND_WALK_POLISH_2026_09_19.md for current settings and verification.

Use

Load an image or select a landscape reference. Choose **Explore**, then **Fly** or **Walk**.

Control	Fly	Walk
Mouse	Look around	Look around
WASD / arrow keys	Move in the viewing direction	Move along the surface; looking up does not lift you
Space / Shift	Rise / descend	No vertical movement
Hold G while moving	3× speed	3× pace
Aim for 2 seconds / left-click	Surface scanner	Same surface scanner
Esc	Exit to Orbit	Exit and restore the pre-walk Orbit view

Walk uses a 1.80 m eye offset above the reconstructed surface for metre-valued scenes. Optional gentle head motion follows actual distance travelled, eases to rest, and is disabled by the system's reduced-motion preference. It can also be turned off in the Explore chooser.

Movement and collision

- Foot movement uses camera yaw only, normalised diagonal input, and gradual acceleration/deceleration.
- Map-scaled scenes use 1.4 m/s walking and 4.2 m/s with G.
- The feet follow the same triangle-interpolated surface used in the visible mesh.
- The body stays within the image boundary. Missing height pixels are not treated as zero ground.
- Building footprints and building-labelled pixels block entry. Tall classified canopy is not assumed to be walkable ground.
- Substepped movement guards against tunnelling through walls. Sliding allows movement alongside obstacles.
- Slopes over 40 degrees and abrupt body-sized height discontinuities are rejected. This is a conservative navigation aid, not a route-safety system.
- Entry searches for an open starting point near the orbit target. If none is found, Walk reports that and suggests Fly rather than placing the camera on a roof.
- Close clipping adapts to human scale, including large hilly scenes. Walk uses a 65-degree field of view; the original camera settings are restored on exit.

Honest limitations

The scene is a predicted/DEM-assisted surface, **not a surveyed pedestrian environment**. Walk does not recover ground underneath tree canopies, interiors, bridges, stairs, or unseen façades. Collision quality depends on predicted classes and geometry. A noisy reconstructed surface can block travel; use Fly to cross areas that cannot support walking.

If horizontal scale is unknown, pace is explicitly labelled approximate. A dimensionless relative-height input gets a human-scale preview, not a metre-scale claim. Display exaggeration still applies; no model heights or exports are changed. The small head bob adds at most 2.4 cm to the nominal eye offset.

Preserved features

Neutral startup, manual reference selection, import reveal, metric-grid transition, first-person target dwell/click scanning, flat roofs and pending-roof-height message, and existing Fly collisions/controls remain intact. Model checkpoints and inference were not modified.

Verification

- TypeScript and focused ESLint checks passed.
- 56 targeted tests passed, including 9 new walking tests: pitch-independent input, diagonal speed, slopes in both directions, cliffs/drops, sprint wall collision/sliding, NoData/borders, safe spawning, anisotropic horizontal scaling, head motion/reduced motion, invalid scale and bounded large deltas.
- Live browser: neutral startup; Urban → Explore → Walk; eye offset 1.80 m at rest; Space/Shift leave position unchanged; 2-second aim and click both show the building scanner; Esc restores the original orbit camera.
- Live hilly reference: ground height changed while walking and the camera retained its eye offset. Normal movement covered about 2.06 m and boosted movement about 6.01 m in comparable intervals (frame timing/acceleration apply).
- Live Fly regression: Space raised the camera about 12.60 scene units, and Space+G about 37.78 in comparable intervals; mouse-look and pointer lock worked.
- Live metric-grid Walk: scanner showed CLICK SCAN, and with walking motion disabled the eye offset stayed exactly 1.80 m during movement. No browser runtime errors were recorded.

Screenshots: `outputs/runtime/explore-options.png`, `walk-urban.png`, `walk-hilly.png`, `walk-metric-grid.png`.

Primary implementation: `viewer/app/walk-navigation.ts`, `viewer/app/terrain-workspace.tsx`, `viewer/app/globals.css`; pure regression tests: `viewer/tests/walk-navigation.test.mjs`.

37 External metric-height holdout: Bay of Plenty v1

SOURCE docs/history/EXTERNAL_METRIC_HEIGHT_HOLDOUT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **downloaded, publisher-hash verified, aligned, and sealed unconsumed; no model inference or score has been run.**

This package closes the missing external-geography evidence path without reusing GAMUS official test data or the known DFC19 showcase. It is a single, fixed New Zealand scene for one post-freeze metric-height evaluation. It must never be used for training, validation, calibration, threshold choice, checkpoint choice, fusion choice, or showcase selection.

Why this source

The source is the official Toitu Te Whenua Land Information New Zealand public AWS archive:

- New Zealand Imagery registry: <https://registry.opendata.aws/nz-imagery/>
- New Zealand Elevation registry: <https://registry.opendata.aws/nz-elevation/>
- LINZ attribution requirements:

<https://www.linz.govt.nz/products-services/data/licensing-and-using-data/attributing-elevation-or-aerial-imagery-data>

The registry describes open aerial imagery and 1 m LiDAR-derived DSM/DEM products with STAC metadata. Each fixed collection says CC-BY-4.0; its STAC provider block identifies BOPLASS as licensor, AAM NZ as producer, and LINZ as host/processor.

Required attribution:

Sourced from the LINZ Data Service and licensed by BOPLASS for re-use under the Creative Commons Attribution 4.0 International licence.

Fixed scene (chosen before pixel inspection)

Selection was mechanical, not visual: the first lexicographic RGB STAC item in the fixed 2018-2019 Bay of Plenty urban collection whose complete footprint is contained by same-season DSM and DEM coverage.

Role	Official collection / item	Capture range	Grid
RGB orthophoto	bay-of-plenty_2018-2019_0.1m, BC36_10000_2314	2018-07-11 to 2019-01-16	EPSG:2193, 0.1 m
LiDAR DSM	bay-of-plenty_2018-2019/dsm_1m, BC36_10000_0302	2018-11-30 to 2019-04-29	EPSG:2193, 1 m
LiDAR DEM	bay-of-plenty_2018-2019/dem_1m, BC36_10000_0302	2018-11-30 to 2019-04-29	EPSG:2193, 1 m

The official capture ranges overlap, but per-pixel dates are unavailable. Real change between the optical and LiDAR acquisitions can therefore contribute to measured error and must be disclosed.

Exact URLs, collection IDs, item bounds, publisher SHA-256 multihashes, file sizes, licence, attribution, CRS, GSD, and dates are frozen in configs/external_metric_height_nz_bop_v1.json. Its repository seal is:

ddf68cdf382366450d29ed433ddb07a737c2fe07581e03b7c32e290a68eb456c

Fixed reference and masks

The metric object-height reference is:

reference nDSM = LiDAR DSM in metres - paired LiDAR DEM in metres

Because DSM and DEM are paired products from the same survey, subtraction removes their shared absolute vertical datum. The GeoTIFFs expose EPSG:2193 as their horizontal CRS but do not embed a compound vertical CRS, so an absolute elevation-datum claim is intentionally not made.

Before any model output existed, three supports were fixed:

1. `all_valid`: valid RGB + finite DSM/DEM + nonnegative DSM-minus-DEM.
2. `object_gt2m`: `all_valid` and reference nDSM strictly above 2 m.
3. `tall_object_gt5m`: `all_valid` and reference nDSM strictly above 5 m.

Only nodata, nonfinite values, and negative DSM-minus-DEM artifacts are excluded. There is no upper-height cutoff. The object supports prevent the large amount of flat ground from hiding structural failures.

Preparation and one-shot use

Acquire, align to the fixed 1 m grid, derive nDSM/masks, and verify hashes:

```
.\venv\Scripts\python.exe scripts\data\prepare_external_metric_height_holdout.py acquire-prepare
.\venv\Scripts\python.exe scripts\data\prepare_external_metric_height_holdout.py verify
```

The compact package lives only at:

D:\MSRData\external_holdouts\nz_bop_2018_2019_bc36_2314_v1

Preparation imports no model and runs no inference. It creates `SEALED_MANIFEST.json`, hashes every raw and derived file, and writes an explicit `NO_TRAIN_OR_TUNE.txt` marker.

Acquisition verification on 2026-09-15 produced:

Check	Sealed result
Raw official files	11 files, 97,123,077 bytes
Prepared aligned rasters	7 files, 3,658,293 bytes
All-valid support	326,925 pixels
Object support, reference nDSM >2 m	133,567 pixels
Tall-object support, reference nDSM >5 m	124,217 pixels
Negative reference artifacts excluded	18,675 pixels
Package-manifest SHA-256	6cff2c60939e8ea5809569bc6f6216dc589728b106420c5ebd0383788fdf7e1b

These are preregistered mask-support counts, not model results. Neither an `EVALUATION_PLAN.json` nor a `CONSUMED.json` exists yet.

Local audit order is also preserved: the final contract was sealed at 11:10:08 UTC, the first raw source file was created at 11:16:22 UTC, and the prepared manifest was sealed at 11:18:52 UTC. Thus the scene, masks, and thresholds were fixed before the first raster was downloaded or decoded.

Only after the candidate, configuration, code, and app pointer are final may a maintainer create `EVALUATION_PLAN.json` with `sealed-plan`. The evaluator must then create `CONSUMED.json` *before* its first model read. A crash still counts as consumption; any exceptional rerun must be disclosed. Complete all-valid, object-only, and tall-object metrics must be published regardless of outcome.

Geography claim limit

A repository inventory on 2026-09-15 found no Bay of Plenty or New Zealand samples in known Monocular Surface Reconstruction training/evaluation manifests. A Christchurch OpenEarthMap holdout exists only as an unacquired design and remains untouched. The permitted wording is:

"One-shot transfer to a preregistered Bay of Plenty metric-height scene, new to known Monocular Surface Reconstruction project data, using an independently hosted LINZ/BOPLASS orthophoto and LiDAR reference."

This does not prove worldwide generalisation or absence from unknown upstream foundation-model pretraining.

38 Two-step fullscreen Escape

SOURCE docs/history/FULLSCREEN_ESCAPE_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Behavior

- Fullscreen + Explore (Fly or Walk): a short Escape press releases mouse capture and returns to Orbit without leaving fullscreen.
- The next Escape press in Orbit exits fullscreen.
- Repeated keydown events from a held key do not cascade through both app actions. The browser's long-hold Escape safety exit remains available.
- Windowed Explore still returns to Orbit with Escape; the fullscreen button still exits directly.

Browser handling

Native fullscreen uses `navigator . keyboard . lock ( [ 'Escape' ] )`. Only Escape is captured; the lock is released on fullscreen exit or component cleanup. Without this browser API, or if its request is rejected, the viewer expands within the app window instead. This fallback preserves the two-step behavior but does not hide browser/desktop chrome. A `ResizeObserver` keeps the canvas aligned with that expanded viewport.

References: [Chrome fullscreen/Keyboard Lock guide](#), [requestFullscreen](#).

Verification

- TypeScript and focused ESLint passed.
- 68 targeted viewer tests passed, including five fullscreen routing/feature-detection tests.
- Chromium browser with real Escape key presses: native fullscreen Fly -> Orbit (still native fullscreen) -> windowed Orbit passed.
- Same native-fullscreen sequence with Walk passed; camera restoration and mouse-capture release checked.
- Browser with Keyboard Lock unavailable: full-window fallback, Fly -> Orbit -> normal layout passed.
- Browser with Keyboard Lock rejecting: fallback activated without leaving native fullscreen stuck; Escape restored normal layout.
- Browser reported no uncaught errors during these checks.

No model weights, inference, prediction values, or dataset files changed.

39 GAMUS direct-height experiment protocol

SOURCE docs/history/GAMUS_DIRECT_HEIGHT_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: predeclared before candidate training.

Purpose

Test whether genuine GAMUS metric-height fine-tuning improves Monocular Surface Reconstruction's final surface prediction. This is not another identification-only router run. The protected application checkpoint remains unchanged throughout.

Data roles

- GAMUS official train: all 5,004 source tiles were audited; 5,001 approved tiles are training data and three all-class-255/no-supervision tiles are quarantined by the immutable QC contract.
- GAMUS official validation: all 859 usable tiles are development evaluation.
- GAMUS official test: all 2,861 source tiles were excluded from this pilot's training, tuning, epoch selection, thresholds, and architecture decisions. The older Stage-3 final audit consumed the official test once; it is now immutable historical evidence and is forbidden for every new V4 decision or repeat evaluation.
- Corrected HighBuild/Open-Canopy validation: regression-safety evidence only; it is not merged into GAMUS validation statistics.
- A newly locked external/geographic reference set is required before an app promotion or an unseen-city claim. Until then, a validation win creates only a candidate checkpoint.

NYC does not satisfy that system-level unseen requirement. A fresh classifier head can exclude NYC while learning on DC+Philadelphia, but the broader system already has New York exposure through earlier GAMUS experiments and its inherited HighBuild-trained checkpoint.

Every accepted GAMUS tile must pass the versioned QC index. The QC report records split counts, file identities, tensor shapes, numeric ranges, semantic classes, unit provenance/assumption, and quarantined records. Files are never deleted by QC.

Model and learning objective

- Warm start from the protected Monocular Surface Reconstruction checkpoint.
- Keep the cached Depth Anything V2 prior fixed for this experiment.
- Directly supervise the final height output against valid GAMUS above-ground heights; semantic identification remains an auxiliary objective.
- Allow only the late encoder/decoder, metric-height path, and required adapter heads to learn at conservative, separately controlled learning rates.
- Use a fresh optimizer and schedule. Do not reuse Stage-1 or Stage-2 optimizer state.
- Use BF16 on the RTX 4070 with a small microbatch and gradient accumulation. Reduce crop size only after a confirmed CUDA out-of-memory failure.
- Save each epoch and a resumable latest checkpoint. Never write to the live pointer automatically.

Fixed validation gates

All comparisons use identical inputs, masks, units, crops/tiles, and metric code for the protected baseline and candidate.

On full GAMUS validation:

- pooled RMSE improves by at least 0.20 m;
- pooled MAE improves by at least 0.10 m;
- correlation drops by no more than 0.01 and R-squared drops by no more than 0.02;
- neither DC nor Philadelphia RMSE regresses by more than 0.10 m, and at least one improves by 0.20 m or more (enforced with a 0.20 m city-macro improvement gate plus the per-city caps);
- ground RMSE regresses by no more than 0.05 m;
- building and vegetation RMSE each regress by no more than 0.10 m, and at least one improves by 0.20 m or more (enforced with a 0.20 m object-domain macro improvement gate plus the per-domain caps);
- tall-building and tall-vegetation RMSE each regress by no more than 0.20 m;
- semantic macro F1 drops by no more than 0.02 and no class F1 drops by more than 0.03;
- no NaN, empty required stratum, split overlap, missing prior, unit mismatch, or artifact-identity mismatch is allowed.

On corrected legacy validation:

- overall, urban, forest, building, and vegetation RMSE each regress by no more than 0.15 m;
- corresponding MAE values regress by no more than 0.10 m;
- HighBuild results are reported separately under `highbuild_all_annotated_positive_v1` and `highbuild_measured_coco_intersection_v1`.

For later comparisons, the canonical corrected full-scene protected baselines are 13.243 m HighBuild strict-measured building-only RMSE, 8.361 m OpenCanopy vegetation-only RMSE, and 6.707 m OpenCanopy all-valid forest RMSE. They have different masks and support and must never be treated as interchangeable scores.

These gates are frozen before results are seen. Failure is recorded honestly; thresholds are not relaxed after the run.

Decision rule

A gate-passing pilot authorizes a longer or correctly masked replay experiment. It does not automatically replace the application model. Promotion additionally requires a newly locked geographic holdout, visual sanity checks, API/viewer regression tests, checkpoint hashing, and explicit human review.

40 GAMUS classifier-head geographic exclusion protocol

SOURCE docs/history/GAMUS_GEOGRAPHIC_HOLDOUT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **sealed for a controlled classifier comparison; not an untouched-system holdout.**

This protocol reserves the entire approved New York City portion of the GAMUS training split before a fresh classifier head is fitted on Washington, DC and Philadelphia. It is useful evidence of transfer by that new head, but it is not a genuinely unseen-system test: earlier GAMUS experiments trained on NYC, and the protected shared/height checkpoint inherits HighBuild training that includes New York examples (data/highbuild_reviewer/manifests/train.csv, including row 132). A separate geography that has not influenced any system component remains required for the final generalisation claim.

The official GAMUS test cannot fill that role. It was consumed once by the older Stage-3 final audit and is now a read-only historical record, forbidden for V4 training, tuning, threshold selection, model selection, visual case selection, or repeat evaluation.

Sealed split

Role	Cities	Approved tiles	Allowed use
Learning	Washington, DC + Philadelphia	3,837	Model fitting and learning-only statistics
Development validation	Washington, DC + Philadelphia	859	Epoch choice, tuning, thresholds and error analysis
Classifier-excluded check	New York City	1,164	One evaluation after the new head and all decisions are frozen
Buffer	Not applicable for a whole-city split	0	Never train
Official GAMUS test	Previously consumed by Stage 3	Not applicable	Historical audit only; never reuse

The choice of NYC for this head-only exclusion uses only the predeclared city prefix in approved GAMUS training tile IDs. No RGB pixel, semantic class, height label, cached prior, model output, or class distribution from this 1,164-tile partition was inspected to choose the city for the new comparison. That clean selection procedure does not erase the system's earlier New York exposure.

Why the old validation cannot support the claim

The existing train and validation IDs do not overlap, but they share the same DC and Philadelphia city domains. For the only released two-dimensional tile index (DC), **350 of 359** validation tiles touch a training tile under a four-neighbour check, and **all 359** touch one under an eight-neighbour check. That validation remains useful for development, but it is not genuinely unseen geography.

NYC is absent from the official validation split. Removing every approved NYC tile from the fresh classifier's learning set therefore gives a useful whole-city head-transfer check, which is stronger and easier to defend than selecting a convenient-looking local block after seeing results. It must be reported as **excluded from this classifier head's training**, not as unseen by the complete Monocular Surface Reconstruction system.

What the local files can and cannot prove

All **5,860** approved train/validation RGB HDF5 containers were inspected for metadata without reading their image pixels. None has file-level or image-level attributes; in particular, none embeds a CRS, affine transform, coordinates, pixel size or acquisition identity.

- DC names contain two released indices, so grid-neighbour checks are possible.
- NYC and Philadelphia names contain scalar release indices. They are treated as opaque IDs, not coordinates.
- City prefix is consequently the strongest geographic grouping provable from the files currently available.
- This does not prove transfer to a different country, satellite, season, ground sampling distance or terrain distribution.

The official GAMUS paper and code describe the multi-city dataset and nominal 0.33 m imagery, but the local containers still lack per-tile geospatial proof:

- <https://arxiv.org/abs/2305.14914>
- <https://github.com/EarthNets/RSI-MMSegmentation>

Fail-closed future training use

Point the next GAMUS experiment at:

`outputs/data_audits/gamus_geographic_holdout_city_nyc_v1/learning_approved_samples.json`

Its SHA-256 is:

`d56764bc9a3cde31760189b1da40cf207de1b9142c7531b24190ed6e2312e5d9`

The derived index is compatible with the existing loader and strict split-count validator. It contains only the 3,837 learning IDs plus the unchanged 859-row development validation inventory. It deliberately contains no official-test split. A direct loader/validator smoke check passed.

Before training:

1. use the derived approved index and its exact hash;
2. regenerate any class-balancing/sampling index from the 3,837 learning IDs only; the existing full-train sampling index is forbidden;
3. calculate normalization, weights and any calibration inputs from learning data only;
4. keep NYC out of training, epoch selection, thresholds, post-processing, error analysis and visual case selection;
5. freeze checkpoint, thresholds, fusion and guards before the one classifier-excluded NYC evaluation;
6. report all six class metrics and height metrics by city/object category;
7. do not promote solely because this classifier-excluded check passes; corrected HighBuild, OpenCanopy and the existing regression guards must also pass.

`assert_training_ids_respect_contract(...)` provides a reusable fail-closed check. It rejects a training inventory containing a holdout, buffer, development-validation or unknown ID, and can require the exact complete learning partition.

Important historical limitation

The current height-focused pilot and the completed six-class candidate were trained on all 5,001 approved official-training tiles. They have already seen NYC and **must not** be presented as unseen-NYC results. Even a newly trained V4 classifier head using the sealed learning index remains attached to shared features and a protected height system with prior New York exposure. Its NYC result is therefore classifier-head-excluded evidence only. A clean system-level unseen-geography claim requires a separately reserved external location. The production app checkpoint and its pointer were not changed by this work.

Reproducible artifacts

The versioned folder `outputs/data_audits/gamus_geographic_holdout_city_nyc_v1/` contains:

- `report.json`: concise audit and protocol result;
- `contract.json`: complete, hash-sealed assignments and use policy;
- `assignments.csv`: human-readable role for every development tile;
- `learning_approved_samples.json`: loader-compatible filtered index;
- `learning_sample_ids.txt` and `geographic_holdout_sample_ids.txt`;
- `buffer_sample_ids.txt`: intentionally empty for whole-city isolation;
- `artifact_hashes.json`: exact byte hashes for every artifact.

The planner refuses to run when the approved-index hash changes, a tile ID is malformed, roles overlap, the holdout city appears in development validation, the learning set loses city coverage, or an existing sealed artifact differs.

Prepared head-only comparison

The separate candidate config is `configs/multidomain_surface_gamus_six_class_head_only_geographic_v1.yaml`. Its model, optimiser, schedule, loss, augmentation, and DC/Philadelphia validation-selection recipe are required by code to match the frozen head-only v2 recipe exactly. The only permitted data changes are the learning-only approved index and the learning-only water sampler.

`outputs/data_audits/gamus_head_only_geographic_candidate_v1/` now contains:

- a 3,837-row DC/Philadelphia sampling index, SHA-256
`32a71cc1cb2d9da75ce8774db5e4db342512991ebb37e4b461f776fe7196eedf`;
- a sampling audit showing 790 water-positive crop candidates and proving that all 1,164 NYC rows were excluded;
- a loader-compatible 1,164-row locked NYC index, SHA-256
`98281b990b5ccddfd4349d0b43f0216455a8bebc085bf1232abab10f689417`;
- a passing strict preflight report.

Preparation decoded learning sampling rows only. It did not construct a GAMUS dataset or open imagery, semantics, heights, priors, or official-test files.

After training, `scripts/evaluate_gamus_locked_nyc.py --freeze-only` seals the validation-selected checkpoint and predeclared thresholds without opening the classifier-excluded NYC partition. The explicit evaluation mode then creates a consumption marker before it constructs that dataset. A second pass is refused. Every per-class result, road-boundary metric, water/dark-pixel proxy, and height metric is reported even if the candidate fails its predeclared targets. Evaluation never promotes or changes the application model, and an NYC pass does not replace the required new external-geography evaluation.

41 GAMUS height-safe v2 pilot

SOURCE docs/history/GAMUS_HEIGHT_SAFE_V2_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: predeclared before training on 2026-09-15.

Why this run exists

The first direct-height pilot showed that GAMUS contains useful height signal: on GAMUS validation its overall RMSE fell from 7.121 m to 4.471 m, building RMSE from 5.250 m to 4.542 m, and vegetation RMSE from 8.897 m to 5.037 m. It was still rejected because ground RMSE worsened from 1.527 m to 2.497 m, building semantic F1 fell from 0.815 to 0.631, and corrected legacy validation regressed severely. The active application model was never changed.

This v2 pilot tests a much narrower hypothesis: can the existing protected model learn useful GAMUS height calibration through only its 33-parameter metric output head and 65-parameter canopy output head while all shared features, routing, building detection, fusion gates, and cached Depth Anything V2 priors remain fixed?

Locked data and supervision contract

- Training uses all 5,001 quality-approved GAMUS train tiles.
- Selection uses all 859 quality-approved GAMUS validation tiles.
- The official GAMUS test split is neither resolved nor evaluated.
- RGB validity, classification validity, and height validity remain separate.
- Height regression requires valid RGB, valid AGL, a valid class, and one of source classes 1, 2, 3, or 6.
- Background/unknown (0), water (4), and road (5) never supervise height.
- Invalid/missing heights may be stored as zero only behind a false regression mask; they never act as zero-height targets.
- Heights above 200 raw units remain excluded as rare, artifact-prone values.
- The local files do not declare units. Raw x1.0 is recorded as `metre_assumed`, so this pilot is development evidence, not a publishable metric claim by itself.
- Cached DAV2 priors are read-only inputs. No DAV2 model is run or updated.

Model safety

Only `base_model.height_head.*` and `canopy_height_head.*` are trainable. The protected checkpoint hash is `e2d507f7be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144`. The runner hashes the live checkpoint before and after training and fails if the live pointer or checkpoint changes.

Decision

The predeclared GAMUS gates remain identical to the first pilot: pooled RMSE and MAE must improve, correlation/R2 must be preserved, DC and Philadelphia must remain safe, ground/building/vegetation and tall-object guards must pass, and three-class identification must not materially regress.

Even a GAMUS gate-passing checkpoint is not promotion eligible. It must first pass the already-locked corrected HighBuild/OpenCanopy development protocols, a genuinely new geographic holdout, visual/API regression checks, and explicit human review. No official test result is permitted for this decision.

42 GAMUS preparation audit v1

SOURCE docs/history/GAMUS_PREPARATION_AUDIT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **passed for the height-focused comparison pilot only**. This is not permission to promote a checkpoint or to tune on the official test split.

Historical clarification: this preparation audit handled the official GAMUS test as inventory-only and did not score it. Separately, the older Stage-3 final evaluation consumed that test once. It is therefore no longer an untouched holdout and is forbidden for V4 tuning, selection, or repeat evaluation.

What was verified

- The immutable GAMUS quality index still has its declared SHA-256 and keeps 5,001 approved training tiles, 859 validation tiles, and 2,860 inventory-only test tiles. The official splits remain disjoint.
- Every approved RGB tile was paired with its cached Depth Anything V2 prior. All 8,720 pairs passed grid, numeric range, recorded model, recorded source path, and unit-tag checks. Current RGB and prior file bytes are locked by per-file SHA-256 in the identity manifest.
- Five deterministic samples spanning every city present in train/validation were regenerated with the locally locked DAV2 model. All 5,242,880 float16 output values matched the cache exactly. This audit did not recompute the official test split; the separate Stage-3 consumption noted above is preserved as historical final-audit evidence.
- The loader exposes separate image-, classification-, and height-validity masks. Real tiles contain thousands of pixels where the image is usable but the height label is not, and those pixels remain excluded from regression. Colour transforms now use image validity rather than height availability.

Units and geographic scale

- The GAMUS paper documents a nominal RGB/nDSM pixel resolution of **0.33 m** and 1024 x 1024 tiles (about 337.92 m wide nominally).
- The local HDF5 files do not embed pixel scale, CRS, transform, or geolocation. The 0.33 m value is therefore dataset-level documentation, not per-file geospatial proof. Precise square-metre output still requires trusted input georeferencing.
- The files and local README do not declare the numeric AGL unit. The official loader reads AGL without rescaling, but that does not prove the unit. For the comparison pilot only, Monocular Surface Reconstruction keeps the explicit assumption that raw AGL x 1.0 is metres. Results must be described as metre-assumed until the publisher or original source products confirm it.

Values above the 200 cutoff

Split	Pixels above 200	Affected tiles	Fraction of nonnegative finite labels	Maximum
Train	3,253	15 / 5,001	0.0000640%	392.60
Validation	43	9 / 859	0.00000482%	272.59
Official test (inventory only)	135,892	8 / 2,860	0.004705%	304.77

Some values above 200 belong to ground, road, low-vegetation, or tree labels, not only tall buildings. That is strong evidence that blindly raising the global cutoff would admit label artefacts. The comparison pilot therefore keeps the predeclared 200 cutoff, while the exact tile/class/city inventory is preserved for later source-level review.

Honest DAV2 provenance limit

The historical cache stores model name, source path and relative_0_1 units, but it did not store source/model/config content hashes when each prior was generated. The v1 audit does not invent that missing history. It locks the current RGB/prior/model/code bytes and adds independent deterministic recomputation evidence. Future cache generation should write those hashes at creation time.

Artifacts

- Machine report: `outputs/data_audits/gamus_preparation_v1/audit_report.json`
- Per-pair DAV2 identity manifest: `outputs/data_audits/gamus_preparation_v1/dav2_prior_identity.jsonl`
- Height-outlier records: `outputs/data_audits/gamus_preparation_v1/height_outliers.jsonl`
- Exact artifact hashes: `outputs/data_audits/gamus_preparation_v1/artifact_hashes.json`

The report, manifests, loader checks, and tests were created without modifying the GAMUS files, starting training, or changing the app's live checkpoint.

43 GAMUS six-class decision-bias calibration

SOURCE docs/history/GAMUS_SIX_CLASS_DECISION_BIAS_CALIBRATION.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **implemented and run once on development data; rejected for app use.**

The sealed default protocol was applied to the completed head-only v2 recipe. It was permitted only after training finished and `audit_gamus_head_only_checkpoint.py` proved that every inherited height/shared tensor is bit-for-bit equal to the protected model.

Why this exists

The head-only GAMUS recipe deliberately oversamples water tiles and gives water a larger training-loss weight. That is useful for teaching the head not to ignore a rare class, but it can also change the class prior learned by its raw logits. A small additive offset per class can correct the final class decision without retraining or touching any height prediction.

The calibrated decision is:

```
chosen class = argmax(raw six-class logits + six fixed biases)
```

The six biases are constrained to have mean zero because adding the same value to every logit would not change the decision.

What it is not

This is **class-decision calibration**, not probability/confidence calibration. It may improve which category wins the argmax, but it does not prove that a displayed value such as 80% means an 80% chance of being correct. Temperature scaling, reliability diagrams, ECE/Brier checks, and an independent calibration set would be required before making that confidence claim.

The utility also does not change:

- any model or height tensor;
- the protected production checkpoint;
- the active application pointer;
- the three-group ground/building/vegetation height-routing machinery (this is not a three-class replacement for the six-class identifier);
- the app's active model.

Data boundary

Fitting and the one exact follow-up evaluation use only the 859 official GAMUS validation tiles from Washington, DC and Philadelphia. The approved index and candidate training configuration are hash-sealed.

- NYC is forbidden during fitting and development evaluation.
- This calibration process never discovers or constructs the official GAMUS test. Project-wide, that test was already consumed by the older Stage-3 final audit and is forbidden for reuse.
- The calibration process never opens the future locked NYC files. The active v2 model was trained on the full approved training set (including NYC), so it cannot be presented as unseen-NYC evidence. Even a later DC+PHL-trained head may describe NYC only as classifier-head-excluded because the inherited HighBuild-trained system also has New York exposure. A separate external geography is required for a system-level generalisation claim.

- Because DC+PHL is development data already used for model selection, its calibrated result is development evidence, not an independent generalization result.

Two-stage procedure

1. Bounded fit

All 859 DC+PHL validation tiles are visited once. At most 64 pixels per class per tile can enter a deterministic priority reservoir, and at most 30,000 pixels per class are retained. Therefore no more than 180,000 six-logit rows can be cached. Each retained class pixel is weighted back toward its observed full-validation class support; the later exact pass remains authoritative.

A deterministic coordinate search tries bounded mean-zero biases. It ranks candidates against gates declared before fitting:

- macro F1 and macro IoU floors;
- minimum water precision, recall, and F1;
- minimum road F1;
- no meaningful regression in ground, buildings, roads, low vegetation, or trees versus the same checkpoint with zero bias;
- a minimum macro-F1 gain.

If any sampled gate fails, the fit is recorded as rejected and the exact pass is refused.

2. One exact DC+PHL pass

An accepted fit can be evaluated once over every valid labelled pixel of all 859 tiles. The evaluator streams confusion, road-boundary, and water/dark-pixel proxy metrics and never stores full logits. It reports both uncalibrated and decision-calibrated results from the same forward pass. Road-boundary quality has its own predeclared floor and retention guard.

A consumption marker is written before the exact dataset is constructed. A crash therefore cannot silently become repeated full-validation tuning.

Reproducible commands

Protocol-only preflight; this does not open imagery or a checkpoint:

```
.\venv\Scripts\python.exe scripts\calibrate_gamus_six_class_decisions.py --validate-protocol
```

After head-only v2 selects `checkpoint_best_landscape.pt`, first run the independent tensor audit. Only after that report passes, fit the external decisions:

```
.\venv\Scripts\python.exe scripts\calibrate_gamus_six_class_decisions.py --fit --candidate-checkpoint <path-to-checkpoint_best_landscape.pt>
```

The fit command independently repeats the state comparison and also requires `outputs/evaluation/gamus_six_class_head_only_v2/checkpoint_best_audit.json` to authenticate the exact same checkpoint hash. It therefore fails before opening the calibration imagery if training/auditing has not finished.

Only if `fit.json` passes every sampled gate, run the one exact development evaluation:

```
.\venv\Scripts\python.exe scripts\calibrate_gamus_six_class_decisions.py --evaluate-full --candidate-checkpoint <path-to-checkpoint_best_landscape.pt> --confirmation RUN_FULL_DC_PHL_DECISION_BIAS_EVALUATION_ONCE
```

Artifacts are written beneath `outputs/evaluation/gamus_six_class_decision_bias_dc_phl_v1/`. Passing these development gates does not promote the checkpoint, authorize app integration, or consume the locked NYC holdout. Those remain separate review decisions.

The Python utility is candidate-path driven. Another compliant six-class head-only candidate can use the same code, but it requires a new versioned YAML protocol with that candidate's exact training-config hash and a separate output folder. An arbitrary checkpoint is intentionally rejected rather than guessed.

Reproducible implementation

- Protocol: `configs/gamus_six_class_decision_bias_v1.yaml`
- Tool: `scripts/calibrate_gamus_six_class_decisions.py`
- Pure search/sampler: `src/msr/evaluation/decision_bias_calibration.py`
- Tests: `tests/test_gamus_decision_bias_calibration.py`

44 GAMUS six-class expansion protocol

SOURCE docs/history/GAMUS_SIX_CLASS_EXPANSION_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: implementation and initial versioned experiments complete; no expanded candidate has been promoted. V4 remains a controlled comparison until its paired report exists and all gates pass.

Objective and comparison

The expanded candidate adds a separate dense identification output for, in this fixed order: ground, buildings, water, roads, low_vegetation, and trees. The existing three-group ground/building/vegetation routing and all height heads remain present and unchanged.

“Six-class” always refers to the final identification output and its exact 6 x 6 confusion matrix. “Three-group” refers only to the coarser ground/building/vegetation **height-routing domains**; it is not a substitute three-class identification result. Merging low vegetation and trees yields a five-group diagnostic, not a six-class macro score.

The height-focused GAMUS pilot is the required warm start and comparison, not an artifact to overwrite. The production checkpoint and outputs/runtime/showcase_checkpoint.txt remain protected. No six-class candidate may become active merely because its classification score improves.

Implemented contract

- GAMUS source class 0, malformed values, and padding use ignore index 255.
- image_valid_mask, classification_valid_mask, and height_valid_mask describe independent availability. Effective semantic supervision uses image-valid AND classification-valid pixels.
- Height regression uses image-valid AND height-valid AND classification-valid pixels from GAMUS ground, building, low-vegetation, or tree classes.
- Water and road pixels teach identification but never height regression.
- Missing or excluded heights are stored as neutral tensor values only after their regression mask is false; they cannot become supervised zero height.
- The optional six-class head is disabled by default, so old configurations and old checkpoint state dictionaries are unchanged.
- Enabling it creates model type domain_gated_surface_v4_six_class. A dedicated compatibility loader allows only the new head tensors to be absent when warm-starting an old checkpoint; every height tensor stays strict.
- Tiled inference can return six aligned probability maps, but the application does not expose them until validation and an explicit model-selection step.

Required evaluation

On the complete authenticated GAMUS validation split, report:

- the 6 x 6 reference-versus-prediction confusion matrix;
- precision, recall, F1, and IoU independently for all six classes;
- macro precision, recall, F1, and IoU;
- a two-pixel-tolerance road-boundary precision, recall, F1, and dilated boundary IoU (tolerance remains in pixels until ground sampling distance is authenticated);

- water false positives on dark non-water RGB pixels as an explicitly labelled proxy only. GAMUS has no shadow class, so this is not shadow ground truth;
- all existing GAMUS, urban, forest, building, canopy, tall-object, city, RMSE, MAE, correlation, and R-squared height guards under identical protocols.

The final report must also use genuinely new external geography. Validation cities may guide development. NYC can be excluded from a newly fitted DC+Philadelphia classifier head, but it is not system-unseen because earlier GAMUS work and the inherited HighBuild-trained checkpoint already exposed New York data. The official GAMUS test was consumed by the older Stage-3 final audit and is permanently forbidden for V4 reuse; it is not a fresh holdout.

Preparation blockers before a longer run

Do not enable colour augmentation or start a long six-class run until the versioned preparation audit has resolved or explicitly bounded:

1. GAMUS AGL height units and per-city pixel scale;
2. the raw values above the current 200-metre cutoff;
3. cached Depth Anything V2 map identity, source-image identity, model identity, alignment, numeric range, and completeness;
4. independent image/class/height availability counts after the new mask contract;
5. train/validation/test and new-geography non-overlap.

HighBuild and OpenCanopy additions

HighBuild per-building instance metrics and boundary checks use the corrected supervision files and report measured versus estimated-height annotations separately. Building counting is a separate validated task, not inferred from pixel RMSE.

OpenCanopy acquisition dates, locations, and source metadata must remain with each sample for stronger geographic and temporal holdouts. Recovering richer vegetation classes is an evaluation/relabeling study, not something inferred from the present binary canopy mask. Infrared inputs and before/after change detection are later extensions requiring separately prepared inputs.

Application outputs after validation

After the candidate passes both identification and height non-regression checks, a review build may expose six-class overlays, building/canopy height summaries, and per-class coverage. Area is reported in square metres only when the input has authenticated geographic scale. Confidence remains a score until it is calibrated on held-out data; it must not be labelled as a probability of correctness beforehand.

Water identification can support a future flood module, but flood simulation also needs ground terrain and hydraulic inputs. Snow cover needs suitable snow labels, and snow depth needs separate metric-depth supervision.

Promotion rule

Publish expanded-candidate results first. Compare them directly with the height-focused pilot and protected production checkpoint on GAMUS plus the corrected urban/forest suites. Only a candidate that passes every predeclared height guard, six-class review, genuinely new external-geography evaluation, visual sanity check, and explicit human approval may change the app's active pointer.

The canonical corrected full-scene protected baselines are:

- HighBuild strict-measured building-only RMSE: **13.243 m**;
- OpenCanopy vegetation-only RMSE: **8.361 m**;
- OpenCanopy all-valid forest RMSE: **6.707 m**.

They use different datasets, masks, and pixel supports and are not interchangeable. Historical crop, zero-filled, inclusive-annotation, pooled, or forest-all-valid values must keep their own protocol label.

Versioned run and reporting

- Height comparison: `configs/multidomain_surface_gamus_direct_height_pilot.yaml`
- Expanded ablation: `configs/multidomain_surface_gamus_six_class_candidate_v1.yaml`
- Frozen gates: `configs/gamus_six_class_comparison_v1.yaml`
- Launcher/watcher: `scripts/run_gamus_six_class_candidate_v1.ps1` and `scripts/watch_gamus_six_class_candidate_v1.ps1`
- Report tool: `scripts/compare_gamus_six_class_candidate.py`

The launcher refuses to start until the newest height-pilot session is terminal and successful. It then seals the exact height metrics file, selected epoch, reproducibly matching checkpoint (if available), terminal checkpoint, and their hashes into the candidate's saved launch config. If a raw-best epoch has no matching saved checkpoint, the report marks that checkpoint unavailable instead of attaching the final epoch's weights to earlier metrics.

The historical run order was height pilot, expanded candidate, identical GAMUS comparison, and corrected HighBuild/OpenCanopy comparison. The controlled V4 order adds paired DC+Philadelphia V3/V4 training, an optional one-shot classifier-excluded NYC check, and finally a separately reserved genuinely new external geography. The already-consumed official GAMUS test is never reused. Expected RTX 4070 time is roughly 75–105 minutes for each six-epoch GAMUS run, plus 20–45 minutes for corrected legacy evaluation; early stopping can shorten this. The initial expanded results are recorded, but no model change is claimed and V4 accuracy remains unproven until its paired evaluation completes.

45 GAMUS six-class experiment status

SOURCE docs/history/GAMUS_SIX_CLASS_EXPERIMENT_STATUS.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Evidence snapshot: 2026-09-13. The live application still points to the protected height checkpoint. No GAMUS experiment has been promoted.

Current decision

- The height-focused pilot is complete and sealed. It is not eligible for promotion because its ground-height and building-identification guards fail.
- The first joint six-class candidate is also complete. It learned five useful categories, but water recall collapsed to effectively zero and its tall-vegetation guard failed.
- The corrected full-scene HighBuild/OpenCanopy triplet benchmark completed successfully as an evaluation job, but its final decision is **failed**: the height pilot and joint six-class candidate are materially worse than the protected model on the corrected forest evidence. This is an acceptance-gate failure, not a crashed benchmark.
- The isolated linear six-class-head correction completed and was rejected for promotion. Its best checkpoint was epoch 3; all 240 inherited tensors (31,428,489 values) were exactly equal to the protected model, but ground, low-vegetation and tree retention gates failed.
- A development-only additive decision calibration materially improved the same frozen-height output, but still missed the original low-vegetation and tree retention thresholds. It is not probability calibration and has not been exposed in the app.
- The sealed spatial-refined six-class head completed after epoch 6. Epoch 2 is the best raw development checkpoint at macro F1 0.554, but the original low-vegetation and tree retention gates still fail. Exact inherited-state and recipe audits pass; it is not promoted.
- The sealed DC+Philadelphia decision-bias check completed. It improved the frozen classifier's development trade-off without changing weights or height output, but still failed the low-vegetation/tree acceptance floors and remains inactive.
- The application model must remain unchanged. The fair next comparison trains V3 and V4 classifier heads on the same DC+Philadelphia split. NYC may then be described only as excluded from those new heads' training—not as system-unseen. A new external geography, corrected height checks, category gates, and explicit human approval are still required before exposure.

Requirement-by-requirement audit

Expanded requirement	Status	Repository evidence and honest limitation
Six GAMUS categories in a separate output	Implemented and evaluated	<code>DomainGatedSurfaceNet.fine_semantic_head</code> outputs ground, buildings, water, roads, low vegetation and trees while old checkpoints remain loadable. The v1 report is <code>outputs/evaluation/gamus_six_class_comparison_v1/report.json</code> .
Preserve three-group height machinery	Implemented and bitwise verified	Here “three-group” means the existing ground/building/vegetation height-routing domains , not a three-class replacement for the six-class identifier. The additive identifier still reports all six classes. Both v2 and v3 audits proved exact equality for all 240 inherited tensors/31,428,489 values. V3 trained only its five <code>fine_semantic_head.*</code> tensors and kept every height loss at zero.
Independent image/classification/height masks	Implemented and audited	<code>GAMUSDataset</code> emits <code>image_valid_mask</code> , <code>classification_valid_mask</code> , <code>height_valid_mask</code> and a separate <code>regression_mask</code> . Water/roads may teach class identity but are excluded from height regression; missing height never becomes zero.
Keep the production height checkpoint safe	Implemented; candidate gates failed	The live pointer and protected checkpoint were hash-identical before/after the corrected triplet, completed v2 run and decision-calibration evaluation. No experimental result has been promoted.
HighBuild individual outlines, heights and boundaries	Implemented for available annotations	<code>src/msr/evaluation/highbuild_instances.py</code> reports per-building matches, height error, outline IoU and boundary scores, separating measured and estimated annotations. The corrected benchmark covered 160 validation tiles and 2,588 source annotations.
Building counting as a separately validated feature	Implemented fail-closed; current evidence unavailable	The evaluator has a distinct per-scene counting section (MAE/RMSE, exact-count rate, over/undercount totals and MAPE when valid). Current positive-only labels cannot support honest precision/count error, so those fields remain null until an exhaustive independent instance reference exists.
OpenCanopy dates, locations and source provenance	Implemented	<code>outputs/data_audits/open_canopy_v2_provenance_v1.json</code> authenticates 600/120/120 accepted samples with region, source image, imagery date, LiDAR date, URL, GSD and metadata hash. It also discloses source-mosaic overlap across the present splits.
Richer vegetation classes	Partial	Low/medium/high upstream vegetation strata can be preserved in versioned sidecars without changing height masks. Full sidecar backfill, training and held-out evaluation remain pending; these are LiDAR height strata, not species labels.
GAMUS units, pixel scale and >200 m outliers	Audited with one unresolved proof	Nominal 0.33 m pixels and all >200 m values are inventoried in <code>outputs/data_audits/gamus_preparation_v1/</code> . Local HDF5 lacks CRS/transform and the publisher has not independently confirmed the raw AGL unit, so results remain metre-assumed rather than absolute-unit-proven.
DAV2 cache/source integrity	Implemented for current cache	All 8,720 approved RGB/prior pairs were hash-locked and five samples reproduced exactly. Historical generation did not embed hashes at creation, so the audit states that limitation rather than inventing provenance.

Expanded requirement	Status	Repository evidence and honest limitation
Per-class precision, recall, IoU and F1	Implemented and reported	Metrics cover all six classes plus a confusion matrix. Road boundaries have a tolerance-aware metric. The dark-pixel check is explicitly only a water/shadow-confusion proxy because GAMUS has no shadow ground truth.
Geographic generalisation	Classifier-head exclusion prepared; genuinely new geography pending	A predeclared NYC partition and DC+Philadelphia learning index exist in <code>outputs/data_audits/gamus_geographic_holdout_city_nyc_v1/</code> . The spatial recipe passed a 3,837-train/859-validation/1,164-NYC preflight without opening NYC for this comparison. NYC is not system-unseen: older GAMUS work and the inherited HighBuild-trained system already exposed New York data. The official GAMUS test was consumed by the older Stage-3 audit and is forbidden for reuse. Final evidence must come from a separately reserved external geography.
Six-class overlays and summaries in the app	Validated-export path implemented; app exposure pending	An inactive fail-closed exporter can write aligned class IDs 0..5/nodata 255, separate validity masks and allowed summaries only from a SHA-bound fully accepted candidate. The active app checkpoint has no six-class head and no overlay is enabled.
Calibrated confidence	Pending	Additive argmax decision biases were fitted and evaluated, but this is explicitly not probability/confidence calibration. Existing confidence-like outputs must not be presented as probability of correctness.
Flood and snow extensions	Explicit future modules, not current claims	Water identification can support flood work, but a flood simulator still needs terrain and hydraulic inputs. Snow cover needs suitable class labels; snow depth needs separate metric supervision. Infrared and before/after change detection also require new preparation.

Completed corrected legacy triplet

The hash-sealed report is `outputs/evaluation/corrected_legacy_triplet_v1/report.json` (SHA-256 0977bc90e034746df74dc6241c6676f15a6ea5f36dc0b422008a75c6767ab351). It used identical raw RGB, cached target-independent relative priors, masks and full native grids for all three models: 160 corrected HighBuild scenes and 120 OpenCanopy scenes. This triplet job did not use the official GAMUS test; that test had already been consumed by the older Stage-3 final audit and cannot be reused by V4.

Primary strict-measured pooled height result:

Model	Pooled RMSE (m)	Pooled MAE (m)	Correlation	R-squared	OpenCanopy all-valid forest RMSE (m)	OpenCanopy v egetation-only RMSE (m)	HighBuild stric t-measured building-only RMSE (m)
Protected live model	10.161	4.911	0.482	0.203	6.707	8.361	13.243
Height-focused pilot	15.460	10.857	-0.139	-0.846	16.919	10.893	13.435
Joint six-class candidate	15.463	10.837	-0.150	-0.847	16.910	10.802	13.457

These are the canonical corrected full-scene baselines. The three protected values answer different questions and are not interchangeable: **13.243 m** is building-only error on strict measured HighBuild pixels, **8.361 m** is vegetation-only error on valid OpenCanopy vegetation pixels, and **6.707 m** is all-valid OpenCanopy forest-scene error including lower/ground pixels. Earlier crop, zero-filled, inclusive-annotation, or mixed pooled scores must retain their own protocol labels and must not be compared as if they measured the same task.

The joint candidate passes the small non-regression deltas versus the already weak height pilot, but fails the predeclared comparison with the protected model: pooled RMSE is +5.302 m, OpenCanopy/forest RMSE is +10.202 m, vegetation RMSE is +2.441 m and corrected HighBuild building RMSE is +0.214 m.

HighBuild instance evidence reinforces the no-promotion decision. The protected model matched 400/2,581 rasterizable reference buildings (annotated recall 0.155), while the joint candidate matched 282/2,581 (0.109). Mean boundary F1 was 0.488 versus 0.456 and mean matched-outline IoU was 0.738 versus 0.709. The candidate's lower RMSE on only its matched subset must not be treated as an overall win because it matched substantially fewer buildings.

Six-class development result

The joint candidate's GAMUS development result was RMSE 4.442 m, MAE 2.604 m, correlation 0.783 and R-squared 0.613. Six-class macro F1/IoU were 0.467/0.330. "Six-class macro" always means the arithmetic mean over the final six classes; merging low vegetation and trees into one vegetation class produces a different five-group statistic and cannot be reported as the six-class score. Per-class F1 was 0.402 ground, 0.690 buildings, effectively 0.000 water, 0.517 roads, 0.573 low vegetation and 0.621 trees; road-boundary F1 was 0.257.

The completed linear head-only correction addressed the concrete imbalance behind the water failure (water is only 3.70% of training pixels) with class-weighted focal loss and a hash-sealed water-aware sampler. Its best epoch 3 achieved macro F1/IoU 0.506/0.362. Per-class F1 was 0.379 ground, 0.833 buildings, 0.220 water, 0.617 roads, 0.518 low vegetation and 0.467 trees; road-boundary F1 was 0.270. The checkpoint audit passed exact equality for all inherited height/shared tensors, but the frozen classification acceptance gate failed ground, low-vegetation and tree retention. It was not promoted.

A one-time full DC+Philadelphia development evaluation of external additive decision biases improved macro F1/IoU from 0.506/0.362 to 0.547/0.395 and accuracy from 0.543 to 0.578. Calibrated per-class F1 was 0.413 ground, 0.842 buildings, 0.324 water, 0.622 roads, 0.522 low vegetation and 0.557 trees; road-boundary F1 was 0.273. The dark-non-water false-water rate fell from 0.371 to 0.126. This is a development-set class-decision adjustment, not probability calibration or unseen-geography evidence. It still misses the original low-vegetation and tree retention targets and remains inactive.

The spatial-refined comparison kept the same data, sampler, losses, seed and gates while replacing only the isolated six-class head with a 1,094-parameter local-context head. It early-stopped after epoch 6 with epoch 2 best: macro F1/IoU 0.554/0.400 and accuracy 0.581. Per-class F1 was 0.409 ground, 0.840 buildings, 0.386 water, 0.630 roads, 0.483 low vegetation and 0.575 trees; road-boundary F1 was 0.287 and the dark-non-water false-water rate was 0.101. All height metrics were identical to the protected initialization and all 31,428,489 inherited values passed exact equality. The raw candidate still misses the original low-vegetation (0.568) and tree (0.616) retention floors, so it remains inactive. Evidence is sealed at `outputs/evaluation/gamus_six_class_spatial_refined_v3/checkpoint_best_landscape_audit.json`.

46 GAMUS six-class v4: isolated hierarchical vegetation head

SOURCE docs/history/GAMUS_SIX_CLASS_V4_HIERARCHICAL_HEAD_DESIGN.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: architecture, loss, configuration builder, and fail-closed safety auditors are implemented and tested. V4 is not yet trained, so none of those checks establish its accuracy. This experiment does not authorize changing the app model or production pointer.

Why a v4 would be justified

The completed experiments show a narrow failure rather than a general collapse:

Candidate	Macro F1	Water F1	Road F1	Low-vegetation F1	Tree F1
Joint v1, epoch 4	0.4673	~0.0000	0.5171	0.5730	0.6211
Frozen linear v2, best macro epoch 3	0.5055	0.2201	0.6165	0.5184	0.4668

At v2 epoch 3, 30.7% of true tree pixels became low vegetation and 14.5% became water. Tree precision was 0.6153 but recall was only 0.3760. At epochs 4 and 5 tree recall fell further, to 0.2791 and 0.2945, while precision stayed high. The classifier is therefore conservative about predicting trees; this is not evidence that the labels contain no vegetation signal. The joint v1 run learned the vegetation split, but its shared-feature changes caused unacceptable height regression. V4 should recover that split without touching shared or height tensors.

Frozen input available to the semantic head

The optional six-class head receives the existing 64-channel adapter_features. Those frozen features are computed from six input planes:

1. RGB image (3 planes),
2. cached Depth Anything V2 relative-depth prior (1 plane),
3. log-normalized protected base height (1 plane), and
4. protected building probability (1 plane).

The current linear v2 head is a 1x1 convolution over these features (390 parameters), so it makes each decision without explicit neighbourhood context. V3 adds a depthwise 3x3 residual refinement and has 1,094 head parameters. If that local context still does not recover both vegetation classes, the smallest useful next change is to factor the same six-class prediction into a coarse decision plus a dedicated vegetation subtype decision.

Proposed architecture

Name: hierarchical_vegetation.

All modules and parameters must exist strictly below the one allowed prefix:

`fine_semantic_head.`

The head is:

1. spatial: depthwise 3x3 convolution, 64 groups, no bias.

2. normalization: GroupNorm with one group, followed by GELU.

3. Residual feature: $h = x + \text{GELU}(\text{GN}(\text{spatial}(x)))$.

4. coarse_classifier: 1x1 convolution from 64 channels to five logits: ground, buildings, water, roads, and combined vegetation.

5. vegetation_split: 1x1 convolution from 64 channels to one logit, giving tree versus low-vegetation probability only inside the vegetation branch.

6. Convert the five-way coarse probabilities and binary split into six aligned probabilities: - the first four probabilities are unchanged; $-P(\text{low vegetation}) = P(\text{vegetation}) * (1 - P(\text{tree} | \text{vegetation}))$; $-P(\text{trees}) = P(\text{vegetation}) * P(\text{tree} | \text{vegetation})$. Their clamped log-probabilities are returned as the six logits, so the existing softmax/export interface remains six-class compatible.

This factorization prevents the low-vegetation/tree objective from competing with water, roads, buildings, and ground for the same single flat decision. It also preserves the total vegetation probability exactly while deciding how to split it between low vegetation and trees.

Parameter count

For 64 frozen input channels:

Tensor group	Parameters
Depthwise 3x3	$64 \times 3 \times 3 = 576$
GroupNorm scale and bias	$64 + 64 = 128$
Five-way 1x1 classifier	$5 \times 64 + 5 = 325$
Vegetation split 1x1 classifier	$1 \times 64 + 1 = 65$
Total	1,094

V4 therefore has the same parameter count as v3 and only 704 more parameters than the v2 linear head. The improvement comes from the task structure, not a larger shared model.

Initialization

- Start again from the protected production height checkpoint for a clean, attributable comparison; do not warm-start or modify any shared tensor.
- Initialize the depthwise convolution to zero, GroupNorm scale to one and bias to zero. The residual path initially behaves as the unchanged frozen feature.
- Initialize both classifier weights with a tiny zero-mean normal distribution (standard deviation 0.001) so the spatial branch can receive gradients on the first optimizer step without making strongly biased predictions.
- Initialize the binary split bias to zero (equal low-vegetation/tree prior).
- Initialize the combined-vegetation coarse bias to $\ln(2)$ and the other four coarse biases to zero. With a zero split logit, this yields an exactly uniform six-class initial probability rather than accidentally giving each vegetation subtype only half the mass of every other class.

Minimal training change

Keep v3's sealed data, raw radiometry, water-aware sampler, six-class focal loss, optimizer, and all height loss weights at zero. Add only one auxiliary loss:

- On pixels whose valid six-class target is low vegetation or trees, apply an equal-weight binary cross-entropy to `vegetation_split`.
- The mask must be `image_valid AND classification_valid AND target in {4,5}`. It must never use `height_valid`, and ignored/background pixels remain absent.
- Suggested auxiliary weight: 0.5 relative to the existing six-class loss.
- Do not change water sampling, class weights, colour augmentation, or decision biases in this architecture comparison. Those would confound the result.

The equal-weight subtype objective directly addresses v2's tree under-recall, which the current global class weights `[0.7, 0.6, 3.0, 0.7, 0.6, 0.4]` can otherwise encourage. If a later decision-bias calibration is used, it must be fitted only after selecting/fixing the model and only on the development geography; it is not confidence calibration.

For the controlled architecture comparison, retain the paired spatial-V3 schedule exactly: 8 epochs, validation every epoch, minimum 4 epochs, and early-stop patience 3. Changing the schedule would confound the comparison.

Exact checkpoint and height-safety contract

The optimizer allowlist must contain exactly `fine_semantic_head..` Every other parameter must have `requires_grad=False`; no other prefix is permitted. GroupNorm is used because it has no running statistics. No BatchNorm, stochastic shared state, or height-dependent post-processing may be added.

After every saved candidate, audit all state entries not starting with `fine_semantic_head.` using exact `torch.equal`, not a floating-point tolerance. The inherited tensor key set, shapes, dtypes, and values must exactly match the protected checkpoint. The production checkpoint file and live pointer hashes must also match their pre-run hashes. Any mismatch makes the artifact invalid, regardless of metrics.

Weight equality alone is insufficient. The separate height-output audit must also prove that canonical preprocessing, tiling, fusion settings, all direct height-path outputs, and deterministic app-level outputs are bit-identical to the protected model. Its sealed pipeline contract is SHA-bound to the V4 configuration.

Selection and acceptance gates

Freeze the paired DC+Philadelphia V3 development metrics before launching V4. The configuration builder derives and seals the numeric gates from that exact audit before training begins. Training-time height guards and macro-F1 selection remain identical to paired V3. The selected candidate is eligible for the classifier-excluded-city check only when all of the following pass:

1. Exact inherited-tensor, height-output, pointer, and checkpoint hash audits pass.
2. Existing GAMUS height guards pass at the current 0.0001 m reporting tolerance; exact output equality is the stronger requirement.
3. Final six-class macro F1 is at least $\max(0.60, \text{paired_v3_macro_f1})$.
4. Ground F1 is at least $\max(0.40, \text{paired_v3_ground_f1} - 0.005)$.
5. Building F1 is at least $\max(0.82, \text{paired_v3_building_f1} - 0.005)$.
6. Water F1 is at least $\max(0.45, \text{paired_v3_water_f1} - 0.005)$.
7. Road F1 is at least $\max(0.65, \text{paired_v3_road_f1} - 0.005)$.
8. Low-vegetation F1 is at least $\max(0.58, \text{paired_v3_low_vegetation_f1} + 0.010)$.
9. Tree F1 is at least $\max(0.62, \text{paired_v3_tree_f1} + 0.010)$.
10. Road-boundary F1 is no more than 0.005 below paired V3, and the dark-non-water false-water rate is no more than 0.010 above paired V3.

Select checkpoints by final six-class macro F1, matching paired V3. The smaller of low-vegetation and tree F1 remains a diagnostic only. Recompute every stored score from the exact 6x6 confusion matrix; a five-group vegetation merge can never satisfy the macro gate.

Report precision, recall, F1, and IoU for all six classes, the full confusion matrix, water/shadow proxy, road-boundary metrics, and city/geography breakdown. Only a fully eligible frozen candidate may receive one NYC check. NYC is excluded from this classifier head's fitting and selection, but it is not system-unseen because earlier GAMUS and inherited HighBuild work exposed New York data. The official GAMUS test was also consumed by an older experiment and is forbidden for reuse. A separately sourced, preregistered geography remains required for a genuine system-level generalisation claim.

Risks and stop rule

- The frozen adapter may simply not encode enough texture/scale information to separate grass from tree canopy across cities. A head cannot recover absent information.
- Water oversampling may still pull dark trees toward water. The hard water/shadow gate prevents hiding that trade-off behind macro F1.
- The hierarchy can improve the vegetation split while worsening the earlier combined-vegetation decision; the per-class and macro gates prevent this.
- Raw RGB radiometry can shift geographically, so DC+Philadelphia success and the classifier-excluded NYC check do not replace a new external geography.
- GAMUS labels identify land cover, not vegetation height. Success here must never be presented as proof of canopy-height accuracy.

If v4 cannot pass both vegetation F1 gates, do not unfreeze the production adapter or height trunk. The next scientifically defensible escalation is a separate semantic-only sidecar encoder whose entire state remains under an isolated prefix; it should be evaluated as a new experiment rather than risking the working height model.

47 Active height-band sampling experiment

SOURCE docs/history/HEIGHT_BAND_SAMPLING_V1_HANDOFF.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Started 19 September 2026 at 17:32:41 UTC. This handoff is a launch record, not an accuracy or completion claim; read the live status and committed reports.

Exact run and saved work

- Run: D:/MSRData/experiments/height_band_sampling_v1/20260919T173241288352Z
- Protocol: docs/HEIGHT_BAND_SAMPLING_V1_PROTOCOL.md
- Configuration: configs/height_band_sampling_v1.json
- Sampler audit/plans: D:/MSRData/experiments/height_band_sampling_preparation_v1/
- Log: outputs/orchestration/height_band_sampling_v1.log
- Errors: outputs/orchestration/height_band_sampling_v1.err.log
- Dashboard registration: outputs/orchestration/class_assisted_height_active.json

The run directory contains copied source, bound configuration, data/cache hashes, immutable sampling plans, the recomputed protected baseline, and versioned checkpoints/atomic commits under control and balanced. Do not edit bound training sources or recipes while this experiment is active.

What this tests

Four epochs per arm with the same model, initialization, original loss and training budget. Both use uniform class inputs. Only training sampling changes: the candidate deliberately includes more labelled tall buildings and short vegetation. All 1,050 training images appear within each two-epoch cycle. The exposure audit is not proof of better accuracy.

Validation uses all 160 urban and 120 forest DEVELOPMENT scenes at native resolution after every epoch. Compare the candidate with the matching control epoch and protected baseline. Safety and improvement are separate requirements; the existing app model must not be replaced automatically. Reserved final-test geography and GAMUS height claims are outside this experiment.

Verified at handoff preparation

- Preparation and preflight passed; original baseline metrics matched exactly.
- Control epochs 1 and 2 committed; epoch 3 training observed.
- Control epochs 1 and 2 did not pass all safety gates. Candidate results were not yet available at that observation.
- 24 targeted sampler, runner, loss, gate, prefetch and dashboard tests passed.
- The native live dashboard was opened and visually checked with both arm progress bars, epoch results and GPU telemetry visible.
- Existing msr-residual-accuracy-check heartbeat reactivated hourly for this exact run. It stays quiet without meaningful changes, reports final evidence and pauses itself after completion. No further experiment or app promotion is authorized by that monitor.

Failure/recovery boundary

Inspect failure/status, logs and actual trainer processes first. A venv wrapper and its Python child count as one job. Do not start a competing GPU job. Only one compatible recovery attempt is permitted by the monitor:

```
.\scripts\run_height_band_sampling.ps1 -Resume  
'D:\MSRData\experiments\height_band_sampling_v1\20260919T173241288352Z'
```

Never bypass hashes, change batch/loss/sampling rules or overwrite orphan artifacts. Active Windows files must be read with `FileShare.ReadWrite` and `FileShare.Delete`. `class_assisted_dashboard.shared_bytes` provides this.

At completion, verify integrity and report best eligible separately from best unrestricted checkpoint. If none passes both safety and benefit, say so plainly and preserve the rejected evidence for error analysis.

48 Sampling comparison: no accepted replacement

SOURCE docs/history/HEIGHT_BAND_SAMPLING_V1_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Completed run: D:/MSRData/experiments/height_band_sampling_v1/20260919T173241288352Z. All eight epochs completed. No candidate passed every safety check. Production and classifier checkpoints/pointer remain unchanged. Final committed artifacts and frozen protected tensors were verified in outputs/diagnostics/height_conditioning_response_v1/report.json.

Native development RMSE (m)	Protected	Control epoch 4	Balanced epoch 4
Buildings	13.243	12.795	13.094
Canopy	8.361	7.807	7.770
Ground	4.894	4.642	4.910
Short vegetation	5.205	4.936	5.215
Tall buildings	29.797	28.834	28.115
Tall canopy	10.486	9.594	9.387

Balanced epoch 4 passes the paired canopy-benefit rule but fails safety. Its additional canopy advantage over matched control is only 0.038 m, while building/ground/short-vegetation aggregates are worse than control. It is not a convincing general solution. Best unrestricted checkpoint by the declared equal building/canopy-RMSE ranking is CONTROL epoch 2; it too fails safety. Best eligible checkpoint: NONE. Do not mix best categories from different epochs.

Diagnostic and next experiment

A same-checkpoint intervention on the original fixed 20 TRAIN examples changed the earlier predicted-class model's input from actual soft classes to uniform classes. Mean absolute height change was 0.18-0.19 m on buildings and 0.22-0.60 m across vegetation bands. Thus class inputs are not disconnected or entirely ignored, but the earlier tiny paired development gains do not show sufficient useful transfer. On these examples, predicted coarse vegetation covered only 26.3% of labelled short-vegetation pixels, versus 84.5% for tall vegetation. This is a small, non-representative training diagnostic, not a segmentation benchmark; zero label height or reference class never enters inference.

Test a different explicit mechanism: six learned height-correction outputs mixed by frozen classifier probabilities at native resolution. Compare with identical architecture and uniform routing. Both use original sampling and original loss, since height-band balancing did not establish an overall gain. This is a hypothesis, not a promised improvement; class mistakes can still hurt. First require a training-only optimization/identity proof before development training. Retain every original safety check, no reference routing, no GAMUS height-unit assumption, no final-test geography, no automatic promotion.

49 Height-band sampling: fixed four-epoch matched experiment

SOURCE docs/history/HEIGHT_BAND_SAMPLING_V1_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Predeclared 19 September 2026, before this run trains. User authorized continuing the accuracy work. All earlier experiments and production remain immutable.

What changes — and what does not

Test sampling, not another classifier or loss weight. Both arms use the original source/domain-balanced Huber + 0.02 MSE loss, fresh identical protected start, residual decoder, uniform six-channel conditioning, bf16, batch 2, accumulation 2, 384 px patches, learning rates and optimizer from the passed feasibility recipe. No weights from failed runs are reused. The rejected low-surface penalty is NOT used in either arm. The current comparison therefore isolates sampling; historical low-surface-loss results remain a separate comparison, not this control.

Both arms run four epochs at 600 urban/forest pairs per epoch (1,200 optimizer updates per arm). This is a newly declared common budget, not a silent extension of a completed two-epoch experiment. No extra epochs or automatic promotion.

Control A reproduces the previous sampler: all 450 urban images plus 150 urban repeats, all 600 forest images once, crops anchored on supported height pixels.

Candidate B uses 300 regular-coverage + 300 height-targeted draws per source:

- Urban: 150 shorter-building (<20 m) and 150 tall-building (≥ 20 m) targeted draws; crop anchors come from supported pixels of the requested band.
- Forest: 75 each for labelled ground, short vegetation ($0 < h \leq 2$ m), medium vegetation ($2 < h < 15$ m) and tall vegetation ($h \geq 15$ m). These chips already have native 384 px size, so balancing changes chip frequency, not crop geometry.
- Targeted image probability is proportional to its fraction of supported pixels in that band, with at least 64 such pixels. No silent missing-band fallback.
- A deterministic coverage cycle ensures ALL 1,050 training images have been shown by epoch 2, and again within the epoch 3–4 cycle. Per-epoch repetition, unique images, regions and actual supported-pixel exposure are recorded.

Height labels choose training observations only. All valid labels within each crop remain supervised; no band-specific loss mask is substituted. No true height/reference class reaches model.forward. Unknown labels remain ignored, and image/class/height validity remain independent. No road/water zero rule.

Measured preparation evidence

The CPU-only preparation read all 1,050 TRAIN images and generated all four epochs before training. It never read development or final-test data.

Epoch 1 actual labelled pixel observations:

Band	Control	Balanced
Short buildings	12,347,770	10,830,097
Tall buildings	1,451,884	4,205,520
Ground	47,710,844	46,158,656

Band	Control	Balanced
Short vegetation	2,275,710	3,242,330
Medium vegetation	19,178,553	20,223,934
Tall vegetation	17,100,459	16,667,717

These are repeated training observations, not new independent labelled pixels or guaranteed accuracy gains. The candidate has fewer distinct images within one epoch but complete coverage by two. Training region frequencies can change; the audit records them. This is a sampling intervention, not a city-holdout claim.

Evaluation, gates and integrity

Recompute the protected native baseline on all 160 urban + 120 forest DEVELOPMENT images, retaining original support and all old groups. Require old baseline metrics/support/semantics to match. Add explicit short-building and medium- vegetation global/region groups; apply original safety limits to these as well. Keep original short vegetation, tall objects, boundaries and region checks.

No original gate is loosened: RMSE +0.15 m, MAE +0.10 m, absolute bias +0.10 m, correlation/R2 decrease tolerance 1e-6. Benefit is ≥ 0.5 m AND $\geq 5\%$ building OR canopy RMSE reduction from protected, beating same-epoch control. Report safety and benefit separately and distinguish unrestricted from eligible checkpoints.

Checkpoint each 50 optimizer updates, atomic commit after save/evaluation, exact optimizer/RNG resume, original-source/software checks, copied immutable sample plans and original data/cache hashes. A shared OS lock prevents overlapping training jobs. Stop on invalid/nonfinite data, gradients, changed bindings or unexpected frozen weights. Do not silently change batch/recipe on recovery.

No GAMUS height units are assumed, no final reserved geography consumed, no all-image accuracy claim, no app checkpoint/pointer changes. New artifacts are on D:. If this succeeds, per-building and real app-path checks precede release.

50 Monocular Surface Reconstruction metric-height scorecard v1

SOURCE docs/history/HEIGHT_PROMOTION_SCORECARD.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **authenticated development baseline; production promotion is blocked until genuinely external metric-height evidence exists.**

This scorecard makes every future height candidate answer one question under one unchanged protocol: *is it materially better than the model currently used by the app, without sacrificing another landscape?* It does not train, select, copy, rename, or activate a checkpoint.

Protected application identity

- Checkpoint: experiments/20260829T203146Z_multidomain_surface_v2_guarded_vegetation/checkpoint_best_guarded_vegetation.pt
- Checkpoint SHA-256: e2d507fbe180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144
- Live pointer SHA-256: a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72baa9
- Canonical full-scene report SHA-256: 0977bc90e034746df74dc6241c6676f15a6ea5f36dc0b422008a75c6767ab351

The scorecard authenticates both the recorded identities and the current bytes on disk. A changed pointer, checkpoint, manifest, contract, input tensor, supervision tensor, sample order, or COCO payload fails closed. The YAML is also checked against configs/height_scorecard_v1.sha256; replacing this version requires an explicit new scorecard and seal rather than silently changing the comparison rules.

Canonical full-scene baseline

These are validation results, not a fresh blind test. HighBuild is evaluated on complete 1024 x 1024 images using the strict measured-height mask. OpenCanopy is evaluated on complete 384 x 384 images. Both use the same raw-RGB and cached Depth Anything V2 prior policy.

Scope	Pixels / objects	RMSE	MAE	Correlation	R2
Pooled strict HighBuild + OpenCanopy	31,011,260 px	10.16 m	4.91 m	0.482	0.203
HighBuild measured building pixels	13,855,690 px	13.24 m	5.36 m	0.186	-0.066
OpenCanopy forest pixels	17,155,570 px	6.71 m	4.55 m	0.659	0.422
OpenCanopy vegetation-only pixels	7,852,823 px	8.36 m	6.51 m	0.411	-0.174
OpenCanopy ground pixels	9,302,747 px	4.89 m	2.90 m	0.502	-0.088
Matched measured buildings	328 objects	11.15 m	4.42 m	0.244	0.037

The pixel and object building scores are different valid questions. The first weights every eligible building pixel; the second takes one median prediction per matched building. Neither may be relabelled as the other.

What the labels actually mean

HighBuild

- `building_height_m.tif` is a sparse building-height label, **not** a complete nDSM.
- The building mask contains all official COCO building outlines and supplies semantic positives.
- The strict validity mask contains only finite positive height pixels whose annotations are not marked `is_estimated_height=true`.
- Pixels outside official footprints are unknown. They are never scored as zero-height ground.
- Estimated heights remain available only in the separately named inclusive protocol. They are excluded from the primary baseline.
- Height values are used as metres according to the source annotation and `building_height_m` encoding. Pixel area is not claimed because projected scale is not independently authenticated for every tile.

OpenCanopy

- LiDAR-derived canopy values were converted from stored uint16 decimetres with `value × 0.1`, and the output rasters declare metres.
- RGB resolution is 1.5 m per pixel.
- Image validity, height validity, and vegetation membership are separate.
- The current validation is region-disjoint, but many larger SPOT acquisitions overlap the learning splits. It is therefore not source-scene independent.

GAMUS

- The loader keeps independent `image_valid_mask`, `classification_valid_mask`, and `height_valid_mask`; missing height never becomes zero.
- Water, road, background, and unknown pixels can teach identification while remaining outside height regression.
- The paper documents nominal 0.33 m pixels, but local HDF5 files have no per-tile CRS, transform, or pixel-scale metadata.
- Local files also lack a numeric AGL unit declaration. Raw AGL x 1.0 may be used only as an explicitly **metre-assumed** development target, not as publisher-verified metric truth.
- Values above the present 200 m cutoff are rare and span several semantic classes, so they stay excluded pending source-level resolution.
- The official GAMUS test has already been consumed by an older experiment. It is legacy evidence and may never be used for further tuning or selection.

Candidate gate

A report is comparable only when it certifies all of the following:

1. Validation-only, complete full scenes; no official test.
2. Exact sample order, model inputs, supervision tensors, COCO data, manifests, mask contracts, radiometry and relative-prior policy.
3. Protected baseline replayed in the same invocation.
4. Checkpoints and the app pointer unchanged before and after evaluation.
5. No overwrite or automatic promotion.

The development gate then requires:

- no RMSE regression above 0.15 m in pooled, urban, forest, building, vegetation, ground, or per-building scopes;
- no MAE regression above 0.10 m;
- no absolute-bias regression above 0.10 m;
- correlation and R2 do not decrease;
- outline IoU, boundary F1, and annotated-building recall do not decrease; and
- at least one of building, vegetation, or ground RMSE improves by either 0.5 m or 5%, while all other guards still pass.

Passing those rules means **development candidate**, not production winner. Production eligibility additionally requires a preregistered, one-shot test on new geography with independent metric-height truth.

External-geography status

No current artifact satisfies that final requirement:

- HighBuild test and OpenCanopy locked test have already been inspected and are legacy regression suites.
- OpenCanopy validation is geographically separated at region level but shares source mosaics with learning data.
- NYC was excluded from the newest classifier's training, but earlier GAMUS work and the inherited HighBuild model have NYC exposure.
- Christchurch OpenEarthMap is only a preregistered six-class segmentation design. It is not downloaded and contains no height truth.

The next scorecard version must be created *before inference* on a genuinely new location with a frozen manifest, independent DSM/nDSM reference, fixed crosswalks and masks, and a no-tuning/no-reselection rule.

Commands

Authenticate the baseline and current app identity:

```
.\.venv\Scripts\python.exe scripts\check_height_candidate_promotion.py
```

Assess a completed same-protocol report without activating its checkpoint:

```
.\.venv\Scripts\python.exe scripts\check_height_candidate_promotion.py `
  --candidate-report outputs\evaluation\<run>\report.json `
  --candidate-model-key <model-key> `
  --output outputs\evaluation\<run>\height_gate_decision.json
```

The output path must be new. The checker refuses to overwrite an existing decision and never writes outputs/runtime/showcase_checkpoint.txt.

51 HighBuild per-building evaluation

SOURCE docs/history/HIGHBUILD_INSTANCE_EVALUATION.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

This evaluator adds object-level evidence without changing the existing pixel height metrics or the corrected HighBuild supervision artifacts.

What is scored

- Every individual COCO building outline is retained as a separate reference object.
- Predicted objects are optimally matched to references at a declared IoU threshold (default 0.5).
- Matched outlines report IoU and boundary precision, recall and F1 at a declared pixel tolerance (default 2 px).
- Matched buildings with eligible height labels report instance-level height RMSE, MAE and signed bias. The predicted height is the median dense model prediction over the reference outline and valid height support.
- Measured and `is_estimated_height=true` annotations are reported separately. The strict protocol excludes estimated heights from regression; the inclusive protocol includes them only in a separate visible slice and combined total. Some upstream cities omit the flag; to remain byte-compatible with the corrected contract those records are treated as "not marked estimated" and their count is disclosed as `implicit_not_estimated_annotations`.
- The original valid-pixel height metrics are included alongside the new object metrics.
- Building counting has its own counting section and is never inferred from pixel accuracy. With exhaustive instance truth it reports per-scene count MAE/RMSE, median absolute error, exact-count rate, overcount/undercount scene totals and percentage error. Per-scene absolute metrics prevent an overcount in one tile from cancelling an undercount in another.
- Valid COCO polygons smaller than the raster grid can cover zero pixel centres. They are not fabricated or silently counted as misses; the report exposes them as `unrasterizable_annotations_excluded`.

The evaluator accepts either a building probability/binary raster or a positive integer instance-label raster. A binary semantic mask is split into connected components. Adjacent buildings can therefore merge; use a separately validated instance postprocessor and `prediction_instance_path` before claiming reliable building counts.

Honest limitation of the current corrected masks

The current corrected HighBuild contract says that COCO polygons are known building positives and pixels outside them are unknown. It does **not** certify every outside pixel as non-building. Consequently:

- annotated-building recall and matched outline quality are valid;
- detection precision and count error are deliberately unavailable;
- a model prediction outside an annotation is not silently called a false positive.

Precision, F1 and all count-accuracy metrics are enabled only when every prediction row supplies an independent `classification_valid_mask_path` that certifies exhaustive annotations in that region. This is distinct from image validity and height-label validity.

Without that independent exhaustive reference, the report still discloses the number of supplied positive outlines and predicted components for auditing, but all comparative count values remain `null`. This is intentionally labelled `unavailable_positive_annotations_are_not_exhaustive`.

Square-metre coverage is emitted only when every evaluated raster has a projected CRS with authenticated linear units. A filename, zoom level or unverified gsd_m value is never used to invent area.

Prediction manifest

Required columns:

- sample_id
- prediction_height_path
- exactly one of prediction_building_path or prediction_instance_path

Optional columns:

- image_valid_mask_path
- classification_valid_mask_path
- classification_validity_contract (must equal exhaustive_building_annotations whenever the classification mask is used)

Relative paths are resolved against the prediction manifest directory. All rasters must use the exact corrected HighBuild shape, transform and CRS. When image_valid_mask_path is absent, image validity comes from the source RGB raster's own band masks—not from class labels or height availability.

Validation command

```
.\.venv\Scripts\python.exe scripts\evaluate_highbuild_instances.py `
  --manifest
data\multidomain_v2\highbuild_measured_coco_intersection_v1\manifests\validation.csv `
  --contract-report
data\multidomain_v2\highbuild_measured_coco_intersection_v1\contract_report.json `
  --source-index data\highbuild_full\msr_splits\validation.csv `
  --split-root data\highbuild_full\msr_splits `
  --prediction-manifest outputs\evaluation\candidate_predictions\manifest.csv `
  --split validation `
  --output outputs\evaluation\candidate_highbuild_instances.json
```

The CLI verifies the immutable corrected-manifest hash in the contract report, checks exact grid alignment, records hashes for every evaluated raster and COCO payload, and refuses to overwrite an existing report. It requires complete manifest coverage unless --allow-subset explicitly labels a development run.

The historical HighBuild test split has already been inspected in this project and remains legacy regression evidence, not a genuinely unseen final holdout.

52 HighBuild classification: what works and what needs improvement

SOURCE docs/history/HIGHBUILD_RGB_CLASSIFICATION_AUDIT_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

What was actually tested

The frozen RGB-only SegFormer six-class V1, guarded epoch 8, was run on all 160 native 1024 x 1024 images in the existing corrected HighBuild validation subset: 40 each from Copenhagen, Lyon, Strasbourg and Munich. This is not the entire HighBuild dataset. These European cities were excluded from this standalone classifier's DC/Philadelphia training, but this is already-used project development/regression data, not an untouched final system benchmark.

The six classes are ground, buildings, water, roads, low vegetation and trees. The classifier is separate from the active application's protected height pipeline. No app model, checkpoint, dataset label or height output was changed. No training was performed.

Measured results

City	Images	Annotated-building pixel recall
Copenhagen	40	88.43%
Lyon	40	87.72%
Strasbourg	40	71.93%
Munich	40	73.86%
Pooled	160	79.21%

This means 14,748,689 of 18,619,411 supplied building-footprint pixels were classified as building. It does NOT mean 79.21% overall accuracy. In particular, it does not penalize invented buildings outside supplied footprints: HighBuild does not certify those unlabelled pixels as non-buildings. Precision, F1, IoU, building-count accuracy and the other five categories' accuracy are therefore deliberately unavailable in this audit.

Of the known building pixels, 12.89% were called road and 6.75% ground. Building-interior recall was 79.85%; the two-pixel inner edge-band recall was 73.08%. Interior errors remain substantial, so this is not just an edge problem.

For 2,581 annotations with rasterizable, image-valid footprint pixels:

- 79.58% had at least half their footprint labelled building.
- 64.59% had at least 80% labelled building.
- 8.95% had less than 10% labelled building.
- Small footprints (<256 pixels; 178 annotations) averaged only **52.86%** building coverage; **39.89%** of those small annotations were nearly missed (<10% coverage).

These are footprint coverage diagnostics, not instance detection/count scores. Seven of the contract's 2,588 supplied annotations contribute no rasterized image-valid pixels and are absent from those object-coverage denominators. Size bins are pixels, not square metres; geographic scale was not authenticated.

What the pictures show (qualitative, not independent six-class ground truth)

Eight of 16 previews selected before inference were inspected across all four cities. Every one of the 160 predictions remains in the gallery, including failures.

- Many large roofs are recognized, but predictions also spill onto adjoining streets/courtyards. For example, Copenhagen grid_00469_z18 has 92.7% building recall yet visibly overpredicts building regions. High recall alone hides this.
- In Copenhagen grid_00003_z18, an obvious sports field is mostly called ground instead of low vegetation. Its 95.2% building recall says nothing about that failure because only the building outlines are scored.
- In Lyon grid_00000_z19, substantial visible canopy is labelled ground, although several building roofs are found. Tree-versus-ground transfer is inconsistent.
- Munich grid_00112_z19 identifies much of a wooded area as trees but breaks some of it into ground and recognizes only 21.4% of the supplied building pixels.
- Copenhagen grid_01943_z18 identifies some open water but also calls a large bright industrial ground area a building. There is no measured water accuracy.
- Strasbourg and Munich examples show fragmented boundaries and roof/road confusion, matching the measured building-error distribution.

Conclusion: the classifier has learned useful recognition, but it is **not yet reliable enough to automatically control all app geometry/heights on HighBuild**. This does not prove a single cause. Geographic appearance, season/shadows, image scale, class definitions and limited detail are hypotheses to test.

Improvement plan, in priority order

1. **Get the right evidence and supervision.** Keep these 160 images as a fixed development regression suite. Obtain independently reviewed six-class labels on representative HighBuild-like imagery, with roofs, asphalt courtyards, shadows, grass and small trees all included. Split by source location/scene, not neighbouring crops. Label background as unknown where it is not verified. HighBuild outlines alone cannot teach or score the other five classes, and positive-only building training alone cannot reliably teach false positives.
 2. **Broaden the learning geography.** Mix the existing GAMUS training data with appropriately mapped, licensed multi-region semantic labels. OpenEarthMap is a candidate, not yet downloaded for this step: it includes imagery from 97 regions in 44 countries, with eight classes requiring an explicit six-class mapping. Do not reuse consumed Christchurch as a fresh final test. Reserve other entire locations before training. Start with a bounded, audited subset.
 3. **Test scale/appearance robustness before a large run.** Current V1 code uses 384-pixel crops and rotations/flips, but no color/scale augmentation. Compare a matched continuation with controlled scale, brightness/contrast/color, JPEG/blur perturbations, while keeping label alignment and independent image validity. Verify actual source pixel scales when available; filenames are not trusted ground sampling distances. Any gain is a hypothesis until measured.
 4. **Then test finer detail.** Small footprints fail disproportionately. Compare a detail-preserving decoder or boundary-supervised training separately from the data/augmentation change. Roof-versus-road and grass-versus-ground need semantic correction too; merely sharpening the existing masks is insufficient.
 5. **Keep six-class and height safety separate.** On exhaustive labels measure every class's precision, recall, F1 and IoU, plus road/roof boundaries, small objects and water/shadow mistakes. Use a fixed baseline and predeclared city regression limits. Leave the production height pipeline frozen; classification can become an overlay only after validation, not silently alter heights.
- Do not simply extend the old sampling run: the controlled RGB sampling V2 experiment failed its primary benefit and safety gates. Its exploratory epoch 2 is preserved but is not an established improvement over its matched control.

Research basis (supports directions, not guaranteed Monocular Surface Reconstruction gains):

- **OpenEarthMap benchmark:** multi-region semantic coverage and explicit geographic/domain-generalization evaluation.
- **SegFormer:** multiscale encoder architecture; architecture properties alone do not remove appearance/domain shifts.

- **AerialFormer**: aerial imagery's small-object challenges and high-resolution, local/global feature aggregation.

Artifacts and checks

- `outputs/evaluation/highbuild_rgb_classifier_20260919_v1/index.html`: all 160 original/prediction pairs, fixed city/sample order, no hidden failure cases.
- Same directory: `report.json`, `protocol.json`, `per_image.jsonl`, `per_object.json`, raw six-class ID PNGs, colored masks and 16 comparison panels.
- `scripts/audit_highbuild_rgb_classifier.py`: reproducible, non-overwriting run.
- Four focused audit tests passed. Known checkpoint hash authenticated before loading; classification source authenticated; corrected manifest/mask/COCO consistency checked; production pointer/checkpoint hashes unchanged afterward.
- Inference plus artifact generation completed in about 51 seconds on the RTX 4070. The evaluation process exited and released its GPU allocation.

Checkpoint SHA-256: 4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021.

53 Inspect class isolation, faster Walk and stable import grid

SOURCE docs/history/INSPECT_CLASSES_AND_WALK_POLISH_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

What changed

- **Walk:** normal pace is now 4.3 m/s (previously 1.4); hold G for 3×. This is game-style exploration, not a travel-time estimate. Starts/stops are more responsive. Footstep bob now includes small vertical dip, side sway, pitch and roll; sprint cadence is limited so boost does not cause frantic head shaking. Bob stops with actual motion and respects the existing toggle and reduced-motion preference. The nominal standing eye height remains 1.80 m.
- **Inspect → Class explorer:** Surface, Buildings and Vegetation show pixel coverage and can be selected individually. The photo transitions to the metric grid, existing geometry flattens, and only the selected class rises. Cyan/amber/green tint and emissive edge lighting highlight the chosen class. Unselected geometry is display-flattened, not re-predicted as zero. All heights restores the complete scene. Compact controls work in fullscreen and at widths where the analysis rail is hidden.
- **Import grid:** the empty-state and processing grids now have identical top, height, opacity and perspective. The loading overlay fades out over 650 ms when a reconstructed scene is ready, instead of jumping directly into a differently projected grid. Reduced motion skips the crossfade/reveal.

Data integrity and limits

Class isolation changes only the presentation mesh, building display scale and materials. Original raster data, collision-height buffers, checkpoints and exports are not modified. Leaving Inspect restores geometry before Fly/Walk or sightline measurements. Selecting Photo view restores all heights. Point inspection continues to report original heights, not the temporary animation height; clicks during the rise animation are ignored.

The categories match the viewer's resolved three-class map and visible building footprints. If semantic labels are absent, the existing RGB-assisted vegetation presentation is explicitly labelled. Unknown/missing-height pixels are not assigned a fabricated named class. Coverage is the fraction of valid-height pixels in the displayed classification map, not a building count or confidence score. Six-class identification is not claimed. Surfaces with near-zero predicted relief remain flat; highlighting does not invent heights.

Verification

- TypeScript and focused ESLint checks passed; no browser runtime or shader errors recorded.
- 63 targeted automated tests: walking, flight, per-pixel inspector, class animation/isolation/coverage, metric materials, import reveal, façades and preserved roof utilities.
- Live Urban reference: three category selectors, flatten/rise phases, rapid switching and All heights restore; final weight vectors were exactly one active class, or all ones on reset.
- Live Strasbourg upload: resolved three-class output with 38.2% surface / 52.5% building footprint / 9.3% vegetation for that result. Building inspection retained the original 14.4 m value in isolation view.
- Live import instrumentation: default, processing and fade-out grids all retained top 128.797 px, height 312.797 px and the identical CSS perspective transform in the test viewport; fade-out opacity was observed below 1 before removal. Import finished normally.

- Live hilly walking: approximately 6.19 m normal movement then 18.35 m boosted movement in comparable intervals. The previous run gave approximately 2.06 m / 6.01 m, respectively. Surface constraints and frame timing affect actual distance. Step bob eased back to the nominal 1.80 m eye offset at rest.
- Live fullscreen: compact selector successfully isolated Vegetation. At 980 px viewport width the analysis rail was hidden, compact controls remained visible, and All heights remained usable.

Implementation: `viewer/app/class-focus.ts`, `surface-query.ts`, `metric-grid.ts`, `walk-navigation.ts`, `terrain-workspace.tsx`, and `globals.css`. Tests: `viewer/tests/class-focus.test.mjs` plus updates to `metric-grid/walking` tests.

54 Metric grid view — 19 September 2026

SOURCE docs/history/METRIC_GRID_VIEW_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Neutral startup

The app starts and reloads with a static neutral grid and “**Upload an image to see the 3D simulation**”. No reference is selected and no demo image/height raster or Three.js canvas is loaded automatically. Urban, Sparse, Hilly and Forest remain explicit reference buttons. A standalone height map loaded before any photograph uses a neutral texture and grid view, never an unrelated urban image.

User-facing behaviour

- The **METRIC GRID** button in the 3D viewport, or **Layers → Metric grid · no photo**, reveals the rendered height surface without optical or façade textures.
- A regular horizontal grid drapes across the surface and flat building tops. Horizontal height bands run across slopes and building walls; these are elevation intervals, **not detected floors**.
- The legend covers the displayed surface plus building tops, states the horizontal cell size and height-band interval, and distinguishes height above ground, absolute elevation and relative units.
- Ordinary images without verified map scale use **pixel** spacing, even if their height estimates are in metres. Projected metric/foot-based raster metadata or explicit metre GSD can establish horizontal scale. Degrees and unreferenced TIFF transforms cannot.
- **PHOTO VIEW** restores the optical surface and original building materials. Fly, Inspect, collisions, the roof-configuration notice and exports remain available. Cosmetic tree proxies are hidden only in grid view.
- Photo/grid changes use a reversible 650 ms eased material blend, including fading façade bump/roughness detail. Geometry and camera position do not change. Reduced-motion preferences bypass the animation.

Import reveal

While a new image is processed, a neutral blank grid and the real elapsed-status text cover the previous scene. The old 3D renderer pauses during inference. No fabricated model percentage or intermediate detection is displayed.

Once the actual image, height products and footprints are ready, a 3.65-second presentation sequence shows **grid → building shapes → height rise → optical wrapping**. This temporarily scales the displayed mesh vertically, then returns exactly to its original display scale. Predictions and raw metadata never change. The card explicitly says the prediction is ready and this is a visual reveal, not live detection.

Skip animation finishes immediately. Reduced-motion users go straight to the finished scene. Fly and inspection are unavailable during the temporary height rise and are restored afterward. Reference-comparison opening is deferred until the reveal ends. Failed inference removes the waiting view and retains the previous usable scene. Replacing a texture does not retrigger the import reveal.

Integrity and implementation

viewer/app/metric-grid.ts supplies bounded grid intervals, scale validation and a texture-free display material. Switching only changes materials/visibility; it does not change mesh positions, raw rasters, predictions, building heights, checkpoints or exports. Colour/contour coordinates account for the current vertical exaggeration without changing the interval labels. Inactive materials are retained for switching and disposed on scene teardown.

The view is a presentation surface, not a bare-earth DEM or an additional model prediction. Smoothing/visual aids can differ from raw height values; use Inspect and exports for those values. Roofs remain flat and roof geometry is unconfigured. Grid axes follow the raster, not an assumed north direction. Horizontal extents are between first and last pixel centres.

Checks

- 47 focused tests passed: scale handling, material transitions, reversible/reduced-motion timing, reveal stages/skip, navigation, inspection, façades and preserved roof modules.
- TypeScript and focused ESLint checks.
- Browser checks cover real-image uploads, optical/grid switching, inspection and hilly absolute-elevation and relative-raster views.
- The Strasbourg example completed every reveal stage without browser errors. Its inspected building read 14.4 m in both photo and grid views, with the same roof-status notice.
- Observed the photo/grid transition across 36 distinct rendered intermediate frames. A real import showed processing, grid, footprints, heights, texture and completion in order; controls re-enabled afterward.
- Confirmed Skip during the height-rise stage, reduced-motion import without the reveal, and recovery from a simulated failed network request in an isolated browser (previous scene retained, upload re-enabled, no uncaught errors).
- Manali's known-scale DSM displayed 500 m horizontal cells and absolute-elevation bands; a relative Chicago raster displayed pixel cells and dimensionless heights, not metre claims.

55 All-six-class mixed-scene test — completed

SOURCE docs/history/OEM_ALL_SIX_CLASS_TEST_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Data and selection

Downloaded 12 new OpenEarthMap v1 RGB/reference-label pairs: four from Duesseldorf (Germany), four from Dhaka (Bangladesh), four from Chiang Mai (Thailand). Each selected reference image contains all six mapped classes, with at least 256 valid labelled pixels per class. Combined RGB/label validity confirmed that every selected image still meets that criterion.

The 12 scenes were chosen by ascending SHA256(filename) within each region, taking the first four with all-six reference support. Nineteen candidate label files were inspected. This selection was locked before RGB decoding or model inference; predictions were not used to choose images. All 12 selected results are published, including failures. This is a deliberately label-enriched mixed-scene challenge, not random city sampling or a global-accuracy estimate.

Only about **39.7 MB** of archive ranges were transferred, not the whole 9.1 GB archive. All selected ZIP entries passed CRC32 and length checks; local source SHA256 hashes are recorded. The full archive's published MD5 was NOT verified.

Data, original images, labels, predictions and the comparison gallery are saved at
D:/MSRData/evaluation/oem_all_six_20260919_v1/.

Official sources: [OpenEarthMap v1](#) and [source attribution](#). Duesseldorf: GeoNRW / North Rhine-Westphalia, DL-DE-BY-2.0. Dhaka: AIGEO Center, CC BY 4.0. Chiang Mai: UR Field Lab Chiang Mai, CC BY 4.0.

Model and reference interpretation

The same frozen RGB SegFormer V1 guarded epoch 8 was used, SHA256 4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021. Inputs were native-resolution raw RGB/255 with internal ImageNet normalization; CUDA bf16 inference, argmax, no augmentation, calibration or postprocessing. No training or app-model replacement occurred. This classifier was trained on DC/Philadelphia, not these three regions; that is not proof about all upstream pretraining data. These downloaded scenes and inspected candidate labels are now consumed diagnostics, not an untouched future final test.

Fixed OEM-to-model mapping: bareland + developed space -> ground; building -> building; water -> water; road -> road; rangeland -> low vegetation; tree -> tree. Unknown and agriculture are ignored in the primary result. Agriculture can include woody crops, so its optional low-vegetation mapping is reported only as a separate sensitivity result. The native 8-by-6 confusion table is preserved. Overall 10.53% of pixels were ignored. These labels are semantic references, **not LiDAR and not height measurements**.

Results

Pooled confusion counts across all 12 images give:

Category	Precision	Recall	F1	IoU
Ground	28.0%	57.4%	37.7%	23.2%
Buildings	80.3%	72.1%	76.0%	61.3%
Water	93.1%	69.7%	79.7%	66.3%
Roads	32.8%	40.7%	36.3%	22.2%
Low vegetation	36.4%	13.4%	19.6%	10.9%

Category	Precision	Recall	F1	IoU
Trees	76.6%	73.1%	74.8%	59.7%

- Six-class macro F1: **54.02%**.
- Macro IoU: **40.60%**.
- Pixel accuracy: **58.46%**. This is influenced by common classes and is not interchangeable with macro F1.
- Two-pixel road-boundary F1: **18.11%**.
- Dark non-water reference pixels called water: **0.746%**. This is a dark-RGB proxy, not accuracy against independently labelled shadows.
- Every category occurs in all 12 tiles and exceeds 10,000 pooled reference pixels. This ensures support for computing the scores, not broad representativeness.

Per-region pooled macro F1: Duesseldorf **55.95%**, Dhaka **40.35%**, Chiang Mai **53.20%**. The global 54.02% comes from pooled per-class confusion counts, not the arithmetic mean of region or image F1 scores.

Plain-language verdict

Water, buildings and trees are the stronger categories on this set, but none is perfect. Grass/low vegetation, ground and roads are unreliable. Only 13.4% of reference low-vegetation pixels are found; around 52.4% are called ground and 23.9% trees. Nearly 47.5% of road reference pixels are called ground.

Visual checks of the first preselected scene from each region confirm useful large-area recognition but blurred/merged roof boundaries and missing narrow roads/grass. Duesseldorf_19 illustrates why pixel accuracy alone can mislead: 81.8% pixel accuracy but only 36.2% macro F1, because a large wooded area is found while several smaller categories are weak.

Black padding may receive predictions when a source TIFF has no optical nodata declaration. Such regions with unknown reference labels are excluded from the reported scores, not treated as ground or water truth. Predictions are not secretly corrected using reference labels for display.

The earlier HighBuild result was 79.2% known-building recall with unknown background, not six-class F1. It is not comparable to this 54.0% macro F1. The weights have not worsened during this test: harder/different imagery, reference definitions and a different metric reveal transfer weaknesses.

Next improvement priority

Prioritize independently reviewed multi-region grass/ground/road training examples with consistent class mapping and geographic splits; then matched scale/appearance augmentation and detail-preserving experiments. Preserve water/building/tree checks and the production height model. Do not claim a reliable improvement from more epochs alone or tune on these scenes while continuing to describe them as an untouched test. New final locations are needed after any model changes motivated by this diagnostic.

Files and verification

- `results/index.html`: all original/reference/prediction comparisons, class table and per-image results; browser checked, all 12 comparison images load.
- `images/*.tif`: original downloaded RGB files.
- `candidate_labels/<region>/*.tif`: original class labels (IDs 0-8, not heights).
- `results/scenes/<id>/original.png`: convenient native-resolution RGB for testing.
- `results/scenes/<id>/reference_ids.png`, `prediction_ids.png`: mapped class IDs.
- `results/report.json`, `per_image.jsonl`, `seal.json`: measurements/provenance.

- `selection_contract.json`, `manifest.json`, receipts and evaluation markers: source integrity and pre-inference selection history.

23 focused selection, range-download, mapping and metric tests passed. Production pointer, protected height checkpoint and classifier weights remained hash-identical. GPU inference finished and released its allocation. No new training process or deployment was started.

56 OpenCanopy richer-label and holdout contract

SOURCE docs/history/OPEN_CANOPY_RICH_LABELS_AND_HOLDOUTS.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: metadata and planning implementation complete; class backfill and model training intentionally not started.

What is already preserved

- Every accepted OpenCanopy record is linked back to its official 1 km cell, EPSG:2154 bounds, 1.5 m ground sampling distance, larger SPOT source image, imagery acquisition date, LiDAR acquisition date, source URLs, licence, and local source .json hash.
- The accepted subset remains 600 training, 120 validation, and 120 locked-test chips. The official test role is unchanged.
- The three accepted splits have no shared 5 km region identifiers.
- Existing canopy heights, binary vegetation masks, validity masks, and CSV manifests are untouched.

The authenticated metadata inventory is outputs/data_audits/open_canopy_v2_provenance_v1.json.

Rich LiDAR classification path

The optional sidecar preserves source-valid upstream byte identifiers exactly and uses 255 only for source-invalid pixels. Known IGN LiDAR HD meanings are:

ID	Meaning	Candidate use
0	never classified / unused	preserve and ignore
1	unclassified	ignore
2	ground	context/audit, not a vegetation subclass
3	low vegetation	richer vegetation target
4	medium vegetation	richer vegetation target
5	high vegetation	richer vegetation target
6	building	exclude from the forest expert
9	water	context/audit, not a vegetation subclass
17	bridge deck	preserve if present; unsupported in the present expert
64	permanent above-ground object	preserve if present; unsupported
65	artifact	preserve if present; ignore
66	synthetic point	preserve if present; unsupported
67	miscellaneous built object	preserve if present; unsupported

These are LiDAR classification/height-stratum labels, not tree species. The three vegetation categories therefore support a low/medium/high vegetation study; they do not justify claims about species, health, snow, or land-cover types absent from the source labels.

The resumable backfill utility is `scripts/data/backfill_open_canopy_classification.py`. It reads only the selected remote windows and writes to a separate versioned root such as `D:/MSRData/data/open_canopy_v2_classification_v1`. It refuses to write inside the prepared binary subset. Train and validation are the default; including locked test requires a separate explicit switch. Each output raster gets an aligned-RGB hash, source URL, class histogram, grid metadata, and its own SHA256 sidecar.

A one-chip train-only smoke artifact has been streamed to `D:/MSRData/data/open_canopy_v2_classification_smoke_v1`. Its aligned 384 x 384 class raster passed its hash/alignment checks and contained IDs 0, 2, 3, 4, 5, and 65. In particular, it recovered 124 low-, 512 medium-, and 146,026 high-vegetation pixels. This verifies the preservation path, not class balance or model quality. The full train/validation backfill has not been downloaded, and no richer-label model has been trained. The smoke raster SHA256 is `e9575207e7202d2f1c236802fedfa61a431d20f88ecb08947cdbeb637fa48f3e`.

Holdout findings

The present split is geographically separated at the existing coarse-region level, but it is not independent by source acquisition. Larger SPOT mosaics shared across accepted splits are:

Pair	shared 5 km regions	shared SPOT source mosaics
train / validation	0	57
train / locked test	0	49
validation / locked test	0	44

Therefore the current forest results may be called region-disjoint, but not source-scene-disjoint or acquisition-disjoint.

`scripts/data/plan_open_canopy_holdout.py` builds a deterministic holdout from accepted official-training metadata only. It groups samples transitively, so a source mosaic that links multiple regions cannot be split across learning and holdout. It can enforce any combination of region, source image, imagery date, and LiDAR date, and it fails closed when required metadata is missing.

Generated versioned plans:

Plan	constraints	components	largest component	learning / holdout
geographic v1	region	507	7	480 / 120
scene-geographic v1	region + source image	64	20	477 / 123
acquisition stress v1	region + source image + both dates	5	459	486 / 114

The scene-geographic plan is the recommended stronger development holdout. The all-acquisition plan technically passes its requested isolation check, but 459 of 600 records collapse into one connected component. That makes it a useful stress test, not a well-balanced primary benchmark.

Artifacts and hashes:

- `outputs/data_audits/open_canopy_v2_holdout_geographic_v1.json` — `c930deb86905f52c4fe9eb0178c1741368022a04dd0264cd2055414515127b08`
- `outputs/data_audits/open_canopy_v2_holdout_scene_geographic_v1.json` — `0ec239e2e7de0a33b7e0a77505ef0d4b8aafe2276cfccebfc040ad5ff8938bab`
- `outputs/data_audits/open_canopy_v2_holdout_scene_acquisition_v1.json` — `8b7e909983479c4d5f7ac4cf11389f433781bc7fe7c8fed881ed17e038b46f5d`

Remaining work before using richer classes

1. Pin an upstream OpenCanopy revision instead of relying on mutable `resolve/main` URLs, then backfill train and validation sidecars.

2. Audit the observed class histogram, spatial coverage, RGB/class alignment, class-specific label sparsity, and the imagery-versus-LiDAR time gap.
3. Decide whether low/medium/high vegetation is sufficiently visible at 1.5 m RGB resolution to justify a new head. Keep the existing binary canopy path as the protected comparison.
4. Train a separate candidate and report each vegetation class independently. Do not promote it unless OpenCanopy, GAMUS, corrected HighBuild, and unseen geography height guards all pass.
5. Open the locked test labels only once the experiment and thresholds are frozen. Metadata-only inventory does not authorize tuning on test pixels.

Source limitations

- OpenCanopy's dataset card explicitly describes 1.5 m VHR imagery, LiDAR-derived canopy height, and a classification raster used to mask class 1.
- The detailed class names above come from the IGN LiDAR HD classification specification used by those source rasters.
- Dates are provenance fields, not proof of simultaneous acquisition. Seasonal and real land-cover change between SPOT and LiDAR dates can create honest label disagreement.
- A France-only source-scene holdout does not prove transfer to a new country, sensor, resolution, or season.

Official references:

- <https://huggingface.co/datasets/AI4Forest/Open-Canopy>
- https://geoservices.ign.fr/sites/default/files/2024-11/Fiche_produit_Nuages-de-points-LiDAR.pdf

57 External six-class holdout: Christchurch OpenEarthMap v1

SOURCE docs/history/OPENEARTHMAP_CHRISTCHURCH_EXTERNAL_HOLDOUT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: **preregistered design only; not downloaded, decoded, or evaluated.**

This holdout is reserved for one post-freeze comparison between the paired DC+Philadelphia spatial-V3 head and an eligible hierarchical-V4 head. It is not part of model selection. Christchurch is absent from the known Monocular Surface Reconstruction GAMUS, HighBuild, OpenCanopy, and demo inventories, but this does not prove absence from generic upstream pretraining corpora.

Pinned source

- Dataset: OpenEarthMap v1
- Region: every released christchurch RGB tile with a same-stem label tile
- DOI: <https://doi.org/10.5281/zenodo.7223446>
- Published archive MD5: 64155d1dc9d3b68536063f79878e1a67
- Source imagery attribution: AIRS / Land Information New Zealand, CC BY 4.0
- OpenEarthMap attribution: <https://open-earth-map.org/attribution.html>
- LINZ attribution guidance:

<https://www.linz.govt.nz/products-services/data/licensing-and-using-data/attributing-elevation-or-aerial-imagery-data>

The repository's MIT software licence is not the imagery licence. Exact source-layer attribution must be retained before redistribution.

Frozen class crosswalk

OEM id	OEM class	Monocular Surface Reconstruction class
0	unknown / no-data	ignore (255)
1	bareland	ground (0)
2	rangeland / grass	low vegetation (4)
3	developed space / pavement	ground (0)
4	road	roads (3)
5	tree	trees (5)
6	water	water (2)
7	agriculture	ignore (255) in the primary result
8	building	buildings (1)

Agriculture is ignored in the primary result because the source class can include woody orchards and vineyards. A separately labelled sensitivity result may map id 7 to low vegetation, but it cannot replace the primary score. The report must also include the native 8-by-6 source-label versus prediction table so merging cannot hide failures.

One-shot procedure

1. First require V4 to pass every paired DC+Philadelphia class gate, state isolation check, independent validation replay, and bit-exact height-output audit.
2. Before acquisition or pixel decoding, write and externally timestamp a contract containing the DOI/version/archive checksum, region inclusion rule, crosswalk, frozen checkpoint/config/code hashes, radiometry, tiling, inference, metrics, gates, and no-tuning/no-reselection rule.
3. Verify the archive MD5, extract only Christchurch, pair files by exact stem, hash every selected RGB/label file, and seal a sorted manifest. Do not selectively omit difficult tiles.
4. Write a CONSUMED.json marker before the first pixel decode. Evaluate the frozen V3 and V4 checkpoints together on identical inputs in one invocation. A crash still counts as consumption and any exceptional rerun must be disclosed.
5. Publish and hash the complete report regardless of outcome. Only then may examples be viewed; select examples by a preregistered hash order rather than visual appeal.

Required report

- Exact 6-by-6 confusion matrix (rows reference, columns prediction).
- Per-class tile/pixel support, precision, recall, F1, and IoU.
- Six-class macro precision, recall, F1, IoU, and overall accuracy.
- V4-minus-V3 deltas with a fixed-seed paired bootstrap over whole tiles.
- Native OEM 8-by-6 table, ignored share, and agriculture sensitivity result.
- Two-pixel road-boundary metrics and the explicitly labelled dark-pixel water false-positive proxy.
- Low-vegetation/tree confusion and combined-vegetation F1 as diagnostics only.
- No height accuracy claim: OpenEarthMap has no height truth. Report only V4-versus-protected height-output identity.

Before calling this a comprehensive six-class holdout, every mapped class must occur in at least five tiles and 10,000 reference pixels. Inadequate coverage is reported honestly and does not authorize swapping locations after results are seen.

Claim limit

Permitted wording: **"One-shot transfer to a preregistered Christchurch benchmark, new to known Monocular Surface Reconstruction project data, under a fixed ontology harmonization."**

This does not prove global generalisation, Christchurch population accuracy, height accuracy, or absence from unknown upstream pretraining data.

58 Monocular Surface Reconstruction progress and recovery record

SOURCE docs/history/PROGRESS.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Last updated: 2026-09-13 (Asia/Calcutta)

This file is the restart point if the active Codex session reaches its usage limit. Source code is also checkpointed in Git; large datasets, model weights, and generated outputs are deliberately ignored by Git but remain on the local disk under the paths documented below.

Verified environment

- Windows, Python 3.10.11
- NVIDIA GeForce RTX 4070 Laptop GPU, 8,188 MiB VRAM, 140 W
- NVIDIA driver 596.36
- Project environment: .venv
- PyTorch 2.7.0+cu128, torchvision 0.22.0+cu128
- CUDA is available and a 2048x2048 matrix-multiplication smoke test passed
- Core test suite: 56 tests passed

Re-run verification:

```
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -c "import torch; print(torch.__version__,
torch.cuda.is_available(), torch.cuda.get_device_name(0))"
```

Implemented

- Explicit CSV sample manifests with mandatory geographic region identifiers
- Geographic train/validation/test leakage rejection
- GeoTIFF RGB/height shape, transform, CRS, nodata, and range audit
- Raster-window PyTorch dataset with no height resampling
- Synchronized rotations/flips and validity masking
- MAE, RMSE, bias, correlation, R-squared, median, P90, and P95 errors
- Exact streaming primary metrics for large validation sets
- Seam-resistant overlapping tile layout and Hann blending
- Clean HeightNet with a timm multiscale encoder, metric height head, and auxiliary building head
- Masked Huber, gradient/edge, and building losses
- CUDA AMP training, accumulation, early stopping, JSONL history, and best/latest checkpoints
- Public predict_height result contract and GeoTIFF prediction CLI
- Isolated adapter for research-only FusedSeg-HE benchmarking

Real smoke data

A public reviewer sample from feifei140729/small-sample is stored at:

data/smoke/highbuild_small/NorthAmerica_USA_Chicago/

Sample grid_00034_z18 contains:

- RGB: 1024x1024, uint8, three bands
- height raster: 1024x1024, float32, range 0-230 m
- 16 building polygons with per-building heights
- no CRS/geotransform or documented GSD

The pair is pixel-aligned and useful for inference plumbing/domain-shift smoke tests. It is not sufficient for final scientific claims because it lacks geospatial metadata and its full-dataset license/provenance still needs review.

Audit output: outputs/smoke_pair_audit.json

Multi-city reviewer training

The public HighBuild reviewer subset is stored at data/highbuild_reviewer/:

- 260 RGB/height pairs from 26 cities
- train: 180 samples / 18 cities
- validation: 40 samples / 4 cities
- test: 40 samples / 4 cities
- all 260 pairs passed the strict alignment/band/validity audit
- raw labels span 0-584 m

This subset declares license: other and is for engineering/reviewer inspection, not final production claims. The two-epoch pilot completed successfully. Validation RMSE improved from 7.91 m to 7.80 m and building-only RMSE from 12.95 m to 12.77 m. The pilot proved BF16 training, metrics, and checkpoint persistence; a resume test also passed after correcting CPU/CUDA RNG-state restoration.

The active preliminary experiment is:

experiments/20260826T135412Z_highbuild_reviewer_train/

At epoch 2 it reached 7.15 m all-pixel RMSE and 11.61 m building-pixel RMSE. Check metrics.jsonl for newer completed epochs. Resume safely with:

```
.\.venv\Scripts\python.exe scripts\train.py `
--config experiments\20260826T135412Z_highbuild_reviewer_train\config.yaml `
--resume experiments\20260826T135412Z_highbuild_reviewer_train\checkpoint_latest.pt
```

Full HighBuild preparation

HighBuild-1M metadata is at data/highbuild_full/; the resumable shard download was started with scripts/data/download_highbuild_full.py --include-shards. The hosted release contains 61,215 benchmark tiles in about 35 GB of TAR shards.

The upstream random split is not acceptable for Monocular Surface Reconstruction because cities leak across train/validation/test. scripts/data/prepare_highbuild_full.py creates a city-disjoint, conservative license-filtered split containing:

- train: 25,446 tiles / 5 cities

- validation: 1,921 tiles / 4 held-out cities
- test: 2,310 tiles / 5 held-out cities

USA, Cape Town, and conditional-license city sources are excluded from the production split by default. `scripts/data/repack_highbuild_full.py` atomically repackages only selected samples and is safe to restart per source shard.

Full HighBuild training and error analysis

The full ConvNeXtV2-tiny run completed normally through epoch 33 and then early-stopped after eight epochs without a new RMSE best. The experiment is:

`experiments/20260826T192229Z_highbuild_full_convnext_tiny/`

Protected checkpoints:

- epoch 25 best RMSE: 4.366 m RMSE, 1.979 m MAE, +1.240 m bias, 0.593 correlation, and 0.093 R-squared
- epoch 32 best MAE: 1.951 m MAE and 4.409 m RMSE

Validation-only error analysis for the epoch-25 checkpoint is saved under:

`outputs/error_analysis/epoch25_best_rmse/`

The main findings are:

- 430.6M of 503.6M validation pixels are below 2 m in the building-height reference; these pixels have an essentially zero target but receive a mean 1.729 m prediction, producing 4.072 m RMSE.
- Building pixels average 11.049 m in the reference and 9.403 m in prediction, with 5.808 m RMSE and -1.646 m bias.
- Tall structures are strongly compressed: the 20-40 m bin has -10.71 m bias, and higher bins degrade sharply.
- Validation is dominated by Copenhagen (1,618/1,921 tiles). Copenhagen reaches 3.611 m RMSE and 0.292 R-squared, while Lyon, Munich, and Strasbourg have 5.668-8.574 m RMSE and negative R-squared.
- The auxiliary building head peaks around 0.477 IoU / 0.646 F1 at threshold 0.3. Hard gating barely changes overall RMSE; soft gating lowers overall RMSE to 3.855 m but damages building RMSE to 7.610 m, so it must not be adopted as the final post-processing rule.
- Two-fold cross-fitted global affine calibration reaches a diagnostic 3.699 m RMSE and 0.349 R-squared, but it is dominated by background pixels. Building- only cross-fitted calibration reaches 5.442 m RMSE and 0.273 R-squared.

The next urban refinement should use city- and height-balanced sampling, explicit tall-building supervision, stronger segmentation/background control, and checkpoint selection for building and cross-city metrics. The untouched test split remains unused.

Targeted refinement and end-to-end raster foundation

The targeted refinement experiment is running at:

`experiments/20260828T044931Z_highbuild_targeted_refinement/`

It starts from the protected epoch-25 full-data checkpoint and uses weighted city streams, foreground/tall-guided crops, explicit foreground MSE and tall Huber terms, and separate checkpoint selection for overall, building-only, and cross-city macro RMSE. The source checkpoint remains untouched, so a failed refinement cannot erase the strongest honest baseline.

The missing raster/calibration foundation has also been added:

- licensed Apache-2.0 Depth Anything V2 Small relative-depth extraction;
- explicit dimensionless rDSM output for non-georeferenced imagery;

- CRS/transform-preserving, atomic float GeoTIFF output;
- DEM reprojection onto the exact RGB grid;
- robust affine GCP calibration with outlier trimming;
- absolute DSM composition as ground terrain plus non-negative object height;
- a unified `scripts/run_pipeline.py` entry point and machine-readable metadata.

The pretrained weights are cached locally under `models/foundation/depth-anything-v2-small-hf/` and intentionally excluded from Git. A real non-georeferenced Chicago sample completed the `relative-depth/rDSM` path successfully under `outputs/pipeline_smoke/`.

Interactive large-raster safety

The local API now enforces a 3072 x 3072 interactive processing budget based on pixel dimensions rather than compressed file size. Rasterio reads an overview directly, preserves the original CRS and full geospatial bounds by updating the affine transform, and records original/processing grids in response metadata. Pixel-coordinate GCPs are scaled onto that overview grid; map-coordinate GCPs continue to use the preserved transform.

The exact 34,713 x 36,360, 202.8 MB GeoTIFF that previously expanded toward 36 GB of process memory completed end-to-end in 23.96 seconds. The bounded path peaked around 2.0 GB RAM, returned HTTP 200, preserved EPSG:31985 and identical map bounds, and emitted a clear overview-resolution confidence warning.

The known-good local app can be restored with:

```
.\scripts\start_prototype.ps1 -OpenBrowser
```

Pretrained research benchmark

The FusedSeg-HE checkpoint and MiT encoder weights were downloaded to:

`checkpoints/benchmarks/fusedhe/`

The upstream implementation was inspected separately at:

`C:/Users/researcher/AppData/Local/Temp/msr-research/FusedHE`

Its license is research/evaluation-only and explicitly non-commercial. None of its source code is copied into the clean Monocular Surface Reconstruction package; it is benchmarked only through `scripts/benchmarks/run_fusedhe.py`.

The first fp16 inference completed but produced non-finite pixels over roughly 44% of the image. Its metrics are invalid and must not be reported as a model score. The adapter now rejects any non-finite tile and defaults to fp32.

The fp32 rerun produced finite values at all 1,048,576 pixels. It is a valid inference run but a failed cross-domain baseline on this sample:

- prediction range: 0-39.44 m; target range: 0-230 m
- all-valid pixels: RMSE 93.89 m, MAE 63.57 m, bias -59.29 m
- positive-height pixels: RMSE 123.01 m, MAE 105.79 m, bias -105.78 m

Metric records are in `outputs/benchmarks/fusedhe/chicago_grid_00034_z18_fp32_*.json`. Do not present these numbers as representative checkpoint quality: the checkpoint was trained for a different sensor/domain and its scale plainly does not transfer to this tall-building reviewer sample. Preserve it as evidence that preprocessing and domain compatibility must be verified before model selection.

Dataset access constraint

The official DFC2023 Track 2 archive is 3.88 GB and requires an IEEE DataPort login plus approval as a competition participant. It cannot be downloaded from an anonymous session. Do not silently substitute an unofficial mirror for the final benchmark. Continue public-data smoke testing while arranging legitimate access or selecting a clearly licensed alternative.

Multi-domain model milestone (2026-08-28)

Implemented and verified the next-stage surface model without overwriting the protected urban checkpoint:

- explicit aligned dsm/ndsm/building_height manifests with ground, building, and vegetation supervision;
- a shared gated adapter whose per-pixel expert weights are mutually exclusive;
- a supervised refinement gate so canopy corrections open over vegetation but remain conservative over protected urban surfaces;
- a frozen-base first training stage and independent guards for all-urban and building-only RMSE;
- landscape-, semantic-domain-, and region-level validation plus signed-error rasters;
- a resumable Open-Canopy range-streamer with official split preservation, 1.5 m output, decimetre-to-metre conversion, and LiDAR-class filtering;
- a city-balanced extractor for the existing conservative HighBuild shards;
- one-time CUDA precomputation of the Depth Anything V2 relative prior;
- a durable preparation/training driver at `scripts/run_multidomain_pilot.ps1`.

The corrected licensed-data pilot used 168 training chips (120 urban and 48 forest), 60 validation chips, and 72 untouched test chips. Forest/urban sampling and semantic loss are class-balanced; repeated crops and augmentation do not get counted as new geographic scenes.

Best eligible held-out validation checkpoint: epoch 9 in `experiments/20260828T175822Z_multidomain_surface_pilot`.

- overall RMSE: 5.67 m;
- landscape-macro RMSE: 6.25 m;
- urban RMSE: 5.21 m, 0.25 m better than the protected urban baseline on the identical pilot validation subset;
- forest RMSE: 7.28 m, improved from the protected baseline's 11.87 m;
- building-only RMSE: 4.34 m, a 0.17 m regression and therefore inside the configured 0.25 m protection guard;
- vegetation-only final RMSE: 10.11 m; canopy-expert RMSE: 8.88 m;
- vegetation routing precision/recall/F1: 52.3% / 94.6% / 67.3%;
- mean fusion strength: 10.3% ground, 18.5% building, 63.5% vegetation.

Epochs 10-13 improved forest RMSE as far as 6.83 m, but exceeded the building regression guard and were correctly rejected. Training early-stopped at epoch 13. The protected urban checkpoint, epoch-9 best checkpoint, and epoch-13 latest diagnostic checkpoint all remain saved. These are pilot results, not a claim of production or industry-wide accuracy; broader geographic training data remains necessary.

Earlier smoke results that proved the plumbing:

- three Open-Canopy chips streamed from the official release and passed raster alignment/unit checks;
- 14 urban samples extracted across 5 train, 4 validation, and 5 test cities;
- combined geographic leakage validation passed;
- a complete BF16 CUDA training/validation/checkpoint epoch ran on the RTX 4070;

- the initial safety blend held sampled urban regression to -0.039 m and building-only regression to -0.017 m relative to the protected base;
- 57 automated tests pass.

The inference stack now loads both legacy HeightNet and domain_gated_surface_v1 checkpoints. It passes the cached Depth Anything prior into the adapter and exports building probability, vegetation probability, and relative prediction confidence alongside nDSM/rDSM. A real CUDA/API filesystem smoke generated all products from the new checkpoint format. The web viewer build passes and exposes the new diagnostic rasters for download.

research showcase integration milestone (2026-08-29)

The viewer/API now cover the remaining high-value prototype interactions while the v2 multidomain model trains independently:

- optional reference nDSM or absolute DSM upload with grid alignment, honest RMSE/MAE/bias/correlation/R2 reporting, and a signed-error GeoTIFF;
- point-height plus metric local-slope inspection and scene mean slope;
- automatic public bare-earth terrain for georeferenced imagery, with source URLs and attribution recorded alongside the derived absolute DSM;
- manual DEM and GCP calibration retained as the accuracy-first alternatives;
- guarded showcase-checkpoint selection so only an explicitly reviewed held-out checkpoint replaces the protected pilot;
- PID-scoped start/stop scripts that do not terminate unrelated Python, browser, GPU, or training processes.

The automatic terrain path was exercised on a real Open-Canopy GeoTIFF and successfully mosaicked and reprojected two public terrain tiles onto the exact RGB grid. This supplies an absolute datum, not high-resolution object truth; building/tree height quality still comes from the trained surface model.

The v2 quality run completed all 24 configured epochs. Epoch 20 was the best guard-eligible validation checkpoint and was then evaluated once on the untouched 320-record test manifest:

- overall final-surface RMSE 6.17 m, MAE 3.38 m, correlation 0.619, R2 0.361;
- forest RMSE 6.96 m versus 11.49 m for the protected urban-only base;
- urban RMSE 5.65 m versus 5.63 m for the base (0.02 m regression);
- vegetation final-surface RMSE 8.79 m versus 16.40 m for the base;
- dedicated canopy expert RMSE 7.04 m, correlation 0.488, R2 0.229;
- building-domain RMSE 10.64 m versus 10.68 m for the base on this harder geographically held-out test set;
- both urban and building regression guards passed.

The test audit is saved at `outputs/analysis/20260829T130607Z_multidomain_surface_v2-test.json`. A real CUDA pipeline smoke also generated rDSM, nDSM, building/vegetation rasters, and metadata under `outputs/showcase/v2_checkpoint_smoke`. The reviewed epoch-20 checkpoint is now the explicit showcase selection; `checkpoint_latest.pt` was not promoted.

GAMUS specialist and guarded routing audit (2026-09-13)

The GAMUS Stage-1 specialist materially improved semantic identification on the authenticated GAMUS validation split, but it was not safe to replace the protected multi-domain model globally. A Stage-2 mixed fine-tune failed its regression guards, so neither checkpoint was promoted directly.

Stage 3 therefore trained a small scene router over frozen shared-model feature descriptors. Its purpose is to select the GAMUS specialist only for scenes that look appropriate and otherwise fall back to the protected model. The final record package contains

7,184 authenticated train/calibration records; test splits were excluded while the router and threshold were chosen. Nine planned GAMUS crops were audited as empty and excluded. Windows-safe manifest replacement retries and full-precision route decisions were added, with fail-closed artifact authentication for the new Stage-3b format.

The selected H128 router passed its calibration requirements at threshold 0.707040: 99.06% source precision, 98.02% GAMUS coverage (842/859), 8/280 legacy selections, and positive routed utility gain. All 230 automated project tests passed before final evaluation.

Dense validation also passed both predeclared hard-guard suites:

- GAMUS identification accuracy: 57.40% protected -> 73.36% routed;
- GAMUS macro F1: 0.6002 -> 0.7495;
- GAMUS vegetation F1: 0.3690 -> 0.7379;
- GAMUS height RMSE: 7.0469 m -> 7.0283 m;
- legacy height RMSE: 6.5768 m -> 6.5767 m;
- legacy macro F1: 0.6326 -> 0.6323 (inside the validation guard);
- the router selected 841/859 GAMUS and 8/280 legacy validation scenes.

After all choices were frozen, the official held-out test was evaluated once. It confirmed strong GAMUS gains but exposed a geographic/style generalisation failure on unseen legacy cities:

- GAMUS test identification accuracy: 55.86% -> 65.45%;
- GAMUS test macro F1: 0.5625 -> 0.6767;
- GAMUS test vegetation F1: 0.2632 -> 0.5591;
- GAMUS test height RMSE: 7.9167 m -> 7.9045 m;
- legacy test height RMSE: 6.1455 m -> 6.1418 m;
- legacy test identification accuracy: 68.50% -> 64.08%;
- legacy test macro F1: 0.6140 -> 0.5837;
- the router selected 102/200 unseen legacy urban scenes and 0/120 legacy forest scenes, despite selecting only 8/160 legacy urban validation scenes.

This router is therefore **not approved for application promotion**. The result suggests it learned dataset/city appearance cues rather than a sufficiently general rule for when the specialist is beneficial. The live application pointer remains on the protected epoch-20 checkpoint, whose SHA-256 is e2d507f7be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144. The immutable final audit is outputs/evaluation/stage3b_routed_final_h128.json; its SHA-256 is 3f251627fadd6a5a1294ec80e84650073127dda978d0baaa62a0ca47ac79dc7e.

The leakage-safe next iteration must use group-held-out cities within the development data, train a content/utility-aware gate rather than a source-label shortcut, and be reviewed on a newly locked external test set. The now-exposed official test remains a final audit record and must not be reused for threshold or model selection, training, architecture comparison, visual case selection, or repeat evaluation.

Stage-4 independent routing audit (2026-09-13)

Stage 4 replaced the coupled source router with two independently guarded endpoint-gain heads: one for metric height and one for semantic surface class. Its authenticated record journal contains 3,347 development scenes (2,208 fit, 1,139 calibration), preserves geographic group separation, excludes both test splits, and retains exact pixel-level SSE/confusion evidence for every frozen endpoint. The generated artifact is valid and reproducible, but both heads were correctly disabled; it was not promoted and the application checkpoint pointer is unchanged.

The audit explains why this is the safe outcome:

- height is already effectively tied between endpoints: exact calibration RMSE is 6.9172 m protected versus 6.9008 m candidate, and only 14/1,131 scenes beat the required 0.25 m material-gain threshold. The predeclared minimum of 32 high-precision selections is therefore mathematically impossible;
- semantic routing has real value on GAMUS mixed scenes (balanced error 0.3657 to 0.2455; 775/859 materially beneficial), but the candidate regresses on legacy forest (0.3747 to 0.4906) and urban scenes (0.1993 to 0.2492);
- the semantic gain regressor learned the sole fit city, NYC, rather than a transferable rule. Its gain correlation changes from +0.711 on fit data to -0.269 on held-out DC/Philadelphia, and no threshold reaches the required 98% beneficial-selection precision while satisfying all non-regression guards.

The next isolated iteration keeps height permanently on the protected endpoint and targets semantic routing only. It uses multiple GAMUS training cities, group-held-out folds, conservative ensemble confidence, out-of-distribution abstention, and a richer label-free content descriptor. Existing validation is development evidence only. Promotion still requires a newly locked external geographic holdout; the previously revealed official test is never reused for tuning.

Corrected supervision and direct-height reset (2026-09-13)

An audit found that every HighBuild row in the multidomain-v2 manifests was tagged as a complete ndsm, even though the raster contains building heights on a zero-filled, unlabelled background. This caused unsupported non-building pixels to be learned and scored as confirmed zero-height ground. Historical scores and checkpoints are preserved, but those urban/ground metrics are now labelled legacy and must not be compared directly with corrected scores.

Two immutable corrected validation protocols are now recorded:

- `highbuild_all_annotated_positive_v1` scores all positive annotated HighBuild height support, including heights marked estimated;
- `highbuild_measured_coco_intersection_v1` scores only COCO footprint pixels with finite positive raster height and `is_estimated_height=false`.

The first corrected-subset pass on the protected live checkpoint reported inclusive overall/building RMSE of 6.8203/7.6652 m and strict measured-only overall/building RMSE of 6.8787/8.2791 m. Its paired forest/vegetation values were 6.6615/8.2742 m. These are preserved as historical, protocol-specific results; they are not interchangeable with the later full-native-scene audit.

The canonical corrected **full-scene** report is now `outputs/evaluation/corrected_legacy_triplet_v1/report.json`. On the protected model it records HighBuild strict-measured building-only RMSE **13.243 m**, OpenCanopy vegetation-only RMSE **8.361 m**, and OpenCanopy all-valid forest RMSE **6.707 m**. These three values use different masks and supports and must remain separately labelled. Earlier 4.366 m overall/5.808 m building crop scores, the preliminary corrected-subset values above, inclusive annotations, and pooled scores remain valid history only under their own protocols; none is a substitute for another.

Further router and head-blend work is paused. The next candidate directly optimizes final metric height on the full QC-approved GAMUS training split while keeping Depth Anything V2 as a fixed cached relative-depth prior. The official training inventory contains 5,004 tiles; 5,001 pass the immutable QC contract and three all-class-255/no-supervision tiles are quarantined. Official validation (859 approved tiles) is development evaluation. The official test (2,861 source tiles) was already consumed by the older Stage-3 final audit and remains excluded from all new tuning, model selection, and repeat evaluation. A separate dataset-QC/quarantine contract and a full-tile paired evaluator gate the experiment. The application pointer and its checkpoint SHA-256 `e2d507f7be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144` remain unchanged until a candidate wins identical-protocol validation plus a newly locked external geographic holdout.

Six-class GAMUS expansion (implemented 2026-09-13; refinement ongoing)

The current three-group height pilot is preserved as the comparison run. Here “three-group” means the ground/building/vegetation **height-routing domains**; it is not a three-class identification output. A separate versioned candidate adds a six-class identification output for ground, building, water, road, low vegetation, and tree while retaining the existing ground/building/vegetation height experts and old-checkpoint loading. Unknown/background stays ignored. Image validity, classification validity, and height validity are now independent; water and road pixels may supervise identification but never become fabricated zero-height targets.

The expansion also adds six-class precision/recall/IoU, targeted water-shadow and road-boundary checks, corrected HighBuild per-building and outline evaluation, OpenCanopy provenance preservation, unit/pixel-scale and >200 m audits, and cached Depth Anything prior authentication. No candidate can change the live application pointer until it beats the protected model and the height-focused pilot on their own identical protocols, passes corrected urban and forest regressions, and succeeds on a genuinely new external geography. Six-class macro metrics must be recomputed across all six final classes; merging low vegetation and trees is a separately labelled five-group diagnostic.

NYC is excluded from the new DC+Philadelphia classifier-head comparison, but it is not system-unseen: earlier GAMUS work and the inherited HighBuild training exposed New York data. Any NYC result must therefore be labelled classifier-head-excluded. Detailed status and gates are in docs/EXPANDED_GAMUS_PLAN.md.

OpenCanopy provenance is now indexed from the accepted manifests rather than from every downloaded candidate folder. The audit passes for 600 train, 120 validation, and 120 locked-test samples with no accepted cross-split coarse region overlap. It preserves imagery and LiDAR acquisition dates, EPSG:2154 location groups, source scenes/URLs, GSD, and metadata hashes in outputs/data_audits/open_canopy_v2_provenance_v1.json. Rejected but cached streaming candidates remain on disk and are explicitly excluded from counts.

Immediate resume commands

```
.\.venv\Scripts\python.exe scripts\evaluate_rasters.py `
  outputs\benchmarks\fusedhe\chicago_grid_00034_z18_fp32.tif `
  data\smoke\highbuild_small\NorthAmerica_USA_Chicago\height\grid_00034_z18.tiff `
  --scope positive-target --height-max-m 300

.\.venv\Scripts\python.exe scripts\data\download_highbuild_full.py `
  --output-root data\highbuild_full --include-shards --workers 6

.\.venv\Scripts\python.exe scripts\data\prepare_highbuild_full.py
.\.venv\Scripts\python.exe scripts\data\repack_highbuild_full.py

.\scripts\run_multidomain_pilot.ps1
```

Usage conservation policy

Long training and download jobs should be self-logging and run locally without continuous Codex polling. Spend agent turns on dataset correctness, failed-run diagnosis, milestone comparisons, and decisions that can change the model.

For unattended continuation, scripts/continue_full_pipeline.ps1 waits for any tracked download/training PIDs, verifies the resumable download, prepares and repackages the conservative full-data split, runs tests, then starts or resumes the full-data experiment. Its durable log is outputs/orchestration/full_pipeline.log.

59 Residual-height evaluation handoff

SOURCE docs/history/RESIDUAL_HEIGHT_POST_RUN.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Active recovery run: experiments/20260915T124056Z_residual_height_v1. Only development validation is permitted during candidate selection. Production is protected; official GAMUS test and the external NZ holdout must not be used for tuning. If no epoch passes guards, report that honestly before planning the next bounded experiment.

Compatibility audit, 15 September 2026

The generic load_predictor and predict_height support residual checkpoints, including cached-prior input and authentication of frozen inherited weights. However, not every older evaluation script accepts the new wrapper:

- evaluate_corrected_legacy_triplet.py seals three historical model identities, their comparison report, and DomainGatedSurfaceNet attributes. It rejects a residual checkpoint. Do not substitute a candidate into the historical plan or label a trainer metrics JSON as a corrected-triplet report.
- height_scorecard.py requires a complete authenticated corrected-triplet report. A new residual-aware report adapter is needed before using it for a formal release decision. Keep existing input/support/annotation digests and historical reports intact.
- evaluate_locked_benchmark.py uses generic tiled inference and accepts the residual model. Use the corrected **validation** manifest with benchmark status development, never relabel it as unseen test data.
- evaluate_highbuild_instances.py is model-independent, but it needs an exact-grid prediction CSV, height GeoTIFFs and protected-building-probability GeoTIFFs. A batch exporter or the residual-aware corrected evaluator must produce these; no exporter has been implemented in this recovery turn.
- audit_height_output_identity.py is for semantic-only candidates whose height output must be unchanged. It must not be applied to an intentional height correction model. Use inherited-weight and semantic-output identity checks.
- audit_protected_inference_protocol_v1.py is a frozen production-only audit, not a residual-candidate evaluator.

Next actions after an eligible checkpoint exists

1. Freeze the selected checkpoint SHA and its training config/source manifest. Read all epoch outcomes, not just the best number.
2. Implement/test a separate residual-aware corrected evaluation adapter. Reuse exact native 1024 HighBuild / 384 OpenCanopy inputs and measured-only COCO masks. Authenticate the protected model embedded in the candidate, unwrap its frozen semantic configuration, and preserve pixel/input/COCO digests. Include per-building and boundary evidence. Do not modify running training source or weaken the old three-model plan's validation.
3. Evaluate the actual 512-tile, 128-overlap app path on the same development manifest, paired with production. This is different from full-native evaluation because spatial normalization/padding changes predictions.
4. Only after development gates and protocol checks pass, freeze a one-shot external evaluation plan and use the preregistered new geography. Never use external results to repeatedly tune this candidate.
5. Keep deployment separate from evaluation; no automatic pointer change.

App-path development benchmark command template

Replace CHECKPOINT, SHA and NEW_OUTPUT with the selected immutable values. Run the same protocol for production when the comparator is needed. The existing foundation model directory is models/foundation/depth-anything-v2-small-hf.

```
.\.venv\Scripts\python.exe scripts\evaluate_locked_benchmark.py --checkpoint CHECKPOINT
--manifest D:\MSRData\data\multidomain_v2\highbuild_measured_coco_intersection_v1\manifests
\validation.csv --deny-manifest
D:\MSRData\data\multidomain_v2\highbuild_measured_coco_intersection_v1\manifests\train.csv
--output NEW_OUTPUT --benchmark-id residual-height-v1-corrected-validation --benchmark-
status development --variants raw --relative-model models\foundation\depth-
anything-v2-small-hf --device cuda --tile-size 512 --overlap 128 --expected-manifest-sha256
2e21024fb7d7562e51d91e219c9434087fadfee58b107a4eacdc2c04cd8fa118 --expected-checkpoint-
sha256 SHA
```

This command is documented but has not yet been run on a learned residual checkpoint. Treat its reports as app-protocol development evidence, not as an automatic replacement for the frozen native/per-building scorecard.

60 Residual-height V1: completed outcome

SOURCE docs/history/RESIDUAL_HEIGHT_V1_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

15 September 2026. Run: experiments/20260915T124056Z_residual_height_v1.

Decision: retain all five checkpoints as rejected development candidates. Do not change the app's active model. The run early-stopped normally after epoch 5; no checkpoint passed every registered guard. Epoch 5 had the lowest selection score, not an eligible "best" checkpoint.

What changed across the run

All values below are final predicted-height RMSE, not expert-head scores. Lower is better. GAMUS units remain assumed metres in the run's contract; this audit has not resolved the publisher-unit question.

Completed epoch	HighBuild measured buildings	OpenCanopy vegetation	OpenCanopy ground	GAMUS domain-macro	Material object improvement	All guards
Protected baseline	13.2426	8.3610	4.8944	5.2248	Comparator	Comparator
1	13.0781	7.9645	5.1504	5.0630	No	Fail
2	13.0710	7.9389	5.0605	5.0054	Yes	Fail
3	13.1037	7.8509	5.0697	4.9677	Yes	Fail
4	13.0162	7.8613	4.9116	4.8916	Yes	Fail
5	13.0121	7.8565	4.8741	4.8495	Yes	Fail

Epoch 5 reduces building RMSE by **1.74%** and vegetation RMSE by **6.03%**. These are error reductions on development validation, not accuracy percentages or unseen-geography results. OpenCanopy vegetation's lowest RMSE was epoch 3, so the last checkpoint is not best for every metric.

The important findings

Buildings: errors are concentrated in tall structures

- Reference heights of at least 20 m account for **17.64% of measured building pixels but 89.31% of baseline squared error**. At epoch 5 they still account for 87.73% of squared error.
- Tall-building RMSE: **29.797 -> 29.017 m**. Correlation: **-0.188807 -> -0.188944**. The correlation regression is tiny, but the performance level itself is poor.
- Overall building MAE barely changes: **5.361 -> 5.328 m**. Bias improves substantially: **-2.788 -> -1.018 m**.
- Decomposing $RMSE^2 = bias^2 + centered\ error\ variance$, centered error RMSE is **12.946 -> 12.972 m**, slightly worse. The aggregate RMSE gain is therefore accounted for by reduced bias, not reduced centered error spread. This is descriptive arithmetic, not proof of what visual features the network learned; no validation-derived correction has been applied.

These are pixel metrics on measured COCO-intersection support, **not per-building instance scores**. Before more height training, audit measured training buildings by height band, city and annotation provenance; verify that tall-reference outliers are aligned and trustworthy, then test height-balanced and per-building supervision separately.

Canopy: genuine aggregate improvement, uneven local behavior

- Vegetation RMSE **8.361** -> **7.857 m**, MAE **6.505** -> **6.102 m**, correlation **0.411** -> **0.458**, R2 **-0.174** -> **-0.036**.
- Ground RMSE **4.894** -> **4.874 m**, MAE **2.896** -> **2.750 m**, correlation **0.502** -> **0.548**, R2 **-0.088** -> **-0.079**.
- **68 of 115 saved OpenCanopy region groups improve RMSE; 47 worsen.** These groups are not 115 independent geographies or individual trees; the validation set contains 120 chips and shares source mosaics with training.
- The largest region-group regression is open_canopy_r143_1270: **3.081** -> **4.437 m RMSE**, with positive bias increasing **1.049** -> **2.079 m**. An improving overall number does not establish consistent improvement everywhere.

GAMUS: object gains coexist with ground damage

- Overall RMSE **7.121** -> **6.189**, correlation **0.366** -> **0.522**, R2 **0.005** -> **0.248**.
- Building RMSE **5.250** -> **4.917**; vegetation **8.897** -> **7.585**.
- Ground RMSE worsens **1.527** -> **2.046**, MAE **0.705** -> **1.105**, bias **+0.458** -> **+0.881**, correlation **0.168** -> **0.124**, R2 **-1.810** -> **-4.044**.
- Tall-vegetation correlation also worsens **0.057** -> **-0.111**, despite lower tall-vegetation RMSE. Raising average heights has not recovered correct ordering within this subset.

Region results expose trade-offs

HighBuild city	Baseline RMSE	Epoch 5 RMSE	Baseline MAE	Epoch 5 MAE
Copenhagen	6.182	5.185	4.283	3.445
Lyon	14.244	14.302	7.072	7.426
Strasbourg	18.563	18.263	5.587	5.590
Munich	9.302	9.118	4.433	4.694

Three cities improve RMSE, but only Copenhagen improves MAE. GAMUS DC improves RMSE **10.998** -> **9.417** and Philadelphia **3.116** -> **2.983**; Philadelphia MAE nevertheless worsens **1.674** -> **1.870**. These are existing development-city aggregates, not fresh external tests.

Why every epoch was rejected

- All five: GAMUS ground RMSE, MAE, absolute bias, correlation and R2 regress beyond their registered limits; GAMUS tall-vegetation correlation regresses.
- Epochs 2-5: HighBuild tall-building correlation also fails the strict no-regression check.
- Epochs 1-3: OpenCanopy ground has additional failed checks. Epoch 4 still fails ground R2. Those OpenCanopy failures recover by epoch 5.
- Epoch 1 does not reach material priority-object improvement; epochs 2-5 do. A material gain alone does not override failed protection checks.

The exact per-epoch failures and paired RMSE/MAE/bias/correlation/R2 for every saved source, domain, tall-object group, region and landscape are preserved in the machine-readable outcome report.

Next measured work, not another blind rerun

1. Keep classification independent. The residual run deliberately preserved identification; all five epochs' saved classification metrics exactly match baseline. A new six-class identifier can be tested separately without using its outputs to gate heights yet.

2. Audit train-only tall buildings and ground/object errors. Check measured annotation reliability, height-band exposure, source pixel scale, image/prior alignment and correction spillover on fixed training examples. Do not inspect the sealed external geography to choose examples or settings.

3. Run a matched height-source ablation, if training evidence warrants it. Compare fresh three-source training with omission of GAMUS's height loss. Keep the same protected start, architecture, seed, optimizer updates, schedule, HighBuild/OpenCanopy examples and numerical loss coefficients. Do not replace omitted GAMUS batches with extra legacy exposure and call the difference a dataset effect. GAMUS identification remains a separate task. This is a hypothesis test, not a conclusion that GAMUS caused the failure.

4. Test one building intervention separately. A measured-building objective or height-balanced sampler needs its own fixed train-only feasibility proof and controlled comparison. Do not simultaneously change sources, model, sampler and loss then attribute any gain to one choice.

5. Evaluate before promotion. Only an eligible development candidate advances to the residual-compatible per-building scorecard and real app-inference path. Freeze the candidate and one-shot protocol before consuming genuinely new geography. The older corrected-triplet evaluator still needs the compatibility adapter described in docs/RESIDUAL_HEIGHT_POST_RUN.md.

Evidence and safety

- This analysis reads saved reports/configuration and hashes protected files only. **No GPU inference, new dataset reads, official-test access or external-holdout access.**
- Source/group keys and all saved support counts match baseline across all five epochs: GAMUS **730,964,838**, HighBuild **13,855,690**, OpenCanopy **17,155,570** valid height pixels. Matching counts alone are not an independent proof of spatial-mask identity; masks were not reread in this audit.
- Both protected SHA-256 values match the experiment's configuration: production checkpoint e2d507fb . . . 0d144; showcase pointer a3d51c68 . . . baa9. No pointer or checkpoint changes were made.
- All five epoch checkpoints, latest checkpoint, source snapshot, configuration, metrics and terminal summary remain saved.
- Derived report: outputs/diagnostics/residual_height_v1_outcome/report.json.
- Reproducible CPU-only summarizer: scripts/analyze_residual_height_outcome.py. It fails on changed protected hashes, changed support counts/classification metrics, unexpected epoch history or a non-rejected outcome. It is a report summarizer, not a release evaluator.

61 Multi-region V3 launched — 19 September 2026

SOURCE docs/history/RGB_MULTIREGION_V3_LAUNCH_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

- Active run: experiments/20260919T142003929194Z_rgb_multiregion_v3.
- Configuration/protocol: configs/rgb_multiregion_v3.yaml, docs/RGB_MULTIREGION_V3_PROTOCOL.md.
- Training is classification-only. Production app/height checkpoint and pointer remain unchanged.
- Download complete: 184 verified image/label pairs, 756,069,754 source-file bytes, on D:. 134 train / 50 development validation. All six classes have support in both splits.
- Data manifest SHA-256: 9c3b12816035cc014ec72df55a654c3920f77a3dd64836742c4d439fb6bdf5cf.
- Preparation files: D:/MSRData/training/oem_multiregion_v3/; original pairs, receipts, manifest, attribution contract, label balance and alignment preview saved.
- 124 targeted tests passed before launch.
- Training-only gradient proof passed: loss 0.6420 -> 0.3276 (48.97% reduction on its fixed training batch). Proof weights discarded. This is not an accuracy result.
- Frozen V1 baseline on the new 50-image OEM development set: pooled six-class macro F1 **54.65%**; Melbourne **40.05%**, Vienna **54.37%**. Equal-region mean **47.21%**. These differ from the earlier all-six challenge and must not be compared as if they were the same benchmark.
- Verified epoch 1 progressing past batch 440/1,375. CUDA RTX 4070 active; sampled temperature 55 C. No error log entries at this check.
- Maximum eight epochs; early stopping and safety/benefit gates remain predeclared. Epoch checkpoint commits include optimizer/scheduler/RNG for compatible resume.
- Logs: outputs/orchestration/rgb_multiregion_v3_20260919.log and .err.log.
- Compact display: scripts/watch_rgb_multiregion_v3.ps1; terminal session 69915 was started and requested in the Codex bottom panel (UI returned queued).
- Existing heartbeat msr-residual-accuracy-check updated to this exact run, hourly. Notify only on new metrics, meaningful stage changes, failures or completion; pause after outcome reporting.

Recovery, only if no trainer is active: scripts/run_rgb_multiregion_v3.ps1 -Resume
.\experiments\20260919T142003929194Z_rgb_multiregion_v3.

Do not change sealed run code/config or silently adjust batch size. Keep V1, earlier V2 experiments, the protected height checkpoint and app pointer intact. Saved development checkpoints require a later review and additional transfer validation before any app integration. No fresh reserved-final geography has been consumed.

62 RGB multi-region V3: completion and error analysis

SOURCE docs/history/RGB_MULTIREGION_V3_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Run: experiments/20260919T142003929194Z_rgb_multiregion_v3. Completed 19 September 2026 at 14:38 UTC after four epochs. The predeclared early-stop criterion fired: three epochs without better equal-region OEM macro F1 after epoch 1. No failure or recovery is indicated; error log is empty and no trainer remains active.

Selection outcome

- **Best safety-passing checkpoint: none.** No epoch meets both benefit and safety; no default model promotion is justified.
- Best unrestricted development checkpoint: **epoch 1**. OEM pooled macro F1 **73.29%**, equal-region macro F1 **69.42%**, GAMUS macro F1 **83.20%**.
- Corresponding V1 baselines: OEM pooled **54.65%**, equal-region **47.21%**, GAMUS **83.83%**. OEM ground/road/low-vegetation F1 changed **30.77/53.63/5.37% → 58.86/64.18/62.98%**.
- Epoch 4 scored higher on pooled OEM F1 (**74.47%**) but lower on the predeclared equal-region selector (**68.56%**). Do not switch selectors after seeing results.

What went wrong / what improved

The added regional training/augmentation package markedly improved OEM development transfer. It did not preserve all GAMUS behavior. At selected epoch 1, pooled ground F1 fell **1.99 percentage points**, low vegetation **1.76 points**, and macro F1 **0.63 points**. The worst city/class F1 regression was Philadelphia ground (**2.23 points**); Philadelphia roads also fell **1.37 points** and its road-boundary guard failed. These are genuine tradeoffs, not a universally improved classifier. Different label taxonomies and changed sampling are plausible contributors, not established causes: this run was not a controlled ablation.

Next steps: inspect fixed development examples for ground/low-vegetation and Philadelphia road edges; explicitly define a matched follow-up experiment preserving V1 behavior while learning the new regions (e.g. controlled replay/teacher regularization). Keep original acceptance thresholds and compare all classes/cities, not just pooled macro scores. Freeze the candidate before evaluating reserved geography. Do not launch a costly run just to extend an early-stopped recipe.

Melbourne/Vienna results are development scores, not final untouched tests. Rosario/Monrovia remain unconsumed. No height-accuracy improvement is established. Later user-authorized opt-in app integration and HighBuild transfer checks are documented separately in docs/SIX_CLASS_APP_PREVIEW_2026_09_19.md; the height/default checkpoint remains protected.

Persistence / monitor completion check

Code snapshot (14 files), configuration, binding, baseline, metrics, all four epoch checkpoints, latest and best-development aliases are saved. Atomic commit points to epoch 4; latest matches that commit. Selected checkpoint SHA256 is 10fbb39c3d916695627e93d49f2115be18ecd163352b4daa96ba691c0090470c; final epoch/latest SHA256 is c732e452b880cb1c9c3e422dee2af94f75d3da7c6475d672cfe26b004278c6b2.

Protected hashes rechecked unchanged: production height

e2d507f7be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144; live pointer

a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72baa9; V1 RGB classifier

4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021.

Completion check GPU snapshot: 39% utilization, 1,893 MiB VRAM, 8.49 W, 48°C. No training process; the app inference server remains running. The snapshot does not attribute all GPU activity to a particular process. No thermal issue is indicated. The hourly monitor should be paused; the app need not be stopped.

63 Multi-region RGB classification V3

SOURCE docs/history/RGB_MULTIREGION_V3_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Purpose and boundaries

Train a standalone six-category classifier after the frozen V1 generalisation audits. The recent 12-scene challenge scored 54.02% macro F1, with particularly weak low vegetation (19.60%) and roads (36.30%). Those are **diagnostic results**, not worldwide accuracy. A prior crop-sampling V2 did not establish a safe benefit over its matched control.

This experiment changes the training-data/augmentation package, not model architecture. It is not an ablation establishing which individual change caused improvement. The production height checkpoint, live pointer, app and earlier experiments remain untouched. No accuracy gain is guaranteed.

Frozen comparison

- Start: RGB SegFormer V1, epoch 8, experiments/20260915T140116654177Z_gamus_rgb_segformer_v1/checkpoint_best_guarded.pt.
- SHA-256: 4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021.
- Independent GAMUS validation replay: 859 full-native DC/PHL tiles, macro F1 83.83%. These are reused development results, not untouched geography.
- Evaluate the exact same V1 checkpoint on the new 50-image OEM development split before any new training updates.

Data and geography

All added downloads are under D:/MSRData/training/oem_multiregion_v3.

Role	Source	Tiles
Training replay	GAMUS DC/Philadelphia	3,837
Added training	OEM Aachen, Chisinau, Accra, Kitsap (30 each), Bogota (5), Niamey (9)	134
Existing development validation	GAMUS DC/Philadelphia	859
New-region development validation	OEM Melbourne and Vienna (25 each)	50
Reserved for later final evaluation	OEM Rosario and Monrovia	Not downloaded or inferred in this run

New paired images are chosen by filename SHA-256 ordering, not model performance or a favourable class-content filter. Entire OEM regions are separated. Reference/RGB alignment, source CRC and SHA-256, class IDs, nonempty labels, duplicate train/validation image content and pooled six-class coverage are checked before training. The selected source files total approximately **756 MB**, not the full 9 GB archive; range requests add network overhead.

Christchurch, Duesseldorf, Dhaka and Chiang Mai are excluded from this run's training and development validation. Their earlier results are preserved as already-consumed diagnostics. Do not relabel them as untouched final tests. The new development regions are excluded from this continuation's training; upstream foundation-model pretraining geography is not comprehensively known.

Inputs, labels and missing data

The six classes remain ground, buildings, water, roads, low vegetation and trees. RGB is raw byte imagery scaled to 0–1, with ImageNet normalisation inside the same model. No height, AGL or DAV2 maps are model inputs here.

OEM mapping: bareland/developed space -> ground; rangeland -> low vegetation; road/tree/water/building retain their identities. Agriculture and unknown -> ignored (255), **not ground or zero height**. This mapping is approximate; OEM and GAMUS annotation definitions are not identical.

Image-validity and classification-validity masks remain independent. Joint validity determines classification loss. Spatial transforms are identical across image, labels and masks; labels/masks use nearest-neighbour resampling. Colour changes do not infer or erase label validity. The dark-pixel water diagnostic remains a proxy, not a labelled shadow test.

Training and resource limits

Maximum eight epochs, with early stopping after at least four and three epochs without improved OEM region-macro F1. Every epoch visits all 3,837 GAMUS images once and adds 1,663 equal-city OEM draws. Random 256–512 px source windows are resampled to 384 px; aligned rotations/flips, modest brightness/contrast/saturation variation and occasional blur augment training only.

Batch 4, two data workers, RTX CUDA bf16. Encoder learning rate 1.5e-5; decoder 1.5e-4; focal supervision and clipping. A 24-update mixed-source overfit/gradient sanity check must reduce its training-only loss by at least 10%; those proof weights are discarded before the real run. This check demonstrates that optimisation works, not predictive accuracy.

Predeclared acceptance

- Evaluate all 859 GAMUS and 50 OEM development images at native resolution after every epoch, with fixed references and no test-time augmentation.
- Protect GAMUS per-class F1 within 1 percentage point, pooled and per city; macro F1 within 0.5 points.
- Protect OEM per-class F1 within 2 points, pooled and per validation region, for classes with reference support.
- Protect road-boundary F1 within 2 points and dark-non-water false-water rate within +0.5 points in each evaluated group.
- Require at least +3 points in OEM region-averaged six-class macro F1, and +5 points in the mean pooled ground/road/low-vegetation F1.
- Save best unrestricted development and best safety-passing checkpoints separately. Safety passing is **not** the same as demonstrated improvement.
- No automatic app promotion. After completion, review errors and HighBuild positive-outline coverage as an additional transfer check. Freeze an accepted candidate before consuming reserved final geography. Classification gains do not establish height gains.

Saved state and recovery

Code/config/data identities are sealed per experiment. Each epoch saves model, optimiser, scheduler, RNG and metrics using an epoch checkpoint and atomic commit. Latest/best files are recoverable aliases. The original V1 checkpoint and height pointer are hash-checked before/throughout the run.

Start: `scripts/run_rgb_multiregion_v3.ps1`. Resume: the same script with `-Resume <exact experiment directory>`. Do not bypass integrity checks, run duplicate trainers or silently change the recipe on CUDA OOM. Incomplete downloads resume from verified receipts.

Compact status: `scripts/watch_rgb_multiregion_v3.ps1`. Training registration: `outputs/orchestration/rgb_multiregion_v3_active.json`.

Sources and licensing

- [OpenEarthMap paper](#), Xia et al., WACV 2023.
- [Published dataset](#) and [official regional attribution](#).
- Aachen: GeoNRW/North Rhine-Westphalia, DL-DE-BY-2.0. Chisinau: HTCD/Lightcyphers, CC BY 4.0. Accra/Niamey: Open Cities AI/GFDRR, CC BY 4.0. Kitsap: USGS, public domain. Bogota: MaptimeBogota, CC BY 4.0. Melbourne: City of Melbourne, CC BY 4.0. Vienna: City of Vienna, public domain.
- Existing MiT-B0 pretrained weight terms remain research/evaluation restrictions; this experiment does not grant commercial deployment rights.

64 RGB sampling V2: completed experiment and decision

SOURCE docs/history/RGB_SAMPLING_V2_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Completed 15 September 2026, 16:09:41 UTC. Pair: experiments/20260915T154725409300Z_rgb_sampling_v2_pair.

Decision

The primary experiment did **not** meet its predefined acceptance rules. Preserve targeted epoch 2 as an exploratory, best-safe candidate; do not promote it to the app or describe this experiment as a proven sampling improvement. No height model, production pointer, or app-active model was changed.

What was compared

Two four-epoch continuations started from the identical V1 classification checkpoint. Control used ordinary crops; targeted used 50% ordinary and 50% ground/low-vegetation-rich crops. Both used the same 3,837 training-tile draws per epoch, model, optimizer recipe and budget. Each epoch was evaluated on the same 859 full native-resolution DC/Philadelphia development tiles.

These validation tiles were used repeatedly for model selection. They are not an untouched geographic test. No external holdout was evaluated in this run, and no height accuracy was measured or improved by this classifier experiment.

Results

All values below are F1 percentages; differences are percentage points (pp).

Measure	Original V1	Control epoch 4	Targeted epoch 4	Exploratory targeted epoch 2
Six-class macro F1	83.83	83.94	84.01	83.98
DC ground F1	68.17	68.39	68.46	68.49
DC low-vegetation F1	22.49	24.38	24.98	28.51
Philadelphia tree F1	85.86	84.62	84.40	85.39
All predefined safety checks versus V1	Baseline	Fail	Fail	Pass

The primary comparison was fixed in advance as epoch 4 versus epoch 4:

- Targeted macro F1 exceeds control by only **0.067 pp**.
- Targeted DC low-vegetation F1 exceeds control by **0.599 pp**, below the required 2 pp. Its gain over V1 is 2.493 pp, below the required 5 pp.
- The mean of DC ground/low-vegetation improves by 1.389 pp over V1 and 0.330 pp over control, below the respective >2.5 pp and >=1 pp rules.
- Philadelphia tree F1 drops 1.465 pp versus V1, exceeding the permitted 1 pp regression. Thus the final candidate fails safety as well as benefit.

Targeted epoch 2 passes the V1 safety and benefit checks, but does not establish that targeted sampling is better: control epoch 2 achieves DC low-vegetation F1 of 29.14%, versus targeted 28.51%. Earlier checkpoints are exploratory, not replacements for the failed primary experiment. Passing safety allows small predeclared regressions; it does not mean every category improved.

Error-analysis conclusions and next experiment

Both arms peak on DC low vegetation at epoch 2 and deteriorate afterward. Additional continuation alone therefore does not reliably resolve this weakness. The training-only sampling audit increased expected low-vegetation pixel exposure by about 12%, but the controlled run does not establish a useful advantage from that change. These metrics identify a pattern, not its cause.

Before more training, audit **training-only** DC/Philadelphia examples for class-definition consistency, small vegetation patches, image scale and boundary detail. Record class-conditioned confusion and size-related failure patterns without using external holdouts to tune the recipe. If fine detail is the confirmed bottleneck, define a matched higher-resolution/detail-preserving comparison with the same baseline and explicit category regression guards. Do not automatically lengthen this failed recipe or relax its acceptance rules. Any eventual candidate still needs independent replay, replication and a fresh geographic evaluation; no universal unseen-image accuracy is established here.

Saved artifacts and completion audit

Each arm retains all four committed epoch checkpoints, metrics, hash receipts, optimizer/scheduler/RNG state and atomic commit records. Pair configuration, source snapshot, proof, binding, status and `comparison_report.json` are saved. The earlier failed preflight pair is preserved separately and must not be resumed.

Authenticated SHA-256 identities:

Artifact	SHA-256
Control epoch 4 / latest	97507b7aeca2238f8da07bf73971137593404ec825d2189c68df543a3bd7e bd9
Targeted epoch 4 / latest	b16822dd10a916357559e2a7314a511d9a89d978833b3708ed64b1c48d769 484
Targeted epoch 2 / best-safe	879ce4cf07b46ec817ec40fcc604ce035dad44d57b757481bb5402e854bfe cbd
Protected production height checkpoint	e2d507fbe180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d 144
Protected production pointer	a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72b aa9

The read-only CPU completion audit authenticated checkpoints before loading. Both latest histories are epochs 1-4 with optimizer/scheduler step 3840; best-safe history is epochs 1-2 with step 1920. All 209 floating model tensors and 615 floating optimizer tensors per inspected checkpoint are finite. Bindings and complete saved RNG keys match. Saved configuration, 12/12 snapshot sources, 12/12 current sources, 27/27 protected artifact identities and 3/3 bound inputs pass verification. No GPU work or dataset decoding was needed for this completion audit. The training process has exited normally.

The hourly completion monitor should now be paused. No new training or app promotion is part of this completion handoff.

65 RGB V2: controlled crop-sampling continuation

SOURCE docs/history/RGB_SAMPLING_V2_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: training launched on 2026-09-15; no accuracy improvement claimed yet.

Verified launch handoff

Active pair: experiments/20260915T154725409300Z_rgb_sampling_v2_pair. The earlier 20260915T154605013891Z_rgb_sampling_v2_pair stopped during the training-only preflight, before its first update: the legacy precision adapter requires torch.device rather than a string. That failed attempt and source snapshot were retained. A new adapter test passes, and the corrected code was sealed into this fresh pair; the failed pair must not be resumed.

- Full pre-launch suite: 734 passed, 12 warnings. After the adapter correction, 52 targeted trainer/dataset/gate tests passed, including the new backward-pass regression test. The entire suite was not rerun during active GPU training.
- Train-only proofs, 24 updates per arm: control loss 0.09201 to 0.04569 (50.34% reduction); targeted 0.08028 to 0.03020 (62.38%). Both had nonzero encoder gradients and their proof weights were discarded. These are optimization checks, not accuracy improvements or a valid comparison between arms.
- Full-native CUDA validation smoke passed before training; approved input checks cover 3,837 train/859 validation images and matched validation content.
- Observed control epoch 1 batch 280/960, RTX4070 utilization 72%, 1385 MiB total VRAM, 64C, 59.82W. These are an instantaneous sample, not sustained averages.
- Epoch-boundary checkpoints, atomic commits, source/config/binding snapshots and metrics are saved separately for each arm. No resumable epoch exists until the first full validation/checkpoint commit completes.
- Existing hourly monitor msr-residual-accuracy-check was updated to this exact pair. It reports meaningful changes, supports one compatible recovery, and pauses after the final comparison/outcome report.

Final training-only index:

outputs/data_audits/rgb_targeted_v2_training/top_quartile/targeted_crop_index_v2.json SHA256 fc439c45e852b65135c791652cb38e460a5a2e1a8295223247c557997374cafa. The 436.6 second scan covered every approved training tile; quantile refinement reused cached counts without decoding more data. Under unchanged tile weights, expected crop ground pixels increase 14.22% and low vegetation 12.15% overall. Water pixel exposure falls 21.86%, which is a real tradeoff guarded in validation; exposure changes are not predicted score improvements. Report is the sibling training_crop_exposure_report_v2.json. No validation/test/external/height pixels were decoded during this training-index audit.

Question and scope

Does selecting richer ground/low-vegetation TRAINING windows improve the known DC identification weakness without materially reducing other class/city scores? This is not a global generalisation claim, a height experiment, or an app release. Christchurch has already been consumed and is excluded from every part of this run. Official GAMUS test, NYC and the separate external height holdout remain excluded.

Fixed paired comparison

Both arms initialize from the exact V1 guarded epoch-eight checkpoint:

4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021. The control uses ordinary random crops. The candidate mixes 50% ordinary crops with 50% target-aware crops from the selected tile. Targets are ground and low vegetation, chosen uniformly among targets with eligible windows. Eligible windows have at least 1% target pixels and are in the target's upper quartile within that tile. The final training-only index/report records exact rules and expected exposure changes before either arm trains.

Unchanged: 3,837 approved DC/PHL training tiles; V1 city-mass-preserving water-hit tile weights; shared epoch-by-epoch tile draws; raw RGB; independent validity masks; 384px training crop size; flips/rotations; architecture; focal-loss weights. Private target RNG preserves ordinary proposals and geometric augmentation RNG. Two data workers are required for the tested epoch-boundary resume contract.

Both get four complete epochs, 960 batches per epoch, batch four, CUDA bf16. Optimizer/scheduler reset identically: encoder LR 2e-5, decoder LR 2e-4, weight decay .01, 50 warmup updates, cosine floor .1. No score-driven early stop that gives one arm more compute. A nonfinite update or runtime failure stops execution and requires compatible checkpoint recovery, not hidden settings changes.

Validation is always the same 859 full native1024 DC/Philadelphia development tiles; file/grid/support identities must match the independently replayed V1. No resizing or favourable-crop evaluation. Sequential arms share one GPU.

Decision rules frozen before training

Against V1 AND the same-epoch control, safety permits at most:

- 1 percentage point lower class F1, for every class overall and in each city;
- 0.5 point lower macro F1 per scope;
- 1 point lower two-pixel road-boundary F1;
- 0.1 point higher dark-non-water false-water rate.

Required benefit versus V1: DC low-vegetation F1 at least +5 points; mean DC ground/low-vegetation F1 strictly greater than +2.5 points. Required benefit versus control: DC low-vegetation F1 at least +2 points and the same joint mean at least +1 point. These are pilot acceptance thresholds, not promises or sufficient production quality.

The PRIMARY experiment decision uses epoch4 against epoch4. Earlier same-epoch comparisons and best-safe checkpoints are exploratory. Safety, useful improvement, and experiment victory are reported separately. A higher mean alone cannot win. This single-seed pilot does not establish statistical reliability; a successful result needs replication, independent replay, and later a fresh geographic test.

Proofs, preservation and recovery

Before full training: CPU tests, source/data/config hashes, 24-step fixed training batch optimization per arm (at least 5% loss reduction and nonzero encoder gradient), plus native CUDA validation smoke. Proof weights are discarded.

New paired experiments live under `experiments/*_rgb_sampling_v2_pair`, with separate control/targeted subfolders. Immutable committed epoch files, hash receipts, optimizer/scheduler/RNG states, source/config snapshots and metrics allow safe epoch-boundary recovery. Atomic commit records are authoritative; latest/best aliases can be rebuilt after interruption. Uncommitted retries are retained. Proof reports are bound to recipe identity.

No sealed V1 source/config or protected height checkpoint/pointer is edited. No six-class output affects heights in this experiment. No automatic app promotion.

Launch: `scripts/run_rgb_sampling_v2.ps1`. Resume: same script with `-ResumePair <exact saved paired experiment>`. Status: `outputs/orchestration/rgb_sampling_v2_active.json` and the paired run's `status.json`. Final results: `comparison_report.json`.

66 RGB segmenter V1: Christchurch external-transfer protocol

SOURCE `docs/history/RGB_SEGMENTER_V1_CHRISTCHURCH_PROTOCOL.md`

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Recorded 2026-09-15, before acquiring or decoding selected image/label files. Status: acquisition permitted; evaluator implementation/seal required before pixel decoding. No external scores exist at the time of writing.

Explicit amendment, not retroactive preregistration

The earlier `OPENEARTHMAP_CHRISTCHURCH_EXTERNAL_HOLDOUT.md` reserved this city for V3/V4. That document remains unchanged. This new protocol evaluates the completed independent RGB segmenter V1 instead. It is a locally recorded pre-evaluation commitment, not an externally timestamped preregistration.

Candidate: epoch 8 of 20260915T140116654177Z_gamus_rgb_segmenter_v1. Checkpoint SHA-256:

4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021. No candidate reselection, calibration, training, or class-threshold tuning on Christchurch. There is no paired V3/V4 comparison in this protocol.

Source, inclusion and integrity

- OpenEarthMap v1, <https://zenodo.org/records/7223446> .
- Official archive size: 9,099,481,727 bytes; published whole-archive MD5: 64155d1dc9d3b68536063f79878e1a67.
- Include EVERY `christchurch/images/*.tif` with its exact same-stem `christchurch/labels/*.tif`, independent of published training/test membership. An orphan label or unexpected format stops preparation; no selective omission of any publicly labelled pair.
- Imagery attribution: AIRS / LINZ, CC BY 4.0; <https://open-earth-map.org/attribution.html> . Retain source attribution.
- Store under `D:/MSRData/external_holdouts/oem_christchurch_rgb_v1`.
- Download official ZIP byte ranges only, with strict HTTP 206 and Content-Range checks. Verify selected entries using ZIP CRC32 and lengths, then record SHA-256 for every extracted file and a sorted complete manifest.
- This selective method does NOT verify the published whole-archive MD5. CRC32 detects accidental corruption, not adversarial tampering. Per-file SHA-256 records retrieved identity; it is not a publisher-provided authenticity proof. HTTPS official source and frozen metadata are additional safeguards.
- ZIP decompression is allowed during staging; TIFF pixel decoding, thumbnails, histograms, and model evaluation are prohibited until the evaluator is sealed.

Fixed ontology

Pre-pixel metadata clarification: the official archive inventory contains 70 Christchurch images but only 49 labels, each paired with an image. Therefore evaluate all 49 publicly labelled pairs and record the other 21 image stems as unscorable (no released label). No image pixels or scores informed this decision. The first staging attempt stopped at the strict pairing check before extracting any files. The amended check rejects orphan labels and duplicate members but allows explicitly inventoried unlabelled images to remain unscored.

OEM IDs map to six-class indices as follows: 0 -> ignore255, 1 -> ground0, 2 -> low_vegetation4, 3 -> ground0, 4 -> roads3, 5 -> trees5, 6 -> water2, 7 -> ignore255, 8 -> buildings1. Agriculture is ambiguous (including

orchards). Report a separate sensitivity table mapping 7 to low vegetation, never replacing the primary result. Report the native OEM 8-by-6 table as well, so merged classes cannot hide errors.

Fixed inference and consumption

Use the sealed V1 model wrapper and strict checkpoint loading; raw uint8 RGB divided by 255, unchanged wrapper ImageNet normalization, native image grid, batch one, CUDA bf16, bilinear logits upsampling as already implemented, argmax, no test-time augmentation or per-scene normalization. No resizing, tuning, height inputs, reference inputs, DEMs, or class-dependent fusion. Unsupported radiometry/dimensions stop the run and must be disclosed, not silently adapted. RGB validity and label validity remain separate; exclude raster no-data masks and OEM 0/7 only in the primary evaluation. Do not assume black RGB is invalid.

Before first TIFF pixel decode, save model/config/code/software/file identities and this protocol's hash, then create CONSUMED.json. Record any execution failure and retry under the same unchanged contract. Do not claim a rerun was the first look. All complete pairs must finish before publishing a headline.

Required output and claim limits

- Exact pooled confusion matrix; per-class P/R/F1/IoU, tile and pixel support; macro scores with absent classes explicitly marked, not called perfect.
- Whole-tile bootstrap (seed 20260915, 2,000 resamples) confidence intervals, labelled as within-this-city sampling uncertainty, not global uncertainty.
- Native OEM 8-by-6 table, ignored share, agriculture sensitivity, low-vegetation/tree confusion and diagnostic combined-vegetation F1.
- Two-pixel road-boundary scores and dark-RGB water false-positive proxy; explicitly not labelled-shadow accuracy.
- Call it comprehensive six-class coverage only if each mapped class appears in at least five tiles and 10,000 reference pixels. Insufficient coverage is reported without swapping locations to obtain a better score.
- Publish the complete aggregate report before inspecting visual examples. Preview order: SHA-256 of UTF-8 tile stem, ascending; first six.
- No app promotion from this result alone. No height-accuracy claim: OEM has no height ground truth. Retain protected production checkpoint and pointer hashes.

Permitted claim: transfer to Christchurch, new to the known project inventories, under this fixed crosswalk. Not proof of global generalisation, absence from unknown upstream pretraining, or performance on every satellite sensor.

67 RGB segmenter V1: completed Christchurch transfer test

SOURCE docs/history/RGB_SEGMENTER_V1_CHRISTCHURCH_RESULTS.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Completed 2026-09-15. The frozen epoch-eight RGB-only classifier was evaluated once on all 49 publicly labelled Christchurch OpenEarthMap pairs at native resolution. No Christchurch training, calibration, checkpoint selection, or reference input to inference occurred. The other 21 regional images lack released labels and were explicitly inventoried before any pixels were decoded.

Results

Class	Precision	Recall	F1	IoU
Ground	22.63%	27.38%	24.78%	14.14%
Buildings	58.69%	95.06%	72.57%	56.95%
Water	95.45%	80.57%	87.38%	77.59%
Roads	49.86%	82.91%	62.27%	45.21%
Low vegetation	68.74%	44.15%	53.77%	36.77%
Trees	79.38%	61.07%	69.03%	52.70%

- Six-class macro F1: **61.63%**; macro IoU: **47.23%**.
- Whole-tile bootstrap macro F1 interval: **58.43-63.83%** (2,000 resamples). This is descriptive within-city sampling uncertainty. Spatial dependence between neighbouring tiles and global geographic uncertainty are not captured.
- All six classes pass the predeclared coverage check: at least five tiles and 10,000 labelled pixels each. This is coverage, not a quality acceptance gate.
- Two-pixel road-boundary F1: **23.07%**. Roads are often identified but their boundaries are not reliably precise.
- Dark non-water pixels predicted as water: **0.554%**. This is a dark-RGB proxy, not labelled-shadow accuracy.
- Agriculture-as-low-vegetation sensitivity macro F1: **61.41%**; it does not replace the primary result. Primary ignored pixel share: **6.37%**.
- Combined-vegetation diagnostic F1: **75.60%**. Combining categories is not an improvement in the six-class score and must never replace it.

What it means

Development F1 was 83.83%, but that was reused DC/Philadelphia validation. The 61.63% result measures transfer to a different city and annotation system. It is not evidence that the model weights degraded, and there is no older-model Christchurch comparison establishing an external improvement percentage.

Building recall is high, but precision is only 58.69%: many non-building pixels are being called buildings. Ground and low vegetation are clear weaknesses. Water transfers considerably better. This is useful experimental performance, not a broadly reliable six-category production claim.

This city is new to the known project inventories, not guaranteed absent from unknown upstream pretraining. The OEM-to-GAMUS ontology mapping is explicit; source-label differences may contribute, but do not excuse incorrect outputs. Christchurch is now consumed. Any future model adjusted in response to these findings needs another genuinely untouched final location.

Safety, integrity and reproducibility

- Candidate SHA-256: 4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021.
- Source/config/software/data identities were sealed before the consumption marker and first pixel decode. Whole-file identities were checked again after evaluation. No evaluation retry was needed.
- 42 targeted tests passed before the run, including crosswalk, masking-related confusion handling, native source matrix, coverage, bootstrap, header checks, download range refusal, cache position, and development replay tests.
- All 98 selected TIFF files passed archive CRC32 and length verification, then were SHA-256 recorded. About 173 MB of network payload including range-cache overhead was used; extracted files total 205.75 MB. The full 9.1 GB archive was not downloaded and its published MD5 was NOT verified.
- Protected production checkpoint and app pointer remain hash-identical. No height pipeline change or height evaluation was performed. This test measures identification only; OpenEarthMap does not supply height truth.
- GPU inference has finished; no new training was launched.

Report wording erratum (numerical results unchanged)

The reused dark-pixel metric helper embeds the word "GAMUS" in its descriptive strings. In this external run the actual reference arrays came exclusively from OpenEarthMap, through the frozen crosswalk. Read that field as "OpenEarthMap reference class is not water" and "OpenEarthMap has no labelled shadow class". The exact counts and rate above are unchanged. The sealed evaluator and original machine report are preserved instead of silently rewriting them after testing.

Saved evidence and next step

- outputs/evaluation/christchurch_rgb_segmenter_v1/report.json
- outputs/evaluation/christchurch_rgb_segmenter_v1/seal.json
- outputs/evaluation/christchurch_rgb_segmenter_v1/per_image_metrics.jsonl
- outputs/evaluation/christchurch_rgb_segmenter_v1/predictions/
- D:/MSRData/external_holdouts/oem_christchurch_rgb_v1/CONSUMED.json
- D:/MSRData/external_holdouts/oem_christchurch_rgb_v1/manifest.json

The staging status/manifest describe the earlier acquisition stage; CONSUMED.json and the evaluator's completed status are authoritative about evaluation use.

Next: audit training-only city/crop coverage, then a controlled small-vegetation sampling experiment against an equal-budget baseline. Protect all six class development guards and the production height pipeline. Do not automatically promote this candidate or start an expensive run without defining that experiment.

Dataset and imagery attribution: OpenEarthMap v1, <https://zenodo.org/records/7223446> ; AIRS / LINZ, CC BY 4.0, <https://open-earth-map.org/attribution.html> .

68 RGB segmenter V1: DC low-vegetation audit

SOURCE docs/history/RGB_SEGMENTER_V1_DC_VEGETATION_AUDIT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

This is development diagnosis, not a new accuracy benchmark or a model change. All 859 independently replayed validation records were checked against the exact pooled city confusion matrices. The six preview label arrays were checked against their hashed categorical PNGs. Three synthetic audit tests pass.

What the evidence says

Low vegetation	Washington DC	Philadelphia
Validation images	359	500
Share of labelled pixels	8.91%	34.43%
F1	22.49%	87.22%
Recall	15.84%	88.30%

DC reference low vegetation is predicted as ground 28.44%, roads 21.90%, trees 22.09%, buildings 10.06%, water 1.66%, and correctly as low vegetation 15.84%. The problem is broader than a simple tree-versus-grass switch.

In the three fixed DC previews, removing a two-pixel boundary leaves only 20-31% of low-vegetation pixels. The corresponding Philadelphia examples retain 65-77%. DC interior recall is still weak (about 6-25% at this radius), so simply sharpening boundaries cannot solve the whole problem. These six preselected examples establish a useful hypothesis, not a city-wide morphology estimate or proof that annotations are wrong.

Controlled follow-up, not an automatic promotion

1. Audit training-only low-vegetation coverage by city and crop. Measure whether small DC vegetation features receive enough supervised training examples.
2. Test city/class-aware crop sampling against an equal-budget continuation baseline. Keep identical development evaluation and six-class regression guards.
3. Only separately test a finer-resolution decoder or boundary loss if sampling does not address the weakness. Do not change several variables and attribute gains to one of them.
4. Preserve the completed V1 checkpoint as the comparison. Never feed the new classification output into production heights without a separate validation.
5. Test the already-frozen V1 on the reserved external city without tuning it on that result. Future models need a different final holdout after this one is used.

Machine-readable evidence: outputs/diagnostics/rgb_segmenter_v1_dc_vegetation_audit/report.json.

No training, app pointer change, or height-pipeline change was performed by this audit.

69 RGB-only six-class pilot: completed outcome

SOURCE docs/history/RGB_SEGMENTER_V1_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Update: the separate fresh-checkpoint replay has now **passed on all 859 validation images with exactly matching counts and scores**. See docs/RGB_SEGMENTER_V1_TEST_REPORT.md and outputs/evaluation/gamus_rgb_segmenter_v1_independent_replay/report.json. This is repeatability evidence, not external-geography validation or app release.

Result and scope

Run experiments/20260915T140116654177Z_gamus_rgb_segmenter_v1 completed eight epochs at 2026-09-15 14:25:30 UTC. Both the best development checkpoint and best checkpoint passing every predeclared gate are **epoch 8**.

On the fixed 859-image DC/Philadelphia development validation, six-class macro F1 increased from **56.48% to 83.83%** (+27.34 percentage points). Macro IoU is **72.47%**. These are pixel-identification metrics, not object-count accuracy, height accuracy, or evidence of generalization to new geography.

Epoch	Six-class macro F1	All 27 checks pass?
1	79.71%	No: DC buildings and road boundaries
2	80.75%	Yes
3	82.02%	Yes
4	82.73%	Yes
5	83.33%	Yes
6	83.35%	Yes
7	83.51%	Yes
8	83.83%	Yes

Best-epoch identification results

Category	Overall F1	DC F1	Philadelphia F1
Ground	77.37%	68.17%	82.53%
Buildings	88.69%	86.38%	91.40%
Water	91.37%	90.01%	92.12%
Roads	81.28%	77.56%	88.88%
Low vegetation	79.87%	22.49%	87.22%
Trees	84.38%	82.86%	85.86%
Macro F1	83.83%	71.24%	88.00%

All overall class F1 scores improved over the fixed V3 replay. DC and Philadelphia macro F1 improved by 19.95 and 32.36 percentage points, respectively. No gate was loosened after observing this run.

Remaining errors: saved-metric analysis, not a new evaluation

- **DC low vegetation remains weak.** F1 is 22.49%, precision 38.74%, and recall 15.84%; most reference low vegetation is still missed. Philadelphia's much stronger performance makes the pooled overall number look better. Passing a guard against an already weak baseline is not production readiness.
- Road-boundary F1 at the fixed two-pixel tolerance is 42.96% overall, 36.52% in DC and 66.62% in Philadelphia. Boundaries remain appreciably less accurate than broad road classification, despite passing the protection gates.
- False water on dark non-water reference pixels is 0.458% overall (DC 0.437%, Philadelphia 0.524%). This is a dark-RGB proxy, not a measurement against independently annotated shadows. Total water false positives remain 3,193,899 pixels; overall water precision is 86.45%.
- Epoch 8 is best on the declared macro-F1 criterion, not every class at once: water F1 peaked at 92.75% in epoch 5 versus 91.37% at epoch 8.
- No height branch was trained or changed. The separate residual-height run remains rejected; its results must not be blended into these percentages.

Saved evidence and next release steps

The exact experiment directory contains epoch 1-8 checkpoints, latest, best_development, best_guarded, metrics, final summary, config, source snapshots, data/software binding and both feasibility reports. The new classifier has **not** replaced the protected app model or pointer. No additional training or holdout evaluation was started while preparing this outcome.

Read-only CPU integrity audit passed: 210 model tensors and all 615 saved optimizer tensors are finite; scheduler/RNG states exist; checkpoint history matches all eight metrics rows. The seven current source files, seven source snapshots, current/saved configs and binding match. Best guarded, best development, latest and epoch 008 checkpoints are byte-identical, SHA-256: 4940b2d4ae8817370384ccb54da9f73aee1b6bf3614435d8c077bf3e3363d021. Protected checkpoint and app pointer hashes still match their original seals. These checks prove artifact integrity, not predictive accuracy.

Next: authenticate an independent checkpoint replay of the exact benchmark; inspect DC low-vegetation confusion and labels without tuning on sealed holdouts; then carry out the separately planned genuinely new-geography evaluation. App integration needs an explicitly tested RGB overlay adapter and regression tests proving that classification cannot silently change height outputs. Height improvement is a separate workstream.

NYC is excluded from this new classifier's training, not historically unseen by the entire system. Official GAMUS test and external holdouts were not opened for this completion check. The NVIDIA MiT-B0 backbone remains restricted to non-commercial research/evaluation under its upstream licence.

The original outcome sections above summarize saved training metrics. The subsequent independent inference replay is documented in the linked test report; neither is a new-geography scientific validation study.

70 Independent RGB six-class identification experiment

SOURCE docs/history/RGB_SEGMENTER_V1_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status: completed eight epochs on 15 September 2026 at 14:25:30 UTC. Best development and guarded checkpoint: epoch 8, six-class macro F1 0.8382524600; all 27 predeclared checks passed. See docs/RGB_SEGMENTER_V1_OUTCOME.md for the remaining city/class weakness and release limitations. Not app-active. Exact run: experiments/20260915T140116654177Z_gamus_rgb_segmenter_v1.

Subsequent verification: a fresh independent-scoring checkpoint replay passed on all 859 development images with exact counts and scores. Diagnostic gallery and limitations are in docs/RGB_SEGMENTER_V1_TEST_REPORT.md. External generalization and app integration are still pending.

Verified launch handoff

- Final CPU suite: **641 passed**, 12 warnings, 44.31 seconds. Includes an interrupted tiny training run followed by epoch-boundary resume, separate validity masks, evaluation contracts, and BOM-aware protected pointer reading.
- Real-data preflight passed for **3,837 training / 859 validation images**. Validation RGB/classification file contents match the fixed independent V3 replay. Training source binding covers approved IDs and path/size/mtime metadata, not a fresh full byte hash of every training file.
- Train-only feasibility: 48 updates on one fixed training batch, loss **0.749761 to 0.114596** (84.7156% reduction), with nonzero encoder gradients. This is optimization evidence only, **not accuracy or generalization**. Proof weights were discarded before starting the full experiment.
- Native 1024-pixel CUDA smoke passed on one development image per city, peak allocated model memory **802,078,208 bytes**. The fresh decoder's smoke scores are not trained-candidate results.
- Epoch 1 was observed advancing through **760/960 training batches**. Initial NVIDIA sample: 47% utilization, 2,418 MiB total GPU memory used, 58 C, 49.69 W. Utilization fluctuates; these are samples, not averages.
- Config, source snapshots, data/software binding and both proof reports are saved in the exact run. Checkpoints are created after each completed validation epoch; before epoch 1 completes there is no resumable checkpoint.
- Existing hourly automation now follows only this exact run, remains quiet for unchanged status, and never promotes it or resumes the rejected residual run. A failed run may resume only the same compatible latest checkpoint.
- All six-class and city gates remain unchanged. First full validation is pending at this handoff; **no classification improvement is claimed yet**.

Purpose

Train an independent image-understanding model for ground, buildings, water, roads, low vegetation and trees. The fair V3 six-class development baseline is macro F1 0.5648374706 on 859 DC/Philadelphia tiles. The older V3/V4 classifiers trained approximately 1,094 head parameters over a frozen adapter. The new SegFormer-B0 encoder and decoder have 3,715,686 trainable parameters and receive RGB only. Higher capacity is a testable hypothesis, not an accuracy claim.

The entire protected height model, its preprocessing/fusion and application pointer remain unchanged. No predicted class is yet used to gate, clip, scale or replace heights. Future overlay integration and full application verification are separate release steps.

Fixed data and comparison rules

- Reuse the approved DC/Philadelphia training index (3,837 images) and the same 859 native 1024-pixel development images as the independently replayed V3.
- Preserve class mapping, unknown/background ignore index 255, independent image/class validity, raw image radiometry, water-aware/city-balanced sampling and the existing focal-loss class weights. Known invalid heights must never remove otherwise valid classification labels.
- The new dataset reads RGB and classification files only. Neither reference height nor a LiDAR image is an inference input; no model predictions are pasted into the supervision. The model's ImageNet normalization is fixed.
- Authenticate the approved index, sampling index, baseline, pretrained files, source code and evaluation content. Check exact validation IDs and class support counts against the old replay before comparing accuracy.
- This is a **new complete model recipe**, not an architecture-only causal ablation: encoder initialization, optimizer groups and normalization differ from V3. The unchanged evaluation makes outcome comparisons useful, but attributing gains to one architectural detail would require another control.
- NYC is excluded from this new classifier's fitting, not unseen by the whole historical system. Official GAMUS test was previously consumed and is not used again. External classification/height holdouts are not opened here.

Preflight and training

The isolated entry point is `scripts/train_rgb_segmenter.py`, configured by `configs/gamus_rgb_segmenter_v1.yaml`. Before the full run:

1. CPU contract and unit tests, pretrained encoder loading with no missing encoder tensors, protected artifact hashes and dataset checks.
2. A small fixed **training-only** learning proof: at least 48 steps, at least 15% loss reduction, finite updates and nonzero encoder gradients.
3. Native 1024x1024 CUDA validation smoke test, class/support/evaluator checks and VRAM measurement. No resized/easier validation substituted for full data.
4. Seal the exact source/config/proofs. Start a fresh model from the pretrained encoder and fresh decoder, not from the memorized feasibility checkpoint.

The pilot is limited to eight epochs with early stopping, batch four and one GPU workload at a time. A different batch/recipe requires explicit evidence and new validated settings rather than an unrecorded resume mutation. Epoch-boundary checkpoints, random states, config/source snapshots, a compact status file and terminal completion/failure summary support safe recovery. No code or configuration changes are allowed during a sealed active run.

Acceptance and interpretation

Report final **six-class** macro F1/IoU and per-class precision, recall, F1 and IoU, separately for DC and Philadelphia. Keep road-boundary tolerance and the dark-pixel water-confusion proxy fixed; dark pixels are not true shadow labels. Carry forward the predeclared V4 class/city acceptance thresholds from its fixed V3 comparison. A better mean score does not excuse a collapsed class. Keep the best macro-score checkpoint distinct from any best fully eligible development checkpoint. Passing development gates does not authorize release or establish generalization to other cities, sensors or landscapes.

Post-run: inspect every epoch and independent per-class evidence. If promising, perform a separately authenticated replay before final external evaluation and future overlay exposure. The protected application cannot load these standalone checkpoints without a separately implemented and verified adapter.

Launch and progress

Launch after all checks pass:

```
.\scripts\run_rgb_segmenter_v1.ps1
```

Safe compatible resume only:

```
.\scripts\run_rgb_segmenter_v1.ps1 -ResumeCheckpoint <exact-run>\checkpoint_latest.pt
```

Compact live viewer: `scripts/watch_rgb_segmenter_v1.ps1`. It reads the exact active run recorded in `outputs/orchestration/rgb_segmenter_v1_active.json`, with Windows delete-sharing enabled so it cannot block checkpoint/status replacement. Closing the viewer does not stop training.

The existing hourly automation `msr-residual-accuracy-check` now tracks the exact run above. It checks milestones, all class/city gates and saved checkpoints, then pauses after reporting completion and saving an outcome summary. No repeated short-interval polling is needed.

Backbone source and licence

- ImageNet-only encoder: <https://huggingface.co/nvidia/mit-b0>
- Pinned revision: `80983a413c30d36a39c20203974ae7807835e2b4`.
- Files and checksums: `D:/MSRData/models/segformer-mit-b0-imagenet/backbone_manifest.json`.
- Download: approximately 14.4 MB of weights; no new large dataset download.
- Upstream NVIDIA SegFormer licence restricts use to non-commercial research or evaluation. Preserve notices and `LICENSE_upstream.txt`. A commercial product needs an appropriate licence or a separately licensed alternative; do not advertise this experiment as unrestricted commercial software.
- Primary model documentation: https://huggingface.co/docs/transformers/model_doc/segformer.

Prior height experiment

Residual-height V1 stopped after five epochs without an eligible checkpoint. It is preserved and must not be restarted by this monitor. Its targeted error analysis is separate from this classifier. Improving identification alone is not evidence that any building or canopy height became more accurate.

71 Independent saved-checkpoint test

SOURCE docs/history/RGB_SEGMENTER_V1_TEST_REPORT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Completed 15 September 2026. The best guarded checkpoint from experiments/20260915T140116654177Z_gamus_rgb_segmenter_v1 was freshly loaded and tested without updating weights or changing the production app.

Outcome

PASS: all 859 development-validation images reproduce epoch 8 exactly.

- Six-class macro F1: **83.8252460040%**; macro IoU: **72.4664564976%**.
- Overall, DC and Philadelphia confusion matrices match exactly, with zero changed integer counts. Every per-class precision, recall, F1 and IoU matches.
- Road-boundary and dark-water proxy counts/scores also match exactly.
- The fixed 27 acceptance checks still pass. Reference-grid hashes, class supports, IDs/order, and validation RGB/CLS source contents were authenticated.
- The replay uses an independent NumPy confusion-count/scoring implementation. It reuses the sealed model, input loader and auxiliary boundary/proxy helpers; it is **not** an independently collected dataset or external scientific audit.
- Source snapshots, config, model pointer and protected height checkpoint remain unchanged. No training, height-label, DAV2, official test, NYC or external-geography rasters were read as part of this replay.
- Replay elapsed time: **148.55 seconds**. New/related targeted CPU tests: **42 passed**, including 18 new evaluator tests. No GPU workload remains.

Artifacts: outputs/evaluation/gamus_rgb_segmenter_v1_independent_replay/.report.json contains the complete scores, identities and exact comparisons; per_image_metrics.jsonl covers every image, including low-vegetation confusion.

Visual test

Local review: <http://127.0.0.1:8766/>. This serves only the diagnostic output directory, not the production app or a new upload/inference endpoint.

Six native 1024x1024 examples were fixed before scoring: first/middle/last lexicographic validation ID in each city. They were not chosen for good scores.

- DC: DC_02_26, DC_37_26, DC_76_32.
- Philadelphia: PHL_6150, PHL_6486, PHL_6925.

For every example, the original RGB, prediction overlay, reference overlay and per-class metrics load correctly. Browser checks confirmed all images are 1024x1024, all six class rows render, opacity 0/65/100% works, and previous/next navigation wraps correctly. No page runtime errors were reported. All 36 PNG/metadata files named in the preview manifest passed their SHA-256 checks. The automated browser session was closed after verification; the lightweight loopback review server was left running for manual testing (launcher PID 4628).

What this does and does not prove

The saved checkpoint truly reproduces the reported improvement; it was not a stale score or an accidental selection of another model. Building and road regions are visibly recognizable in the inspected examples, but boundaries are smoother than the references and low vegetation is still frequently missed.

This does not establish accuracy on a new city, sensor, resolution or season. These validation images were not used for gradient updates, but their results were used for checkpoint selection. A separately preregistered external test is still needed. There is no new height-accuracy claim or automatic app release.

The important unresolved case is DC low vegetation: F1 **22.49%**, recall **15.84%**. Of its reference pixels, 28.44% were predicted ground, 21.90% roads, 22.09% trees, 10.06% buildings and 1.66% water. In the inspected DC example, some low-vegetation labels form narrow strips around other classes; the Philadelphia example has broad low-vegetation areas. This is a reason to investigate appearance, scale and annotation consistency, **not proof that the reference labels are wrong**. Do not retune on an external holdout.

Next steps: audit that DC failure pattern on development data, establish the external identification protocol, and test an explicit overlay adapter before allowing this classifier into the app. Keep heights protected.

72 Roof reconstruction: stage 1, 19 September 2026

SOURCE docs/history/ROOF_RECONSTRUCTION_STAGE1_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Current viewer decision — flat roofs restored

At the user's request, roof relief and experimental fitting are no longer connected to the viewer. Buildings retain flat, optical-textured roofs and their existing model-derived building-height estimates. Fly metadata, the Inspect popover and selected-building analysis show **“Roof height estimation yet to be configured”**. Raw predictions, exports, checkpoints and facade textures are unchanged. The experimental modules, tests and prepared data remain saved but inactive; no further roof training or downloads were started. The stage-1 notes below describe the earlier experiment, not currently enabled viewer features.

Status — foundational implementation, NOT a completed roof-accuracy fix

The viewer can now represent a sharp two-plane gable with lower eaves and vertical walls instead of placing rounded relief on top of a retained flat slab. This is an **experimental geometric fitter**, not a newly trained roof detector. It does not yet separate attached buildings, predict arbitrary roof parts, reconstruct hipped/complex roofs, or reliably identify rooftop rooms and dormers.

The exact user example, Strasbourg grid_00074_z19.jpg, still has **0 accepted fits out of 12 existing connected footprints**: eight are edge-clipped and four are complex/merged/undersized under the current gate. The original view remains. This is a rejection/coverage result, not a height-accuracy score. The source image was already inspected and is a development example, not a fresh test.

Implemented

- viewer/app/roof-reconstruction.ts: bounded orientation search, robust two-plane fitting, spatially distributed internal fit-check samples, comparison against constant/tilted-plane alternatives, optical ridge support and rejection for insufficient/missing/ambiguous evidence. These heuristics are not calibrated probabilities of roof correctness. Agreement with model heights is not agreement with independently surveyed heights.
- Sharp mesh joins, image-projected roof UVs, separate illustrative wall UVs and full-height walls. Accepted replacements hide the old solid/relief and lower only the underlying presentation mesh under that footprint. Switching off restores the original geometry. No export raster is changed.
- Viewer layer **Fitted roof shapes · experimental**, off by default; disabled when no candidate passes. The collapsed report gives rejection reasons and a downloadable JSON audit. Fitted roof inspection distinguishes raw source height from fitted display height. Collision retains a conservative maximum envelope.
- src/msr/data/roof3d.py: strict decoding of the release's list-wrapped uncompressed COCO RLE, independent image/classification/boundary/height-validity masks, overlap exclusion, and cropped-border safeguards. Unknown background is not a negative class; missing height is never a zero-height target.
- Versioned data preparation and acquisition scripts; all new datasets on **D**.

Data acquired and audited

Roof3D v2 training-source pilot

Path: D:/MSRData/data/roof3d_training_pilot_v1.

Official record: <https://zenodo.org/records/8300629>. Authors' dataset repository: <https://github.com/dlrPHS/GPUB>. Licence in the publisher's API: **CC-BY-4.0**; attribution is saved in the manifest.

- Published archive: 4,724,970,465 bytes. **Not downloaded in full.**
- Read its inventory and selectively fetched two training annotation files and 16 RGB/DSM pairs using strict HTTP byte ranges: 44,252,029 bytes transferred.
- Selected ZIP members checked by CRC32 and length; local SHA256 recorded. The full-archive MD5 cannot be claimed verified with a partial download.
- The archive lists 3,337 training RGB images. Only 16 evenly spaced filenames were selected for this **schema/preparation pilot**, not a spatially independent evaluation set. No v2 validation pixels were accessed by this staging script.
- Selected examples contain 1,705 roof-plane annotations and 850 building-section annotations. These are patch annotation counts, not unique physical roof counts.
- Plane supervision: 1,133,803 valid labelled pixels, 123,159 valid boundary pixels; 164,394 overlapping pixels excluded. Section supervision: 1,279,733 valid pixels, 95,921 boundary pixels; 5,990 overlapping pixels excluded.
- RGB and DSM grids match within all 16 selected pairs, but the chips lack useful georeferencing/source-location tags. Numeric names alone do not identify city, original tile, or real versus synthetic source. Recover this provenance before constructing geography-level validation claims.
- Both JSON files use the generic category name `individual_building`; their **file role** (`annotation_plane` vs `annotation_sec`) distinguishes supervision. Segmentation is a list of uncompressed RLE dictionaries, not ordinary polygons.

Prepared targets: `D:/MSRData/data/roof3d_training_pilot_v1/prepared_positive_only_v1`. All `height_valid` masks are deliberately false pending a verified height contract.

Roof3D v3 small reference tile

Path: `D:/MSRData/data/roof3d_reference_pilot_v1`. Official record: <https://zenodo.org/records/10910492>.

59,299,694 bytes, all four released file MD5 values verified. This record contains the publisher's test area; it is marked as a **development reference diagnostic**, not training data or our untouched final holdout.

The grid audit did NOT pass exact agreement: GT has no CRS and has a slightly different transform; RGB and DSM/DTM use differently described CRS metadata. DTM is constant 45 in this tile, and GT values are not on the same numerical range as the absolute DSM. Do not guess the GT's datum or subtract DTM twice. Verify the product definition, alignment and vertical reference before metrics.

Verification

- 35 viewer tests passed (14 new reconstruction tests plus 21 existing navigation, inspection, relief and facade checks).
- 21 Python tests passed (10 new Roof3D label tests and 11 range/staging regressions).
- TypeScript and targeted ESLint passed.
- Fresh isolated-browser upload of the exact Strasbourg image: normal rendering, expected rejection report, no page errors.
- A clearly named **synthetic** gable fixture tested the full UI replacement toggle, removal/restoration of the underlying slab, and Fly click inspection. It is not a real satellite prediction or evaluation result. Its mock API response was confined to the isolated test browser and removed by reload.
- Screenshots: `outputs/runtime/roof-fit-synthetic-on.png`, `roof-fit-synthetic-off.png`, `roof-fit-synthetic-inspection.png`, and `roof-fit-strasbourg-fallback.png`. Only the last is the actual user scene.

The running development browser briefly accumulated errors during an intermediate hot reload when the builder's return signature changed. After completing both sides of that change, a fresh session passed without page errors. Refresh an old app tab if it retained an interrupted development render.

Next implementation stages / release gates

1. Audit plane overlaps and annotation/image alignment visually; recover source geography/type and freeze a non-overlapping split. Preserve the official split as published evidence, not a claim that the whole existing system never saw those cities.
2. Train an **isolated RGB roof-part/ridge model**. Preserve production height and semantic checkpoints and preprocessing. Use RGB only at inference; reference DSM/roof labels are supervision/evaluation, never hidden inference inputs.
3. Decompose merged footprints, preserve courtyard holes, and predict roof parts. Extend geometry to supported hip/flat/complex roofs and vertical rooftop steps. A predicted ridge alone does not establish its metric height.
4. Resolve roof-level height supervision before learning local roof residuals. Distinguish ridge/eave height and roof-plane error from whole-building RMSE.
5. Report roof-part/boundary precision, recall and IoU; ridge-position error; roof-only height errors; accepted coverage and failures, separately by roof type, material, size and geography. Include confidently wrong fits, not only accepted attractive examples. Check building, forest and terrain regressions.
6. Promote only after independent reference evaluation and app regression tests. **No roof-model training has started and no production model was changed here.**

Research direction

Building decomposition plus ridge detection is an established direction, but no paper's results are claimed for our geometric baseline:

- Zhang & Aliaga, *Procedural Roof Generation From a Single Satellite Image*, 2022: <https://doi.org/10.1111/cgf.14472>.
- Schuegraf et al., *Roof3D*, 2023: <https://doi.org/10.5194/isprs-annals-X-1-W1-2023-971-2023>.

Reproduction

```
.env/Scripts/python.exe scripts/data/prepare_roof3d_reference_pilot.py
.env/Scripts/python.exe scripts/data/stage_roof3d_training_pilot.py
.env/Scripts/python.exe scripts/data/prepare_roof3d_training_masks.py
.env/Scripts/python.exe -m pytest tests/test_roof3d.py
tests/test_openearthmap_archive_ranges.py tests/test_stage_christchurch_holdout.py -q
```

From viewer:

```
node --experimental-strip-types --test tests/roof-reconstruction.test.mjs tests/roof-
geometry.test.mjs tests/building-facades.test.mjs tests/flight-navigation.test.mjs
tests/surface-query.test.mjs
npx tsc --noEmit
npx eslint app/roof-reconstruction.ts app/terrain-workspace.tsx app/surface-query.ts
```

73 Predicted rooftop relief — 19 September 2026

SOURCE docs/history/ROOFTOP_RELIEF_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

What was fixed

The semantic viewer used one percentile-derived height per connected building footprint. This intentionally clean display discarded smaller raised regions in the raw prediction. The new **Predicted roof relief** layer retains supported raised regions above that existing summary roof. The original optical image is projected onto both the base and the raised geometry; illustrative facade textures remain on the outer walls.

The source is the existing raw metric nDSM, not image brightness, invented roof templates, a reference raster, or a new model. Export rasters, model checkpoints and evaluation metrics are unchanged. This is a rendering improvement, not a claim of better predictive accuracy.

Safeguards and implementation

- Bounded regular-grid samples inside each existing footprint. Semantic building labels (or the existing 0.3 building-probability threshold for legacy scenes) gate support.
- A 3x3 spatial median rejects isolated height spikes. At least nine supported samples and 65% supported coverage are required.
- At least five samples / 1% of supported samples must exceed the summary roof by the larger of 0.5 m or 7% of its height. These are conservative display heuristics, not calibrated correctness probabilities.
- Lower edge estimates do not pull down the existing solid. Detail is clipped inside the footprint and meets the retained roof at unsupported boundaries. Relative and absolute-DSM input scenes do not silently become metric rooftop nDSM detail.
- Regular-grid triangles avoid long thin triangles across irregular footprints. New roof geometry is bounded to roughly 30,000 sampling points per scene, with per-building budgets of 160–1,400 and a minimum spacing of two source pixels.
- The checkbox toggles the extra roof geometry; existing building shapes and wall textures remain available. Raycasts ignore disabled roof meshes.
- Fly scanning identifies a raised-roof hit as **ROOF POINT · ABOVE GROUND** and reads its raw source pixel. Displayed relief can differ from the raw value due to the baseline floor, median filtering and interpolation. The parent building ID is unchanged, so small aim movements do not restart dwell.
- Collision uses the conservative maximum of the base solid and the roof-detail bounds. It prevents passage through raised roofs, but may keep the camera above lower parts of the same building.

Exact user-image check

Input: data/highbuild_reviewer/images/Europe_France_Strasbourg/grid_00074_z19.jpg.

The existing production model generated job b3c8c9df2b6d4783b5cde2a6fcde648e through the normal upload API. It contained **12 building components**; **11** had enough supported raised-height variation to enable this display layer. These are component/rendering counts, not validated building counts or eleven proven rooftop reconstructions.

The same saved inference response was reused in the isolated browser test while editing the viewer, avoiding repeated GPU inference. On/off screenshots used the identical prediction. A Fly click on raised geometry reported a raw roof-pixel estimate (22.0 m in the sampled view) rather than repeating the whole-building summary. That point has not been checked against surveyed truth.

Screenshots: `outputs/runtime/strasbourg-roofs-detailed.png`, `strasbourg-roofs-flat.png`, and `strasbourg-roof-inspect.png`.

Verification

- 21 targeted viewer tests passed, including coherent rooftop-block retention, isolated-spike rejection, raw-data immutability, unknown/vegetation/NoData handling, relative-input rejection, low-edge protection, concave-footprint clipping, facade/roof UV preservation and navigation regressions.
- TypeScript and targeted ESLint passed.
- Exact-image browser upload/render, relief toggle and Fly roof-pixel inspection checked; no browser page errors reported.

What remains unresolved

This is not a dormer, chimney, HVAC-unit or rooftop-room detector. It cannot recover shapes missing in the height prediction, resolve all roof slopes, separate touching buildings, restore courtyard holes missing from existing footprints, or infer hidden walls from one overhead image. Model errors can still produce incorrect relief. It also deliberately does not reconstruct lower roof tiers beneath the existing summary height. Fine rooftop accuracy needs appropriate labels, better-resolution predictions and independent validation.

74 Original-image reference in the 3D viewport

SOURCE docs/history/SCENE_IMAGE_REFERENCE_2026_09_27.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

The upper-right **Original image** card displays the RGB texture paired with the active 3D scene. It is a 2D input reference, not a prediction, classification overlay, second inference, or accuracy claim.

Behaviour

- Uses `sceneInput.textureUrl` for reference scenes, imported imagery and replacement textures.
- TIFF/GeoTIFF and other server-decoded formats use the same browser-readable RGB preview returned by inference and used on the mesh.
- Shows the complete image with preserved aspect ratio (no crop), plus the current scene label.
- Collapsible with an accessible button; a new image resets the card to expanded.
- Remains in colour during metric, elevation and classification views.
- Shrinks during Inspect/Explore; the reference and height readout share a vertical dock, avoiding overlap.
- Remains inside native or window fullscreen. Narrow views use a smaller preview.
- Hidden in the neutral empty workspace and while a new scene is processing/revealing; failed image loads have an explicit placeholder.
- Source heights, inference, exports, mesh geometry and selection/glow logic are unchanged.

Verification

- Existing 142 viewer tests passed.
- TypeScript and targeted ESLint passed.
- Dedicated browser: all four Urban/Sparse/Hilly/Forest references; collapse and scene reset; metric view; Inspect; Fly; fullscreen; layout at 390/820/1280/1600 px.
- Real PNG, JPG and TIFF uploads: each displayed image URL matched its own API response's `texture_url`. Old-image previews were absent during processing.
- No browser page errors captured.

Repeatable browser check: `scripts/verify_scene_image_reference.mjs`. Evidence and screenshots: `outputs/runtime/scene_image_reference/verification.json`.

75 Active inspection boundary

SOURCE docs/history/SELECTION_GLOW_2026_09_27.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

The active Inspect / Fly / Walk readout now has a mint boundary with a soft halo. It uses three thin screen-space strokes, depth testing and a short fade-in (instant with reduced motion), not a full-screen bloom pass.

Behavior

- Inspect click pins the selected boundary with the existing metadata. Pointer movement alone does not unpin it.
- Selecting a different object replaces the old boundary; outlines do not accumulate.
- Explore's two-second aim and click scanner share the same selection/readout source. Moving off the target, changing mode, exiting Explore, starting an import or changing scenes clears the highlight with the active inspection.
- Building solids use their actual transformed mesh edges, including nonrectangular footprints. Vertical display exaggeration remains aligned with the selected solid.
- Raster-only surfaces use the connected region under the selected sample. Disconnected patches of the same class are different dwell targets. Adjacent trees are not falsely claimed to be separate tree instances.
- Region outlines use a bounded, at-most-512-pixel display grid and follow the displayed terrain triangles. Holes and ignored labels stay separate. Missing or undersized regions receive a small local sample locator, not a fabricated object boundary.
- Photo/metric switching preserves the selected boundary and numeric height. The rendering overlay does not change model weights, source heights, collisions or exports.

Verification

- 142 viewer tests passed, including seven new tests covering disconnected regions, holes, ignored labels, bounded sampling, local locators, terrain draping, transformed building edges, resource cleanup and noninterfering picking.
- TypeScript and ESLint checks passed for the changed TypeScript files.
- Browser verification passed: two different building IDs, connected surface selection, persistence, photo/metric switching with identical readout, mode clearing, empty-space clearing, Fly and Walk dwell/click selection, looking away into empty sky, and scene switching.
- No page errors were captured during the final browser verification.
- This was a display/interaction change, not a new model accuracy evaluation.

Evidence: outputs/runtime/selection_glow/verification.json and screenshots in the same folder. Reproducible browser harness: scripts/verify_selection_glow.mjs, connected only to a dedicated agent-browser session.

Files: viewer/app/selection-region.ts, viewer/app/selection-glow.ts, viewer/app/terrain-workspace.tsx, viewer/tests/selection-glow.test.mjs.

76 Active six-class guided height experiment

SOURCE docs/history/SEMANTIC_EXPERT_HEIGHT_V1_HANDOFF.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Launched 19 September 2026 at 18:07:01 UTC. This is a launch record, not a claim that height accuracy has improved or that the app model has changed.

- Exact run: D:/MSRData/experiments/semantic_expert_height_v1/20260919T180701956748Z
- Protocol: docs/SEMANTIC_EXPERT_HEIGHT_V1_PROTOCOL.md
- Configuration: configs/semantic_expert_height_v1.json
- Log/error log: outputs/orchestration/semantic_expert_height_v1.log and .err.log
- Live window: scripts/class_assisted_dashboard.py
- Registration: outputs/orchestration/class_assisted_height_active.json
- Launcher: scripts/run_semantic_expert_height.ps1

Completed work and evidence

The preceding height-band sampling run finished with NO safety-and-benefit eligible candidate. Its outcome and diagnostic are documented in docs/HEIGHT_BAND_SAMPLING_V1_OUTCOME.md and outputs/diagnostics/height_conditioning_response_v1/report.json. Both final committed checkpoints/evaluations were hash verified; all protected tensors were compared with production and matched. Protected pointer, V1 and V3 hashes were verified too. Old experiments remain intact.

New model: src/msr/models/semantic_expert_height.py. Six correction outputs, mixed at native pixels by actual frozen classifier probabilities in B. A has identical capacity but neutral mixing. No class reference masks or target heights enter inference; no class forces zero height. Both keep the ORIGINAL sampler/loss/initialization/budget, four epochs each, 600 source pairs per epoch.

The bounded train-only proof at D:/MSRData/experiments/semantic_expert_height_proof_v1/20260919T180155574728Z passed initialization/semantic/frozen-state checks and finite learning. Training loss decreased 54.13% in A and 53.30% in B on 20 fixed training examples. This is NOT accuracy or generalisation. Proof weights were not reused. Actual-vs-uniform class routing changed the learned B proof heights by mean 0.361m on that sample.

29 targeted tests passed; 24 training source files were bound at preflight. The protected full-development baseline matched exactly, and uniform/control epoch 1 training was observed at handoff preparation. Data validation covers all source and cached-prior/classifier receipts. Only one actual GPU trainer is allowed. Files/checkpoints and source snapshots are saved on D:.

Monitoring and recovery

The existing msr-residual-accuracy-check heartbeat is ACTIVE hourly for this exact run, quiet without meaningful changes. It must compare matched epochs, preserve all original category/region/bias/correlation/R2 guards, and report best eligible separately from unrestricted results. 160 HighBuild +120 OpenCanopy images are DEVELOPMENT, not untouched final tests. Six correction routes are not six independently validated height categories. No GAMUS height claim or automatic app promotion.

On a visible failure, inspect actual processes/logs first. At most one compatible resume after inspection, without source/hash/batch changes:

```
.\scripts\run_semantic_expert_height.ps1 -Resume
'D:\MSRData\experiments\semantic_expert_height_v1\20260919T180701956748Z'
```

Use `FileShare.ReadWrite` plus `FileShare.Delete` for active Windows files; the dashboard's `shared_bytes/read_json` helps implement this. Never rerun a failed proof or overwrite orphan checkpoints. At completion verify final hashes, produce error analysis, pause the monitor and do not automatically start another expensive run. Current prototype models remain unchanged.

77 Six-class-guided height experiment: completion and error analysis

SOURCE docs/history/SEMANTIC_EXPERT_HEIGHT_V1_OUTCOME.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Run: D:/MSRData/experiments/semantic_expert_height_v1/20260919T180701956748Z. Completed 19 September 2026 at 18:25:34 UTC, four epochs per arm. This closeout supersedes the active-run status in the original launch handoff; the launch record and bound training files remain unchanged.

Verdict

Actual six-class probabilities successfully guide the candidate's six height correction outputs. However, this experiment does not demonstrate a clear overall advantage over the equally sized, equally trained neutral-routing control. The classifier was frozen: these are height results, not a new classification F1 result.

No candidate met both the original safety and paired-benefit rules. Best eligible candidate: **none**. Best unrestricted checkpoint under the predeclared equal-building/canopy RMSE ranking: **uniform/control epoch 2**. That checkpoint is not release-approved. The application remains unchanged.

Matched final-epoch comparison

RMSE in metres; lower is better. All values below refer to the same development evaluation: 160 HighBuild scenes and 120 OpenCanopy scenes, not an untouched final geographical test.

Group	Protected app	Neutral control, epoch 4	Six-class candidate, epoch 4
Buildings	13.2426	12.7649	12.7558
Canopy	8.3610	7.7913	7.8238
Ground	4.8944	4.5556	4.6083
Short vegetation	5.2049	4.6974	4.8969
Short buildings	4.7701	4.6421	4.5103
Medium vegetation	5.8957	5.6938	5.5906
Tall buildings	29.7967	28.6888	28.7640
Tall canopy	10.4863	9.6631	9.7688

What improved, and what did not

- Both final arms improve the four headline RMSE groups versus the protected app. This supports useful height adaptation, not a clear added benefit from classification guidance.
- Class guidance reduces final building RMSE by just 0.0091 m versus control. There is no repeated-seed evidence that this tiny difference is reliable.
- The candidate improves short-building and medium-vegetation RMSE versus control, but worsens overall canopy, ground, short vegetation, tall buildings and tall canopy. Its class-specific benefit is mixed.

- No candidate epoch passes paired benefit. The final canopy gain versus the protected model exceeds 0.5 m and 5%, but the neutral control does better. Final building gain does not meet the required minimum improvement.
- All epochs also fail the unchanged safety rules. Aggregate RMSE alone does not establish acceptable bias, MAE, shape agreement, or regional behaviour. Conversely, an extremely strict correlation/R2 tolerance is not by itself a practical measure of the size or reliability of a regression.
- Best unrestricted control epoch 2 has building RMSE 12.7711 m and canopy RMSE 7.6911 m, but ground/short-vegetation RMSE 5.2373/5.8443 m. Do not combine different checkpoints' best group scores into a fictional model.

Evidence and integrity checks

The run's `DEVELOPMENT_REPORT.md`, per-epoch evaluations, atomic commits, `outcome.json`, configuration and source snapshots remain saved. Closeout verification authenticated the source/config/data binding, both final epoch-4 checkpoint and evaluation hashes, and matching checkpoint bindings. All 240 protected model tensors match production in each final checkpoint. Production checkpoint/pointer and V1/V3 classifier bindings match their protected hashes. No trainer remained active at completion inspection.

The hourly `msr-residual-accuracy-check` monitor is now **PAUSED**. No model promotion, new training, recipe change, or final-test consumption was performed during this closeout.

Justified next steps, not launched

1. Inspect the existing scene-level predictions and subgroup errors together with class-routing mistakes, especially short vegetation and tall objects. Determine whether errors arise from classification, height supervision, image scale, or the height model rather than assuming more classes fix them.
2. Predefine practical regression limits and uncertainty reporting for the next experiment. Keep these original failures recorded; do not retroactively change the rules to promote this candidate.
3. Choose one evidence-led modification and retain the matched neutral control. A longer run needs a stated reason and a bounded budget, not just more epochs.
4. Recheck classification independently if its inputs or weights change, and reserve genuinely new geography for a deliberately defined final evaluation.

Six routes do not certify six independent height categories. These results provide no new road/water height validation, universal unseen-data accuracy, or accuracy percentage.

78 Six-class soft-routed height corrections: prospective experiment

SOURCE docs/history/SEMANTIC_EXPERT_HEIGHT_V1_PROTOCOL.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

19 September 2026. Prompted by completed sampler and conditioning diagnostics. No earlier candidate is promoted; earlier experiments and protected app remain.

Explicit hypothesis

The previous model produced one correction field from concatenated class inputs. It used classes somewhat, but achieved only tiny additional development gains. Test six learned correction outputs with frozen RGB class probabilities mixing them at each native pixel. This gives classification a direct, inspectable path to height without forcing any category to a hand-written height or using labels at inference. Soft routing can still fail when classification is wrong. A small training diagnostic found weak short-vegetation identification; this limitation is not solved merely by adding the router.

Uniform-router A and predicted-six-class-router B have identical parameter counts and initial states. Spatial features receive identical RGB, DAV2 and protected-height context; ONLY the final mixing probabilities differ. Sum six bounded signed corrections, then add to the frozen model's height and clamp the result to nonnegative. Invalid imagery has no correction. Independent RGB classifier and all protected semantic/height weights remain frozen. Semantic labels are not height labels, missing height is ignored, roads/water are never assigned zero by a category rule.

Bounded training-only proof first

Reuse the authenticated fixed 20 TRAIN feasibility examples, source-bound DAV2 and classifier caches. Fresh model, 64 updates per arm, original loss/batch/LRs. Require identical starting states/sample sequences, zero-start height identity, unchanged protected tensors/semantic outputs, finite gradients, at least 10% training-loss reduction per arm, and measurable actual-vs-uniform routing effect after learning. Proof weights are discarded for the development run. No retry or recipe change to turn a failed proof into a claimed pass.

Development run only if proof passes

Four epochs per arm, 600 urban/forest pairs per epoch. ORIGINAL sampler in both, not the unsuccessful height-band intervention: same sealed original plans/crops, same original source-balanced loss, seed, batch 2, accumulation 2, learning rates, 384px patches and optimizer updates. Start both from the protected weights and new zero correction heads. No augmentation. All 450 HighBuild +600 OpenCanopy training images included each epoch. No training on development or final geography.

Recompute/check protected baseline over all 160 urban +120 forest native-size DEVELOPMENT scenes, preserving exact support, original semantics, short/medium/ tall height bands, boundaries and per-region checks. Same strict safety rules as previous run; do not loosen them after seeing results. Paired benefit requires building OR canopy RMSE reduction $\geq 0.5\text{m}$ AND $\geq 5\%$ from protected, beating the same-epoch uniform-router control. Passing benefit alone is not acceptance. Report best eligible separately from unrestricted equal building/canopy RMSE.

Hash-bound source/config/data/runtime, optimizer/RNG checkpoints each 50 updates, atomic commit, source/input verification before and after. No automatic OOM recipe/batch changes or competing GPU job. One compatible resume after inspection is allowed for development, not a failed feasibility proof. No automatic app promotion; per-building, app-path and a deliberately reserved final geography check remain later release gates. GAMUS metric height, broad plains/hills and road/water height accuracy remain uncertified. Six routes do not mean six newly validated height categories.

79 Six-class application preview — 19 September 2026

SOURCE docs/history/SIX_CLASS_APP_PREVIEW_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

What is available

Open <http://localhost:3000>, upload an RGB image (or choose a reference scene), then select **Identify six classes** under **Scene intelligence** in the right panel.

- Pinned V3 epoch-1 RGB model identifies ground, buildings, water, roads, low vegetation and trees.
- A separate color overlay is projected onto the existing 3D surface and building solids. Class filtering dims other classes; it does not flatten or raise geometry.
- The miniature classification map, pixel coverage, label GeoTIFF and provenance JSON are available. Unknown image pixels remain 255/transparent and are excluded from coverage.
- Projected map areas are reported only for a projected CRS with known units. Ordinary JPEGs and bundled JPEG demos report pixels, not invented square metres.
- Inspection and flight readouts can show an additional **RGB preview class**, separately from the protected height model's classification and height.
- Disable **Show on 3D surface** to restore the previous appearance. Three-group height-isolation controls are disabled while this overlay is visible, to prevent confusing two different features.
- Switching scenes clears the previous classification. Requests use the original uploaded RGB, not a stretched display texture. Bundled demos use their explicitly identified display RGB.

Boundaries

This is an opt-in experimental classifier, **not a new production height model**. No default checkpoint/pointer promotion, retraining, height correction, mesh-mask replacement, collision change, object-count claim or calibrated confidence claim occurred.

The preview currently accepts 8-bit RGB only. It uses native-resolution tiles up to 1024 pixels, 128-pixel overlaps and mean probabilities, with the app's existing 3072-pixel processing cap. Larger scenes therefore use a tiled preview recipe; development scores are not measured accuracy for arbitrary uploads. High-bit-depth imagery requires a separately validated classification radiometry pipeline. The hilly 10-metre demo is explicitly outside the high-resolution training domain.

Fresh HighBuild comparison

Same 160 corrected development images, same native CUDA/bfloat16 inference, four cities, all supplied building outlines. These positive-only annotations support building-pixel recall/coverage, **not precision, F1, counting accuracy or height accuracy**.

Diagnostic	V1	V3 epoch 1
Annotated building-pixel recall	79.21%	82.60%
Building interior recall	79.85%	83.64%
Building boundary recall	73.08%	72.52%

Diagnostic	V1	V3 epoch 1
Copenhagen recall	88.43%	86.73%
Lyon recall	87.72%	87.01%
Strasbourg recall	71.93%	78.24%
Munich recall	73.86%	80.50%
Outlines at least 80% covered	64.59%	69.82%
Outlines less than 10% covered	8.95%	6.32%

Result: useful overall gains, **not universal improvement**. Existing GAMUS ground/low-vegetation and road-boundary regression failures still block default promotion. These HighBuild results are development evidence; reserved final geographies remain unconsumed.

Full audit: `outputs/evaluation/highbuild_rgb_classifier_v3_preview_20260919/report.json`; complete image gallery in the same directory, `index.html`. Earlier evidence is preserved.

Verification

- 40 targeted Python tests passed, covering API, preview tiling, validity, projected area, source-path restrictions, exports, HighBuild audit, existing prediction and mesh behavior.
- 77 viewer tests passed; TypeScript and edited-file ESLint passed.
- Real browser → API → CUDA → class-raster → 3D overlay exercised on urban and forest demos and an uploaded 1024×1024 GeoTIFF.
- Class filters work; unknown pixels stay transparent; scene changes clear the overlay; inspection shows the new class separately. Browser showed no framework error overlay or console error logs during these flows.
- All original files in uploaded job `f0163ba3aed24b19b28eec01bf9627b7` remained byte-identical after classification. Re-uploading the exact same source into job `1bc8a7a5a6624755aa7907ec23735f61` produced **identical pixel values** in raw heights, presentation heights and three-group labels (maximum difference 0).
- Protected height checkpoint SHA256 remains `e2d507f9be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144`; pointer SHA256 remains `a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72baa9`.
- Classifier SHA256: `10fbb39c3d916695627e93d49f2115be18ecd163352b4daa96ba691c0090470c`; verified before loading.

Next accuracy work

Fix the measured city/class/boundary regressions on development data using the same baseline and preregistered acceptance rules. Do not compensate by quietly lowering acceptance thresholds. Freeze a candidate before one reserved-geography evaluation. Only an accepted classification release should become the default; any semantic-guided height candidate requires its own height-regression evaluation.

80 Six-class default integration — 20 September 2026

SOURCE docs/history/SIX_CLASS_DEFAULT_INTEGRATION_2026_09_20.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Delivered today

User-authorized integration: make six-class identification the normal app experience, replace the three-category Class Explorer, and remove repeated experimental badges. This is an interface/inference integration, not a new training run or a height-model promotion.

- Supported scenes are identified automatically after reconstruction finishes. The neutral start screen does not trigger inference.
- Class Explorer offers **Ground, Buildings, Water, Roads, Low vegetation and Trees**, with coverage statistics and class-map exports. Absent categories remain visible but cannot be selected.
- Inspect and the Fly/Walk scanner use the same six-class map. Both two-second aim and click scanning remain available.
- Category selection highlights the selected class without changing surface heights or flattening other geometry. The photo remains visible by default until the user requests a classification overlay.
- Compact/expanded inspection controls contain all six categories plus All classes.
- Main badges now say Automatic / 6 categories. Known model weaknesses remain visible under **Model details & accuracy limits**; they have not been removed from provenance or release evidence.
- Pending, failed and unsupported identification are explicit. They never silently fall back to a three-group label presented as a six-class prediction. Height inspection remains available independently.
- Retry repeats identification, not reconstruction. Repeated requests for the same source share one request, and up to four successful maps are cached in the browser session. Changing scenes immediately discards the old scene's displayed labels.

Protected height pipeline

The existing three-group height machinery remains internal. The RGB classifier controls identity/overlay only, not the height model, building geometry, collision masks or exported height values. A class label can disagree with the protected geometry; inspection explains that disagreement rather than changing the height to fit the label.

Water labels do not establish water depth. Relative scores are not metric heights. Ground/road labels do not establish absolute terrain elevation. Areas are only reported when supported geographic scale is available. Unsupported higher-bit-depth inputs retain explicit classifier limitations; the separate height workflow is not relabelled as supported classification.

V3 epoch 1 remains a development classifier with class/city and road-boundary regression failures. Its automatic integration is authorized product behavior, **not accuracy certification**, and no new accuracy improvement is claimed here.

Verification

Check	Result
Viewer Node tests	107 passed
Targeted Python API, prediction, mesh and RGB-preview tests	37 passed
TypeScript check	Passed
ESLint on changed TypeScript components/helpers	Passed
Automated UI flow	Passed; no page errors
Real uploaded RGB: reconstruction → automatic classification	Passed; no page errors
Raw/export/presentation artifact preservation	All 12 original job files byte-identical before and after classification
Fly two-second aim and click scanner	Manually verified with browser controls
Walk two-second aim and pointer-event scanner	Manually verified with browser controls/event handler
Expanded six-category selector layout	Visually checked

Automated UI checks include neutral startup, automatic classification, all six controls, stable inspected height across filtering, scene switching without stale labels, cached revisit, simulated classifier failure, successful retry, unsupported-input handling and the hilly scene's out-of-domain notice. The Python test additionally rejects a real synthetic uint16 GeoTIFF before classifier loading.

The navigation automation initially failed because pointer-capture automation rotated the camera away from its target. The saved navigation-verification.json records that failure and is not a passed test. Manual checks subsequently verified the actual scanner behavior: Fly displayed Ground with its independent approximately 0.4 m height; Walk displayed Low vegetation with an approximately 0.1 m height and the geometry-disagreement warning. These example values are UI observations, not accuracy measurements. The navigation harness was adjusted afterward but was not rerun. Native fullscreen/Escape behavior was not newly certified in this turn; the expanded selector layout was checked separately. The final compact-selector CSS adjustment was visually checked after the automated UI suite.

Saved evidence

Folder: outputs/runtime/six_class_day1/

- ui-verification.json
- upload-verification.json
- urban-inspection.png, classifier-failure.png, six-class-upload.png
- fly-scanner.png, walk-scanner.png, fullscreen-six-classes.png
- Repeatable browser harness: scripts/verify_six_class_integration.mjs

The real upload used an existing Duesseldorf diagnostic image, **not an untouched final test**. Job: outputs/web_jobs/5f1991e16d8641f9b68161a023badae8. The upload report records SHA-256 values for all 12 unchanged artifacts, including raw height, presentation mesh height and building solids.

Protected hashes checked and unchanged:

Artifact	SHA-256
Protected height checkpoint	e2d507fbe180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144

Artifact	SHA-256
outputs/runtime/showcase_checkpoint.txt	a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72baa9
Pinned V3 classifier checkpoint	10fbb39c3d916695627e93d49f2115be18ecd163352b4daa96ba691c0090470c

Try it

1. Open or refresh <http://localhost:3000/>.
2. Upload an RGB image or select a reference scene. The 3D reconstruction appears before automatic identification completes.
3. Use Class Explorer for six-category highlights, or Inspect / Explore for point labels and the separately predicted height.
4. Open Model details for current accuracy limitations.

No training was started, no protected checkpoint was replaced, and no final reserved geography was consumed. Remaining model accuracy work needs its own controlled experiment and evaluation; this integration does not resolve those limitations.

81 Offline six-class export

SOURCE docs/history/SIX_CLASS_OFFLINE_EXPORT.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Status and safety boundary

This is an **offline review tool**, not an application feature. It does not change `outputs/runtime/showcase_checkpoint.txt`, the protected production checkpoint, an API route, or the viewer. A candidate cannot be exported merely because training finished: the command fails closed unless a separate JSON report authenticates the exact checkpoint SHA-256 and records all required validation gates as passed.

The accepted classes and raster IDs are fixed:

ID	Class
0	ground
1	buildings
2	water
3	roads
4	low_vegetation
5	trees
255	invalid / unclassified

No probability map or confidence percentage is exported. The class raster is only the categorical argmax decision. This is not presented as a calibrated probability of correctness.

Required validation contract

The independent evaluator/reviewer must create a report with this exact minimum contract only after the candidate has passed the applicable checks:

```
{
  "schema": "msr.validated_six_class_candidate.v1",
  "candidate_checkpoint": {
    "path": "C:/absolute/path/to/candidate.pt",
    "sha256": "exact checkpoint SHA-256"
  },
  "offline_export": {
    "eligible": true,
    "six_class_validation_passed": true,
    "height_regression_guards_passed": true,
    "unseen_geography_evaluated": true
  }
}
```

The checkpoint path and hash are checked again before inference. A normal training log, an incomplete evaluator output, a failed gate, or a mismatched checkpoint is rejected. Offline export eligibility does not promote the model into the app.

For this legacy schema, `unseen_geography_evaluated` means a genuinely new external geography that did not influence any shared model, height pipeline, classifier head, threshold, or case selection. NYC excluded only from a fresh DC+Philadelphia head

does not satisfy it because earlier GAMUS experiments and the inherited HighBuild-trained system have New York exposure. The official GAMUS test also cannot satisfy it: Stage 3 already consumed that test, so it is forbidden for reuse.

Command

```
.\.venv\Scripts\python.exe scripts\export_validated_six_class.py `
D:\input\scene.tif `
C:\path\to\validated_candidate.pt `
C:\path\to\validated_candidate_report.json `
outputs\offline_six_class\scene_01
```

Optional exact-grid sidecars are supported:

```
--image-valid-mask image_valid.tif
--classification-valid-mask classification_valid.tif
--height-valid-mask height_valid.tif
--relative-prior relative_depth.tif
```

Sidecars must already match the RGB raster's dimensions, transform, and CRS; the exporter never silently resamples them. If no classification-valid mask is provided, a separate copy of image validity is used because dense classification exists wherever RGB inference is valid. If no height-valid mask is provided, finite predicted height on image-valid pixels is used. The chosen origin of every mask is recorded in the summary.

Output bundle

The destination must not already exist. It is staged and then published as a single new directory containing:

- `six_class_ids.tif`: aligned uint8 class IDs, nodata/invalid 255;
- `height_m.tif`: predicted height, with invalid height pixels kept as NaN;
- `image_valid_mask.tif`, `classification_valid_mask.tif`, and `height_valid_mask.tif`: three separate 0/1 masks;
- `summary.json`: checkpoint/validation identities, artifact hashes, per-class pixel counts and shares, area availability, and height summaries.

Height summaries are produced only for buildings, low vegetation, and trees, after intersecting image validity, classification validity, the selected class, and height validity. Missing heights are excluded rather than converted to zero. Water and road heights are never reported.

Square-metre coverage is reported only for a projected CRS with a finite, positive linear pixel area. For PNG/JPG, missing CRS, geographic/angular CRS, or invalid transforms, every class receives `coverage_m2: null` and the summary states the reason. This prevents approximate or invented area claims.

82 Six-class scanning and height-training audit — 19 September 2026

SOURCE docs/history/SIX_CLASS_SCANNER_AND_HEIGHT_AUDIT_20260919.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Shipped in the running app

Choose a scene/upload, click **Identify six classes**, then Explore → Fly or Walk. Aim at a surface for two seconds or click to scan immediately. The primary scanner label can now be Ground, Buildings, Water, Roads, Low vegetation or Trees. Uncheck **Show on 3D surface** to keep the original photograph; classification remains available to the scanner. Changing scenes clears the previous labels.

The scanner and Inspect panel distinguish classification source from height source. An explicit disagreement warning appears when the RGB class conflicts with the protected three-group height geometry. Classification never gates, rescales, zeroes or replaces heights, collisions or exported rasters.

- nDSM: estimated surface height above local ground, not altitude.
- Building solid: protected model's building-object height summary.
- Calibrated DSM: surface elevation in its datum, not object height.
- Relative rDSM: dimensionless score, no metre claim.
- Water: surface value only; never water depth and not validated water height.
- Roads/ground: not automatically zero or sea-level elevation.

V3 epoch 1 remains explicitly **experimental**, because its development safety gates failed. It has not become the production height model. A six-class label does not establish class-specific height accuracy.

Verification

- 89 viewer unit tests passed, including all six scanner labels, no-data, class disagreements, hidden overlays, relative units and DSM/object-height distinction.
- 25 targeted backend/audit tests passed; TypeScript and edited-file ESLint passed.
- Browser: actual CUDA classification of the Urban reference; colors turned off; Fly two-second aim and real mouse click both produce metadata; Walk aim produces the building class and 6.4 m object estimate. These are UI examples, not accuracy scores.
- Forest scene switch removed old class data; no browser errors recorded.
- Screenshot evidence: outputs/runtime/six_class_scanner_click_20260919.png and outputs/runtime/six_class_scanner_walk_20260919.png.
- Production checkpoint and live pointer SHA-256 values remain unchanged:
e2d507f7be180988e000a91a873db047407fc15c0fac90446f7d5f05990c0d144 and
a3d51c686ca2abee9cb7e622c600f5ded9eca3332638ccf56f6f3d8b2f72baa9.

Next.js/React review kept the scanner client-side, retained asynchronous API inference, preserved effect cleanup and kept display visibility independent of the semantic data. Browser verification tested the actual workflow, not just rendering.

New training-only evidence

CPU audit of the pinned corrected training manifest inspected 450 HighBuild tiles from five cities and 9,841 measured-flag building annotations. No new validation, official test or reserved external images were consumed.

Check	Finding
Corrected semantic/height masks vs COCO contract	All 450 match
Cached-prior grid, finite values, [0,1] range	All 450 pass
Supported training height pixels above 200 m	0
Supported pixels with height >20 m	11.73%
Training annotations with height >20 m	626 / 9,841 (6.36%)
Explicit GSD in these training rows	Missing in all 450
Prior source-image + model-weight hash tags	Missing in all 450
Annotation/raster footprint agreement below 95%	363 instances flagged
Annotation vs footprint median differs by >1 m	13 instances flagged

The large-disagreement review shows five fully overlapped footprints, three partly overlapped footprints whose exclusive-support median matches their own annotation, and five tiny supports of 1–7 pixels. This does **not** justify removing labels or declaring them corrupt. Overlap-aware per-building support is necessary before instance-height loss/evaluation.

To check actual prior reproducibility, two lexicographically selected training images per city were replayed through the saved DAV2 model on CUDA. **All ten matched their cached priors exactly** (maximum absolute difference 0). The report records RGB, cache, model weights and preprocessing hashes. This is a sampled reproducibility check, not certification of every cache or an accuracy test.

Evidence:

- `outputs/diagnostics/height_train_support_20260919/report.json`
- `outputs/diagnostics/height_train_support_20260919/measured_instances.csv`
- `outputs/diagnostics/height_train_support_20260919/overlap_review.json`
- `outputs/diagnostics/height_training_prior_replay_20260919/report.json`

Decision

Do not start another long six-class height run simply because identification looks better. The earlier residual-height pilot improved some aggregate errors but failed ground/tall-object protections; the saved development analysis found 89.31% of baseline building squared error in references ≥ 20 m.

Keep its evidence and the protected baseline. The next proposal is a **matched, bounded class-conditioning experiment**, defined in `CLASS_ASSISTED_HEIGHT_PILOT_PROTOCOL.md`. It is not trained or promoted yet. Resolve the specified preparation gates, run a train-only feasibility check, then a paired development comparison. No new height-accuracy improvement is claimed from this turn's UI and data-audit work.

83 Smoothness and accuracy review — 26

September 2026

SOURCE docs/history/SMOOTHNESS_AND_ACCURACY_REVIEW_2026_09_26.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Evidence reviewed

- User recording: D:/Downloads/Recording_2026-09-26_141802.mp4, 157 frames, approximately 5.21 seconds, 1252 × 812. The full frame sequence was decoded and inspected at half-second intervals, including the last frame. Its visible label reads **Absolute DSM terrain · DA3MONO-LARGE prior · rendered LOD 0 · 1.5× display Z**.
- Our same downloaded Joshimath RGB GeoTIFF and generated job e87ecbe2eebb4ae3a77767ff04ebb070. Its active presentation multiplier was 1×, not the 12× automatic boost used on some other low-relief regional imports.
- Our source and rendered-mesh sampling code. The initial investigation left rendering unchanged. The subsequent visual improvements below change presentation only; prediction and source elevation values remain unchanged.

Current implementation

The stronger reconstruction and consistent display scale supersede the earlier limited filter. See [Terrain video-style implementation and verification](#) for current defaults, quantitative changes and all-region browser evidence.

The earlier filter capped each vertex displacement and retained a sharp source residual. Its Soft setting could even increase local high-frequency roughness. The replacement removes that limiter, reconstructs a continuous terrain reference and preserves object-height residuals separately. The brief lighting experiment was removed; subsequent work concerns geometry and display scale.

Findings

The recorded scene looks smoother, especially around its lower valley surfaces. The label establishes a different depth prior and display multiplier; it does not expose its DEM source, smoothing parameters, mesh dimensions, calibration, exported heights, or validation error. LOD 0 alone does not establish a particular mesh resolution. The same RGB TIFF does not guarantee the same height field.

Our surface on this file is dominated by the terrain reference, not learned building or vegetation heights. It spans 1139.491–6702.545 m, while added object heights range from 0.060 to 6.793 m, with a median of 0.412 m. A one-pixel Gaussian high-pass diagnostic gives RMS amplitudes of 5.106 m for the terrain reference, 5.106 m for the final DSM, and 0.030 m for added object heights. For source locations with gradient slope below 5°, the corresponding amplitudes are 3.327 m, 3.327 m, and 0.024 m. These are roughness diagnostics, not ground-truth errors: genuine terrain features also contribute to a high-pass residual.

Before this change, the browser limited the 1024 × 1024 raster to 512 × 512 mesh cells (513 × 513 vertices). Vertex spacing was approximately 65.79 × 65.46 m, versus source sample spacing of 32.92 × 32.76 m. Reconstructing that mesh's bilinear vertex sampling and triangular faces at all source centres produced these differences from the source DSM:

Metric	Difference
RMS, all source centres	4.986 m
95th percentile absolute difference	10.320 m
Maximum absolute difference	75.006 m

Metric	Difference
RMS where source slope is below 5°	3.153 m

This is a measurable approximation in the displayed geometry, not a demonstrated error against actual ground. The raw exports and inspection samples retain the original raster values. Finite mesh resolution can cause visible facets, but finer geometry alone cannot prove which small terrain variations are genuine.

A HEAD request for the reference tile <https://s3.amazonaws.com/elevation-tiles-prod/geotiff/12/2953/1681.tif> returned `x-amz-meta-x-imagery-sources: srtm/N30E079.tif`. This identifies SRTM for that sampled tile, not necessarily every tile in the mosaic. [Mapzen's source documentation](#) describes mixed sources and oversampling. [USGS explains that SRTM elevations can contain canopy/building effects](#). Consequently, treating every public terrain tile as bare earth and adding object heights can double-count surface features; the existing "bare-earth" assumption is not generally justified.

[DA3's official model description](#) identifies DA3MONO-LARGE as relative monocular depth. A downstream pipeline may calibrate it to a DEM, but the recording alone does not show whether or how that was done. A larger model and smoother rendering do not establish better geographic elevation accuracy.

What should improve next

1. Extend the bounded source-resolution mesh to adaptive level of detail for much larger rasters, with filtered reduction. The current implementation improves the tested 1024-pixel imports; it does not claim unlimited resolution or universal source quality.
2. Track whether an elevation source is a bare-earth DTM, surface DSM, or unspecified mixture. Only add ground-relative building/canopy heights to an appropriate ground reference. Retain provider, resolution, date, datum and uncertainty metadata.
3. Use conservative, terrain-aware artifact handling only when justified. A smoother preview may be useful, but must not silently replace analytical elevations or erase real ridgelines, buildings, and vegetation.
4. Compare both exported height products against independent, aligned reference elevations (survey/LiDAR where available, or a documented independent DEM comparison). Match CRS, vertical datum, resolution, extent and product type; report bias, MAE and RMSE separately for flatter terrain, hills and surface objects. Agreement with the DEM used to generate a result is not independent validation.

The recording supports a visual-quality comparison. A defensible accuracy ranking requires the other pipeline's height output and an independent reference. No accuracy improvement from smoothing or a DA3 model swap has been established here.

Local analysis artifacts: `outputs/runtime/smoothness-review/contact-sheet.jpg`, `roughness-analysis.json`, `mesh-sampling-analysis.json`, and `mesh-fidelity.png`.

84 Terrain imports and measured profiles — 26 September 2026

SOURCE docs/history/TERRAIN_IMPORT_AND_PROFILE_2026_09_26.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

The shared import and rendering paths now preserve terrain relief instead of normalizing every mountain scene into a fixed-height mesh. These changes apply across imported scenes; no filename, place or demo identifier selects special terrain behavior.

Import and rendering behavior

- Georeferenced RGB TIFF imports request a public terrain reference by default. The decision uses GeoTIFF CRS and extent metadata. Uploaded DEMs or GCPs take priority. Users can explicitly disable automatic calibration.
- Ordinary imagery without map coordinates retains the existing surface-height workflow. TIFF is a container format, not evidence of an elevation datum. Missing terrain or scale is labeled; metric terrain cannot be recovered from format alone.
- Automatic lookup keeps the preferred source zoom for small scenes and chooses a coarser source for larger extents within the tile budget. The output metadata records actual source URLs, zoom and tile count. Unsupported extents or download failures show an error; users can crop or supply a DEM.
- A calibrated DSM is rendered directly. The above-ground height product is no longer substituted for a mountainous absolute surface. Standalone height-map imports remain available.
- Metric meshes use a consistent scale across horizontal distance and elevation, with independent X/Y pixel spacing. $Z \times 1$ gives source proportions; exaggeration changes display only. Web Mercator output metadata now accounts for local ground scale instead of treating projected map metres as ground metres.
- Bilinear mesh sampling retains continuous terrain. Nodata remains missing and is excluded from triangles rather than becoming a zero-height cliff.
- Optical, elevation, shaded-relief, slope, contour and metric-grid views share the same geometry. Fit-terrain, top-down and reset-exaggeration controls frame the actual extent and relief.

Measurements

Point inspection reports source surface elevation, the available aligned terrain reference, their difference, local slope and raster pixel. The elevation legend shows the source range and relief. Profile mode accepts two clicks and displays a measured cross-section with distance, endpoint elevations, elevation change, ascent/descent and CSV export. Profiles remain available when the right sidebar is hidden. Nodata gaps are retained, and measurements do not change with visual exaggeration.

The public terrain reference is coarse and can contain source artifacts or vegetation/structures. Its datum and the predicted object heights determine the resulting DSM's accuracy. This work does not establish survey accuracy or improve the existing semantic classifier. Horizontal measurements use local centre-grid spacing; they are not a full geodesic measurement across arbitrary continental extents.

Regional presentation update

Eight additional region TIFFs were prepared in Downloads/Monocular Surface Reconstruction_Region_TIFFs: New Delhi urban, Punjab plains, Dandeli forest, Munnar hills, Thar desert, Goa coast, Sundarbans wetlands and Ladakh rocky mountains. These are RGB GeoTIFF mosaics of public EOX Sentinel-2 cloudless 2024 map imagery, with source URLs, hashes and previews. They are not elevation ground truth.

Testing exposed weak visual depth in low-relief scenes at true scale. Georeferenced absolute DSMs select a clearly labeled automatic display exaggeration from the central 96% of source heights and map extent, bounded to 1–12×. A constant raster never receives invented relief. The physical mesh scale and all source samples remain unchanged. Users can select True scale, adjust the slider or return to Auto relief. Valid outer edges of absolute DSMs extend down to the scene minimum to make terrain volume visible; these edges do not represent measured subsurface geology. An Elevation colours shortcut makes the source elevation ramp accessible directly from the legend.

Unreferenced imagery retains the original urban/object presentation: $\min(0.36, 12 / \text{analysisRange})$ world units per predicted metre, with a labeled 1.7× display multiplier. Both vegetation and building solids use that same scale. The complete analysis range determines the scale, rather than a display mesh that may have removed buildings. This corrects the inflated object-height regression without changing predicted heights. The original closer camera is retained for these scenes, with portrait fitting; georeferenced terrain keeps extent-based framing. Absolute terrain sidewalls are not added to nDSM object scenes. TIFFs without a CRS and usable transform receive the same unreferenced treatment as ordinary images.

The viewport toolbar now uses content-sized rows and container queries, keeping navigation, relief controls and actions inside their available width. Mobile header controls stay in layout instead of overlaying the toolbar; sidebar scrollbars use the application's dark palette.

Verification on the user's downloaded TIFF

Final browser import used `C:\Users\researcher\Downloads\joshimath_s2cloudless_2024.tif`, with automatic terrain left at its default setting. The uploaded bytes match the original file:

```
989d7ec01634bed96e211ebdc47d395871533a238b20b7314389e670b1a9b6f7
```

- Source and generated rasters: 1024 × 1024, EPSG:3857, original transform retained.
- Ground spacing: 32.924868805 m × 32.761287028 m; independently checked using transformed geographic coordinates.
- Rendered surface range: 1139.491–6702.545 m; relief: 5563.054 m.
- DSM equals terrain reference plus predicted height to within 0.000245 m of float32 rounding.
- Inspected pixel (561, 565): surface 3349.638672 m, terrain reference 3349.218750 m, difference 0.419922 m. Displayed values agree after rounding.
- Profile: 812 samples over 26,673.780 m. Independent bilinear resampling agrees within 0.041 m, including the CSV's six-decimal UV rounding. CSV download completed and all 37,067 bytes matched the displayed profile export.
- Browser verified unchanged measurements at Z × 4, all terrain styles, contour switching, top-down framing and a 900 × 760 compact viewport. No browser errors were reported.
- TypeScript and targeted ESLint checks passed; 120 viewer tests passed after the presentation update; the prior 18 relevant backend tests remain applicable to the unchanged backend. Production build passed, with the existing large-bundle warning and vinext route-classification notice.

Reproducible checks:

```
node scripts/verify_terrain_analysis.mjs <agent-browser-executable>
C:/Users/researcher/Downloads/joshimath_s2cloudless_2024.tif
./venv/Scripts/python.exe scripts/verify_terrain_source.py
```

Screenshots, downloaded CSV and machine-readable browser/source verification reports are under `outputs/runtime/terrain-final/` (local runtime artifacts). Browser download verification uses the Chrome protocol with a Windows-native directory because the agent-browser download helper canceled its downloads on this host.

File drop, clipboard and visible scale follow-up

The main image import supports file selection, dragging files anywhere in the workspace, Ctrl+V / Command+V image paste, and a Paste image button. All routes use one validator and the existing inference/calibration pipeline. Supported RGB formats are TIFF/GeoTIFF, PNG, JPEG, WebP, BMP and JPEG 2000. Multiple supplied images open an explicit chooser; importing is limited to one active scene at a time. Unsupported, empty and over-512 MB files receive feedback before inference. Normal text-field paste is unaffected. Clipboard denial or text-only content has fallback guidance, and drop feedback also works in fullscreen.

Original file drops preserve GeoTIFF bytes and metadata. Clipboard image pixels commonly lack the original map coordinates; the UI therefore recommends dropping the original GeoTIFF for geospatial work. The API handles ordinary grayscale/palette PNG images as RGB, while single-band elevation TIFFs remain the separate **Load existing height map** workflow. JP2 can request terrain conditionally after the backend checks its CRS; manual DEM/GCP inputs keep priority.

The previous decorative, unnumbered scale bar is replaced with a numeric ruler that updates when the camera moves or zooms. It measures horizontal distance on a plane through the view centre, using metres when ground scale is known and pixels otherwise. The adjacent readout includes X/Y extents, calibrated grid spacing, the full rendered Z range including building solids, and display exaggeration. Photo and analytical views use the same calibrated floor grid. Unknown map scale is stated rather than invented.

Follow-up validation: 125 viewer tests, 31 relevant backend tests, TypeScript, targeted ESLint and production build pass. Twenty end-to-end browser checks include real OS file-drop events, actual clipboard PNG paste by keyboard and button, all supported formats, multiple-file selection, busy import handling, fullscreen, and ten widths from 320–1920 px. Additional checks cover clipboard-denial feedback, mobile camera-button separation and real wheel zoom; the ruler agrees with an independent camera projection within 0.5%.

The final imported source was again the original downloaded Joshimath TIFF, now through drag-and-drop. Job e87ecbe2eebb4ae3a77767ff04ebb070 preserves its SHA-256, grid and CRS, the 1139.491–6702.545 m surface range, and terrain-plus-height composition within 0.000245 m. Reports and screenshots are in `outputs/runtime/import-review/`, including `verification.json`, `final-ui-checks.json`, `source-check.json`, `05-urban-final.png` and `09-final-verified-joshimath.png`.

```
./venv/Scripts/python.exe scripts/data/create_import_test_fixtures.py
node scripts/verify_image_imports.mjs <agent-browser-executable>
C:/Users/researcher/Downloads/joshimath_s2cloudless_2024.tif
```

Reference video

Reviewed the full available transcript and storyboard frames spanning the 4:49 reference demo. YouTube playback itself returned an error, so uninterrupted audiovisual playback was unavailable. The implemented changes address visible terrain relief, explicit elevation/depth context and measured profiles; this is not a claim of complete feature parity with that demo.

85 Terrain presentation matched toward the supplied video

SOURCE docs/history/TERRAIN_VIDEO_STYLE_2026_09_26.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

The request concerns continuous, rounded slopes and valley floors across imported TIFF regions. The supplied recording shows a different height-generation pipeline and 1.5x display Z. Its precise elevation field is unavailable, so an exact geometric match or an elevation-accuracy ranking cannot be established from the video.

Final behavior

- **Smooth / Soft is the default** for compatible terrain rasters. Fine and Balanced remain available; Source restores the original rendered heights.
- The terrain reference uses symmetric Gaussian reconstruction with 2, 6 or 10 source-cell standard deviation, bounded to 360 m physical support. Independent axis spacing handles rectangular pixels. Positive normalized weights avoid overshooting contributing terrain samples.
- The old per-vertex displacement limiter was removed: it retained sharp residuals and could make stronger smoothing locally rougher. Supplied ground references are reconstructed continuously instead of treating steep slopes as structural edges.
- Surface-minus-ground object heights are added back unchanged. Object demos keep their established mesh and display scale. Fine-resolution standalone DSMs below 5 m without a ground reference retain their geometry to protect buildings; standalone coarse rasters protect abrupt breaks. Missing data remains missing.
- Imported absolute terrain now defaults to a gentle **1.5x** display multiplier, matching the visible setting in the reference video. The earlier adaptive 4-12x boost made forests and low-relief areas look disproportionately steep. Truly flat/unknown surfaces receive no automatic boost. True scale and the manual Z slider remain available.
- The implementation is shared by terrain imports, with no filename, location or region-specific logic. Unknown-scale/relative imagery cannot be treated as measured metric terrain. TIFF is a container: image-only TIFFs still need inferred heights or a terrain reference.
- Inspection, slope analysis, profile CSV and exported elevations continue to use the original source data. Mesh boundaries and collision sampling follow the displayed surface. Lighting is unchanged by this revision.

Actual downloaded Joshimath TIFF

Input SHA256: 989d7ec01634bed96e211ebdc47d395871533a238b20b7314389e670b1a9b6f7.

The final fresh import is job 2f49ba86b9ed45a5a5679317f3427a14. It retains all 1024 x 1024 source samples in the mesh (1,048,576 vertices). Same-camera screenshots use identical lighting and 1.5x Z.

Geometry	Pixel-scale roughness RMS	Display shift RMS	Maximum display shift
Source	5.115 m	0 m	0 m
Fine	2.231 m	12.76 m	118.57 m
Balanced	0.909 m	49.05 m	293.09 m
Soft (default)	0.544 m	85.87 m	375.68 m

Roughness is the RMS of a one-pixel Gaussian high-pass residual, excluding a 20-pixel border. The default reduces this diagnostic by **89.4%**; it is a presentation metric, not a ground-truth error. Source relief is 5563.05 m; Soft displayed relief is 5337.21 m. This deliberately removes local terrain detail to achieve the requested rounded appearance. Measurements and exports do not use these adjusted heights.

Verification

135 viewer tests pass, including continuous smoothing without retained sharp residuals, broad hill preservation, no overshoot, nodata, object residuals, source immutability and gentle automatic relief. TypeScript, targeted ESLint and the production build pass. The build retains its existing large-chunk/static-route-classification notices.

The browser test imported all eight downloaded regional RGB GeoTIFFs through the API, then a standalone float elevation GeoTIFF, and finally the original Joshimath TIFF. Every regional import selected Soft automatically and used 1.5x display Z:

Region	Source unchanged	Display adjustment RMS
Urban, New Delhi	Yes	2.39 m
Plains, Punjab	Yes	1.28 m
Forest, Dandeli	Yes	6.10 m
Hills, Munnar	Yes	11.11 m
Desert, Thar	Yes	1.40 m
Coast, Goa	Yes	2.75 m
Wetlands, Sundarbans	Yes	1.06 m
Rocky, Ladakh	Yes	11.69 m

Urban/Sparse/Forest object demos bypass terrain smoothing, with the previous Urban 0.36 vertical scale and 1.7x display multiplier intact. Source/Smooth and strength changes retain camera and source elevation range. All visual layers retain source scale. Browser and shader checks report no errors.

The original TIFF's uploaded bytes are identical to the download, its exported DSM matches the earlier source exactly, and its inspected height matches the source raster. All 744 profile samples agree with source interpolation within 0.043 m, including rounded CSV coordinates. Manual display exaggeration leaves the profile CSV unchanged.

Evidence: `outputs/runtime/video-match-review/` contains per-region screenshots, `final-verification.json`, `regional-source-check.json`, `final-source-check.json`, `geometry-metrics.json`, `same-camera-comparison.json`, `source-close.png`, `soft-close.png`, and the orbit preview. Reproduce the browser run with `scripts/verify_terrain_presentation.mjs <agent-browser.exe> <original.tif> <regional.tif> --all-regions`.

This covers the downloaded regional set and supported terrain/height import paths. It does not establish identical appearance for every possible TIFF encoding, unknown scale, source quality or missing elevation field.

86 Viewer controls update — 19 September 2026

SOURCE docs/history/VIEWER_CONTROLS_2026_09_19.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Changes

- Sightline accepts one observer/target pair and then disarms selection. Reset starts another measurement. Clicking Sightline again, Escape, switching modes or loading another scene clears its graphics/state.
- Fly keeps mouse look, WASD/arrows, Space up, Shift down and G for 3x speed. Normalized movement avoids faster diagonal flight.
- Collision uses the displayed terrain triangles and building footprints, with swept substeps and clearance. Gentle slopes raise the camera; walls/cliffs block or allow sliding. Space climbs over obstacles. It follows vertical exaggeration and safely lifts a camera initially embedded by an orbit view.
- Fly only: aiming the centre crosshair at the same building or continuous surface class for two seconds reveals a fixed scanner HUD. Left-click reveals it immediately. Small mouse movements within the target/class preserve the dwell and update values; changing targets, aiming at the sky, or leaving Fly clears it. No confidence percentages are presented.
- Readouts sample full-resolution source pixels rather than downsampled mesh-triangle averages. NoData remains unavailable. Unknown class pixels remain unclassified; RGB assistance does not override an existing classification map.
- The flight HUD includes slope when metric pixel scale exists. DSMs are labelled surface elevation (ASL), not tree/building height. Relative products retain dimensionless units. Building solids report their model-derived object summary; vegetation retains RGB-assisted/model basis. Collision feedback prompts Space to climb when movement is blocked.
- Building roofs retain optical projection, including corrected box-roof UVs. Walls now use three coordinated procedural facade palettes, plaster grain, inset-window patterns, panel joints, floor bands and soft ground-contact shading. Their colors are gently matched to each roof. These are illustrative architectural patterns, not observed facade photographs, detected windows or measured floors; the UI explicitly says so.
- Wall textures follow each wall's true tangent, including diagonal outlines and rectangular scene scaling. Optical roof UVs and all vertex positions are preserved. Color, bump and roughness maps are shared per palette and disposed on scene replacement; no extra geometry or external texture downloads are required.

Checks

- `npx tsc --noEmit`: passed.
- `npx eslint app/terrain-workspace.tsx app/flight-navigation.ts app/surface-query.ts app/building-facades.ts`: passed.
- `node --experimental-strip-types --test tests/flight-navigation.test.mjs tests/surface-query.test.mjs tests/building-facades.test.mjs`: 14 tests passed (collision/navigation, source-pixel values, class identity during movement, NoData, raster alignment, unknown labels, metric slope, facade tiling and geometry/roof preservation).
- Facade browser checks: urban overview and closer orbit screenshots inspected; new textures render, optical roofs remain intact, disclosure is present, and no page errors were reported.
- Browser: urban, forest and hilly scenes loaded. The HUD remained empty before dwell, appeared despite repeated small mouse movements, switched to CLICK SCAN on an immediate click, and stayed visible during further jitter. Sky aiming cleared the target; leaving Fly removed the HUD. The hill readout explicitly used ASL. Sightline completed and graphics cleared on mode switch. No browser page errors observed.

- Existing development-server log warns about multiple concurrent renderers; no associated browser failure observed during these checks.

Model checkpoints, exported prediction rasters, evaluation scores and backend inference were not changed. Collision protects navigation through the reconstructed scene; it is not evidence that the estimated geometry matches the real world. Experimental decorative tree objects are not physical colliders; their underlying terrain/canopy heightfield is.

87 Compact scene details and original-image zoom

SOURCE docs/history/VIEWPORT_DETAILS_ZOOM_2026_09_27.md

Historical development record. Read alongside the current [release status](#). Public names and local path labels have been normalized; dated measurements retain their original scope. Referenced training archives are not bundled unless listed in the evidence index.

Delivered

- Outside fullscreen, height information and map scale are collapsed by default, with compact **Height info** / **Map scale** buttons at the bottom of the 3D view.
- Click a button to reveal its panel; click again or use its close button to dismiss. Only one opens at a time. New scenes and navigation mode changes reset the panels.
- Fullscreen keeps the original expanded readouts. Leaving fullscreen collapses them. Heights, source rasters and the live perspective scale calculation are unchanged.
- Clicking the **Original image** thumbnail opens an accessible modal with the current scene's RGB preview.
- Zoom from fit to 8× with buttons, the mouse wheel or +/-; drag to pan within the image edges; **Fit image** or 0 restores the complete image.
- Escape closes the image first, without also leaving fullscreen or clearing an underlying inspection. The native dialog provides modal focus handling; a close button is also available.
- Image dimensions refer to the displayed RGB preview. Zoom does not add source detail or change any height/model output.

Verification

- TypeScript and targeted ESLint passed.
- All 146 viewer tests passed, including four new tests covering pan bounds, aspect ratio, zoom anchor and limits.
- Browser checks passed for windowed toggles, panel exclusivity, live ruler, image identity, zoom/pan/reset/keyboard handling, fullscreen Escape ordering, scene changes and 390px layout. No page errors captured.
- Evidence: outputs/runtime/viewport_details_zoom/verification.json and adjacent screenshots.
- Repeatable browser harness: scripts/verify_viewport_details_zoom.mjs using a dedicated agent-browser CDP session.